AEROSPACE OEM ENGINEERING COMPENDIUM — VOLUME 3

Covers 25 Certified Machine Models (Models #51 to #75)
350 Full-Page Fault SOPs | 125 Pages PLC Registers | 75 Pages BOM & PM Matrices

INCLUDED MACHINE MODELS IN THIS VOLUME:

• Model #51: Trumpf-TruLaser-L150 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #52: Trumpf-TruLaser-L200 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #53: Trumpf-TruLaser-L250 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #54: Trumpf-TruLaser-L300 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #55: Trumpf-TruLaser-L350 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #56: Trumpf-TruLaser-L400 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #57: Trumpf-TruLaser-L450 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #58: Trumpf-TruLaser-L500 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #59: Trumpf-TruLaser-L550 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #60: Trumpf-TruLaser-L600 (5-Axis Fuselage Skin Laser Cutter) — Controller: Siemens 840D sl Laser — Prefix: LAS-CELL-XXXX
• Model #61: Schuler-HPD-P150 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #62: Schuler-HPD-P200 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #63: Schuler-HPD-P250 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #64: Schuler-HPD-P300 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #65: Schuler-HPD-P350 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #66: Schuler-HPD-P400 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #67: Schuler-HPD-P450 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #68: Schuler-HPD-P500 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #69: Schuler-HPD-P550 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #70: Schuler-HPD-P600 (Isothermal Landing Gear Forging Press) — Controller: Beckhoff TwinCAT 3 — Prefix: HYD-PRESS-XXXX
• Model #71: Grob-Aerospace-G150 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #72: Grob-Aerospace-G200 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #73: Grob-Aerospace-G250 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #74: Grob-Aerospace-G300 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-MILL-XXXX
• Model #75: Grob-Aerospace-G350 (Horizontal Swivel-Table Structural Mill) — Controller: Heidenhain TNC 640 — Prefix: UNI-

MILL-XXXX

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 1/9)

MACHINE MODEL: Trumpf-TruLaser-L150 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L150"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Trumpf-TruLaser-L200 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L200"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Trumpf-TruLaser-L250 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L250"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 2/9)

MACHINE MODEL: Trumpf-TruLaser-L300 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L300"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: Trumpf-TruLaser-L350 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L350"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: Trumpf-TruLaser-L400 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L400"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 3/9)

MACHINE MODEL: Trumpf-TruLaser-L450 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L450"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Trumpf-TruLaser-L500 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L500"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: Trumpf-TruLaser-L550 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L550"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 4/9)

MACHINE MODEL: Trumpf-TruLaser-L600 (Asset Prefix: LAS-CELL)

• RAG Metadata Filter Value: "Trumpf-TruLaser-L600"
• Equipment Classification: 5-Axis Fuselage Skin Laser Cutter | Manufacturer: Trumpf-TruLaser
• CNC / PLC Control System: Siemens 840D sl Laser
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Schuler-HPD-P150 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P150"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Schuler-HPD-P200 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P200"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 5/9)

MACHINE MODEL: Schuler-HPD-P250 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P250"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Schuler-HPD-P300 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P300"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: Schuler-HPD-P350 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P350"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 6/9)

MACHINE MODEL: Schuler-HPD-P400 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P400"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Schuler-HPD-P450 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P450"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Schuler-HPD-P500 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P500"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 7/9)

MACHINE MODEL: Schuler-HPD-P550 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P550"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Schuler-HPD-P600 (Asset Prefix: HYD-PRESS)

• RAG Metadata Filter Value: "Schuler-HPD-P600"
• Equipment Classification: Isothermal Landing Gear Forging Press | Manufacturer: Schuler-HPD
• CNC / PLC Control System: Beckhoff TwinCAT 3
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Grob-Aerospace-G150 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G150"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 8/9)

MACHINE MODEL: Grob-Aerospace-G200 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G200"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Grob-Aerospace-G250 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G250"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Grob-Aerospace-G300 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G300"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 9/9)

MACHINE MODEL: Grob-Aerospace-G350 (Asset Prefix: UNI-MILL)

• RAG Metadata Filter Value: "Grob-Aerospace-G350"
• Equipment Classification: Horizontal Swivel-Table Structural Mill | Manufacturer: Grob-Aerospace
• CNC / PLC Control System: Heidenhain TNC 640
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-SPN-1

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU150-1
- PLC Hex Register Address: DB25.DBW_0xBF68
- Nominal Operating Range: 15 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-BRG-CER-5101
- Overhaul Seal & Consumable Kit: KIT-TRU150-SPN-1
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-SPN-1 at PLC register 0xBF68.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-BRG-CER-5101) with seal kit (KIT-TRU150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-SPN-1 on Trumpf-TruLaser-L150 by installing PN-TRU150-BRG-CER-5101 & kit KIT-TRU150-SPN-1. Fasteners torqued to 139 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-ATC-2

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU150-2
- PLC Hex Register Address: DB25.DBW_0xBF70
- Nominal Operating Range: 18 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-CAM-GRIP-5102
- Overhaul Seal & Consumable Kit: KIT-TRU150-ATC-2
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-ATC-2 at PLC register 0xBF70.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-CAM-GRIP-5102) with seal kit (KIT-TRU150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-ATC-2 on Trumpf-TruLaser-L150 by installing PN-TRU150-CAM-GRIP-5102 & kit KIT-TRU150-ATC-2. Fasteners torqued to 148 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-HYD-3

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU150-3
- PLC Hex Register Address: DB25.DBW_0xBF78
- Nominal Operating Range: 21 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-VLV-PROP-5103
- Overhaul Seal & Consumable Kit: KIT-TRU150-HYD-3
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-HYD-3 at PLC register 0xBF78.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-VLV-PROP-5103) with seal kit (KIT-TRU150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-HYD-3 on Trumpf-TruLaser-L150 by installing PN-TRU150-VLV-PROP-5103 & kit KIT-TRU150-HYD-3. Fasteners torqued to 157 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-CLN-4

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU150-4
- PLC Hex Register Address: DB25.DBW_0xBF80
- Nominal Operating Range: 24 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-PMP-PLG-5104
- Overhaul Seal & Consumable Kit: KIT-TRU150-CLN-4
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-CLN-4 at PLC register 0xBF80.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-PMP-PLG-5104) with seal kit (KIT-TRU150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-CLN-4 on Trumpf-TruLaser-L150 by installing PN-TRU150-PMP-PLG-5104 & kit KIT-TRU150-CLN-4. Fasteners torqued to 166 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-XAX-5

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU150-5
- PLC Hex Register Address: DB25.DBW_0xBF88
- Nominal Operating Range: 27 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-LIN-COIL-5105
- Overhaul Seal & Consumable Kit: KIT-TRU150-XAX-5
- Critical Fastener Torque Value: 175 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-XAX-5 at PLC register 0xBF88.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-LIN-COIL-5105) with seal kit (KIT-TRU150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 175 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-XAX-5 on Trumpf-TruLaser-L150 by installing PN-TRU150-LIN-COIL-5105 & kit KIT-TRU150-XAX-5. Fasteners torqued to 175 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-YAX-6

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU150-6
- PLC Hex Register Address: DB25.DBW_0xBF90
- Nominal Operating Range: 30 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-BSW-NUT-5106
- Overhaul Seal & Consumable Kit: KIT-TRU150-YAX-6
- Critical Fastener Torque Value: 184 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-YAX-6 at PLC register 0xBF90.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-BSW-NUT-5106) with seal kit (KIT-TRU150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 184 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-YAX-6 on Trumpf-TruLaser-L150 by installing PN-TRU150-BSW-NUT-5106 & kit KIT-TRU150-YAX-6. Fasteners torqued to 184 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-ZAX-7

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU150-7
- PLC Hex Register Address: DB25.DBW_0xBF98
- Nominal Operating Range: 33 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-CYL-SEAL-5107
- Overhaul Seal & Consumable Kit: KIT-TRU150-ZAX-7
- Critical Fastener Torque Value: 193 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-ZAX-7 at PLC register 0xBF98.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-CYL-SEAL-5107) with seal kit (KIT-TRU150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 193 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-ZAX-7 on Trumpf-TruLaser-L150 by installing PN-TRU150-CYL-SEAL-5107 & kit KIT-TRU150-ZAX-7. Fasteners torqued to 193 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-BAX-8

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU150-8
- PLC Hex Register Address: DB25.DBW_0xBFA0
- Nominal Operating Range: 36 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-TRQ-STAT-5108
- Overhaul Seal & Consumable Kit: KIT-TRU150-BAX-8
- Critical Fastener Torque Value: 202 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-BAX-8 at PLC register 0xBFA0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-TRQ-STAT-5108) with seal kit (KIT-TRU150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 202 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-BAX-8 on Trumpf-TruLaser-L150 by installing PN-TRU150-TRQ-STAT-5108 & kit KIT-TRU150-BAX-8. Fasteners torqued to 202 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-CAX-9

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU150-9
- PLC Hex Register Address: DB25.DBW_0xBFA8
- Nominal Operating Range: 39 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-BRK-DISC-5109
- Overhaul Seal & Consumable Kit: KIT-TRU150-CAX-9
- Critical Fastener Torque Value: 21 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-CAX-9 at PLC register 0xBFA8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-BRK-DISC-5109) with seal kit (KIT-TRU150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 21 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-CAX-9 on Trumpf-TruLaser-L150 by installing PN-TRU150-BRK-DISC-5109 & kit KIT-TRU150-CAX-9. Fasteners torqued to 21 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-OPT-10

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU150-10
- PLC Hex Register Address: DB25.DBW_0xBFB0
- Nominal Operating Range: 42 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-GLASS-RD-5110
- Overhaul Seal & Consumable Kit: KIT-TRU150-OPT-10
- Critical Fastener Torque Value: 30 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-OPT-10 at PLC register 0xBFB0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-GLASS-RD-5110) with seal kit (KIT-TRU150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 30 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-OPT-10 on Trumpf-TruLaser-L150 by installing PN-TRU150-GLASS-RD-5110 & kit KIT-TRU150-OPT-10. Fasteners torqued to 30 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-PLC-11

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU150-11
- PLC Hex Register Address: DB25.DBW_0xBFB8
- Nominal Operating Range: 45 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-IGBT-MOD-5111
- Overhaul Seal & Consumable Kit: KIT-TRU150-PLC-11
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-PLC-11 at PLC register 0xBFB8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-IGBT-MOD-5111) with seal kit (KIT-TRU150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-PLC-11 on Trumpf-TruLaser-L150 by installing PN-TRU150-IGBT-MOD-5111 & kit KIT-TRU150-PLC-11. Fasteners torqued to 39 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-LUB-12

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU150-12
- PLC Hex Register Address: DB25.DBW_0xBFC0
- Nominal Operating Range: 48 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-INJ-MET-5112
- Overhaul Seal & Consumable Kit: KIT-TRU150-LUB-12
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-LUB-12 at PLC register 0xBFC0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-INJ-MET-5112) with seal kit (KIT-TRU150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-LUB-12 on Trumpf-TruLaser-L150 by installing PN-TRU150-INJ-MET-5112 & kit KIT-TRU150-LUB-12. Fasteners torqued to 48 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-VAC-13

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU150-13
- PLC Hex Register Address: DB25.DBW_0xBFC8
- Nominal Operating Range: 51 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-GASK-VIT-5113
- Overhaul Seal & Consumable Kit: KIT-TRU150-VAC-13
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-VAC-13 at PLC register 0xBFC8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-GASK-VIT-5113) with seal kit (KIT-TRU150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-VAC-13 on Trumpf-TruLaser-L150 by installing PN-TRU150-GASK-VIT-5113 & kit KIT-TRU150-VAC-13. Fasteners torqued to 57 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU150-THM-14

Target Machine Model: Trumpf-TruLaser-L150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L150 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU150-14
- PLC Hex Register Address: DB25.DBW_0xBFD0
- Nominal Operating Range: 54 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU150-HEX-CORE-5114
- Overhaul Seal & Consumable Kit: KIT-TRU150-THM-14
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L150 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU150-THM-14 at PLC register 0xBFD0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU150-HEX-CORE-5114) with seal kit (KIT-TRU150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L150, technician must log: "Resolved ERR-TRU150-THM-14 on Trumpf-TruLaser-L150 by installing PN-TRU150-HEX-CORE-5114 & kit KIT-TRU150-THM-14. Fasteners torqued to 66 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-SPN-1

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU200-1
- PLC Hex Register Address: DB26.DBW_0xC288
- Nominal Operating Range: 15 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-BRG-CER-5201
- Overhaul Seal & Consumable Kit: KIT-TRU200-SPN-1
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-SPN-1 at PLC register 0xC288.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-BRG-CER-5201) with seal kit (KIT-TRU200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-SPN-1 on Trumpf-TruLaser-L200 by installing PN-TRU200-BRG-CER-5201 & kit KIT-TRU200-SPN-1. Fasteners torqued to 89 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-ATC-2

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU200-2
- PLC Hex Register Address: DB26.DBW_0xC290
- Nominal Operating Range: 18 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-CAM-GRIP-5202
- Overhaul Seal & Consumable Kit: KIT-TRU200-ATC-2
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-ATC-2 at PLC register 0xC290.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-CAM-GRIP-5202) with seal kit (KIT-TRU200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-ATC-2 on Trumpf-TruLaser-L200 by installing PN-TRU200-CAM-GRIP-5202 & kit KIT-TRU200-ATC-2. Fasteners torqued to 98 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-HYD-3

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU200-3
- PLC Hex Register Address: DB26.DBW_0xC298
- Nominal Operating Range: 21 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-VLV-PROP-5203
- Overhaul Seal & Consumable Kit: KIT-TRU200-HYD-3
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-HYD-3 at PLC register 0xC298.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-VLV-PROP-5203) with seal kit (KIT-TRU200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-HYD-3 on Trumpf-TruLaser-L200 by installing PN-TRU200-VLV-PROP-5203 & kit KIT-TRU200-HYD-3. Fasteners torqued to 107 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-CLN-4

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU200-4
- PLC Hex Register Address: DB26.DBW_0xC2A0
- Nominal Operating Range: 24 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-PMP-PLG-5204
- Overhaul Seal & Consumable Kit: KIT-TRU200-CLN-4
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-CLN-4 at PLC register 0xC2A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-PMP-PLG-5204) with seal kit (KIT-TRU200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-CLN-4 on Trumpf-TruLaser-L200 by installing PN-TRU200-PMP-PLG-5204 & kit KIT-TRU200-CLN-4. Fasteners torqued to 116 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-XAX-5

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU200-5
- PLC Hex Register Address: DB26.DBW_0xC2A8
- Nominal Operating Range: 27 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-LIN-COIL-5205
- Overhaul Seal & Consumable Kit: KIT-TRU200-XAX-5
- Critical Fastener Torque Value: 125 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-XAX-5 at PLC register 0xC2A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-LIN-COIL-5205) with seal kit (KIT-TRU200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 125 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-XAX-5 on Trumpf-TruLaser-L200 by installing PN-TRU200-LIN-COIL-5205 & kit KIT-TRU200-XAX-5. Fasteners torqued to 125 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-YAX-6

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU200-6
- PLC Hex Register Address: DB26.DBW_0xC2B0
- Nominal Operating Range: 30 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-BSW-NUT-5206
- Overhaul Seal & Consumable Kit: KIT-TRU200-YAX-6
- Critical Fastener Torque Value: 134 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-YAX-6 at PLC register 0xC2B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-BSW-NUT-5206) with seal kit (KIT-TRU200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 134 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-YAX-6 on Trumpf-TruLaser-L200 by installing PN-TRU200-BSW-NUT-5206 & kit KIT-TRU200-YAX-6. Fasteners torqued to 134 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-ZAX-7

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU200-7
- PLC Hex Register Address: DB26.DBW_0xC2B8
- Nominal Operating Range: 33 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-CYL-SEAL-5207
- Overhaul Seal & Consumable Kit: KIT-TRU200-ZAX-7
- Critical Fastener Torque Value: 143 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-ZAX-7 at PLC register 0xC2B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-CYL-SEAL-5207) with seal kit (KIT-TRU200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 143 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-ZAX-7 on Trumpf-TruLaser-L200 by installing PN-TRU200-CYL-SEAL-5207 & kit KIT-TRU200-ZAX-7. Fasteners torqued to 143 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-BAX-8

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU200-8
- PLC Hex Register Address: DB26.DBW_0xC2C0
- Nominal Operating Range: 36 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-TRQ-STAT-5208
- Overhaul Seal & Consumable Kit: KIT-TRU200-BAX-8
- Critical Fastener Torque Value: 152 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-BAX-8 at PLC register 0xC2C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-TRQ-STAT-5208) with seal kit (KIT-TRU200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 152 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-BAX-8 on Trumpf-TruLaser-L200 by installing PN-TRU200-TRQ-STAT-5208 & kit KIT-TRU200-BAX-8. Fasteners torqued to 152 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-CAX-9

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU200-9
- PLC Hex Register Address: DB26.DBW_0xC2C8
- Nominal Operating Range: 39 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-BRK-DISC-5209
- Overhaul Seal & Consumable Kit: KIT-TRU200-CAX-9
- Critical Fastener Torque Value: 161 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-CAX-9 at PLC register 0xC2C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-BRK-DISC-5209) with seal kit (KIT-TRU200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 161 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-CAX-9 on Trumpf-TruLaser-L200 by installing PN-TRU200-BRK-DISC-5209 & kit KIT-TRU200-CAX-9. Fasteners torqued to 161 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-OPT-10

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU200-10
- PLC Hex Register Address: DB26.DBW_0xC2D0
- Nominal Operating Range: 42 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-GLASS-RD-5210
- Overhaul Seal & Consumable Kit: KIT-TRU200-OPT-10
- Critical Fastener Torque Value: 170 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-OPT-10 at PLC register 0xC2D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-GLASS-RD-5210) with seal kit (KIT-TRU200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 170 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-OPT-10 on Trumpf-TruLaser-L200 by installing PN-TRU200-GLASS-RD-5210 & kit KIT-TRU200-OPT-10. Fasteners torqued to 170 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-PLC-11

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU200-11
- PLC Hex Register Address: DB26.DBW_0xC2D8
- Nominal Operating Range: 45 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-IGBT-MOD-5211
- Overhaul Seal & Consumable Kit: KIT-TRU200-PLC-11
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-PLC-11 at PLC register 0xC2D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-IGBT-MOD-5211) with seal kit (KIT-TRU200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-PLC-11 on Trumpf-TruLaser-L200 by installing PN-TRU200-IGBT-MOD-5211 & kit KIT-TRU200-PLC-11. Fasteners torqued to 179 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-LUB-12

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU200-12
- PLC Hex Register Address: DB26.DBW_0xC2E0
- Nominal Operating Range: 48 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-INJ-MET-5212
- Overhaul Seal & Consumable Kit: KIT-TRU200-LUB-12
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-LUB-12 at PLC register 0xC2E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-INJ-MET-5212) with seal kit (KIT-TRU200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-LUB-12 on Trumpf-TruLaser-L200 by installing PN-TRU200-INJ-MET-5212 & kit KIT-TRU200-LUB-12. Fasteners torqued to 188 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-VAC-13

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU200-13
- PLC Hex Register Address: DB26.DBW_0xC2E8
- Nominal Operating Range: 51 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-GASK-VIT-5213
- Overhaul Seal & Consumable Kit: KIT-TRU200-VAC-13
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-VAC-13 at PLC register 0xC2E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-GASK-VIT-5213) with seal kit (KIT-TRU200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-VAC-13 on Trumpf-TruLaser-L200 by installing PN-TRU200-GASK-VIT-5213 & kit KIT-TRU200-VAC-13. Fasteners torqued to 197 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU200-THM-14

Target Machine Model: Trumpf-TruLaser-L200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L200 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU200-14
- PLC Hex Register Address: DB26.DBW_0xC2F0
- Nominal Operating Range: 54 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU200-HEX-CORE-5214
- Overhaul Seal & Consumable Kit: KIT-TRU200-THM-14
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L200 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU200-THM-14 at PLC register 0xC2F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU200-HEX-CORE-5214) with seal kit (KIT-TRU200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L200, technician must log: "Resolved ERR-TRU200-THM-14 on Trumpf-TruLaser-L200 by installing PN-TRU200-HEX-CORE-5214 & kit KIT-TRU200-THM-14. Fasteners torqued to 206 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-SPN-1

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU250-1
- PLC Hex Register Address: DB27.DBW_0xC5A8
- Nominal Operating Range: 15 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-BRG-CER-5301
- Overhaul Seal & Consumable Kit: KIT-TRU250-SPN-1
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-SPN-1 at PLC register 0xC5A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-BRG-CER-5301) with seal kit (KIT-TRU250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-SPN-1 on Trumpf-TruLaser-L250 by installing PN-TRU250-BRG-CER-5301 & kit KIT-TRU250-SPN-1. Fasteners torqued to 39 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-ATC-2

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU250-2
- PLC Hex Register Address: DB27.DBW_0xC5B0
- Nominal Operating Range: 18 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-CAM-GRIP-5302
- Overhaul Seal & Consumable Kit: KIT-TRU250-ATC-2
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-ATC-2 at PLC register 0xC5B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-CAM-GRIP-5302) with seal kit (KIT-TRU250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-ATC-2 on Trumpf-TruLaser-L250 by installing PN-TRU250-CAM-GRIP-5302 & kit KIT-TRU250-ATC-2. Fasteners torqued to 48 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-HYD-3

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU250-3
- PLC Hex Register Address: DB27.DBW_0xC5B8
- Nominal Operating Range: 21 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-VLV-PROP-5303
- Overhaul Seal & Consumable Kit: KIT-TRU250-HYD-3
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-HYD-3 at PLC register 0xC5B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-VLV-PROP-5303) with seal kit (KIT-TRU250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-HYD-3 on Trumpf-TruLaser-L250 by installing PN-TRU250-VLV-PROP-5303 & kit KIT-TRU250-HYD-3. Fasteners torqued to 57 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-CLN-4

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU250-4
- PLC Hex Register Address: DB27.DBW_0xC5C0
- Nominal Operating Range: 24 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-PMP-PLG-5304
- Overhaul Seal & Consumable Kit: KIT-TRU250-CLN-4
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-CLN-4 at PLC register 0xC5C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-PMP-PLG-5304) with seal kit (KIT-TRU250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-CLN-4 on Trumpf-TruLaser-L250 by installing PN-TRU250-PMP-PLG-5304 & kit KIT-TRU250-CLN-4. Fasteners torqued to 66 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-XAX-5

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU250-5
- PLC Hex Register Address: DB27.DBW_0xC5C8
- Nominal Operating Range: 27 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-LIN-COIL-5305
- Overhaul Seal & Consumable Kit: KIT-TRU250-XAX-5
- Critical Fastener Torque Value: 75 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-XAX-5 at PLC register 0xC5C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-LIN-COIL-5305) with seal kit (KIT-TRU250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 75 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-XAX-5 on Trumpf-TruLaser-L250 by installing PN-TRU250-LIN-COIL-5305 & kit KIT-TRU250-XAX-5. Fasteners torqued to 75 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-YAX-6

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU250-6
- PLC Hex Register Address: DB27.DBW_0xC5D0
- Nominal Operating Range: 30 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-BSW-NUT-5306
- Overhaul Seal & Consumable Kit: KIT-TRU250-YAX-6
- Critical Fastener Torque Value: 84 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-YAX-6 at PLC register 0xC5D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-BSW-NUT-5306) with seal kit (KIT-TRU250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 84 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-YAX-6 on Trumpf-TruLaser-L250 by installing PN-TRU250-BSW-NUT-5306 & kit KIT-TRU250-YAX-6. Fasteners torqued to 84 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-ZAX-7

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU250-7
- PLC Hex Register Address: DB27.DBW_0xC5D8
- Nominal Operating Range: 33 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-CYL-SEAL-5307
- Overhaul Seal & Consumable Kit: KIT-TRU250-ZAX-7
- Critical Fastener Torque Value: 93 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-ZAX-7 at PLC register 0xC5D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-CYL-SEAL-5307) with seal kit (KIT-TRU250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 93 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-ZAX-7 on Trumpf-TruLaser-L250 by installing PN-TRU250-CYL-SEAL-5307 & kit KIT-TRU250-ZAX-7. Fasteners torqued to 93 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-BAX-8

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU250-8
- PLC Hex Register Address: DB27.DBW_0xC5E0
- Nominal Operating Range: 36 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-TRQ-STAT-5308
- Overhaul Seal & Consumable Kit: KIT-TRU250-BAX-8
- Critical Fastener Torque Value: 102 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-BAX-8 at PLC register 0xC5E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-TRQ-STAT-5308) with seal kit (KIT-TRU250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 102 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-BAX-8 on Trumpf-TruLaser-L250 by installing PN-TRU250-TRQ-STAT-5308 & kit KIT-TRU250-BAX-8. Fasteners torqued to 102 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-CAX-9

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU250-9
- PLC Hex Register Address: DB27.DBW_0xC5E8
- Nominal Operating Range: 39 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-BRK-DISC-5309
- Overhaul Seal & Consumable Kit: KIT-TRU250-CAX-9
- Critical Fastener Torque Value: 111 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-CAX-9 at PLC register 0xC5E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-BRK-DISC-5309) with seal kit (KIT-TRU250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 111 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-CAX-9 on Trumpf-TruLaser-L250 by installing PN-TRU250-BRK-DISC-5309 & kit KIT-TRU250-CAX-9. Fasteners torqued to 111 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-OPT-10

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU250-10
- PLC Hex Register Address: DB27.DBW_0xC5F0
- Nominal Operating Range: 42 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-GLASS-RD-5310
- Overhaul Seal & Consumable Kit: KIT-TRU250-OPT-10
- Critical Fastener Torque Value: 120 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-OPT-10 at PLC register 0xC5F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-GLASS-RD-5310) with seal kit (KIT-TRU250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 120 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-OPT-10 on Trumpf-TruLaser-L250 by installing PN-TRU250-GLASS-RD-5310 & kit KIT-TRU250-OPT-10. Fasteners torqued to 120 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-PLC-11

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU250-11
- PLC Hex Register Address: DB27.DBW_0xC5F8
- Nominal Operating Range: 45 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-IGBT-MOD-5311
- Overhaul Seal & Consumable Kit: KIT-TRU250-PLC-11
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-PLC-11 at PLC register 0xC5F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-IGBT-MOD-5311) with seal kit (KIT-TRU250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-PLC-11 on Trumpf-TruLaser-L250 by installing PN-TRU250-IGBT-MOD-5311 & kit KIT-TRU250-PLC-11. Fasteners torqued to 129 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-LUB-12

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU250-12
- PLC Hex Register Address: DB27.DBW_0xC600
- Nominal Operating Range: 48 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-INJ-MET-5312
- Overhaul Seal & Consumable Kit: KIT-TRU250-LUB-12
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-LUB-12 at PLC register 0xC600.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-INJ-MET-5312) with seal kit (KIT-TRU250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-LUB-12 on Trumpf-TruLaser-L250 by installing PN-TRU250-INJ-MET-5312 & kit KIT-TRU250-LUB-12. Fasteners torqued to 138 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-VAC-13

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU250-13
- PLC Hex Register Address: DB27.DBW_0xC608
- Nominal Operating Range: 51 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-GASK-VIT-5313
- Overhaul Seal & Consumable Kit: KIT-TRU250-VAC-13
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-VAC-13 at PLC register 0xC608.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-GASK-VIT-5313) with seal kit (KIT-TRU250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-VAC-13 on Trumpf-TruLaser-L250 by installing PN-TRU250-GASK-VIT-5313 & kit KIT-TRU250-VAC-13. Fasteners torqued to 147 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU250-THM-14

Target Machine Model: Trumpf-TruLaser-L250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L250 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU250-14
- PLC Hex Register Address: DB27.DBW_0xC610
- Nominal Operating Range: 54 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU250-HEX-CORE-5314
- Overhaul Seal & Consumable Kit: KIT-TRU250-THM-14
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L250 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU250-THM-14 at PLC register 0xC610.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU250-HEX-CORE-5314) with seal kit (KIT-TRU250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L250, technician must log: "Resolved ERR-TRU250-THM-14 on Trumpf-TruLaser-L250 by installing PN-TRU250-HEX-CORE-5314 & kit KIT-TRU250-THM-14. Fasteners torqued to 156 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-SPN-1

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU300-1
- PLC Hex Register Address: DB28.DBW_0xC8C8
- Nominal Operating Range: 15 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-BRG-CER-5401
- Overhaul Seal & Consumable Kit: KIT-TRU300-SPN-1
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-SPN-1 at PLC register 0xC8C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-BRG-CER-5401) with seal kit (KIT-TRU300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-SPN-1 on Trumpf-TruLaser-L300 by installing PN-TRU300-BRG-CER-5401 & kit KIT-TRU300-SPN-1. Fasteners torqued to 179 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-ATC-2

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU300-2
- PLC Hex Register Address: DB28.DBW_0xC8D0
- Nominal Operating Range: 18 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-CAM-GRIP-5402
- Overhaul Seal & Consumable Kit: KIT-TRU300-ATC-2
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-ATC-2 at PLC register 0xC8D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-CAM-GRIP-5402) with seal kit (KIT-TRU300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-ATC-2 on Trumpf-TruLaser-L300 by installing PN-TRU300-CAM-GRIP-5402 & kit KIT-TRU300-ATC-2. Fasteners torqued to 188 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-HYD-3

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU300-3
- PLC Hex Register Address: DB28.DBW_0xC8D8
- Nominal Operating Range: 21 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-VLV-PROP-5403
- Overhaul Seal & Consumable Kit: KIT-TRU300-HYD-3
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-HYD-3 at PLC register 0xC8D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-VLV-PROP-5403) with seal kit (KIT-TRU300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-HYD-3 on Trumpf-TruLaser-L300 by installing PN-TRU300-VLV-PROP-5403 & kit KIT-TRU300-HYD-3. Fasteners torqued to 197 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-CLN-4

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU300-4
- PLC Hex Register Address: DB28.DBW_0xC8E0
- Nominal Operating Range: 24 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-PMP-PLG-5404
- Overhaul Seal & Consumable Kit: KIT-TRU300-CLN-4
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-CLN-4 at PLC register 0xC8E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-PMP-PLG-5404) with seal kit (KIT-TRU300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-CLN-4 on Trumpf-TruLaser-L300 by installing PN-TRU300-PMP-PLG-5404 & kit KIT-TRU300-CLN-4. Fasteners torqued to 206 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-XAX-5

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU300-5
- PLC Hex Register Address: DB28.DBW_0xC8E8
- Nominal Operating Range: 27 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-LIN-COIL-5405
- Overhaul Seal & Consumable Kit: KIT-TRU300-XAX-5
- Critical Fastener Torque Value: 25 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-XAX-5 at PLC register 0xC8E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-LIN-COIL-5405) with seal kit (KIT-TRU300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 25 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-XAX-5 on Trumpf-TruLaser-L300 by installing PN-TRU300-LIN-COIL-5405 & kit KIT-TRU300-XAX-5. Fasteners torqued to 25 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-YAX-6

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU300-6
- PLC Hex Register Address: DB28.DBW_0xC8F0
- Nominal Operating Range: 30 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-BSW-NUT-5406
- Overhaul Seal & Consumable Kit: KIT-TRU300-YAX-6
- Critical Fastener Torque Value: 34 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-YAX-6 at PLC register 0xC8F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-BSW-NUT-5406) with seal kit (KIT-TRU300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 34 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-YAX-6 on Trumpf-TruLaser-L300 by installing PN-TRU300-BSW-NUT-5406 & kit KIT-TRU300-YAX-6. Fasteners torqued to 34 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-ZAX-7

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU300-7
- PLC Hex Register Address: DB28.DBW_0xC8F8
- Nominal Operating Range: 33 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-CYL-SEAL-5407
- Overhaul Seal & Consumable Kit: KIT-TRU300-ZAX-7
- Critical Fastener Torque Value: 43 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-ZAX-7 at PLC register 0xC8F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-CYL-SEAL-5407) with seal kit (KIT-TRU300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 43 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-ZAX-7 on Trumpf-TruLaser-L300 by installing PN-TRU300-CYL-SEAL-5407 & kit KIT-TRU300-ZAX-7. Fasteners torqued to 43 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-BAX-8

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU300-8
- PLC Hex Register Address: DB28.DBW_0xC900
- Nominal Operating Range: 36 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-TRQ-STAT-5408
- Overhaul Seal & Consumable Kit: KIT-TRU300-BAX-8
- Critical Fastener Torque Value: 52 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-BAX-8 at PLC register 0xC900.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-TRQ-STAT-5408) with seal kit (KIT-TRU300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 52 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-BAX-8 on Trumpf-TruLaser-L300 by installing PN-TRU300-TRQ-STAT-5408 & kit KIT-TRU300-BAX-8. Fasteners torqued to 52 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-CAX-9

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU300-9
- PLC Hex Register Address: DB28.DBW_0xC908
- Nominal Operating Range: 39 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-BRK-DISC-5409
- Overhaul Seal & Consumable Kit: KIT-TRU300-CAX-9
- Critical Fastener Torque Value: 61 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-CAX-9 at PLC register 0xC908.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-BRK-DISC-5409) with seal kit (KIT-TRU300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 61 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-CAX-9 on Trumpf-TruLaser-L300 by installing PN-TRU300-BRK-DISC-5409 & kit KIT-TRU300-CAX-9. Fasteners torqued to 61 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-OPT-10

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU300-10
- PLC Hex Register Address: DB28.DBW_0xC910
- Nominal Operating Range: 42 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-GLASS-RD-5410
- Overhaul Seal & Consumable Kit: KIT-TRU300-OPT-10
- Critical Fastener Torque Value: 70 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-OPT-10 at PLC register 0xC910.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-GLASS-RD-5410) with seal kit (KIT-TRU300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 70 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-OPT-10 on Trumpf-TruLaser-L300 by installing PN-TRU300-GLASS-RD-5410 & kit KIT-TRU300-OPT-10. Fasteners torqued to 70 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-PLC-11

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU300-11
- PLC Hex Register Address: DB28.DBW_0xC918
- Nominal Operating Range: 45 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-IGBT-MOD-5411
- Overhaul Seal & Consumable Kit: KIT-TRU300-PLC-11
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-PLC-11 at PLC register 0xC918.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-IGBT-MOD-5411) with seal kit (KIT-TRU300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-PLC-11 on Trumpf-TruLaser-L300 by installing PN-TRU300-IGBT-MOD-5411 & kit KIT-TRU300-PLC-11. Fasteners torqued to 79 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-LUB-12

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU300-12
- PLC Hex Register Address: DB28.DBW_0xC920
- Nominal Operating Range: 48 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-INJ-MET-5412
- Overhaul Seal & Consumable Kit: KIT-TRU300-LUB-12
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-LUB-12 at PLC register 0xC920.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-INJ-MET-5412) with seal kit (KIT-TRU300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-LUB-12 on Trumpf-TruLaser-L300 by installing PN-TRU300-INJ-MET-5412 & kit KIT-TRU300-LUB-12. Fasteners torqued to 88 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-VAC-13

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU300-13
- PLC Hex Register Address: DB28.DBW_0xC928
- Nominal Operating Range: 51 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-GASK-VIT-5413
- Overhaul Seal & Consumable Kit: KIT-TRU300-VAC-13
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-VAC-13 at PLC register 0xC928.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-GASK-VIT-5413) with seal kit (KIT-TRU300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-VAC-13 on Trumpf-TruLaser-L300 by installing PN-TRU300-GASK-VIT-5413 & kit KIT-TRU300-VAC-13. Fasteners torqued to 97 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU300-THM-14

Target Machine Model: Trumpf-TruLaser-L300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L300 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU300-14
- PLC Hex Register Address: DB28.DBW_0xC930
- Nominal Operating Range: 54 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU300-HEX-CORE-5414
- Overhaul Seal & Consumable Kit: KIT-TRU300-THM-14
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L300 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU300-THM-14 at PLC register 0xC930.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU300-HEX-CORE-5414) with seal kit (KIT-TRU300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L300, technician must log: "Resolved ERR-TRU300-THM-14 on Trumpf-TruLaser-L300 by installing PN-TRU300-HEX-CORE-5414 & kit KIT-TRU300-THM-14. Fasteners torqued to 106 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-SPN-1

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU350-1
- PLC Hex Register Address: DB29.DBW_0xCBE8
- Nominal Operating Range: 15 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-BRG-CER-5501
- Overhaul Seal & Consumable Kit: KIT-TRU350-SPN-1
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-SPN-1 at PLC register 0xCBE8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-BRG-CER-5501) with seal kit (KIT-TRU350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-SPN-1 on Trumpf-TruLaser-L350 by installing PN-TRU350-BRG-CER-5501 & kit KIT-TRU350-SPN-1. Fasteners torqued to 129 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-ATC-2

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU350-2
- PLC Hex Register Address: DB29.DBW_0xCBF0
- Nominal Operating Range: 18 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-CAM-GRIP-5502
- Overhaul Seal & Consumable Kit: KIT-TRU350-ATC-2
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-ATC-2 at PLC register 0xCBF0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-CAM-GRIP-5502) with seal kit (KIT-TRU350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-ATC-2 on Trumpf-TruLaser-L350 by installing PN-TRU350-CAM-GRIP-5502 & kit KIT-TRU350-ATC-2. Fasteners torqued to 138 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-HYD-3

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU350-3
- PLC Hex Register Address: DB29.DBW_0xCBF8
- Nominal Operating Range: 21 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-VLV-PROP-5503
- Overhaul Seal & Consumable Kit: KIT-TRU350-HYD-3
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-HYD-3 at PLC register 0xCBF8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-VLV-PROP-5503) with seal kit (KIT-TRU350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-HYD-3 on Trumpf-TruLaser-L350 by installing PN-TRU350-VLV-PROP-5503 & kit KIT-TRU350-HYD-3. Fasteners torqued to 147 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-CLN-4

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU350-4
- PLC Hex Register Address: DB29.DBW_0xCC00
- Nominal Operating Range: 24 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-PMP-PLG-5504
- Overhaul Seal & Consumable Kit: KIT-TRU350-CLN-4
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-CLN-4 at PLC register 0xCC00.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-PMP-PLG-5504) with seal kit (KIT-TRU350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-CLN-4 on Trumpf-TruLaser-L350 by installing PN-TRU350-PMP-PLG-5504 & kit KIT-TRU350-CLN-4. Fasteners torqued to 156 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-XAX-5

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU350-5
- PLC Hex Register Address: DB29.DBW_0xCC08
- Nominal Operating Range: 27 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-LIN-COIL-5505
- Overhaul Seal & Consumable Kit: KIT-TRU350-XAX-5
- Critical Fastener Torque Value: 165 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-XAX-5 at PLC register 0xCC08.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-LIN-COIL-5505) with seal kit (KIT-TRU350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 165 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-XAX-5 on Trumpf-TruLaser-L350 by installing PN-TRU350-LIN-COIL-5505 & kit KIT-TRU350-XAX-5. Fasteners torqued to 165 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-YAX-6

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU350-6
- PLC Hex Register Address: DB29.DBW_0xCC10
- Nominal Operating Range: 30 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-BSW-NUT-5506
- Overhaul Seal & Consumable Kit: KIT-TRU350-YAX-6
- Critical Fastener Torque Value: 174 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-YAX-6 at PLC register 0xCC10.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-BSW-NUT-5506) with seal kit (KIT-TRU350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 174 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-YAX-6 on Trumpf-TruLaser-L350 by installing PN-TRU350-BSW-NUT-5506 & kit KIT-TRU350-YAX-6. Fasteners torqued to 174 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-ZAX-7

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU350-7
- PLC Hex Register Address: DB29.DBW_0xCC18
- Nominal Operating Range: 33 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-CYL-SEAL-5507
- Overhaul Seal & Consumable Kit: KIT-TRU350-ZAX-7
- Critical Fastener Torque Value: 183 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-ZAX-7 at PLC register 0xCC18.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-CYL-SEAL-5507) with seal kit (KIT-TRU350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 183 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-ZAX-7 on Trumpf-TruLaser-L350 by installing PN-TRU350-CYL-SEAL-5507 & kit KIT-TRU350-ZAX-7. Fasteners torqued to 183 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-BAX-8

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU350-8
- PLC Hex Register Address: DB29.DBW_0xCC20
- Nominal Operating Range: 36 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-TRQ-STAT-5508
- Overhaul Seal & Consumable Kit: KIT-TRU350-BAX-8
- Critical Fastener Torque Value: 192 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-BAX-8 at PLC register 0xCC20.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-TRQ-STAT-5508) with seal kit (KIT-TRU350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 192 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-BAX-8 on Trumpf-TruLaser-L350 by installing PN-TRU350-TRQ-STAT-5508 & kit KIT-TRU350-BAX-8. Fasteners torqued to 192 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-CAX-9

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU350-9
- PLC Hex Register Address: DB29.DBW_0xCC28
- Nominal Operating Range: 39 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-BRK-DISC-5509
- Overhaul Seal & Consumable Kit: KIT-TRU350-CAX-9
- Critical Fastener Torque Value: 201 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-CAX-9 at PLC register 0xCC28.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-BRK-DISC-5509) with seal kit (KIT-TRU350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 201 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-CAX-9 on Trumpf-TruLaser-L350 by installing PN-TRU350-BRK-DISC-5509 & kit KIT-TRU350-CAX-9. Fasteners torqued to 201 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-OPT-10

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU350-10
- PLC Hex Register Address: DB29.DBW_0xCC30
- Nominal Operating Range: 42 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-GLASS-RD-5510
- Overhaul Seal & Consumable Kit: KIT-TRU350-OPT-10
- Critical Fastener Torque Value: 20 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-OPT-10 at PLC register 0xCC30.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-GLASS-RD-5510) with seal kit (KIT-TRU350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 20 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-OPT-10 on Trumpf-TruLaser-L350 by installing PN-TRU350-GLASS-RD-5510 & kit KIT-TRU350-OPT-10. Fasteners torqued to 20 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-PLC-11

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU350-11
- PLC Hex Register Address: DB29.DBW_0xCC38
- Nominal Operating Range: 45 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-IGBT-MOD-5511
- Overhaul Seal & Consumable Kit: KIT-TRU350-PLC-11
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-PLC-11 at PLC register 0xCC38.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-IGBT-MOD-5511) with seal kit (KIT-TRU350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-PLC-11 on Trumpf-TruLaser-L350 by installing PN-TRU350-IGBT-MOD-5511 & kit KIT-TRU350-PLC-11. Fasteners torqued to 29 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-LUB-12

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU350-12
- PLC Hex Register Address: DB29.DBW_0xCC40
- Nominal Operating Range: 48 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-INJ-MET-5512
- Overhaul Seal & Consumable Kit: KIT-TRU350-LUB-12
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-LUB-12 at PLC register 0xCC40.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-INJ-MET-5512) with seal kit (KIT-TRU350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-LUB-12 on Trumpf-TruLaser-L350 by installing PN-TRU350-INJ-MET-5512 & kit KIT-TRU350-LUB-12. Fasteners torqued to 38 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-VAC-13

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU350-13
- PLC Hex Register Address: DB29.DBW_0xCC48
- Nominal Operating Range: 51 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-GASK-VIT-5513
- Overhaul Seal & Consumable Kit: KIT-TRU350-VAC-13
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-VAC-13 at PLC register 0xCC48.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-GASK-VIT-5513) with seal kit (KIT-TRU350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-VAC-13 on Trumpf-TruLaser-L350 by installing PN-TRU350-GASK-VIT-5513 & kit KIT-TRU350-VAC-13. Fasteners torqued to 47 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU350-THM-14

Target Machine Model: Trumpf-TruLaser-L350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L350 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU350-14
- PLC Hex Register Address: DB29.DBW_0xCC50
- Nominal Operating Range: 54 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU350-HEX-CORE-5514
- Overhaul Seal & Consumable Kit: KIT-TRU350-THM-14
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L350 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU350-THM-14 at PLC register 0xCC50.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU350-HEX-CORE-5514) with seal kit (KIT-TRU350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L350, technician must log: "Resolved ERR-TRU350-THM-14 on Trumpf-TruLaser-L350 by installing PN-TRU350-HEX-CORE-5514 & kit KIT-TRU350-THM-14. Fasteners torqued to 56 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-SPN-1

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU400-1
- PLC Hex Register Address: DB30.DBW_0xCF08
- Nominal Operating Range: 15 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-BRG-CER-5601
- Overhaul Seal & Consumable Kit: KIT-TRU400-SPN-1
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-SPN-1 at PLC register 0xCF08.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-BRG-CER-5601) with seal kit (KIT-TRU400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-SPN-1 on Trumpf-TruLaser-L400 by installing PN-TRU400-BRG-CER-5601 & kit KIT-TRU400-SPN-1. Fasteners torqued to 79 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-ATC-2

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU400-2
- PLC Hex Register Address: DB30.DBW_0xCF10
- Nominal Operating Range: 18 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-CAM-GRIP-5602
- Overhaul Seal & Consumable Kit: KIT-TRU400-ATC-2
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-ATC-2 at PLC register 0xCF10.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-CAM-GRIP-5602) with seal kit (KIT-TRU400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-ATC-2 on Trumpf-TruLaser-L400 by installing PN-TRU400-CAM-GRIP-5602 & kit KIT-TRU400-ATC-2. Fasteners torqued to 88 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-HYD-3

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU400-3
- PLC Hex Register Address: DB30.DBW_0xCF18
- Nominal Operating Range: 21 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-VLV-PROP-5603
- Overhaul Seal & Consumable Kit: KIT-TRU400-HYD-3
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-HYD-3 at PLC register 0xCF18.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-VLV-PROP-5603) with seal kit (KIT-TRU400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-HYD-3 on Trumpf-TruLaser-L400 by installing PN-TRU400-VLV-PROP-5603 & kit KIT-TRU400-HYD-3. Fasteners torqued to 97 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-CLN-4

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU400-4
- PLC Hex Register Address: DB30.DBW_0xCF20
- Nominal Operating Range: 24 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-PMP-PLG-5604
- Overhaul Seal & Consumable Kit: KIT-TRU400-CLN-4
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-CLN-4 at PLC register 0xCF20.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-PMP-PLG-5604) with seal kit (KIT-TRU400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-CLN-4 on Trumpf-TruLaser-L400 by installing PN-TRU400-PMP-PLG-5604 & kit KIT-TRU400-CLN-4. Fasteners torqued to 106 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-XAX-5

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU400-5
- PLC Hex Register Address: DB30.DBW_0xCF28
- Nominal Operating Range: 27 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-LIN-COIL-5605
- Overhaul Seal & Consumable Kit: KIT-TRU400-XAX-5
- Critical Fastener Torque Value: 115 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-XAX-5 at PLC register 0xCF28.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-LIN-COIL-5605) with seal kit (KIT-TRU400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 115 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-XAX-5 on Trumpf-TruLaser-L400 by installing PN-TRU400-LIN-COIL-5605 & kit KIT-TRU400-XAX-5. Fasteners torqued to 115 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-YAX-6

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU400-6
- PLC Hex Register Address: DB30.DBW_0xCF30
- Nominal Operating Range: 30 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-BSW-NUT-5606
- Overhaul Seal & Consumable Kit: KIT-TRU400-YAX-6
- Critical Fastener Torque Value: 124 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-YAX-6 at PLC register 0xCF30.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-BSW-NUT-5606) with seal kit (KIT-TRU400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 124 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-YAX-6 on Trumpf-TruLaser-L400 by installing PN-TRU400-BSW-NUT-5606 & kit KIT-TRU400-YAX-6. Fasteners torqued to 124 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-ZAX-7

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU400-7
- PLC Hex Register Address: DB30.DBW_0xCF38
- Nominal Operating Range: 33 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-CYL-SEAL-5607
- Overhaul Seal & Consumable Kit: KIT-TRU400-ZAX-7
- Critical Fastener Torque Value: 133 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-ZAX-7 at PLC register 0xCF38.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-CYL-SEAL-5607) with seal kit (KIT-TRU400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 133 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-ZAX-7 on Trumpf-TruLaser-L400 by installing PN-TRU400-CYL-SEAL-5607 & kit KIT-TRU400-ZAX-7. Fasteners torqued to 133 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-BAX-8

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU400-8
- PLC Hex Register Address: DB30.DBW_0xCF40
- Nominal Operating Range: 36 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-TRQ-STAT-5608
- Overhaul Seal & Consumable Kit: KIT-TRU400-BAX-8
- Critical Fastener Torque Value: 142 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-BAX-8 at PLC register 0xCF40.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-TRQ-STAT-5608) with seal kit (KIT-TRU400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 142 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-BAX-8 on Trumpf-TruLaser-L400 by installing PN-TRU400-TRQ-STAT-5608 & kit KIT-TRU400-BAX-8. Fasteners torqued to 142 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-CAX-9

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU400-9
- PLC Hex Register Address: DB30.DBW_0xCF48
- Nominal Operating Range: 39 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-BRK-DISC-5609
- Overhaul Seal & Consumable Kit: KIT-TRU400-CAX-9
- Critical Fastener Torque Value: 151 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-CAX-9 at PLC register 0xCF48.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-BRK-DISC-5609) with seal kit (KIT-TRU400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 151 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-CAX-9 on Trumpf-TruLaser-L400 by installing PN-TRU400-BRK-DISC-5609 & kit KIT-TRU400-CAX-9. Fasteners torqued to 151 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-OPT-10

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU400-10
- PLC Hex Register Address: DB30.DBW_0xCF50
- Nominal Operating Range: 42 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-GLASS-RD-5610
- Overhaul Seal & Consumable Kit: KIT-TRU400-OPT-10
- Critical Fastener Torque Value: 160 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-OPT-10 at PLC register 0xCF50.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-GLASS-RD-5610) with seal kit (KIT-TRU400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 160 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-OPT-10 on Trumpf-TruLaser-L400 by installing PN-TRU400-GLASS-RD-5610 & kit KIT-TRU400-OPT-10. Fasteners torqued to 160 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-PLC-11

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU400-11
- PLC Hex Register Address: DB30.DBW_0xCF58
- Nominal Operating Range: 45 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-IGBT-MOD-5611
- Overhaul Seal & Consumable Kit: KIT-TRU400-PLC-11
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-PLC-11 at PLC register 0xCF58.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-IGBT-MOD-5611) with seal kit (KIT-TRU400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-PLC-11 on Trumpf-TruLaser-L400 by installing PN-TRU400-IGBT-MOD-5611 & kit KIT-TRU400-PLC-11. Fasteners torqued to 169 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-LUB-12

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU400-12
- PLC Hex Register Address: DB30.DBW_0xCF60
- Nominal Operating Range: 48 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-INJ-MET-5612
- Overhaul Seal & Consumable Kit: KIT-TRU400-LUB-12
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-LUB-12 at PLC register 0xCF60.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-INJ-MET-5612) with seal kit (KIT-TRU400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-LUB-12 on Trumpf-TruLaser-L400 by installing PN-TRU400-INJ-MET-5612 & kit KIT-TRU400-LUB-12. Fasteners torqued to 178 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-VAC-13

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU400-13
- PLC Hex Register Address: DB30.DBW_0xCF68
- Nominal Operating Range: 51 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-GASK-VIT-5613
- Overhaul Seal & Consumable Kit: KIT-TRU400-VAC-13
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-VAC-13 at PLC register 0xCF68.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-GASK-VIT-5613) with seal kit (KIT-TRU400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-VAC-13 on Trumpf-TruLaser-L400 by installing PN-TRU400-GASK-VIT-5613 & kit KIT-TRU400-VAC-13. Fasteners torqued to 187 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU400-THM-14

Target Machine Model: Trumpf-TruLaser-L400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L400 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU400-14
- PLC Hex Register Address: DB30.DBW_0xCF70
- Nominal Operating Range: 54 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 120 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU400-HEX-CORE-5614
- Overhaul Seal & Consumable Kit: KIT-TRU400-THM-14
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L400 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU400-THM-14 at PLC register 0xCF70.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU400-HEX-CORE-5614) with seal kit (KIT-TRU400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L400, technician must log: "Resolved ERR-TRU400-THM-14 on Trumpf-TruLaser-L400 by installing PN-TRU400-HEX-CORE-5614 & kit KIT-TRU400-THM-14. Fasteners torqued to 196 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-SPN-1

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU450-1
- PLC Hex Register Address: DB31.DBW_0xD228
- Nominal Operating Range: 15 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-BRG-CER-5701
- Overhaul Seal & Consumable Kit: KIT-TRU450-SPN-1
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-SPN-1 at PLC register 0xD228.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-BRG-CER-5701) with seal kit (KIT-TRU450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-SPN-1 on Trumpf-TruLaser-L450 by installing PN-TRU450-BRG-CER-5701 & kit KIT-TRU450-SPN-1. Fasteners torqued to 29 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-ATC-2

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU450-2
- PLC Hex Register Address: DB31.DBW_0xD230
- Nominal Operating Range: 18 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-CAM-GRIP-5702
- Overhaul Seal & Consumable Kit: KIT-TRU450-ATC-2
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-ATC-2 at PLC register 0xD230.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-CAM-GRIP-5702) with seal kit (KIT-TRU450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-ATC-2 on Trumpf-TruLaser-L450 by installing PN-TRU450-CAM-GRIP-5702 & kit KIT-TRU450-ATC-2. Fasteners torqued to 38 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-HYD-3

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU450-3
- PLC Hex Register Address: DB31.DBW_0xD238
- Nominal Operating Range: 21 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-VLV-PROP-5703
- Overhaul Seal & Consumable Kit: KIT-TRU450-HYD-3
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-HYD-3 at PLC register 0xD238.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-VLV-PROP-5703) with seal kit (KIT-TRU450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-HYD-3 on Trumpf-TruLaser-L450 by installing PN-TRU450-VLV-PROP-5703 & kit KIT-TRU450-HYD-3. Fasteners torqued to 47 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-CLN-4

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU450-4
- PLC Hex Register Address: DB31.DBW_0xD240
- Nominal Operating Range: 24 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-PMP-PLG-5704
- Overhaul Seal & Consumable Kit: KIT-TRU450-CLN-4
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-CLN-4 at PLC register 0xD240.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-PMP-PLG-5704) with seal kit (KIT-TRU450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-CLN-4 on Trumpf-TruLaser-L450 by installing PN-TRU450-PMP-PLG-5704 & kit KIT-TRU450-CLN-4. Fasteners torqued to 56 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-XAX-5

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU450-5
- PLC Hex Register Address: DB31.DBW_0xD248
- Nominal Operating Range: 27 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-LIN-COIL-5705
- Overhaul Seal & Consumable Kit: KIT-TRU450-XAX-5
- Critical Fastener Torque Value: 65 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-XAX-5 at PLC register 0xD248.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-LIN-COIL-5705) with seal kit (KIT-TRU450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 65 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-XAX-5 on Trumpf-TruLaser-L450 by installing PN-TRU450-LIN-COIL-5705 & kit KIT-TRU450-XAX-5. Fasteners torqued to 65 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-YAX-6

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU450-6
- PLC Hex Register Address: DB31.DBW_0xD250
- Nominal Operating Range: 30 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-BSW-NUT-5706
- Overhaul Seal & Consumable Kit: KIT-TRU450-YAX-6
- Critical Fastener Torque Value: 74 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-YAX-6 at PLC register 0xD250.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-BSW-NUT-5706) with seal kit (KIT-TRU450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 74 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-YAX-6 on Trumpf-TruLaser-L450 by installing PN-TRU450-BSW-NUT-5706 & kit KIT-TRU450-YAX-6. Fasteners torqued to 74 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-ZAX-7

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU450-7
- PLC Hex Register Address: DB31.DBW_0xD258
- Nominal Operating Range: 33 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-CYL-SEAL-5707
- Overhaul Seal & Consumable Kit: KIT-TRU450-ZAX-7
- Critical Fastener Torque Value: 83 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-ZAX-7 at PLC register 0xD258.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-CYL-SEAL-5707) with seal kit (KIT-TRU450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 83 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-ZAX-7 on Trumpf-TruLaser-L450 by installing PN-TRU450-CYL-SEAL-5707 & kit KIT-TRU450-ZAX-7. Fasteners torqued to 83 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-BAX-8

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU450-8
- PLC Hex Register Address: DB31.DBW_0xD260
- Nominal Operating Range: 36 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-TRQ-STAT-5708
- Overhaul Seal & Consumable Kit: KIT-TRU450-BAX-8
- Critical Fastener Torque Value: 92 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-BAX-8 at PLC register 0xD260.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-TRQ-STAT-5708) with seal kit (KIT-TRU450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 92 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-BAX-8 on Trumpf-TruLaser-L450 by installing PN-TRU450-TRQ-STAT-5708 & kit KIT-TRU450-BAX-8. Fasteners torqued to 92 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-CAX-9

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU450-9
- PLC Hex Register Address: DB31.DBW_0xD268
- Nominal Operating Range: 39 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-BRK-DISC-5709
- Overhaul Seal & Consumable Kit: KIT-TRU450-CAX-9
- Critical Fastener Torque Value: 101 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-CAX-9 at PLC register 0xD268.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-BRK-DISC-5709) with seal kit (KIT-TRU450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 101 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-CAX-9 on Trumpf-TruLaser-L450 by installing PN-TRU450-BRK-DISC-5709 & kit KIT-TRU450-CAX-9. Fasteners torqued to 101 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-OPT-10

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU450-10
- PLC Hex Register Address: DB31.DBW_0xD270
- Nominal Operating Range: 42 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-GLASS-RD-5710
- Overhaul Seal & Consumable Kit: KIT-TRU450-OPT-10
- Critical Fastener Torque Value: 110 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-OPT-10 at PLC register 0xD270.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-GLASS-RD-5710) with seal kit (KIT-TRU450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 110 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-OPT-10 on Trumpf-TruLaser-L450 by installing PN-TRU450-GLASS-RD-5710 & kit KIT-TRU450-OPT-10. Fasteners torqued to 110 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-PLC-11

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU450-11
- PLC Hex Register Address: DB31.DBW_0xD278
- Nominal Operating Range: 45 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-IGBT-MOD-5711
- Overhaul Seal & Consumable Kit: KIT-TRU450-PLC-11
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-PLC-11 at PLC register 0xD278.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-IGBT-MOD-5711) with seal kit (KIT-TRU450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-PLC-11 on Trumpf-TruLaser-L450 by installing PN-TRU450-IGBT-MOD-5711 & kit KIT-TRU450-PLC-11. Fasteners torqued to 119 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-LUB-12

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU450-12
- PLC Hex Register Address: DB31.DBW_0xD280
- Nominal Operating Range: 48 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-INJ-MET-5712
- Overhaul Seal & Consumable Kit: KIT-TRU450-LUB-12
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-LUB-12 at PLC register 0xD280.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-INJ-MET-5712) with seal kit (KIT-TRU450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-LUB-12 on Trumpf-TruLaser-L450 by installing PN-TRU450-INJ-MET-5712 & kit KIT-TRU450-LUB-12. Fasteners torqued to 128 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-VAC-13

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU450-13
- PLC Hex Register Address: DB31.DBW_0xD288
- Nominal Operating Range: 51 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-GASK-VIT-5713
- Overhaul Seal & Consumable Kit: KIT-TRU450-VAC-13
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-VAC-13 at PLC register 0xD288.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-GASK-VIT-5713) with seal kit (KIT-TRU450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-VAC-13 on Trumpf-TruLaser-L450 by installing PN-TRU450-GASK-VIT-5713 & kit KIT-TRU450-VAC-13. Fasteners torqued to 137 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU450-THM-14

Target Machine Model: Trumpf-TruLaser-L450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L450 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU450-14
- PLC Hex Register Address: DB31.DBW_0xD290
- Nominal Operating Range: 54 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 121 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU450-HEX-CORE-5714
- Overhaul Seal & Consumable Kit: KIT-TRU450-THM-14
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L450 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU450-THM-14 at PLC register 0xD290.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU450-HEX-CORE-5714) with seal kit (KIT-TRU450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L450, technician must log: "Resolved ERR-TRU450-THM-14 on Trumpf-TruLaser-L450 by installing PN-TRU450-HEX-CORE-5714 & kit KIT-TRU450-THM-14. Fasteners torqued to 146 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-SPN-1

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU500-1
- PLC Hex Register Address: DB32.DBW_0xD548
- Nominal Operating Range: 15 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-BRG-CER-5801
- Overhaul Seal & Consumable Kit: KIT-TRU500-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-SPN-1 at PLC register 0xD548.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-BRG-CER-5801) with seal kit (KIT-TRU500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-SPN-1 on Trumpf-TruLaser-L500 by installing PN-TRU500-BRG-CER-5801 & kit KIT-TRU500-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-ATC-2

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU500-2
- PLC Hex Register Address: DB32.DBW_0xD550
- Nominal Operating Range: 18 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-CAM-GRIP-5802
- Overhaul Seal & Consumable Kit: KIT-TRU500-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-ATC-2 at PLC register 0xD550.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-CAM-GRIP-5802) with seal kit (KIT-TRU500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-ATC-2 on Trumpf-TruLaser-L500 by installing PN-TRU500-CAM-GRIP-5802 & kit KIT-TRU500-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-HYD-3

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU500-3
- PLC Hex Register Address: DB32.DBW_0xD558
- Nominal Operating Range: 21 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-VLV-PROP-5803
- Overhaul Seal & Consumable Kit: KIT-TRU500-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-HYD-3 at PLC register 0xD558.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-VLV-PROP-5803) with seal kit (KIT-TRU500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-HYD-3 on Trumpf-TruLaser-L500 by installing PN-TRU500-VLV-PROP-5803 & kit KIT-TRU500-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-CLN-4

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU500-4
- PLC Hex Register Address: DB32.DBW_0xD560
- Nominal Operating Range: 24 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-PMP-PLG-5804
- Overhaul Seal & Consumable Kit: KIT-TRU500-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-CLN-4 at PLC register 0xD560.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-PMP-PLG-5804) with seal kit (KIT-TRU500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-CLN-4 on Trumpf-TruLaser-L500 by installing PN-TRU500-PMP-PLG-5804 & kit KIT-TRU500-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-XAX-5

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU500-5
- PLC Hex Register Address: DB32.DBW_0xD568
- Nominal Operating Range: 27 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-LIN-COIL-5805
- Overhaul Seal & Consumable Kit: KIT-TRU500-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-XAX-5 at PLC register 0xD568.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-LIN-COIL-5805) with seal kit (KIT-TRU500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-XAX-5 on Trumpf-TruLaser-L500 by installing PN-TRU500-LIN-COIL-5805 & kit KIT-TRU500-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-YAX-6

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU500-6
- PLC Hex Register Address: DB32.DBW_0xD570
- Nominal Operating Range: 30 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-BSW-NUT-5806
- Overhaul Seal & Consumable Kit: KIT-TRU500-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-YAX-6 at PLC register 0xD570.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-BSW-NUT-5806) with seal kit (KIT-TRU500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-YAX-6 on Trumpf-TruLaser-L500 by installing PN-TRU500-BSW-NUT-5806 & kit KIT-TRU500-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-ZAX-7

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU500-7
- PLC Hex Register Address: DB32.DBW_0xD578
- Nominal Operating Range: 33 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-CYL-SEAL-5807
- Overhaul Seal & Consumable Kit: KIT-TRU500-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-ZAX-7 at PLC register 0xD578.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-CYL-SEAL-5807) with seal kit (KIT-TRU500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-ZAX-7 on Trumpf-TruLaser-L500 by installing PN-TRU500-CYL-SEAL-5807 & kit KIT-TRU500-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-BAX-8

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU500-8
- PLC Hex Register Address: DB32.DBW_0xD580
- Nominal Operating Range: 36 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-TRQ-STAT-5808
- Overhaul Seal & Consumable Kit: KIT-TRU500-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-BAX-8 at PLC register 0xD580.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-TRQ-STAT-5808) with seal kit (KIT-TRU500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-BAX-8 on Trumpf-TruLaser-L500 by installing PN-TRU500-TRQ-STAT-5808 & kit KIT-TRU500-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-CAX-9

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU500-9
- PLC Hex Register Address: DB32.DBW_0xD588
- Nominal Operating Range: 39 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-BRK-DISC-5809
- Overhaul Seal & Consumable Kit: KIT-TRU500-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-CAX-9 at PLC register 0xD588.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-BRK-DISC-5809) with seal kit (KIT-TRU500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-CAX-9 on Trumpf-TruLaser-L500 by installing PN-TRU500-BRK-DISC-5809 & kit KIT-TRU500-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-OPT-10

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU500-10
- PLC Hex Register Address: DB32.DBW_0xD590
- Nominal Operating Range: 42 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-GLASS-RD-5810
- Overhaul Seal & Consumable Kit: KIT-TRU500-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-OPT-10 at PLC register 0xD590.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-GLASS-RD-5810) with seal kit (KIT-TRU500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-OPT-10 on Trumpf-TruLaser-L500 by installing PN-TRU500-GLASS-RD-5810 & kit KIT-TRU500-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-PLC-11

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU500-11
- PLC Hex Register Address: DB32.DBW_0xD598
- Nominal Operating Range: 45 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-IGBT-MOD-5811
- Overhaul Seal & Consumable Kit: KIT-TRU500-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-PLC-11 at PLC register 0xD598.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-IGBT-MOD-5811) with seal kit (KIT-TRU500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-PLC-11 on Trumpf-TruLaser-L500 by installing PN-TRU500-IGBT-MOD-5811 & kit KIT-TRU500-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-LUB-12

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU500-12
- PLC Hex Register Address: DB32.DBW_0xD5A0
- Nominal Operating Range: 48 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-INJ-MET-5812
- Overhaul Seal & Consumable Kit: KIT-TRU500-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-LUB-12 at PLC register 0xD5A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-INJ-MET-5812) with seal kit (KIT-TRU500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-LUB-12 on Trumpf-TruLaser-L500 by installing PN-TRU500-INJ-MET-5812 & kit KIT-TRU500-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-VAC-13

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU500-13
- PLC Hex Register Address: DB32.DBW_0xD5A8
- Nominal Operating Range: 51 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-GASK-VIT-5813
- Overhaul Seal & Consumable Kit: KIT-TRU500-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-VAC-13 at PLC register 0xD5A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-GASK-VIT-5813) with seal kit (KIT-TRU500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-VAC-13 on Trumpf-TruLaser-L500 by installing PN-TRU500-GASK-VIT-5813 & kit KIT-TRU500-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU500-THM-14

Target Machine Model: Trumpf-TruLaser-L500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L500 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU500-14
- PLC Hex Register Address: DB32.DBW_0xD5B0
- Nominal Operating Range: 54 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 122 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU500-HEX-CORE-5814
- Overhaul Seal & Consumable Kit: KIT-TRU500-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L500 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU500-THM-14 at PLC register 0xD5B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU500-HEX-CORE-5814) with seal kit (KIT-TRU500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L500, technician must log: "Resolved ERR-TRU500-THM-14 on Trumpf-TruLaser-L500 by installing PN-TRU500-HEX-CORE-5814 & kit KIT-TRU500-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-SPN-1

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU550-1
- PLC Hex Register Address: DB33.DBW_0xD868
- Nominal Operating Range: 15 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-BRG-CER-5901
- Overhaul Seal & Consumable Kit: KIT-TRU550-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-SPN-1 at PLC register 0xD868.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-BRG-CER-5901) with seal kit (KIT-TRU550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-SPN-1 on Trumpf-TruLaser-L550 by installing PN-TRU550-BRG-CER-5901 & kit KIT-TRU550-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-ATC-2

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU550-2
- PLC Hex Register Address: DB33.DBW_0xD870
- Nominal Operating Range: 18 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-CAM-GRIP-5902
- Overhaul Seal & Consumable Kit: KIT-TRU550-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-ATC-2 at PLC register 0xD870.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-CAM-GRIP-5902) with seal kit (KIT-TRU550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-ATC-2 on Trumpf-TruLaser-L550 by installing PN-TRU550-CAM-GRIP-5902 & kit KIT-TRU550-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-HYD-3

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU550-3
- PLC Hex Register Address: DB33.DBW_0xD878
- Nominal Operating Range: 21 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-VLV-PROP-5903
- Overhaul Seal & Consumable Kit: KIT-TRU550-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-HYD-3 at PLC register 0xD878.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-VLV-PROP-5903) with seal kit (KIT-TRU550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-HYD-3 on Trumpf-TruLaser-L550 by installing PN-TRU550-VLV-PROP-5903 & kit KIT-TRU550-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-CLN-4

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU550-4
- PLC Hex Register Address: DB33.DBW_0xD880
- Nominal Operating Range: 24 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-PMP-PLG-5904
- Overhaul Seal & Consumable Kit: KIT-TRU550-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-CLN-4 at PLC register 0xD880.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-PMP-PLG-5904) with seal kit (KIT-TRU550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-CLN-4 on Trumpf-TruLaser-L550 by installing PN-TRU550-PMP-PLG-5904 & kit KIT-TRU550-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-XAX-5

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU550-5
- PLC Hex Register Address: DB33.DBW_0xD888
- Nominal Operating Range: 27 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-LIN-COIL-5905
- Overhaul Seal & Consumable Kit: KIT-TRU550-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-XAX-5 at PLC register 0xD888.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-LIN-COIL-5905) with seal kit (KIT-TRU550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-XAX-5 on Trumpf-TruLaser-L550 by installing PN-TRU550-LIN-COIL-5905 & kit KIT-TRU550-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-YAX-6

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU550-6
- PLC Hex Register Address: DB33.DBW_0xD890
- Nominal Operating Range: 30 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-BSW-NUT-5906
- Overhaul Seal & Consumable Kit: KIT-TRU550-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-YAX-6 at PLC register 0xD890.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-BSW-NUT-5906) with seal kit (KIT-TRU550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-YAX-6 on Trumpf-TruLaser-L550 by installing PN-TRU550-BSW-NUT-5906 & kit KIT-TRU550-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-ZAX-7

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU550-7
- PLC Hex Register Address: DB33.DBW_0xD898
- Nominal Operating Range: 33 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-CYL-SEAL-5907
- Overhaul Seal & Consumable Kit: KIT-TRU550-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-ZAX-7 at PLC register 0xD898.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-CYL-SEAL-5907) with seal kit (KIT-TRU550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-ZAX-7 on Trumpf-TruLaser-L550 by installing PN-TRU550-CYL-SEAL-5907 & kit KIT-TRU550-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-BAX-8

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU550-8
- PLC Hex Register Address: DB33.DBW_0xD8A0
- Nominal Operating Range: 36 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-TRQ-STAT-5908
- Overhaul Seal & Consumable Kit: KIT-TRU550-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-BAX-8 at PLC register 0xD8A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-TRQ-STAT-5908) with seal kit (KIT-TRU550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-BAX-8 on Trumpf-TruLaser-L550 by installing PN-TRU550-TRQ-STAT-5908 & kit KIT-TRU550-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-CAX-9

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU550-9
- PLC Hex Register Address: DB33.DBW_0xD8A8
- Nominal Operating Range: 39 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-BRK-DISC-5909
- Overhaul Seal & Consumable Kit: KIT-TRU550-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-CAX-9 at PLC register 0xD8A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-BRK-DISC-5909) with seal kit (KIT-TRU550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-CAX-9 on Trumpf-TruLaser-L550 by installing PN-TRU550-BRK-DISC-5909 & kit KIT-TRU550-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-OPT-10

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU550-10
- PLC Hex Register Address: DB33.DBW_0xD8B0
- Nominal Operating Range: 42 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-GLASS-RD-5910
- Overhaul Seal & Consumable Kit: KIT-TRU550-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-OPT-10 at PLC register 0xD8B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-GLASS-RD-5910) with seal kit (KIT-TRU550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-OPT-10 on Trumpf-TruLaser-L550 by installing PN-TRU550-GLASS-RD-5910 & kit KIT-TRU550-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-PLC-11

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU550-11
- PLC Hex Register Address: DB33.DBW_0xD8B8
- Nominal Operating Range: 45 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-IGBT-MOD-5911
- Overhaul Seal & Consumable Kit: KIT-TRU550-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-PLC-11 at PLC register 0xD8B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-IGBT-MOD-5911) with seal kit (KIT-TRU550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-PLC-11 on Trumpf-TruLaser-L550 by installing PN-TRU550-IGBT-MOD-5911 & kit KIT-TRU550-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-LUB-12

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU550-12
- PLC Hex Register Address: DB33.DBW_0xD8C0
- Nominal Operating Range: 48 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-INJ-MET-5912
- Overhaul Seal & Consumable Kit: KIT-TRU550-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-LUB-12 at PLC register 0xD8C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-INJ-MET-5912) with seal kit (KIT-TRU550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-LUB-12 on Trumpf-TruLaser-L550 by installing PN-TRU550-INJ-MET-5912 & kit KIT-TRU550-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-VAC-13

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU550-13
- PLC Hex Register Address: DB33.DBW_0xD8C8
- Nominal Operating Range: 51 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-GASK-VIT-5913
- Overhaul Seal & Consumable Kit: KIT-TRU550-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-VAC-13 at PLC register 0xD8C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-GASK-VIT-5913) with seal kit (KIT-TRU550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-VAC-13 on Trumpf-TruLaser-L550 by installing PN-TRU550-GASK-VIT-5913 & kit KIT-TRU550-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU550-THM-14

Target Machine Model: Trumpf-TruLaser-L550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L550 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU550-14
- PLC Hex Register Address: DB33.DBW_0xD8D0
- Nominal Operating Range: 54 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 123 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU550-HEX-CORE-5914
- Overhaul Seal & Consumable Kit: KIT-TRU550-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L550 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU550-THM-14 at PLC register 0xD8D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU550-HEX-CORE-5914) with seal kit (KIT-TRU550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L550, technician must log: "Resolved ERR-TRU550-THM-14 on Trumpf-TruLaser-L550 by installing PN-TRU550-HEX-CORE-5914 & kit KIT-TRU550-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-SPN-1

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PT-SPN-TRU600-1
- PLC Hex Register Address: DB34.DBW_0xDB88
- Nominal Operating Range: 15 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-BRG-CER-6001
- Overhaul Seal & Consumable Kit: KIT-TRU600-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-SPN-1 at PLC register 0xDB88.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-BRG-CER-6001) with seal kit (KIT-TRU600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-SPN-1 on Trumpf-TruLaser-L600 by installing PN-TRU600-BRG-CER-6001 & kit KIT-TRU600-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-ATC-2

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PX-ATC-TRU600-2
- PLC Hex Register Address: DB34.DBW_0xDB90
- Nominal Operating Range: 18 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-CAM-GRIP-6002
- Overhaul Seal & Consumable Kit: KIT-TRU600-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-ATC-2 at PLC register 0xDB90.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-CAM-GRIP-6002) with seal kit (KIT-TRU600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-ATC-2 on Trumpf-TruLaser-L600 by installing PN-TRU600-CAM-GRIP-6002 & kit KIT-TRU600-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-HYD-3

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-HYD-TRU600-3
- PLC Hex Register Address: DB34.DBW_0xDB98
- Nominal Operating Range: 21 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-VLV-PROP-6003
- Overhaul Seal & Consumable Kit: KIT-TRU600-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-HYD-3 at PLC register 0xDB98.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-VLV-PROP-6003) with seal kit (KIT-TRU600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-HYD-3 on Trumpf-TruLaser-L600 by installing PN-TRU600-VLV-PROP-6003 & kit KIT-TRU600-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-CLN-4

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-CLN-TRU600-4
- PLC Hex Register Address: DB34.DBW_0xDBA0
- Nominal Operating Range: 24 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-PMP-PLG-6004
- Overhaul Seal & Consumable Kit: KIT-TRU600-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-CLN-4 at PLC register 0xDBA0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-PMP-PLG-6004) with seal kit (KIT-TRU600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-CLN-4 on Trumpf-TruLaser-L600 by installing PN-TRU600-PMP-PLG-6004 & kit KIT-TRU600-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-XAX-5

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-XAX-TRU600-5
- PLC Hex Register Address: DB34.DBW_0xDBA8
- Nominal Operating Range: 27 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-LIN-COIL-6005
- Overhaul Seal & Consumable Kit: KIT-TRU600-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-XAX-5 at PLC register 0xDBA8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-LIN-COIL-6005) with seal kit (KIT-TRU600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-XAX-5 on Trumpf-TruLaser-L600 by installing PN-TRU600-LIN-COIL-6005 & kit KIT-TRU600-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-YAX-6

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-YAX-TRU600-6
- PLC Hex Register Address: DB34.DBW_0xDBB0
- Nominal Operating Range: 30 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-BSW-NUT-6006
- Overhaul Seal & Consumable Kit: KIT-TRU600-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-YAX-6 at PLC register 0xDBB0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-BSW-NUT-6006) with seal kit (KIT-TRU600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-YAX-6 on Trumpf-TruLaser-L600 by installing PN-TRU600-BSW-NUT-6006 & kit KIT-TRU600-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-ZAX-7

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-ZAX-TRU600-7
- PLC Hex Register Address: DB34.DBW_0xDBB8
- Nominal Operating Range: 33 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-CYL-SEAL-6007
- Overhaul Seal & Consumable Kit: KIT-TRU600-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-ZAX-7 at PLC register 0xDBB8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-CYL-SEAL-6007) with seal kit (KIT-TRU600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-ZAX-7 on Trumpf-TruLaser-L600 by installing PN-TRU600-CYL-SEAL-6007 & kit KIT-TRU600-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-BAX-8

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: EN-BAX-TRU600-8
- PLC Hex Register Address: DB34.DBW_0xDBC0
- Nominal Operating Range: 36 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-TRQ-STAT-6008
- Overhaul Seal & Consumable Kit: KIT-TRU600-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-BAX-8 at PLC register 0xDBC0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-TRQ-STAT-6008) with seal kit (KIT-TRU600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-BAX-8 on Trumpf-TruLaser-L600 by installing PN-TRU600-TRQ-STAT-6008 & kit KIT-TRU600-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-CAX-9

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: PR-CAX-TRU600-9
- PLC Hex Register Address: DB34.DBW_0xDBC8
- Nominal Operating Range: 39 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-BRK-DISC-6009
- Overhaul Seal & Consumable Kit: KIT-TRU600-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-CAX-9 at PLC register 0xDBC8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-BRK-DISC-6009) with seal kit (KIT-TRU600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-CAX-9 on Trumpf-TruLaser-L600 by installing PN-TRU600-BRK-DISC-6009 & kit KIT-TRU600-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-OPT-10

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: OP-SCL-TRU600-10
- PLC Hex Register Address: DB34.DBW_0xDBD0
- Nominal Operating Range: 42 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-GLASS-RD-6010
- Overhaul Seal & Consumable Kit: KIT-TRU600-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-OPT-10 at PLC register 0xDBD0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-GLASS-RD-6010) with seal kit (KIT-TRU600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-OPT-10 on Trumpf-TruLaser-L600 by installing PN-TRU600-GLASS-RD-6010 & kit KIT-TRU600-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-PLC-11

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VT-DRV-TRU600-11
- PLC Hex Register Address: DB34.DBW_0xDBD8
- Nominal Operating Range: 45 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-IGBT-MOD-6011
- Overhaul Seal & Consumable Kit: KIT-TRU600-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-PLC-11 at PLC register 0xDBD8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-IGBT-MOD-6011) with seal kit (KIT-TRU600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-PLC-11 on Trumpf-TruLaser-L600 by installing PN-TRU600-IGBT-MOD-6011 & kit KIT-TRU600-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-LUB-12

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: FL-LUB-TRU600-12
- PLC Hex Register Address: DB34.DBW_0xDBE0
- Nominal Operating Range: 48 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-INJ-MET-6012
- Overhaul Seal & Consumable Kit: KIT-TRU600-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-LUB-12 at PLC register 0xDBE0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-INJ-MET-6012) with seal kit (KIT-TRU600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-LUB-12 on Trumpf-TruLaser-L600 by installing PN-TRU600-INJ-MET-6012 & kit KIT-TRU600-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-VAC-13

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: VC-MNF-TRU600-13
- PLC Hex Register Address: DB34.DBW_0xDBE8
- Nominal Operating Range: 51 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-GASK-VIT-6013
- Overhaul Seal & Consumable Kit: KIT-TRU600-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-VAC-13 at PLC register 0xDBE8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-GASK-VIT-6013) with seal kit (KIT-TRU600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-VAC-13 on Trumpf-TruLaser-L600 by installing PN-TRU600-GASK-VIT-6013 & kit KIT-TRU600-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-TRU600-THM-14

Target Machine Model: Trumpf-TruLaser-L600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Trumpf-TruLaser-L600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Trumpf-TruLaser-L600 (Siemens 840D sl Laser)
- Telemetry Sensor ID: TM-CHL-TRU600-14
- PLC Hex Register Address: DB34.DBW_0xDBF0
- Nominal Operating Range: 54 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 124 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-TRU600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-TRU600-HEX-CORE-6014
- Overhaul Seal & Consumable Kit: KIT-TRU600-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Trumpf-TruLaser-L600 (Siemens 840D sl Laser) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Siemens 840D sl Laser and verify fault latch ERR-TRU600-THM-14 at PLC register 0xDBF0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Trumpf-TruLaser-L600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-TRU600-HEX-CORE-6014) with seal kit (KIT-TRU600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-TRU600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Trumpf-TruLaser-L600, technician must log: "Resolved ERR-TRU600-THM-14 on Trumpf-TruLaser-L600 by installing PN-TRU600-HEX-CORE-6014 & kit KIT-TRU600-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-SPN-1

Target Machine Model: Schuler-HPD-P150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH150-1
- PLC Hex Register Address: DB20.DBW_0xDEA8
- Nominal Operating Range: 15 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-BRG-CER-6101
- Overhaul Seal & Consumable Kit: KIT-SCH150-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-SPN-1 at PLC register 0xDEA8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-BRG-CER-6101) with seal kit (KIT-SCH150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-SPN-1 on Schuler-HPD-P150 by installing PN-SCH150-BRG-CER-6101 & kit KIT-SCH150-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-ATC-2

Target Machine Model: Schuler-HPD-P150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH150-2
- PLC Hex Register Address: DB20.DBW_0xDEB0
- Nominal Operating Range: 18 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-CAM-GRIP-6102
- Overhaul Seal & Consumable Kit: KIT-SCH150-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-ATC-2 at PLC register 0xDEB0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-CAM-GRIP-6102) with seal kit (KIT-SCH150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-ATC-2 on Schuler-HPD-P150 by installing PN-SCH150-CAM-GRIP-6102 & kit KIT-SCH150-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-HYD-3

Target Machine Model: Schuler-HPD-P150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH150-3
- PLC Hex Register Address: DB20.DBW_0xDEB8
- Nominal Operating Range: 21 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-VLV-PROP-6103
- Overhaul Seal & Consumable Kit: KIT-SCH150-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-HYD-3 at PLC register 0xDEB8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-VLV-PROP-6103) with seal kit (KIT-SCH150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-HYD-3 on Schuler-HPD-P150 by installing PN-SCH150-VLV-PROP-6103 & kit KIT-SCH150-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-CLN-4

Target Machine Model: Schuler-HPD-P150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH150-4
- PLC Hex Register Address: DB20.DBW_0xDEC0
- Nominal Operating Range: 24 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-PMP-PLG-6104
- Overhaul Seal & Consumable Kit: KIT-SCH150-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-CLN-4 at PLC register 0xDEC0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-PMP-PLG-6104) with seal kit (KIT-SCH150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-CLN-4 on Schuler-HPD-P150 by installing PN-SCH150-PMP-PLG-6104 & kit KIT-SCH150-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-XAX-5

Target Machine Model: Schuler-HPD-P150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH150-5
- PLC Hex Register Address: DB20.DBW_0xDEC8
- Nominal Operating Range: 27 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-LIN-COIL-6105
- Overhaul Seal & Consumable Kit: KIT-SCH150-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-XAX-5 at PLC register 0xDEC8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-LIN-COIL-6105) with seal kit (KIT-SCH150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-XAX-5 on Schuler-HPD-P150 by installing PN-SCH150-LIN-COIL-6105 & kit KIT-SCH150-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-YAX-6

Target Machine Model: Schuler-HPD-P150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH150-6
- PLC Hex Register Address: DB20.DBW_0xDED0
- Nominal Operating Range: 30 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-BSW-NUT-6106
- Overhaul Seal & Consumable Kit: KIT-SCH150-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-YAX-6 at PLC register 0xDED0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-BSW-NUT-6106) with seal kit (KIT-SCH150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-YAX-6 on Schuler-HPD-P150 by installing PN-SCH150-BSW-NUT-6106 & kit KIT-SCH150-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-ZAX-7

Target Machine Model: Schuler-HPD-P150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH150-7
- PLC Hex Register Address: DB20.DBW_0xDED8
- Nominal Operating Range: 33 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-CYL-SEAL-6107
- Overhaul Seal & Consumable Kit: KIT-SCH150-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-ZAX-7 at PLC register 0xDED8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-CYL-SEAL-6107) with seal kit (KIT-SCH150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-ZAX-7 on Schuler-HPD-P150 by installing PN-SCH150-CYL-SEAL-6107 & kit KIT-SCH150-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-BAX-8

Target Machine Model: Schuler-HPD-P150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH150-8
- PLC Hex Register Address: DB20.DBW_0xDEE0
- Nominal Operating Range: 36 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-TRQ-STAT-6108
- Overhaul Seal & Consumable Kit: KIT-SCH150-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-BAX-8 at PLC register 0xDEE0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-TRQ-STAT-6108) with seal kit (KIT-SCH150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-BAX-8 on Schuler-HPD-P150 by installing PN-SCH150-TRQ-STAT-6108 & kit KIT-SCH150-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-CAX-9

Target Machine Model: Schuler-HPD-P150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH150-9
- PLC Hex Register Address: DB20.DBW_0xDEE8
- Nominal Operating Range: 39 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-BRK-DISC-6109
- Overhaul Seal & Consumable Kit: KIT-SCH150-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-CAX-9 at PLC register 0xDEE8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-BRK-DISC-6109) with seal kit (KIT-SCH150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-CAX-9 on Schuler-HPD-P150 by installing PN-SCH150-BRK-DISC-6109 & kit KIT-SCH150-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-OPT-10

Target Machine Model: Schuler-HPD-P150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH150-10
- PLC Hex Register Address: DB20.DBW_0xDEF0
- Nominal Operating Range: 42 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-GLASS-RD-6110
- Overhaul Seal & Consumable Kit: KIT-SCH150-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-OPT-10 at PLC register 0xDEF0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-GLASS-RD-6110) with seal kit (KIT-SCH150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-OPT-10 on Schuler-HPD-P150 by installing PN-SCH150-GLASS-RD-6110 & kit KIT-SCH150-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-PLC-11

Target Machine Model: Schuler-HPD-P150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH150-11
- PLC Hex Register Address: DB20.DBW_0xDEF8
- Nominal Operating Range: 45 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-IGBT-MOD-6111
- Overhaul Seal & Consumable Kit: KIT-SCH150-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-PLC-11 at PLC register 0xDEF8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-IGBT-MOD-6111) with seal kit (KIT-SCH150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-PLC-11 on Schuler-HPD-P150 by installing PN-SCH150-IGBT-MOD-6111 & kit KIT-SCH150-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-LUB-12

Target Machine Model: Schuler-HPD-P150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH150-12
- PLC Hex Register Address: DB20.DBW_0xDF00
- Nominal Operating Range: 48 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-INJ-MET-6112
- Overhaul Seal & Consumable Kit: KIT-SCH150-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-LUB-12 at PLC register 0xDF00.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-INJ-MET-6112) with seal kit (KIT-SCH150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-LUB-12 on Schuler-HPD-P150 by installing PN-SCH150-INJ-MET-6112 & kit KIT-SCH150-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-VAC-13

Target Machine Model: Schuler-HPD-P150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH150-13
- PLC Hex Register Address: DB20.DBW_0xDF08
- Nominal Operating Range: 51 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-GASK-VIT-6113
- Overhaul Seal & Consumable Kit: KIT-SCH150-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-VAC-13 at PLC register 0xDF08.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-GASK-VIT-6113) with seal kit (KIT-SCH150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-VAC-13 on Schuler-HPD-P150 by installing PN-SCH150-GASK-VIT-6113 & kit KIT-SCH150-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH150-THM-14

Target Machine Model: Schuler-HPD-P150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P150 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH150-14
- PLC Hex Register Address: DB20.DBW_0xDF10
- Nominal Operating Range: 54 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH150-HEX-CORE-6114
- Overhaul Seal & Consumable Kit: KIT-SCH150-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P150 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH150-THM-14 at PLC register 0xDF10.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH150-HEX-CORE-6114) with seal kit (KIT-SCH150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P150, technician must log: "Resolved ERR-SCH150-THM-14 on Schuler-HPD-P150 by installing PN-SCH150-HEX-CORE-6114 & kit KIT-SCH150-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-SPN-1

Target Machine Model: Schuler-HPD-P200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH200-1
- PLC Hex Register Address: DB21.DBW_0xE1C8
- Nominal Operating Range: 15 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-BRG-CER-6201
- Overhaul Seal & Consumable Kit: KIT-SCH200-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-SPN-1 at PLC register 0xE1C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-BRG-CER-6201) with seal kit (KIT-SCH200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-SPN-1 on Schuler-HPD-P200 by installing PN-SCH200-BRG-CER-6201 & kit KIT-SCH200-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-ATC-2

Target Machine Model: Schuler-HPD-P200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH200-2
- PLC Hex Register Address: DB21.DBW_0xE1D0
- Nominal Operating Range: 18 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-CAM-GRIP-6202
- Overhaul Seal & Consumable Kit: KIT-SCH200-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-ATC-2 at PLC register 0xE1D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-CAM-GRIP-6202) with seal kit (KIT-SCH200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-ATC-2 on Schuler-HPD-P200 by installing PN-SCH200-CAM-GRIP-6202 & kit KIT-SCH200-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-HYD-3

Target Machine Model: Schuler-HPD-P200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH200-3
- PLC Hex Register Address: DB21.DBW_0xE1D8
- Nominal Operating Range: 21 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-VLV-PROP-6203
- Overhaul Seal & Consumable Kit: KIT-SCH200-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-HYD-3 at PLC register 0xE1D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-VLV-PROP-6203) with seal kit (KIT-SCH200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-HYD-3 on Schuler-HPD-P200 by installing PN-SCH200-VLV-PROP-6203 & kit KIT-SCH200-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-CLN-4

Target Machine Model: Schuler-HPD-P200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH200-4
- PLC Hex Register Address: DB21.DBW_0xE1E0
- Nominal Operating Range: 24 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-PMP-PLG-6204
- Overhaul Seal & Consumable Kit: KIT-SCH200-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-CLN-4 at PLC register 0xE1E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-PMP-PLG-6204) with seal kit (KIT-SCH200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-CLN-4 on Schuler-HPD-P200 by installing PN-SCH200-PMP-PLG-6204 & kit KIT-SCH200-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-XAX-5

Target Machine Model: Schuler-HPD-P200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH200-5
- PLC Hex Register Address: DB21.DBW_0xE1E8
- Nominal Operating Range: 27 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-LIN-COIL-6205
- Overhaul Seal & Consumable Kit: KIT-SCH200-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-XAX-5 at PLC register 0xE1E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-LIN-COIL-6205) with seal kit (KIT-SCH200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-XAX-5 on Schuler-HPD-P200 by installing PN-SCH200-LIN-COIL-6205 & kit KIT-SCH200-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-YAX-6

Target Machine Model: Schuler-HPD-P200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH200-6
- PLC Hex Register Address: DB21.DBW_0xE1F0
- Nominal Operating Range: 30 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-BSW-NUT-6206
- Overhaul Seal & Consumable Kit: KIT-SCH200-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-YAX-6 at PLC register 0xE1F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-BSW-NUT-6206) with seal kit (KIT-SCH200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-YAX-6 on Schuler-HPD-P200 by installing PN-SCH200-BSW-NUT-6206 & kit KIT-SCH200-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-ZAX-7

Target Machine Model: Schuler-HPD-P200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH200-7
- PLC Hex Register Address: DB21.DBW_0xE1F8
- Nominal Operating Range: 33 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-CYL-SEAL-6207
- Overhaul Seal & Consumable Kit: KIT-SCH200-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-ZAX-7 at PLC register 0xE1F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-CYL-SEAL-6207) with seal kit (KIT-SCH200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-ZAX-7 on Schuler-HPD-P200 by installing PN-SCH200-CYL-SEAL-6207 & kit KIT-SCH200-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-BAX-8

Target Machine Model: Schuler-HPD-P200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH200-8
- PLC Hex Register Address: DB21.DBW_0xE200
- Nominal Operating Range: 36 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-TRQ-STAT-6208
- Overhaul Seal & Consumable Kit: KIT-SCH200-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-BAX-8 at PLC register 0xE200.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-TRQ-STAT-6208) with seal kit (KIT-SCH200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-BAX-8 on Schuler-HPD-P200 by installing PN-SCH200-TRQ-STAT-6208 & kit KIT-SCH200-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-CAX-9

Target Machine Model: Schuler-HPD-P200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH200-9
- PLC Hex Register Address: DB21.DBW_0xE208
- Nominal Operating Range: 39 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-BRK-DISC-6209
- Overhaul Seal & Consumable Kit: KIT-SCH200-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-CAX-9 at PLC register 0xE208.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-BRK-DISC-6209) with seal kit (KIT-SCH200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-CAX-9 on Schuler-HPD-P200 by installing PN-SCH200-BRK-DISC-6209 & kit KIT-SCH200-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-OPT-10

Target Machine Model: Schuler-HPD-P200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH200-10
- PLC Hex Register Address: DB21.DBW_0xE210
- Nominal Operating Range: 42 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-GLASS-RD-6210
- Overhaul Seal & Consumable Kit: KIT-SCH200-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-OPT-10 at PLC register 0xE210.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-GLASS-RD-6210) with seal kit (KIT-SCH200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-OPT-10 on Schuler-HPD-P200 by installing PN-SCH200-GLASS-RD-6210 & kit KIT-SCH200-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-PLC-11

Target Machine Model: Schuler-HPD-P200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH200-11
- PLC Hex Register Address: DB21.DBW_0xE218
- Nominal Operating Range: 45 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-IGBT-MOD-6211
- Overhaul Seal & Consumable Kit: KIT-SCH200-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-PLC-11 at PLC register 0xE218.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-IGBT-MOD-6211) with seal kit (KIT-SCH200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-PLC-11 on Schuler-HPD-P200 by installing PN-SCH200-IGBT-MOD-6211 & kit KIT-SCH200-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-LUB-12

Target Machine Model: Schuler-HPD-P200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH200-12
- PLC Hex Register Address: DB21.DBW_0xE220
- Nominal Operating Range: 48 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-INJ-MET-6212
- Overhaul Seal & Consumable Kit: KIT-SCH200-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-LUB-12 at PLC register 0xE220.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-INJ-MET-6212) with seal kit (KIT-SCH200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-LUB-12 on Schuler-HPD-P200 by installing PN-SCH200-INJ-MET-6212 & kit KIT-SCH200-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-VAC-13

Target Machine Model: Schuler-HPD-P200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH200-13
- PLC Hex Register Address: DB21.DBW_0xE228
- Nominal Operating Range: 51 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-GASK-VIT-6213
- Overhaul Seal & Consumable Kit: KIT-SCH200-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-VAC-13 at PLC register 0xE228.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-GASK-VIT-6213) with seal kit (KIT-SCH200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-VAC-13 on Schuler-HPD-P200 by installing PN-SCH200-GASK-VIT-6213 & kit KIT-SCH200-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH200-THM-14

Target Machine Model: Schuler-HPD-P200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P200 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH200-14
- PLC Hex Register Address: DB21.DBW_0xE230
- Nominal Operating Range: 54 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH200-HEX-CORE-6214
- Overhaul Seal & Consumable Kit: KIT-SCH200-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P200 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH200-THM-14 at PLC register 0xE230.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH200-HEX-CORE-6214) with seal kit (KIT-SCH200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P200, technician must log: "Resolved ERR-SCH200-THM-14 on Schuler-HPD-P200 by installing PN-SCH200-HEX-CORE-6214 & kit KIT-SCH200-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-SPN-1

Target Machine Model: Schuler-HPD-P250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH250-1
- PLC Hex Register Address: DB22.DBW_0xE4E8
- Nominal Operating Range: 15 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-BRG-CER-6301
- Overhaul Seal & Consumable Kit: KIT-SCH250-SPN-1
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-SPN-1 at PLC register 0xE4E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-BRG-CER-6301) with seal kit (KIT-SCH250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-SPN-1 on Schuler-HPD-P250 by installing PN-SCH250-BRG-CER-6301 & kit KIT-SCH250-SPN-1. Fasteners torqued to 109 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-ATC-2

Target Machine Model: Schuler-HPD-P250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH250-2
- PLC Hex Register Address: DB22.DBW_0xE4F0
- Nominal Operating Range: 18 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-CAM-GRIP-6302
- Overhaul Seal & Consumable Kit: KIT-SCH250-ATC-2
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-ATC-2 at PLC register 0xE4F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-CAM-GRIP-6302) with seal kit (KIT-SCH250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-ATC-2 on Schuler-HPD-P250 by installing PN-SCH250-CAM-GRIP-6302 & kit KIT-SCH250-ATC-2. Fasteners torqued to 118 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-HYD-3

Target Machine Model: Schuler-HPD-P250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH250-3
- PLC Hex Register Address: DB22.DBW_0xE4F8
- Nominal Operating Range: 21 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-VLV-PROP-6303
- Overhaul Seal & Consumable Kit: KIT-SCH250-HYD-3
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-HYD-3 at PLC register 0xE4F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-VLV-PROP-6303) with seal kit (KIT-SCH250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-HYD-3 on Schuler-HPD-P250 by installing PN-SCH250-VLV-PROP-6303 & kit KIT-SCH250-HYD-3. Fasteners torqued to 127 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-CLN-4

Target Machine Model: Schuler-HPD-P250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH250-4
- PLC Hex Register Address: DB22.DBW_0xE500
- Nominal Operating Range: 24 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-PMP-PLG-6304
- Overhaul Seal & Consumable Kit: KIT-SCH250-CLN-4
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-CLN-4 at PLC register 0xE500.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-PMP-PLG-6304) with seal kit (KIT-SCH250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-CLN-4 on Schuler-HPD-P250 by installing PN-SCH250-PMP-PLG-6304 & kit KIT-SCH250-CLN-4. Fasteners torqued to 136 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-XAX-5

Target Machine Model: Schuler-HPD-P250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH250-5
- PLC Hex Register Address: DB22.DBW_0xE508
- Nominal Operating Range: 27 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-LIN-COIL-6305
- Overhaul Seal & Consumable Kit: KIT-SCH250-XAX-5
- Critical Fastener Torque Value: 145 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-XAX-5 at PLC register 0xE508.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-LIN-COIL-6305) with seal kit (KIT-SCH250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 145 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-XAX-5 on Schuler-HPD-P250 by installing PN-SCH250-LIN-COIL-6305 & kit KIT-SCH250-XAX-5. Fasteners torqued to 145 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-YAX-6

Target Machine Model: Schuler-HPD-P250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH250-6
- PLC Hex Register Address: DB22.DBW_0xE510
- Nominal Operating Range: 30 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-BSW-NUT-6306
- Overhaul Seal & Consumable Kit: KIT-SCH250-YAX-6
- Critical Fastener Torque Value: 154 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-YAX-6 at PLC register 0xE510.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-BSW-NUT-6306) with seal kit (KIT-SCH250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 154 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-YAX-6 on Schuler-HPD-P250 by installing PN-SCH250-BSW-NUT-6306 & kit KIT-SCH250-YAX-6. Fasteners torqued to 154 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-ZAX-7

Target Machine Model: Schuler-HPD-P250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH250-7
- PLC Hex Register Address: DB22.DBW_0xE518
- Nominal Operating Range: 33 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-CYL-SEAL-6307
- Overhaul Seal & Consumable Kit: KIT-SCH250-ZAX-7
- Critical Fastener Torque Value: 163 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-ZAX-7 at PLC register 0xE518.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-CYL-SEAL-6307) with seal kit (KIT-SCH250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 163 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-ZAX-7 on Schuler-HPD-P250 by installing PN-SCH250-CYL-SEAL-6307 & kit KIT-SCH250-ZAX-7. Fasteners torqued to 163 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-BAX-8

Target Machine Model: Schuler-HPD-P250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH250-8
- PLC Hex Register Address: DB22.DBW_0xE520
- Nominal Operating Range: 36 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-TRQ-STAT-6308
- Overhaul Seal & Consumable Kit: KIT-SCH250-BAX-8
- Critical Fastener Torque Value: 172 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-BAX-8 at PLC register 0xE520.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-TRQ-STAT-6308) with seal kit (KIT-SCH250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 172 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-BAX-8 on Schuler-HPD-P250 by installing PN-SCH250-TRQ-STAT-6308 & kit KIT-SCH250-BAX-8. Fasteners torqued to 172 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-CAX-9

Target Machine Model: Schuler-HPD-P250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH250-9
- PLC Hex Register Address: DB22.DBW_0xE528
- Nominal Operating Range: 39 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-BRK-DISC-6309
- Overhaul Seal & Consumable Kit: KIT-SCH250-CAX-9
- Critical Fastener Torque Value: 181 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-CAX-9 at PLC register 0xE528.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-BRK-DISC-6309) with seal kit (KIT-SCH250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 181 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-CAX-9 on Schuler-HPD-P250 by installing PN-SCH250-BRK-DISC-6309 & kit KIT-SCH250-CAX-9. Fasteners torqued to 181 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-OPT-10

Target Machine Model: Schuler-HPD-P250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH250-10
- PLC Hex Register Address: DB22.DBW_0xE530
- Nominal Operating Range: 42 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-GLASS-RD-6310
- Overhaul Seal & Consumable Kit: KIT-SCH250-OPT-10
- Critical Fastener Torque Value: 190 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-OPT-10 at PLC register 0xE530.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-GLASS-RD-6310) with seal kit (KIT-SCH250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 190 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-OPT-10 on Schuler-HPD-P250 by installing PN-SCH250-GLASS-RD-6310 & kit KIT-SCH250-OPT-10. Fasteners torqued to 190 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-PLC-11

Target Machine Model: Schuler-HPD-P250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH250-11
- PLC Hex Register Address: DB22.DBW_0xE538
- Nominal Operating Range: 45 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-IGBT-MOD-6311
- Overhaul Seal & Consumable Kit: KIT-SCH250-PLC-11
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-PLC-11 at PLC register 0xE538.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-IGBT-MOD-6311) with seal kit (KIT-SCH250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-PLC-11 on Schuler-HPD-P250 by installing PN-SCH250-IGBT-MOD-6311 & kit KIT-SCH250-PLC-11. Fasteners torqued to 199 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-LUB-12

Target Machine Model: Schuler-HPD-P250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH250-12
- PLC Hex Register Address: DB22.DBW_0xE540
- Nominal Operating Range: 48 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-INJ-MET-6312
- Overhaul Seal & Consumable Kit: KIT-SCH250-LUB-12
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-LUB-12 at PLC register 0xE540.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-INJ-MET-6312) with seal kit (KIT-SCH250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-LUB-12 on Schuler-HPD-P250 by installing PN-SCH250-INJ-MET-6312 & kit KIT-SCH250-LUB-12. Fasteners torqued to 208 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-VAC-13

Target Machine Model: Schuler-HPD-P250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH250-13
- PLC Hex Register Address: DB22.DBW_0xE548
- Nominal Operating Range: 51 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-GASK-VIT-6313
- Overhaul Seal & Consumable Kit: KIT-SCH250-VAC-13
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-VAC-13 at PLC register 0xE548.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-GASK-VIT-6313) with seal kit (KIT-SCH250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-VAC-13 on Schuler-HPD-P250 by installing PN-SCH250-GASK-VIT-6313 & kit KIT-SCH250-VAC-13. Fasteners torqued to 27 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH250-THM-14

Target Machine Model: Schuler-HPD-P250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P250 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH250-14
- PLC Hex Register Address: DB22.DBW_0xE550
- Nominal Operating Range: 54 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH250-HEX-CORE-6314
- Overhaul Seal & Consumable Kit: KIT-SCH250-THM-14
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P250 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH250-THM-14 at PLC register 0xE550.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH250-HEX-CORE-6314) with seal kit (KIT-SCH250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P250, technician must log: "Resolved ERR-SCH250-THM-14 on Schuler-HPD-P250 by installing PN-SCH250-HEX-CORE-6314 & kit KIT-SCH250-THM-14. Fasteners torqued to 36 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-SPN-1

Target Machine Model: Schuler-HPD-P300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH300-1
- PLC Hex Register Address: DB23.DBW_0xE808
- Nominal Operating Range: 15 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-BRG-CER-6401
- Overhaul Seal & Consumable Kit: KIT-SCH300-SPN-1
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-SPN-1 at PLC register 0xE808.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-BRG-CER-6401) with seal kit (KIT-SCH300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-SPN-1 on Schuler-HPD-P300 by installing PN-SCH300-BRG-CER-6401 & kit KIT-SCH300-SPN-1. Fasteners torqued to 59 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-ATC-2

Target Machine Model: Schuler-HPD-P300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH300-2
- PLC Hex Register Address: DB23.DBW_0xE810
- Nominal Operating Range: 18 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-CAM-GRIP-6402
- Overhaul Seal & Consumable Kit: KIT-SCH300-ATC-2
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-ATC-2 at PLC register 0xE810.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-CAM-GRIP-6402) with seal kit (KIT-SCH300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-ATC-2 on Schuler-HPD-P300 by installing PN-SCH300-CAM-GRIP-6402 & kit KIT-SCH300-ATC-2. Fasteners torqued to 68 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-HYD-3

Target Machine Model: Schuler-HPD-P300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH300-3
- PLC Hex Register Address: DB23.DBW_0xE818
- Nominal Operating Range: 21 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-VLV-PROP-6403
- Overhaul Seal & Consumable Kit: KIT-SCH300-HYD-3
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-HYD-3 at PLC register 0xE818.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-VLV-PROP-6403) with seal kit (KIT-SCH300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-HYD-3 on Schuler-HPD-P300 by installing PN-SCH300-VLV-PROP-6403 & kit KIT-SCH300-HYD-3. Fasteners torqued to 77 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-CLN-4

Target Machine Model: Schuler-HPD-P300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH300-4
- PLC Hex Register Address: DB23.DBW_0xE820
- Nominal Operating Range: 24 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-PMP-PLG-6404
- Overhaul Seal & Consumable Kit: KIT-SCH300-CLN-4
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-CLN-4 at PLC register 0xE820.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-PMP-PLG-6404) with seal kit (KIT-SCH300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-CLN-4 on Schuler-HPD-P300 by installing PN-SCH300-PMP-PLG-6404 & kit KIT-SCH300-CLN-4. Fasteners torqued to 86 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-XAX-5

Target Machine Model: Schuler-HPD-P300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH300-5
- PLC Hex Register Address: DB23.DBW_0xE828
- Nominal Operating Range: 27 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-LIN-COIL-6405
- Overhaul Seal & Consumable Kit: KIT-SCH300-XAX-5
- Critical Fastener Torque Value: 95 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-XAX-5 at PLC register 0xE828.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-LIN-COIL-6405) with seal kit (KIT-SCH300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 95 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-XAX-5 on Schuler-HPD-P300 by installing PN-SCH300-LIN-COIL-6405 & kit KIT-SCH300-XAX-5. Fasteners torqued to 95 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-YAX-6

Target Machine Model: Schuler-HPD-P300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH300-6
- PLC Hex Register Address: DB23.DBW_0xE830
- Nominal Operating Range: 30 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-BSW-NUT-6406
- Overhaul Seal & Consumable Kit: KIT-SCH300-YAX-6
- Critical Fastener Torque Value: 104 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-YAX-6 at PLC register 0xE830.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-BSW-NUT-6406) with seal kit (KIT-SCH300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 104 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-YAX-6 on Schuler-HPD-P300 by installing PN-SCH300-BSW-NUT-6406 & kit KIT-SCH300-YAX-6. Fasteners torqued to 104 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-ZAX-7

Target Machine Model: Schuler-HPD-P300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH300-7
- PLC Hex Register Address: DB23.DBW_0xE838
- Nominal Operating Range: 33 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-CYL-SEAL-6407
- Overhaul Seal & Consumable Kit: KIT-SCH300-ZAX-7
- Critical Fastener Torque Value: 113 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-ZAX-7 at PLC register 0xE838.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-CYL-SEAL-6407) with seal kit (KIT-SCH300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 113 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-ZAX-7 on Schuler-HPD-P300 by installing PN-SCH300-CYL-SEAL-6407 & kit KIT-SCH300-ZAX-7. Fasteners torqued to 113 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-BAX-8

Target Machine Model: Schuler-HPD-P300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH300-8
- PLC Hex Register Address: DB23.DBW_0xE840
- Nominal Operating Range: 36 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-TRQ-STAT-6408
- Overhaul Seal & Consumable Kit: KIT-SCH300-BAX-8
- Critical Fastener Torque Value: 122 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-BAX-8 at PLC register 0xE840.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-TRQ-STAT-6408) with seal kit (KIT-SCH300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 122 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-BAX-8 on Schuler-HPD-P300 by installing PN-SCH300-TRQ-STAT-6408 & kit KIT-SCH300-BAX-8. Fasteners torqued to 122 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-CAX-9

Target Machine Model: Schuler-HPD-P300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH300-9
- PLC Hex Register Address: DB23.DBW_0xE848
- Nominal Operating Range: 39 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-BRK-DISC-6409
- Overhaul Seal & Consumable Kit: KIT-SCH300-CAX-9
- Critical Fastener Torque Value: 131 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-CAX-9 at PLC register 0xE848.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-BRK-DISC-6409) with seal kit (KIT-SCH300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 131 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-CAX-9 on Schuler-HPD-P300 by installing PN-SCH300-BRK-DISC-6409 & kit KIT-SCH300-CAX-9. Fasteners torqued to 131 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-OPT-10

Target Machine Model: Schuler-HPD-P300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH300-10
- PLC Hex Register Address: DB23.DBW_0xE850
- Nominal Operating Range: 42 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-GLASS-RD-6410
- Overhaul Seal & Consumable Kit: KIT-SCH300-OPT-10
- Critical Fastener Torque Value: 140 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-OPT-10 at PLC register 0xE850.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-GLASS-RD-6410) with seal kit (KIT-SCH300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 140 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-OPT-10 on Schuler-HPD-P300 by installing PN-SCH300-GLASS-RD-6410 & kit KIT-SCH300-OPT-10. Fasteners torqued to 140 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-PLC-11

Target Machine Model: Schuler-HPD-P300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH300-11
- PLC Hex Register Address: DB23.DBW_0xE858
- Nominal Operating Range: 45 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-IGBT-MOD-6411
- Overhaul Seal & Consumable Kit: KIT-SCH300-PLC-11
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-PLC-11 at PLC register 0xE858.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-IGBT-MOD-6411) with seal kit (KIT-SCH300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-PLC-11 on Schuler-HPD-P300 by installing PN-SCH300-IGBT-MOD-6411 & kit KIT-SCH300-PLC-11. Fasteners torqued to 149 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-LUB-12

Target Machine Model: Schuler-HPD-P300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH300-12
- PLC Hex Register Address: DB23.DBW_0xE860
- Nominal Operating Range: 48 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-INJ-MET-6412
- Overhaul Seal & Consumable Kit: KIT-SCH300-LUB-12
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-LUB-12 at PLC register 0xE860.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-INJ-MET-6412) with seal kit (KIT-SCH300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-LUB-12 on Schuler-HPD-P300 by installing PN-SCH300-INJ-MET-6412 & kit KIT-SCH300-LUB-12. Fasteners torqued to 158 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-VAC-13

Target Machine Model: Schuler-HPD-P300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH300-13
- PLC Hex Register Address: DB23.DBW_0xE868
- Nominal Operating Range: 51 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-GASK-VIT-6413
- Overhaul Seal & Consumable Kit: KIT-SCH300-VAC-13
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-VAC-13 at PLC register 0xE868.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-GASK-VIT-6413) with seal kit (KIT-SCH300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-VAC-13 on Schuler-HPD-P300 by installing PN-SCH300-GASK-VIT-6413 & kit KIT-SCH300-VAC-13. Fasteners torqued to 167 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH300-THM-14

Target Machine Model: Schuler-HPD-P300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P300 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH300-14
- PLC Hex Register Address: DB23.DBW_0xE870
- Nominal Operating Range: 54 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH300-HEX-CORE-6414
- Overhaul Seal & Consumable Kit: KIT-SCH300-THM-14
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P300 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH300-THM-14 at PLC register 0xE870.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH300-HEX-CORE-6414) with seal kit (KIT-SCH300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P300, technician must log: "Resolved ERR-SCH300-THM-14 on Schuler-HPD-P300 by installing PN-SCH300-HEX-CORE-6414 & kit KIT-SCH300-THM-14. Fasteners torqued to 176 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-SPN-1

Target Machine Model: Schuler-HPD-P350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH350-1
- PLC Hex Register Address: DB24.DBW_0xEB28
- Nominal Operating Range: 15 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-BRG-CER-6501
- Overhaul Seal & Consumable Kit: KIT-SCH350-SPN-1
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-SPN-1 at PLC register 0xEB28.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-BRG-CER-6501) with seal kit (KIT-SCH350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-SPN-1 on Schuler-HPD-P350 by installing PN-SCH350-BRG-CER-6501 & kit KIT-SCH350-SPN-1. Fasteners torqued to 199 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-ATC-2

Target Machine Model: Schuler-HPD-P350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH350-2
- PLC Hex Register Address: DB24.DBW_0xEB30
- Nominal Operating Range: 18 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-CAM-GRIP-6502
- Overhaul Seal & Consumable Kit: KIT-SCH350-ATC-2
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-ATC-2 at PLC register 0xEB30.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-CAM-GRIP-6502) with seal kit (KIT-SCH350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-ATC-2 on Schuler-HPD-P350 by installing PN-SCH350-CAM-GRIP-6502 & kit KIT-SCH350-ATC-2. Fasteners torqued to 208 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-HYD-3

Target Machine Model: Schuler-HPD-P350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH350-3
- PLC Hex Register Address: DB24.DBW_0xEB38
- Nominal Operating Range: 21 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-VLV-PROP-6503
- Overhaul Seal & Consumable Kit: KIT-SCH350-HYD-3
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-HYD-3 at PLC register 0xEB38.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-VLV-PROP-6503) with seal kit (KIT-SCH350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-HYD-3 on Schuler-HPD-P350 by installing PN-SCH350-VLV-PROP-6503 & kit KIT-SCH350-HYD-3. Fasteners torqued to 27 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-CLN-4

Target Machine Model: Schuler-HPD-P350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH350-4
- PLC Hex Register Address: DB24.DBW_0xEB40
- Nominal Operating Range: 24 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-PMP-PLG-6504
- Overhaul Seal & Consumable Kit: KIT-SCH350-CLN-4
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-CLN-4 at PLC register 0xEB40.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-PMP-PLG-6504) with seal kit (KIT-SCH350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-CLN-4 on Schuler-HPD-P350 by installing PN-SCH350-PMP-PLG-6504 & kit KIT-SCH350-CLN-4. Fasteners torqued to 36 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-XAX-5

Target Machine Model: Schuler-HPD-P350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH350-5
- PLC Hex Register Address: DB24.DBW_0xEB48
- Nominal Operating Range: 27 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-LIN-COIL-6505
- Overhaul Seal & Consumable Kit: KIT-SCH350-XAX-5
- Critical Fastener Torque Value: 45 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-XAX-5 at PLC register 0xEB48.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-LIN-COIL-6505) with seal kit (KIT-SCH350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 45 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-XAX-5 on Schuler-HPD-P350 by installing PN-SCH350-LIN-COIL-6505 & kit KIT-SCH350-XAX-5. Fasteners torqued to 45 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-YAX-6

Target Machine Model: Schuler-HPD-P350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH350-6
- PLC Hex Register Address: DB24.DBW_0xEB50
- Nominal Operating Range: 30 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-BSW-NUT-6506
- Overhaul Seal & Consumable Kit: KIT-SCH350-YAX-6
- Critical Fastener Torque Value: 54 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-YAX-6 at PLC register 0xEB50.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-BSW-NUT-6506) with seal kit (KIT-SCH350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 54 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-YAX-6 on Schuler-HPD-P350 by installing PN-SCH350-BSW-NUT-6506 & kit KIT-SCH350-YAX-6. Fasteners torqued to 54 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-ZAX-7

Target Machine Model: Schuler-HPD-P350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH350-7
- PLC Hex Register Address: DB24.DBW_0xEB58
- Nominal Operating Range: 33 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-CYL-SEAL-6507
- Overhaul Seal & Consumable Kit: KIT-SCH350-ZAX-7
- Critical Fastener Torque Value: 63 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-ZAX-7 at PLC register 0xEB58.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-CYL-SEAL-6507) with seal kit (KIT-SCH350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 63 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-ZAX-7 on Schuler-HPD-P350 by installing PN-SCH350-CYL-SEAL-6507 & kit KIT-SCH350-ZAX-7. Fasteners torqued to 63 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-BAX-8

Target Machine Model: Schuler-HPD-P350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH350-8
- PLC Hex Register Address: DB24.DBW_0xEB60
- Nominal Operating Range: 36 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-TRQ-STAT-6508
- Overhaul Seal & Consumable Kit: KIT-SCH350-BAX-8
- Critical Fastener Torque Value: 72 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-BAX-8 at PLC register 0xEB60.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-TRQ-STAT-6508) with seal kit (KIT-SCH350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 72 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-BAX-8 on Schuler-HPD-P350 by installing PN-SCH350-TRQ-STAT-6508 & kit KIT-SCH350-BAX-8. Fasteners torqued to 72 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-CAX-9

Target Machine Model: Schuler-HPD-P350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH350-9
- PLC Hex Register Address: DB24.DBW_0xEB68
- Nominal Operating Range: 39 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-BRK-DISC-6509
- Overhaul Seal & Consumable Kit: KIT-SCH350-CAX-9
- Critical Fastener Torque Value: 81 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-CAX-9 at PLC register 0xEB68.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-BRK-DISC-6509) with seal kit (KIT-SCH350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 81 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-CAX-9 on Schuler-HPD-P350 by installing PN-SCH350-BRK-DISC-6509 & kit KIT-SCH350-CAX-9. Fasteners torqued to 81 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-OPT-10

Target Machine Model: Schuler-HPD-P350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH350-10
- PLC Hex Register Address: DB24.DBW_0xEB70
- Nominal Operating Range: 42 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-GLASS-RD-6510
- Overhaul Seal & Consumable Kit: KIT-SCH350-OPT-10
- Critical Fastener Torque Value: 90 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-OPT-10 at PLC register 0xEB70.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-GLASS-RD-6510) with seal kit (KIT-SCH350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 90 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-OPT-10 on Schuler-HPD-P350 by installing PN-SCH350-GLASS-RD-6510 & kit KIT-SCH350-OPT-10. Fasteners torqued to 90 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-PLC-11

Target Machine Model: Schuler-HPD-P350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH350-11
- PLC Hex Register Address: DB24.DBW_0xEB78
- Nominal Operating Range: 45 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-IGBT-MOD-6511
- Overhaul Seal & Consumable Kit: KIT-SCH350-PLC-11
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-PLC-11 at PLC register 0xEB78.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-IGBT-MOD-6511) with seal kit (KIT-SCH350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-PLC-11 on Schuler-HPD-P350 by installing PN-SCH350-IGBT-MOD-6511 & kit KIT-SCH350-PLC-11. Fasteners torqued to 99 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-LUB-12

Target Machine Model: Schuler-HPD-P350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH350-12
- PLC Hex Register Address: DB24.DBW_0xEB80
- Nominal Operating Range: 48 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-INJ-MET-6512
- Overhaul Seal & Consumable Kit: KIT-SCH350-LUB-12
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-LUB-12 at PLC register 0xEB80.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-INJ-MET-6512) with seal kit (KIT-SCH350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-LUB-12 on Schuler-HPD-P350 by installing PN-SCH350-INJ-MET-6512 & kit KIT-SCH350-LUB-12. Fasteners torqued to 108 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-VAC-13

Target Machine Model: Schuler-HPD-P350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH350-13
- PLC Hex Register Address: DB24.DBW_0xEB88
- Nominal Operating Range: 51 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-GASK-VIT-6513
- Overhaul Seal & Consumable Kit: KIT-SCH350-VAC-13
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-VAC-13 at PLC register 0xEB88.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-GASK-VIT-6513) with seal kit (KIT-SCH350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-VAC-13 on Schuler-HPD-P350 by installing PN-SCH350-GASK-VIT-6513 & kit KIT-SCH350-VAC-13. Fasteners torqued to 117 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH350-THM-14

Target Machine Model: Schuler-HPD-P350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P350 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH350-14
- PLC Hex Register Address: DB24.DBW_0xEB90
- Nominal Operating Range: 54 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH350-HEX-CORE-6514
- Overhaul Seal & Consumable Kit: KIT-SCH350-THM-14
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P350 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH350-THM-14 at PLC register 0xEB90.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH350-HEX-CORE-6514) with seal kit (KIT-SCH350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P350, technician must log: "Resolved ERR-SCH350-THM-14 on Schuler-HPD-P350 by installing PN-SCH350-HEX-CORE-6514 & kit KIT-SCH350-THM-14. Fasteners torqued to 126 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-SPN-1

Target Machine Model: Schuler-HPD-P400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH400-1
- PLC Hex Register Address: DB25.DBW_0xEE48
- Nominal Operating Range: 15 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-BRG-CER-6601
- Overhaul Seal & Consumable Kit: KIT-SCH400-SPN-1
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-SPN-1 at PLC register 0xEE48.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-BRG-CER-6601) with seal kit (KIT-SCH400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-SPN-1 on Schuler-HPD-P400 by installing PN-SCH400-BRG-CER-6601 & kit KIT-SCH400-SPN-1. Fasteners torqued to 149 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-ATC-2

Target Machine Model: Schuler-HPD-P400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH400-2
- PLC Hex Register Address: DB25.DBW_0xEE50
- Nominal Operating Range: 18 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-CAM-GRIP-6602
- Overhaul Seal & Consumable Kit: KIT-SCH400-ATC-2
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-ATC-2 at PLC register 0xEE50.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-CAM-GRIP-6602) with seal kit (KIT-SCH400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-ATC-2 on Schuler-HPD-P400 by installing PN-SCH400-CAM-GRIP-6602 & kit KIT-SCH400-ATC-2. Fasteners torqued to 158 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-HYD-3

Target Machine Model: Schuler-HPD-P400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH400-3
- PLC Hex Register Address: DB25.DBW_0xEE58
- Nominal Operating Range: 21 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-VLV-PROP-6603
- Overhaul Seal & Consumable Kit: KIT-SCH400-HYD-3
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-HYD-3 at PLC register 0xEE58.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-VLV-PROP-6603) with seal kit (KIT-SCH400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-HYD-3 on Schuler-HPD-P400 by installing PN-SCH400-VLV-PROP-6603 & kit KIT-SCH400-HYD-3. Fasteners torqued to 167 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-CLN-4

Target Machine Model: Schuler-HPD-P400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH400-4
- PLC Hex Register Address: DB25.DBW_0xEE60
- Nominal Operating Range: 24 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-PMP-PLG-6604
- Overhaul Seal & Consumable Kit: KIT-SCH400-CLN-4
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-CLN-4 at PLC register 0xEE60.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-PMP-PLG-6604) with seal kit (KIT-SCH400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-CLN-4 on Schuler-HPD-P400 by installing PN-SCH400-PMP-PLG-6604 & kit KIT-SCH400-CLN-4. Fasteners torqued to 176 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-XAX-5

Target Machine Model: Schuler-HPD-P400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH400-5
- PLC Hex Register Address: DB25.DBW_0xEE68
- Nominal Operating Range: 27 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-LIN-COIL-6605
- Overhaul Seal & Consumable Kit: KIT-SCH400-XAX-5
- Critical Fastener Torque Value: 185 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-XAX-5 at PLC register 0xEE68.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-LIN-COIL-6605) with seal kit (KIT-SCH400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 185 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-XAX-5 on Schuler-HPD-P400 by installing PN-SCH400-LIN-COIL-6605 & kit KIT-SCH400-XAX-5. Fasteners torqued to 185 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-YAX-6

Target Machine Model: Schuler-HPD-P400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH400-6
- PLC Hex Register Address: DB25.DBW_0xEE70
- Nominal Operating Range: 30 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-BSW-NUT-6606
- Overhaul Seal & Consumable Kit: KIT-SCH400-YAX-6
- Critical Fastener Torque Value: 194 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-YAX-6 at PLC register 0xEE70.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-BSW-NUT-6606) with seal kit (KIT-SCH400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 194 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-YAX-6 on Schuler-HPD-P400 by installing PN-SCH400-BSW-NUT-6606 & kit KIT-SCH400-YAX-6. Fasteners torqued to 194 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-ZAX-7

Target Machine Model: Schuler-HPD-P400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH400-7
- PLC Hex Register Address: DB25.DBW_0xEE78
- Nominal Operating Range: 33 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-CYL-SEAL-6607
- Overhaul Seal & Consumable Kit: KIT-SCH400-ZAX-7
- Critical Fastener Torque Value: 203 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-ZAX-7 at PLC register 0xEE78.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-CYL-SEAL-6607) with seal kit (KIT-SCH400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 203 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-ZAX-7 on Schuler-HPD-P400 by installing PN-SCH400-CYL-SEAL-6607 & kit KIT-SCH400-ZAX-7. Fasteners torqued to 203 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-BAX-8

Target Machine Model: Schuler-HPD-P400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH400-8
- PLC Hex Register Address: DB25.DBW_0xEE80
- Nominal Operating Range: 36 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-TRQ-STAT-6608
- Overhaul Seal & Consumable Kit: KIT-SCH400-BAX-8
- Critical Fastener Torque Value: 22 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-BAX-8 at PLC register 0xEE80.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-TRQ-STAT-6608) with seal kit (KIT-SCH400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 22 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-BAX-8 on Schuler-HPD-P400 by installing PN-SCH400-TRQ-STAT-6608 & kit KIT-SCH400-BAX-8. Fasteners torqued to 22 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-CAX-9

Target Machine Model: Schuler-HPD-P400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH400-9
- PLC Hex Register Address: DB25.DBW_0xEE88
- Nominal Operating Range: 39 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-BRK-DISC-6609
- Overhaul Seal & Consumable Kit: KIT-SCH400-CAX-9
- Critical Fastener Torque Value: 31 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-CAX-9 at PLC register 0xEE88.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-BRK-DISC-6609) with seal kit (KIT-SCH400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 31 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-CAX-9 on Schuler-HPD-P400 by installing PN-SCH400-BRK-DISC-6609 & kit KIT-SCH400-CAX-9. Fasteners torqued to 31 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-OPT-10

Target Machine Model: Schuler-HPD-P400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH400-10
- PLC Hex Register Address: DB25.DBW_0xEE90
- Nominal Operating Range: 42 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-GLASS-RD-6610
- Overhaul Seal & Consumable Kit: KIT-SCH400-OPT-10
- Critical Fastener Torque Value: 40 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-OPT-10 at PLC register 0xEE90.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-GLASS-RD-6610) with seal kit (KIT-SCH400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 40 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-OPT-10 on Schuler-HPD-P400 by installing PN-SCH400-GLASS-RD-6610 & kit KIT-SCH400-OPT-10. Fasteners torqued to 40 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-PLC-11

Target Machine Model: Schuler-HPD-P400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH400-11
- PLC Hex Register Address: DB25.DBW_0xEE98
- Nominal Operating Range: 45 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-IGBT-MOD-6611
- Overhaul Seal & Consumable Kit: KIT-SCH400-PLC-11
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-PLC-11 at PLC register 0xEE98.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-IGBT-MOD-6611) with seal kit (KIT-SCH400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-PLC-11 on Schuler-HPD-P400 by installing PN-SCH400-IGBT-MOD-6611 & kit KIT-SCH400-PLC-11. Fasteners torqued to 49 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-LUB-12

Target Machine Model: Schuler-HPD-P400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH400-12
- PLC Hex Register Address: DB25.DBW_0xEEA0
- Nominal Operating Range: 48 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-INJ-MET-6612
- Overhaul Seal & Consumable Kit: KIT-SCH400-LUB-12
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-LUB-12 at PLC register 0xEEA0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-INJ-MET-6612) with seal kit (KIT-SCH400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-LUB-12 on Schuler-HPD-P400 by installing PN-SCH400-INJ-MET-6612 & kit KIT-SCH400-LUB-12. Fasteners torqued to 58 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-VAC-13

Target Machine Model: Schuler-HPD-P400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH400-13
- PLC Hex Register Address: DB25.DBW_0xEEA8
- Nominal Operating Range: 51 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-GASK-VIT-6613
- Overhaul Seal & Consumable Kit: KIT-SCH400-VAC-13
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-VAC-13 at PLC register 0xEEA8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-GASK-VIT-6613) with seal kit (KIT-SCH400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-VAC-13 on Schuler-HPD-P400 by installing PN-SCH400-GASK-VIT-6613 & kit KIT-SCH400-VAC-13. Fasteners torqued to 67 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH400-THM-14

Target Machine Model: Schuler-HPD-P400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P400 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH400-14
- PLC Hex Register Address: DB25.DBW_0xEEB0
- Nominal Operating Range: 54 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH400-HEX-CORE-6614
- Overhaul Seal & Consumable Kit: KIT-SCH400-THM-14
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P400 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH400-THM-14 at PLC register 0xEEB0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH400-HEX-CORE-6614) with seal kit (KIT-SCH400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P400, technician must log: "Resolved ERR-SCH400-THM-14 on Schuler-HPD-P400 by installing PN-SCH400-HEX-CORE-6614 & kit KIT-SCH400-THM-14. Fasteners torqued to 76 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-SPN-1

Target Machine Model: Schuler-HPD-P450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH450-1
- PLC Hex Register Address: DB26.DBW_0xF168
- Nominal Operating Range: 15 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-BRG-CER-6701
- Overhaul Seal & Consumable Kit: KIT-SCH450-SPN-1
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-SPN-1 at PLC register 0xF168.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-BRG-CER-6701) with seal kit (KIT-SCH450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-SPN-1 on Schuler-HPD-P450 by installing PN-SCH450-BRG-CER-6701 & kit KIT-SCH450-SPN-1. Fasteners torqued to 99 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-ATC-2

Target Machine Model: Schuler-HPD-P450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH450-2
- PLC Hex Register Address: DB26.DBW_0xF170
- Nominal Operating Range: 18 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-CAM-GRIP-6702
- Overhaul Seal & Consumable Kit: KIT-SCH450-ATC-2
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-ATC-2 at PLC register 0xF170.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-CAM-GRIP-6702) with seal kit (KIT-SCH450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-ATC-2 on Schuler-HPD-P450 by installing PN-SCH450-CAM-GRIP-6702 & kit KIT-SCH450-ATC-2. Fasteners torqued to 108 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-HYD-3

Target Machine Model: Schuler-HPD-P450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH450-3
- PLC Hex Register Address: DB26.DBW_0xF178
- Nominal Operating Range: 21 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-VLV-PROP-6703
- Overhaul Seal & Consumable Kit: KIT-SCH450-HYD-3
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-HYD-3 at PLC register 0xF178.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-VLV-PROP-6703) with seal kit (KIT-SCH450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-HYD-3 on Schuler-HPD-P450 by installing PN-SCH450-VLV-PROP-6703 & kit KIT-SCH450-HYD-3. Fasteners torqued to 117 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-CLN-4

Target Machine Model: Schuler-HPD-P450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH450-4
- PLC Hex Register Address: DB26.DBW_0xF180
- Nominal Operating Range: 24 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-PMP-PLG-6704
- Overhaul Seal & Consumable Kit: KIT-SCH450-CLN-4
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-CLN-4 at PLC register 0xF180.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-PMP-PLG-6704) with seal kit (KIT-SCH450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-CLN-4 on Schuler-HPD-P450 by installing PN-SCH450-PMP-PLG-6704 & kit KIT-SCH450-CLN-4. Fasteners torqued to 126 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-XAX-5

Target Machine Model: Schuler-HPD-P450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH450-5
- PLC Hex Register Address: DB26.DBW_0xF188
- Nominal Operating Range: 27 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-LIN-COIL-6705
- Overhaul Seal & Consumable Kit: KIT-SCH450-XAX-5
- Critical Fastener Torque Value: 135 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-XAX-5 at PLC register 0xF188.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-LIN-COIL-6705) with seal kit (KIT-SCH450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 135 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-XAX-5 on Schuler-HPD-P450 by installing PN-SCH450-LIN-COIL-6705 & kit KIT-SCH450-XAX-5. Fasteners torqued to 135 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-YAX-6

Target Machine Model: Schuler-HPD-P450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH450-6
- PLC Hex Register Address: DB26.DBW_0xF190
- Nominal Operating Range: 30 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-BSW-NUT-6706
- Overhaul Seal & Consumable Kit: KIT-SCH450-YAX-6
- Critical Fastener Torque Value: 144 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-YAX-6 at PLC register 0xF190.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-BSW-NUT-6706) with seal kit (KIT-SCH450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 144 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-YAX-6 on Schuler-HPD-P450 by installing PN-SCH450-BSW-NUT-6706 & kit KIT-SCH450-YAX-6. Fasteners torqued to 144 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-ZAX-7

Target Machine Model: Schuler-HPD-P450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH450-7
- PLC Hex Register Address: DB26.DBW_0xF198
- Nominal Operating Range: 33 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-CYL-SEAL-6707
- Overhaul Seal & Consumable Kit: KIT-SCH450-ZAX-7
- Critical Fastener Torque Value: 153 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-ZAX-7 at PLC register 0xF198.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-CYL-SEAL-6707) with seal kit (KIT-SCH450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 153 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-ZAX-7 on Schuler-HPD-P450 by installing PN-SCH450-CYL-SEAL-6707 & kit KIT-SCH450-ZAX-7. Fasteners torqued to 153 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-BAX-8

Target Machine Model: Schuler-HPD-P450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH450-8
- PLC Hex Register Address: DB26.DBW_0xF1A0
- Nominal Operating Range: 36 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-TRQ-STAT-6708
- Overhaul Seal & Consumable Kit: KIT-SCH450-BAX-8
- Critical Fastener Torque Value: 162 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-BAX-8 at PLC register 0xF1A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-TRQ-STAT-6708) with seal kit (KIT-SCH450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 162 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-BAX-8 on Schuler-HPD-P450 by installing PN-SCH450-TRQ-STAT-6708 & kit KIT-SCH450-BAX-8. Fasteners torqued to 162 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-CAX-9

Target Machine Model: Schuler-HPD-P450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH450-9
- PLC Hex Register Address: DB26.DBW_0xF1A8
- Nominal Operating Range: 39 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-BRK-DISC-6709
- Overhaul Seal & Consumable Kit: KIT-SCH450-CAX-9
- Critical Fastener Torque Value: 171 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-CAX-9 at PLC register 0xF1A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-BRK-DISC-6709) with seal kit (KIT-SCH450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 171 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-CAX-9 on Schuler-HPD-P450 by installing PN-SCH450-BRK-DISC-6709 & kit KIT-SCH450-CAX-9. Fasteners torqued to 171 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-OPT-10

Target Machine Model: Schuler-HPD-P450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH450-10
- PLC Hex Register Address: DB26.DBW_0xF1B0
- Nominal Operating Range: 42 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-GLASS-RD-6710
- Overhaul Seal & Consumable Kit: KIT-SCH450-OPT-10
- Critical Fastener Torque Value: 180 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-OPT-10 at PLC register 0xF1B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-GLASS-RD-6710) with seal kit (KIT-SCH450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 180 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-OPT-10 on Schuler-HPD-P450 by installing PN-SCH450-GLASS-RD-6710 & kit KIT-SCH450-OPT-10. Fasteners torqued to 180 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-PLC-11

Target Machine Model: Schuler-HPD-P450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH450-11
- PLC Hex Register Address: DB26.DBW_0xF1B8
- Nominal Operating Range: 45 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-IGBT-MOD-6711
- Overhaul Seal & Consumable Kit: KIT-SCH450-PLC-11
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-PLC-11 at PLC register 0xF1B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-IGBT-MOD-6711) with seal kit (KIT-SCH450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-PLC-11 on Schuler-HPD-P450 by installing PN-SCH450-IGBT-MOD-6711 & kit KIT-SCH450-PLC-11. Fasteners torqued to 189 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-LUB-12

Target Machine Model: Schuler-HPD-P450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH450-12
- PLC Hex Register Address: DB26.DBW_0xF1C0
- Nominal Operating Range: 48 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-INJ-MET-6712
- Overhaul Seal & Consumable Kit: KIT-SCH450-LUB-12
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-LUB-12 at PLC register 0xF1C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-INJ-MET-6712) with seal kit (KIT-SCH450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-LUB-12 on Schuler-HPD-P450 by installing PN-SCH450-INJ-MET-6712 & kit KIT-SCH450-LUB-12. Fasteners torqued to 198 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-VAC-13

Target Machine Model: Schuler-HPD-P450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH450-13
- PLC Hex Register Address: DB26.DBW_0xF1C8
- Nominal Operating Range: 51 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-GASK-VIT-6713
- Overhaul Seal & Consumable Kit: KIT-SCH450-VAC-13
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-VAC-13 at PLC register 0xF1C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-GASK-VIT-6713) with seal kit (KIT-SCH450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-VAC-13 on Schuler-HPD-P450 by installing PN-SCH450-GASK-VIT-6713 & kit KIT-SCH450-VAC-13. Fasteners torqued to 207 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH450-THM-14

Target Machine Model: Schuler-HPD-P450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P450 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH450-14
- PLC Hex Register Address: DB26.DBW_0xF1D0
- Nominal Operating Range: 54 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH450-HEX-CORE-6714
- Overhaul Seal & Consumable Kit: KIT-SCH450-THM-14
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P450 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH450-THM-14 at PLC register 0xF1D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH450-HEX-CORE-6714) with seal kit (KIT-SCH450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P450, technician must log: "Resolved ERR-SCH450-THM-14 on Schuler-HPD-P450 by installing PN-SCH450-HEX-CORE-6714 & kit KIT-SCH450-THM-14. Fasteners torqued to 26 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-SPN-1

Target Machine Model: Schuler-HPD-P500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH500-1
- PLC Hex Register Address: DB27.DBW_0xF488
- Nominal Operating Range: 15 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-BRG-CER-6801
- Overhaul Seal & Consumable Kit: KIT-SCH500-SPN-1
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-SPN-1 at PLC register 0xF488.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-BRG-CER-6801) with seal kit (KIT-SCH500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-SPN-1 on Schuler-HPD-P500 by installing PN-SCH500-BRG-CER-6801 & kit KIT-SCH500-SPN-1. Fasteners torqued to 49 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-ATC-2

Target Machine Model: Schuler-HPD-P500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH500-2
- PLC Hex Register Address: DB27.DBW_0xF490
- Nominal Operating Range: 18 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-CAM-GRIP-6802
- Overhaul Seal & Consumable Kit: KIT-SCH500-ATC-2
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-ATC-2 at PLC register 0xF490.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-CAM-GRIP-6802) with seal kit (KIT-SCH500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-ATC-2 on Schuler-HPD-P500 by installing PN-SCH500-CAM-GRIP-6802 & kit KIT-SCH500-ATC-2. Fasteners torqued to 58 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-HYD-3

Target Machine Model: Schuler-HPD-P500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH500-3
- PLC Hex Register Address: DB27.DBW_0xF498
- Nominal Operating Range: 21 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-VLV-PROP-6803
- Overhaul Seal & Consumable Kit: KIT-SCH500-HYD-3
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-HYD-3 at PLC register 0xF498.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-VLV-PROP-6803) with seal kit (KIT-SCH500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-HYD-3 on Schuler-HPD-P500 by installing PN-SCH500-VLV-PROP-6803 & kit KIT-SCH500-HYD-3. Fasteners torqued to 67 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-CLN-4

Target Machine Model: Schuler-HPD-P500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH500-4
- PLC Hex Register Address: DB27.DBW_0xF4A0
- Nominal Operating Range: 24 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-PMP-PLG-6804
- Overhaul Seal & Consumable Kit: KIT-SCH500-CLN-4
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-CLN-4 at PLC register 0xF4A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-PMP-PLG-6804) with seal kit (KIT-SCH500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-CLN-4 on Schuler-HPD-P500 by installing PN-SCH500-PMP-PLG-6804 & kit KIT-SCH500-CLN-4. Fasteners torqued to 76 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-XAX-5

Target Machine Model: Schuler-HPD-P500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH500-5
- PLC Hex Register Address: DB27.DBW_0xF4A8
- Nominal Operating Range: 27 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-LIN-COIL-6805
- Overhaul Seal & Consumable Kit: KIT-SCH500-XAX-5
- Critical Fastener Torque Value: 85 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-XAX-5 at PLC register 0xF4A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-LIN-COIL-6805) with seal kit (KIT-SCH500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 85 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-XAX-5 on Schuler-HPD-P500 by installing PN-SCH500-LIN-COIL-6805 & kit KIT-SCH500-XAX-5. Fasteners torqued to 85 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-YAX-6

Target Machine Model: Schuler-HPD-P500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH500-6
- PLC Hex Register Address: DB27.DBW_0xF4B0
- Nominal Operating Range: 30 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-BSW-NUT-6806
- Overhaul Seal & Consumable Kit: KIT-SCH500-YAX-6
- Critical Fastener Torque Value: 94 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-YAX-6 at PLC register 0xF4B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-BSW-NUT-6806) with seal kit (KIT-SCH500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 94 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-YAX-6 on Schuler-HPD-P500 by installing PN-SCH500-BSW-NUT-6806 & kit KIT-SCH500-YAX-6. Fasteners torqued to 94 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-ZAX-7

Target Machine Model: Schuler-HPD-P500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH500-7
- PLC Hex Register Address: DB27.DBW_0xF4B8
- Nominal Operating Range: 33 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-CYL-SEAL-6807
- Overhaul Seal & Consumable Kit: KIT-SCH500-ZAX-7
- Critical Fastener Torque Value: 103 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-ZAX-7 at PLC register 0xF4B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-CYL-SEAL-6807) with seal kit (KIT-SCH500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 103 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-ZAX-7 on Schuler-HPD-P500 by installing PN-SCH500-CYL-SEAL-6807 & kit KIT-SCH500-ZAX-7. Fasteners torqued to 103 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-BAX-8

Target Machine Model: Schuler-HPD-P500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH500-8
- PLC Hex Register Address: DB27.DBW_0xF4C0
- Nominal Operating Range: 36 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-TRQ-STAT-6808
- Overhaul Seal & Consumable Kit: KIT-SCH500-BAX-8
- Critical Fastener Torque Value: 112 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-BAX-8 at PLC register 0xF4C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-TRQ-STAT-6808) with seal kit (KIT-SCH500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 112 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-BAX-8 on Schuler-HPD-P500 by installing PN-SCH500-TRQ-STAT-6808 & kit KIT-SCH500-BAX-8. Fasteners torqued to 112 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-CAX-9

Target Machine Model: Schuler-HPD-P500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH500-9
- PLC Hex Register Address: DB27.DBW_0xF4C8
- Nominal Operating Range: 39 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-BRK-DISC-6809
- Overhaul Seal & Consumable Kit: KIT-SCH500-CAX-9
- Critical Fastener Torque Value: 121 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-CAX-9 at PLC register 0xF4C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-BRK-DISC-6809) with seal kit (KIT-SCH500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 121 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-CAX-9 on Schuler-HPD-P500 by installing PN-SCH500-BRK-DISC-6809 & kit KIT-SCH500-CAX-9. Fasteners torqued to 121 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-OPT-10

Target Machine Model: Schuler-HPD-P500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH500-10
- PLC Hex Register Address: DB27.DBW_0xF4D0
- Nominal Operating Range: 42 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-GLASS-RD-6810
- Overhaul Seal & Consumable Kit: KIT-SCH500-OPT-10
- Critical Fastener Torque Value: 130 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-OPT-10 at PLC register 0xF4D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-GLASS-RD-6810) with seal kit (KIT-SCH500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 130 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-OPT-10 on Schuler-HPD-P500 by installing PN-SCH500-GLASS-RD-6810 & kit KIT-SCH500-OPT-10. Fasteners torqued to 130 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-PLC-11

Target Machine Model: Schuler-HPD-P500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH500-11
- PLC Hex Register Address: DB27.DBW_0xF4D8
- Nominal Operating Range: 45 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-IGBT-MOD-6811
- Overhaul Seal & Consumable Kit: KIT-SCH500-PLC-11
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-PLC-11 at PLC register 0xF4D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-IGBT-MOD-6811) with seal kit (KIT-SCH500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-PLC-11 on Schuler-HPD-P500 by installing PN-SCH500-IGBT-MOD-6811 & kit KIT-SCH500-PLC-11. Fasteners torqued to 139 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-LUB-12

Target Machine Model: Schuler-HPD-P500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH500-12
- PLC Hex Register Address: DB27.DBW_0xF4E0
- Nominal Operating Range: 48 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-INJ-MET-6812
- Overhaul Seal & Consumable Kit: KIT-SCH500-LUB-12
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-LUB-12 at PLC register 0xF4E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-INJ-MET-6812) with seal kit (KIT-SCH500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-LUB-12 on Schuler-HPD-P500 by installing PN-SCH500-INJ-MET-6812 & kit KIT-SCH500-LUB-12. Fasteners torqued to 148 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-VAC-13

Target Machine Model: Schuler-HPD-P500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH500-13
- PLC Hex Register Address: DB27.DBW_0xF4E8
- Nominal Operating Range: 51 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-GASK-VIT-6813
- Overhaul Seal & Consumable Kit: KIT-SCH500-VAC-13
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-VAC-13 at PLC register 0xF4E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-GASK-VIT-6813) with seal kit (KIT-SCH500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-VAC-13 on Schuler-HPD-P500 by installing PN-SCH500-GASK-VIT-6813 & kit KIT-SCH500-VAC-13. Fasteners torqued to 157 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH500-THM-14

Target Machine Model: Schuler-HPD-P500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P500 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH500-14
- PLC Hex Register Address: DB27.DBW_0xF4F0
- Nominal Operating Range: 54 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH500-HEX-CORE-6814
- Overhaul Seal & Consumable Kit: KIT-SCH500-THM-14
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P500 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH500-THM-14 at PLC register 0xF4F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH500-HEX-CORE-6814) with seal kit (KIT-SCH500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P500, technician must log: "Resolved ERR-SCH500-THM-14 on Schuler-HPD-P500 by installing PN-SCH500-HEX-CORE-6814 & kit KIT-SCH500-THM-14. Fasteners torqued to 166 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-SPN-1

Target Machine Model: Schuler-HPD-P550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH550-1
- PLC Hex Register Address: DB28.DBW_0xF7A8
- Nominal Operating Range: 15 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-BRG-CER-6901
- Overhaul Seal & Consumable Kit: KIT-SCH550-SPN-1
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-SPN-1 at PLC register 0xF7A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-BRG-CER-6901) with seal kit (KIT-SCH550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-SPN-1 on Schuler-HPD-P550 by installing PN-SCH550-BRG-CER-6901 & kit KIT-SCH550-SPN-1. Fasteners torqued to 189 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-ATC-2

Target Machine Model: Schuler-HPD-P550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH550-2
- PLC Hex Register Address: DB28.DBW_0xF7B0
- Nominal Operating Range: 18 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-CAM-GRIP-6902
- Overhaul Seal & Consumable Kit: KIT-SCH550-ATC-2
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-ATC-2 at PLC register 0xF7B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-CAM-GRIP-6902) with seal kit (KIT-SCH550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-ATC-2 on Schuler-HPD-P550 by installing PN-SCH550-CAM-GRIP-6902 & kit KIT-SCH550-ATC-2. Fasteners torqued to 198 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-HYD-3

Target Machine Model: Schuler-HPD-P550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH550-3
- PLC Hex Register Address: DB28.DBW_0xF7B8
- Nominal Operating Range: 21 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-VLV-PROP-6903
- Overhaul Seal & Consumable Kit: KIT-SCH550-HYD-3
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-HYD-3 at PLC register 0xF7B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-VLV-PROP-6903) with seal kit (KIT-SCH550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-HYD-3 on Schuler-HPD-P550 by installing PN-SCH550-VLV-PROP-6903 & kit KIT-SCH550-HYD-3. Fasteners torqued to 207 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-CLN-4

Target Machine Model: Schuler-HPD-P550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH550-4
- PLC Hex Register Address: DB28.DBW_0xF7C0
- Nominal Operating Range: 24 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-PMP-PLG-6904
- Overhaul Seal & Consumable Kit: KIT-SCH550-CLN-4
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-CLN-4 at PLC register 0xF7C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-PMP-PLG-6904) with seal kit (KIT-SCH550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-CLN-4 on Schuler-HPD-P550 by installing PN-SCH550-PMP-PLG-6904 & kit KIT-SCH550-CLN-4. Fasteners torqued to 26 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-XAX-5

Target Machine Model: Schuler-HPD-P550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH550-5
- PLC Hex Register Address: DB28.DBW_0xF7C8
- Nominal Operating Range: 27 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-LIN-COIL-6905
- Overhaul Seal & Consumable Kit: KIT-SCH550-XAX-5
- Critical Fastener Torque Value: 35 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-XAX-5 at PLC register 0xF7C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-LIN-COIL-6905) with seal kit (KIT-SCH550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 35 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-XAX-5 on Schuler-HPD-P550 by installing PN-SCH550-LIN-COIL-6905 & kit KIT-SCH550-XAX-5. Fasteners torqued to 35 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-YAX-6

Target Machine Model: Schuler-HPD-P550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH550-6
- PLC Hex Register Address: DB28.DBW_0xF7D0
- Nominal Operating Range: 30 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-BSW-NUT-6906
- Overhaul Seal & Consumable Kit: KIT-SCH550-YAX-6
- Critical Fastener Torque Value: 44 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-YAX-6 at PLC register 0xF7D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-BSW-NUT-6906) with seal kit (KIT-SCH550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 44 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-YAX-6 on Schuler-HPD-P550 by installing PN-SCH550-BSW-NUT-6906 & kit KIT-SCH550-YAX-6. Fasteners torqued to 44 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-ZAX-7

Target Machine Model: Schuler-HPD-P550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH550-7
- PLC Hex Register Address: DB28.DBW_0xF7D8
- Nominal Operating Range: 33 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-CYL-SEAL-6907
- Overhaul Seal & Consumable Kit: KIT-SCH550-ZAX-7
- Critical Fastener Torque Value: 53 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-ZAX-7 at PLC register 0xF7D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-CYL-SEAL-6907) with seal kit (KIT-SCH550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 53 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-ZAX-7 on Schuler-HPD-P550 by installing PN-SCH550-CYL-SEAL-6907 & kit KIT-SCH550-ZAX-7. Fasteners torqued to 53 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-BAX-8

Target Machine Model: Schuler-HPD-P550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH550-8
- PLC Hex Register Address: DB28.DBW_0xF7E0
- Nominal Operating Range: 36 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-TRQ-STAT-6908
- Overhaul Seal & Consumable Kit: KIT-SCH550-BAX-8
- Critical Fastener Torque Value: 62 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-BAX-8 at PLC register 0xF7E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-TRQ-STAT-6908) with seal kit (KIT-SCH550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 62 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-BAX-8 on Schuler-HPD-P550 by installing PN-SCH550-TRQ-STAT-6908 & kit KIT-SCH550-BAX-8. Fasteners torqued to 62 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-CAX-9

Target Machine Model: Schuler-HPD-P550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH550-9
- PLC Hex Register Address: DB28.DBW_0xF7E8
- Nominal Operating Range: 39 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-BRK-DISC-6909
- Overhaul Seal & Consumable Kit: KIT-SCH550-CAX-9
- Critical Fastener Torque Value: 71 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-CAX-9 at PLC register 0xF7E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-BRK-DISC-6909) with seal kit (KIT-SCH550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 71 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-CAX-9 on Schuler-HPD-P550 by installing PN-SCH550-BRK-DISC-6909 & kit KIT-SCH550-CAX-9. Fasteners torqued to 71 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-OPT-10

Target Machine Model: Schuler-HPD-P550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH550-10
- PLC Hex Register Address: DB28.DBW_0xF7F0
- Nominal Operating Range: 42 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-GLASS-RD-6910
- Overhaul Seal & Consumable Kit: KIT-SCH550-OPT-10
- Critical Fastener Torque Value: 80 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-OPT-10 at PLC register 0xF7F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-GLASS-RD-6910) with seal kit (KIT-SCH550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 80 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-OPT-10 on Schuler-HPD-P550 by installing PN-SCH550-GLASS-RD-6910 & kit KIT-SCH550-OPT-10. Fasteners torqued to 80 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-PLC-11

Target Machine Model: Schuler-HPD-P550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH550-11
- PLC Hex Register Address: DB28.DBW_0xF7F8
- Nominal Operating Range: 45 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-IGBT-MOD-6911
- Overhaul Seal & Consumable Kit: KIT-SCH550-PLC-11
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-PLC-11 at PLC register 0xF7F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-IGBT-MOD-6911) with seal kit (KIT-SCH550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-PLC-11 on Schuler-HPD-P550 by installing PN-SCH550-IGBT-MOD-6911 & kit KIT-SCH550-PLC-11. Fasteners torqued to 89 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-LUB-12

Target Machine Model: Schuler-HPD-P550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH550-12
- PLC Hex Register Address: DB28.DBW_0xF800
- Nominal Operating Range: 48 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-INJ-MET-6912
- Overhaul Seal & Consumable Kit: KIT-SCH550-LUB-12
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-LUB-12 at PLC register 0xF800.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-INJ-MET-6912) with seal kit (KIT-SCH550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-LUB-12 on Schuler-HPD-P550 by installing PN-SCH550-INJ-MET-6912 & kit KIT-SCH550-LUB-12. Fasteners torqued to 98 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-VAC-13

Target Machine Model: Schuler-HPD-P550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH550-13
- PLC Hex Register Address: DB28.DBW_0xF808
- Nominal Operating Range: 51 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-GASK-VIT-6913
- Overhaul Seal & Consumable Kit: KIT-SCH550-VAC-13
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-VAC-13 at PLC register 0xF808.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-GASK-VIT-6913) with seal kit (KIT-SCH550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-VAC-13 on Schuler-HPD-P550 by installing PN-SCH550-GASK-VIT-6913 & kit KIT-SCH550-VAC-13. Fasteners torqued to 107 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH550-THM-14

Target Machine Model: Schuler-HPD-P550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P550 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH550-14
- PLC Hex Register Address: DB28.DBW_0xF810
- Nominal Operating Range: 54 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH550-HEX-CORE-6914
- Overhaul Seal & Consumable Kit: KIT-SCH550-THM-14
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P550 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH550-THM-14 at PLC register 0xF810.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH550-HEX-CORE-6914) with seal kit (KIT-SCH550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P550, technician must log: "Resolved ERR-SCH550-THM-14 on Schuler-HPD-P550 by installing PN-SCH550-HEX-CORE-6914 & kit KIT-SCH550-THM-14. Fasteners torqued to 116 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-SPN-1

Target Machine Model: Schuler-HPD-P600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PT-SPN-SCH600-1
- PLC Hex Register Address: DB29.DBW_0xFAC8
- Nominal Operating Range: 15 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-BRG-CER-7001
- Overhaul Seal & Consumable Kit: KIT-SCH600-SPN-1
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-SPN-1 at PLC register 0xFAC8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-BRG-CER-7001) with seal kit (KIT-SCH600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-SPN-1 on Schuler-HPD-P600 by installing PN-SCH600-BRG-CER-7001 & kit KIT-SCH600-SPN-1. Fasteners torqued to 139 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-ATC-2

Target Machine Model: Schuler-HPD-P600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PX-ATC-SCH600-2
- PLC Hex Register Address: DB29.DBW_0xFAD0
- Nominal Operating Range: 18 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-CAM-GRIP-7002
- Overhaul Seal & Consumable Kit: KIT-SCH600-ATC-2
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-ATC-2 at PLC register 0xFAD0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-CAM-GRIP-7002) with seal kit (KIT-SCH600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-ATC-2 on Schuler-HPD-P600 by installing PN-SCH600-CAM-GRIP-7002 & kit KIT-SCH600-ATC-2. Fasteners torqued to 148 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-HYD-3

Target Machine Model: Schuler-HPD-P600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-HYD-SCH600-3
- PLC Hex Register Address: DB29.DBW_0xFAD8
- Nominal Operating Range: 21 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-VLV-PROP-7003
- Overhaul Seal & Consumable Kit: KIT-SCH600-HYD-3
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-HYD-3 at PLC register 0xFAD8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-VLV-PROP-7003) with seal kit (KIT-SCH600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-HYD-3 on Schuler-HPD-P600 by installing PN-SCH600-VLV-PROP-7003 & kit KIT-SCH600-HYD-3. Fasteners torqued to 157 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-CLN-4

Target Machine Model: Schuler-HPD-P600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-CLN-SCH600-4
- PLC Hex Register Address: DB29.DBW_0xFAE0
- Nominal Operating Range: 24 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-PMP-PLG-7004
- Overhaul Seal & Consumable Kit: KIT-SCH600-CLN-4
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-CLN-4 at PLC register 0xFAE0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-PMP-PLG-7004) with seal kit (KIT-SCH600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-CLN-4 on Schuler-HPD-P600 by installing PN-SCH600-PMP-PLG-7004 & kit KIT-SCH600-CLN-4. Fasteners torqued to 166 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-XAX-5

Target Machine Model: Schuler-HPD-P600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-XAX-SCH600-5
- PLC Hex Register Address: DB29.DBW_0xFAE8
- Nominal Operating Range: 27 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-LIN-COIL-7005
- Overhaul Seal & Consumable Kit: KIT-SCH600-XAX-5
- Critical Fastener Torque Value: 175 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-XAX-5 at PLC register 0xFAE8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-LIN-COIL-7005) with seal kit (KIT-SCH600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 175 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-XAX-5 on Schuler-HPD-P600 by installing PN-SCH600-LIN-COIL-7005 & kit KIT-SCH600-XAX-5. Fasteners torqued to 175 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-YAX-6

Target Machine Model: Schuler-HPD-P600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-YAX-SCH600-6
- PLC Hex Register Address: DB29.DBW_0xFAF0
- Nominal Operating Range: 30 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-BSW-NUT-7006
- Overhaul Seal & Consumable Kit: KIT-SCH600-YAX-6
- Critical Fastener Torque Value: 184 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-YAX-6 at PLC register 0xFAF0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-BSW-NUT-7006) with seal kit (KIT-SCH600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 184 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-YAX-6 on Schuler-HPD-P600 by installing PN-SCH600-BSW-NUT-7006 & kit KIT-SCH600-YAX-6. Fasteners torqued to 184 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-ZAX-7

Target Machine Model: Schuler-HPD-P600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-ZAX-SCH600-7
- PLC Hex Register Address: DB29.DBW_0xFAF8
- Nominal Operating Range: 33 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-CYL-SEAL-7007
- Overhaul Seal & Consumable Kit: KIT-SCH600-ZAX-7
- Critical Fastener Torque Value: 193 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-ZAX-7 at PLC register 0xFAF8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-CYL-SEAL-7007) with seal kit (KIT-SCH600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 193 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-ZAX-7 on Schuler-HPD-P600 by installing PN-SCH600-CYL-SEAL-7007 & kit KIT-SCH600-ZAX-7. Fasteners torqued to 193 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-BAX-8

Target Machine Model: Schuler-HPD-P600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: EN-BAX-SCH600-8
- PLC Hex Register Address: DB29.DBW_0xFB00
- Nominal Operating Range: 36 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-TRQ-STAT-7008
- Overhaul Seal & Consumable Kit: KIT-SCH600-BAX-8
- Critical Fastener Torque Value: 202 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-BAX-8 at PLC register 0xFB00.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-TRQ-STAT-7008) with seal kit (KIT-SCH600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 202 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-BAX-8 on Schuler-HPD-P600 by installing PN-SCH600-TRQ-STAT-7008 & kit KIT-SCH600-BAX-8. Fasteners torqued to 202 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-CAX-9

Target Machine Model: Schuler-HPD-P600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: PR-CAX-SCH600-9
- PLC Hex Register Address: DB29.DBW_0xFB08
- Nominal Operating Range: 39 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-BRK-DISC-7009
- Overhaul Seal & Consumable Kit: KIT-SCH600-CAX-9
- Critical Fastener Torque Value: 21 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-CAX-9 at PLC register 0xFB08.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-BRK-DISC-7009) with seal kit (KIT-SCH600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 21 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-CAX-9 on Schuler-HPD-P600 by installing PN-SCH600-BRK-DISC-7009 & kit KIT-SCH600-CAX-9. Fasteners torqued to 21 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-OPT-10

Target Machine Model: Schuler-HPD-P600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: OP-SCL-SCH600-10
- PLC Hex Register Address: DB29.DBW_0xFB10
- Nominal Operating Range: 42 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-GLASS-RD-7010
- Overhaul Seal & Consumable Kit: KIT-SCH600-OPT-10
- Critical Fastener Torque Value: 30 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-OPT-10 at PLC register 0xFB10.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-GLASS-RD-7010) with seal kit (KIT-SCH600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 30 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-OPT-10 on Schuler-HPD-P600 by installing PN-SCH600-GLASS-RD-7010 & kit KIT-SCH600-OPT-10. Fasteners torqued to 30 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-PLC-11

Target Machine Model: Schuler-HPD-P600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VT-DRV-SCH600-11
- PLC Hex Register Address: DB29.DBW_0xFB18
- Nominal Operating Range: 45 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-IGBT-MOD-7011
- Overhaul Seal & Consumable Kit: KIT-SCH600-PLC-11
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-PLC-11 at PLC register 0xFB18.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-IGBT-MOD-7011) with seal kit (KIT-SCH600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-PLC-11 on Schuler-HPD-P600 by installing PN-SCH600-IGBT-MOD-7011 & kit KIT-SCH600-PLC-11. Fasteners torqued to 39 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-LUB-12

Target Machine Model: Schuler-HPD-P600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: FL-LUB-SCH600-12
- PLC Hex Register Address: DB29.DBW_0xFB20
- Nominal Operating Range: 48 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-INJ-MET-7012
- Overhaul Seal & Consumable Kit: KIT-SCH600-LUB-12
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-LUB-12 at PLC register 0xFB20.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-INJ-MET-7012) with seal kit (KIT-SCH600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-LUB-12 on Schuler-HPD-P600 by installing PN-SCH600-INJ-MET-7012 & kit KIT-SCH600-LUB-12. Fasteners torqued to 48 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-VAC-13

Target Machine Model: Schuler-HPD-P600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: VC-MNF-SCH600-13
- PLC Hex Register Address: DB29.DBW_0xFB28
- Nominal Operating Range: 51 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-GASK-VIT-7013
- Overhaul Seal & Consumable Kit: KIT-SCH600-VAC-13
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-VAC-13 at PLC register 0xFB28.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-GASK-VIT-7013) with seal kit (KIT-SCH600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-VAC-13 on Schuler-HPD-P600 by installing PN-SCH600-GASK-VIT-7013 & kit KIT-SCH600-VAC-13. Fasteners torqued to 57 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SCH600-THM-14

Target Machine Model: Schuler-HPD-P600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Schuler-HPD-P600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Schuler-HPD-P600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Schuler-HPD-P600 (Beckhoff TwinCAT 3)
- Telemetry Sensor ID: TM-CHL-SCH600-14
- PLC Hex Register Address: DB29.DBW_0xFB30
- Nominal Operating Range: 54 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SCH600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SCH600-HEX-CORE-7014
- Overhaul Seal & Consumable Kit: KIT-SCH600-THM-14
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Schuler-HPD-P600 (Beckhoff TwinCAT 3) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Beckhoff TwinCAT 3 and verify fault latch ERR-SCH600-THM-14 at PLC register 0xFB30.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Schuler-HPD-P600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SCH600-HEX-CORE-7014) with seal kit (KIT-SCH600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SCH600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Schuler-HPD-P600, technician must log: "Resolved ERR-SCH600-THM-14 on Schuler-HPD-P600 by installing PN-SCH600-HEX-CORE-7014 & kit KIT-SCH600-THM-14. Fasteners torqued to 66 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-SPN-1

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO150-1
- PLC Hex Register Address: DB30.DBW_0xFDE8
- Nominal Operating Range: 15 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-BRG-CER-7101
- Overhaul Seal & Consumable Kit: KIT-GRO150-SPN-1
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-SPN-1 at PLC register 0xFDE8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-BRG-CER-7101) with seal kit (KIT-GRO150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-SPN-1 on Grob-Aerospace-G150 by installing PN-GRO150-BRG-CER-7101 & kit KIT-GRO150-SPN-1. Fasteners torqued to 89 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-ATC-2

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO150-2
- PLC Hex Register Address: DB30.DBW_0xFDF0
- Nominal Operating Range: 18 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-CAM-GRIP-7102
- Overhaul Seal & Consumable Kit: KIT-GRO150-ATC-2
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-ATC-2 at PLC register 0xFDF0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-CAM-GRIP-7102) with seal kit (KIT-GRO150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-ATC-2 on Grob-Aerospace-G150 by installing PN-GRO150-CAM-GRIP-7102 & kit KIT-GRO150-ATC-2. Fasteners torqued to 98 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-HYD-3

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO150-3
- PLC Hex Register Address: DB30.DBW_0xFDF8
- Nominal Operating Range: 21 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-VLV-PROP-7103
- Overhaul Seal & Consumable Kit: KIT-GRO150-HYD-3
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-HYD-3 at PLC register 0xFDF8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-VLV-PROP-7103) with seal kit (KIT-GRO150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-HYD-3 on Grob-Aerospace-G150 by installing PN-GRO150-VLV-PROP-7103 & kit KIT-GRO150-HYD-3. Fasteners torqued to 107 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-CLN-4

Target Machine Model: Grob-Aerospace-G150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO150-4
- PLC Hex Register Address: DB30.DBW_0xFE00
- Nominal Operating Range: 24 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-PMP-PLG-7104
- Overhaul Seal & Consumable Kit: KIT-GRO150-CLN-4
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-CLN-4 at PLC register 0xFE00.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-PMP-PLG-7104) with seal kit (KIT-GRO150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-CLN-4 on Grob-Aerospace-G150 by installing PN-GRO150-PMP-PLG-7104 & kit KIT-GRO150-CLN-4. Fasteners torqued to 116 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-XAX-5

Target Machine Model: Grob-Aerospace-G150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO150-5
- PLC Hex Register Address: DB30.DBW_0xFE08
- Nominal Operating Range: 27 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-LIN-COIL-7105
- Overhaul Seal & Consumable Kit: KIT-GRO150-XAX-5
- Critical Fastener Torque Value: 125 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-XAX-5 at PLC register 0xFE08.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-LIN-COIL-7105) with seal kit (KIT-GRO150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 125 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-XAX-5 on Grob-Aerospace-G150 by installing PN-GRO150-LIN-COIL-7105 & kit KIT-GRO150-XAX-5. Fasteners torqued to 125 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-YAX-6

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO150-6
- PLC Hex Register Address: DB30.DBW_0xFE10
- Nominal Operating Range: 30 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-BSW-NUT-7106
- Overhaul Seal & Consumable Kit: KIT-GRO150-YAX-6
- Critical Fastener Torque Value: 134 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-YAX-6 at PLC register 0xFE10.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-BSW-NUT-7106) with seal kit (KIT-GRO150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 134 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-YAX-6 on Grob-Aerospace-G150 by installing PN-GRO150-BSW-NUT-7106 & kit KIT-GRO150-YAX-6. Fasteners torqued to 134 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-ZAX-7

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO150-7
- PLC Hex Register Address: DB30.DBW_0xFE18
- Nominal Operating Range: 33 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-CYL-SEAL-7107
- Overhaul Seal & Consumable Kit: KIT-GRO150-ZAX-7
- Critical Fastener Torque Value: 143 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-ZAX-7 at PLC register 0xFE18.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-CYL-SEAL-7107) with seal kit (KIT-GRO150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 143 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-ZAX-7 on Grob-Aerospace-G150 by installing PN-GRO150-CYL-SEAL-7107 & kit KIT-GRO150-ZAX-7. Fasteners torqued to 143 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-BAX-8

Target Machine Model: Grob-Aerospace-G150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO150-8
- PLC Hex Register Address: DB30.DBW_0xFE20
- Nominal Operating Range: 36 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-TRQ-STAT-7108
- Overhaul Seal & Consumable Kit: KIT-GRO150-BAX-8
- Critical Fastener Torque Value: 152 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-BAX-8 at PLC register 0xFE20.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-TRQ-STAT-7108) with seal kit (KIT-GRO150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 152 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-BAX-8 on Grob-Aerospace-G150 by installing PN-GRO150-TRQ-STAT-7108 & kit KIT-GRO150-BAX-8. Fasteners torqued to 152 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-CAX-9

Target Machine Model: Grob-Aerospace-G150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO150-9
- PLC Hex Register Address: DB30.DBW_0xFE28
- Nominal Operating Range: 39 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-BRK-DISC-7109
- Overhaul Seal & Consumable Kit: KIT-GRO150-CAX-9
- Critical Fastener Torque Value: 161 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-CAX-9 at PLC register 0xFE28.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-BRK-DISC-7109) with seal kit (KIT-GRO150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 161 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-CAX-9 on Grob-Aerospace-G150 by installing PN-GRO150-BRK-DISC-7109 & kit KIT-GRO150-CAX-9. Fasteners torqued to 161 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-OPT-10

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO150-10
- PLC Hex Register Address: DB30.DBW_0xFE30
- Nominal Operating Range: 42 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-GLASS-RD-7110
- Overhaul Seal & Consumable Kit: KIT-GRO150-OPT-10
- Critical Fastener Torque Value: 170 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-OPT-10 at PLC register 0xFE30.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-GLASS-RD-7110) with seal kit (KIT-GRO150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 170 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-OPT-10 on Grob-Aerospace-G150 by installing PN-GRO150-GLASS-RD-7110 & kit KIT-GRO150-OPT-10. Fasteners torqued to 170 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-PLC-11

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO150-11
- PLC Hex Register Address: DB30.DBW_0xFE38
- Nominal Operating Range: 45 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-IGBT-MOD-7111
- Overhaul Seal & Consumable Kit: KIT-GRO150-PLC-11
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-PLC-11 at PLC register 0xFE38.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-IGBT-MOD-7111) with seal kit (KIT-GRO150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-PLC-11 on Grob-Aerospace-G150 by installing PN-GRO150-IGBT-MOD-7111 & kit KIT-GRO150-PLC-11. Fasteners torqued to 179 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-LUB-12

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO150-12
- PLC Hex Register Address: DB30.DBW_0xFE40
- Nominal Operating Range: 48 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-INJ-MET-7112
- Overhaul Seal & Consumable Kit: KIT-GRO150-LUB-12
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-LUB-12 at PLC register 0xFE40.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-INJ-MET-7112) with seal kit (KIT-GRO150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-LUB-12 on Grob-Aerospace-G150 by installing PN-GRO150-INJ-MET-7112 & kit KIT-GRO150-LUB-12. Fasteners torqued to 188 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-VAC-13

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO150-13
- PLC Hex Register Address: DB30.DBW_0xFE48
- Nominal Operating Range: 51 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-GASK-VIT-7113
- Overhaul Seal & Consumable Kit: KIT-GRO150-VAC-13
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-VAC-13 at PLC register 0xFE48.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-GASK-VIT-7113) with seal kit (KIT-GRO150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-VAC-13 on Grob-Aerospace-G150 by installing PN-GRO150-GASK-VIT-7113 & kit KIT-GRO150-VAC-13. Fasteners torqued to 197 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO150-THM-14

Target Machine Model: Grob-Aerospace-G150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G150 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO150-14
- PLC Hex Register Address: DB30.DBW_0xFE50
- Nominal Operating Range: 54 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO150-HEX-CORE-7114
- Overhaul Seal & Consumable Kit: KIT-GRO150-THM-14
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G150 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO150-THM-14 at PLC register 0xFE50.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO150-HEX-CORE-7114) with seal kit (KIT-GRO150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G150, technician must log: "Resolved ERR-GRO150-THM-14 on Grob-Aerospace-G150 by installing PN-GRO150-HEX-CORE-7114 & kit KIT-GRO150-THM-14. Fasteners torqued to 206 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-SPN-1

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO200-1
- PLC Hex Register Address: DB31.DBW_0x10108
- Nominal Operating Range: 15 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-BRG-CER-7201
- Overhaul Seal & Consumable Kit: KIT-GRO200-SPN-1
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-SPN-1 at PLC register 0x10108.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-BRG-CER-7201) with seal kit (KIT-GRO200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-SPN-1 on Grob-Aerospace-G200 by installing PN-GRO200-BRG-CER-7201 & kit KIT-GRO200-SPN-1. Fasteners torqued to 39 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-ATC-2

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO200-2
- PLC Hex Register Address: DB31.DBW_0x10110
- Nominal Operating Range: 18 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-CAM-GRIP-7202
- Overhaul Seal & Consumable Kit: KIT-GRO200-ATC-2
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-ATC-2 at PLC register 0x10110.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-CAM-GRIP-7202) with seal kit (KIT-GRO200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-ATC-2 on Grob-Aerospace-G200 by installing PN-GRO200-CAM-GRIP-7202 & kit KIT-GRO200-ATC-2. Fasteners torqued to 48 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-HYD-3

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO200-3
- PLC Hex Register Address: DB31.DBW_0x10118
- Nominal Operating Range: 21 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-VLV-PROP-7203
- Overhaul Seal & Consumable Kit: KIT-GRO200-HYD-3
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-HYD-3 at PLC register 0x10118.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-VLV-PROP-7203) with seal kit (KIT-GRO200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-HYD-3 on Grob-Aerospace-G200 by installing PN-GRO200-VLV-PROP-7203 & kit KIT-GRO200-HYD-3. Fasteners torqued to 57 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-CLN-4

Target Machine Model: Grob-Aerospace-G200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO200-4
- PLC Hex Register Address: DB31.DBW_0x10120
- Nominal Operating Range: 24 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-PMP-PLG-7204
- Overhaul Seal & Consumable Kit: KIT-GRO200-CLN-4
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-CLN-4 at PLC register 0x10120.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-PMP-PLG-7204) with seal kit (KIT-GRO200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-CLN-4 on Grob-Aerospace-G200 by installing PN-GRO200-PMP-PLG-7204 & kit KIT-GRO200-CLN-4. Fasteners torqued to 66 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-XAX-5

Target Machine Model: Grob-Aerospace-G200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO200-5
- PLC Hex Register Address: DB31.DBW_0x10128
- Nominal Operating Range: 27 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-LIN-COIL-7205
- Overhaul Seal & Consumable Kit: KIT-GRO200-XAX-5
- Critical Fastener Torque Value: 75 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-XAX-5 at PLC register 0x10128.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-LIN-COIL-7205) with seal kit (KIT-GRO200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 75 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-XAX-5 on Grob-Aerospace-G200 by installing PN-GRO200-LIN-COIL-7205 & kit KIT-GRO200-XAX-5. Fasteners torqued to 75 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-YAX-6

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO200-6
- PLC Hex Register Address: DB31.DBW_0x10130
- Nominal Operating Range: 30 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-BSW-NUT-7206
- Overhaul Seal & Consumable Kit: KIT-GRO200-YAX-6
- Critical Fastener Torque Value: 84 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-YAX-6 at PLC register 0x10130.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-BSW-NUT-7206) with seal kit (KIT-GRO200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 84 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-YAX-6 on Grob-Aerospace-G200 by installing PN-GRO200-BSW-NUT-7206 & kit KIT-GRO200-YAX-6. Fasteners torqued to 84 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-ZAX-7

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO200-7
- PLC Hex Register Address: DB31.DBW_0x10138
- Nominal Operating Range: 33 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-CYL-SEAL-7207
- Overhaul Seal & Consumable Kit: KIT-GRO200-ZAX-7
- Critical Fastener Torque Value: 93 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-ZAX-7 at PLC register 0x10138.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-CYL-SEAL-7207) with seal kit (KIT-GRO200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 93 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-ZAX-7 on Grob-Aerospace-G200 by installing PN-GRO200-CYL-SEAL-7207 & kit KIT-GRO200-ZAX-7. Fasteners torqued to 93 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-BAX-8

Target Machine Model: Grob-Aerospace-G200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO200-8
- PLC Hex Register Address: DB31.DBW_0x10140
- Nominal Operating Range: 36 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-TRQ-STAT-7208
- Overhaul Seal & Consumable Kit: KIT-GRO200-BAX-8
- Critical Fastener Torque Value: 102 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-BAX-8 at PLC register 0x10140.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-TRQ-STAT-7208) with seal kit (KIT-GRO200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 102 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-BAX-8 on Grob-Aerospace-G200 by installing PN-GRO200-TRQ-STAT-7208 & kit KIT-GRO200-BAX-8. Fasteners torqued to 102 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-CAX-9

Target Machine Model: Grob-Aerospace-G200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO200-9
- PLC Hex Register Address: DB31.DBW_0x10148
- Nominal Operating Range: 39 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-BRK-DISC-7209
- Overhaul Seal & Consumable Kit: KIT-GRO200-CAX-9
- Critical Fastener Torque Value: 111 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-CAX-9 at PLC register 0x10148.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-BRK-DISC-7209) with seal kit (KIT-GRO200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 111 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-CAX-9 on Grob-Aerospace-G200 by installing PN-GRO200-BRK-DISC-7209 & kit KIT-GRO200-CAX-9. Fasteners torqued to 111 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-OPT-10

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO200-10
- PLC Hex Register Address: DB31.DBW_0x10150
- Nominal Operating Range: 42 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-GLASS-RD-7210
- Overhaul Seal & Consumable Kit: KIT-GRO200-OPT-10
- Critical Fastener Torque Value: 120 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-OPT-10 at PLC register 0x10150.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-GLASS-RD-7210) with seal kit (KIT-GRO200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 120 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-OPT-10 on Grob-Aerospace-G200 by installing PN-GRO200-GLASS-RD-7210 & kit KIT-GRO200-OPT-10. Fasteners torqued to 120 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-PLC-11

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO200-11
- PLC Hex Register Address: DB31.DBW_0x10158
- Nominal Operating Range: 45 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-IGBT-MOD-7211
- Overhaul Seal & Consumable Kit: KIT-GRO200-PLC-11
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-PLC-11 at PLC register 0x10158.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-IGBT-MOD-7211) with seal kit (KIT-GRO200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-PLC-11 on Grob-Aerospace-G200 by installing PN-GRO200-IGBT-MOD-7211 & kit KIT-GRO200-PLC-11. Fasteners torqued to 129 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-LUB-12

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO200-12
- PLC Hex Register Address: DB31.DBW_0x10160
- Nominal Operating Range: 48 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-INJ-MET-7212
- Overhaul Seal & Consumable Kit: KIT-GRO200-LUB-12
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-LUB-12 at PLC register 0x10160.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-INJ-MET-7212) with seal kit (KIT-GRO200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-LUB-12 on Grob-Aerospace-G200 by installing PN-GRO200-INJ-MET-7212 & kit KIT-GRO200-LUB-12. Fasteners torqued to 138 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-VAC-13

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO200-13
- PLC Hex Register Address: DB31.DBW_0x10168
- Nominal Operating Range: 51 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-GASK-VIT-7213
- Overhaul Seal & Consumable Kit: KIT-GRO200-VAC-13
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-VAC-13 at PLC register 0x10168.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-GASK-VIT-7213) with seal kit (KIT-GRO200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-VAC-13 on Grob-Aerospace-G200 by installing PN-GRO200-GASK-VIT-7213 & kit KIT-GRO200-VAC-13. Fasteners torqued to 147 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO200-THM-14

Target Machine Model: Grob-Aerospace-G200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G200 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO200-14
- PLC Hex Register Address: DB31.DBW_0x10170
- Nominal Operating Range: 54 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO200-HEX-CORE-7214
- Overhaul Seal & Consumable Kit: KIT-GRO200-THM-14
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G200 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO200-THM-14 at PLC register 0x10170.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO200-HEX-CORE-7214) with seal kit (KIT-GRO200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G200, technician must log: "Resolved ERR-GRO200-THM-14 on Grob-Aerospace-G200 by installing PN-GRO200-HEX-CORE-7214 & kit KIT-GRO200-THM-14. Fasteners torqued to 156 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-SPN-1

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO250-1
- PLC Hex Register Address: DB32.DBW_0x10428
- Nominal Operating Range: 15 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-BRG-CER-7301
- Overhaul Seal & Consumable Kit: KIT-GRO250-SPN-1
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-SPN-1 at PLC register 0x10428.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-BRG-CER-7301) with seal kit (KIT-GRO250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-SPN-1 on Grob-Aerospace-G250 by installing PN-GRO250-BRG-CER-7301 & kit KIT-GRO250-SPN-1. Fasteners torqued to 179 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-ATC-2

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO250-2
- PLC Hex Register Address: DB32.DBW_0x10430
- Nominal Operating Range: 18 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-CAM-GRIP-7302
- Overhaul Seal & Consumable Kit: KIT-GRO250-ATC-2
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-ATC-2 at PLC register 0x10430.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-CAM-GRIP-7302) with seal kit (KIT-GRO250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-ATC-2 on Grob-Aerospace-G250 by installing PN-GRO250-CAM-GRIP-7302 & kit KIT-GRO250-ATC-2. Fasteners torqued to 188 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-HYD-3

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO250-3
- PLC Hex Register Address: DB32.DBW_0x10438
- Nominal Operating Range: 21 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-VLV-PROP-7303
- Overhaul Seal & Consumable Kit: KIT-GRO250-HYD-3
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-HYD-3 at PLC register 0x10438.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-VLV-PROP-7303) with seal kit (KIT-GRO250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-HYD-3 on Grob-Aerospace-G250 by installing PN-GRO250-VLV-PROP-7303 & kit KIT-GRO250-HYD-3. Fasteners torqued to 197 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-CLN-4

Target Machine Model: Grob-Aerospace-G250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO250-4
- PLC Hex Register Address: DB32.DBW_0x10440
- Nominal Operating Range: 24 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-PMP-PLG-7304
- Overhaul Seal & Consumable Kit: KIT-GRO250-CLN-4
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-CLN-4 at PLC register 0x10440.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-PMP-PLG-7304) with seal kit (KIT-GRO250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-CLN-4 on Grob-Aerospace-G250 by installing PN-GRO250-PMP-PLG-7304 & kit KIT-GRO250-CLN-4. Fasteners torqued to 206 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-XAX-5

Target Machine Model: Grob-Aerospace-G250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO250-5
- PLC Hex Register Address: DB32.DBW_0x10448
- Nominal Operating Range: 27 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-LIN-COIL-7305
- Overhaul Seal & Consumable Kit: KIT-GRO250-XAX-5
- Critical Fastener Torque Value: 25 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-XAX-5 at PLC register 0x10448.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-LIN-COIL-7305) with seal kit (KIT-GRO250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 25 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-XAX-5 on Grob-Aerospace-G250 by installing PN-GRO250-LIN-COIL-7305 & kit KIT-GRO250-XAX-5. Fasteners torqued to 25 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-YAX-6

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO250-6
- PLC Hex Register Address: DB32.DBW_0x10450
- Nominal Operating Range: 30 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-BSW-NUT-7306
- Overhaul Seal & Consumable Kit: KIT-GRO250-YAX-6
- Critical Fastener Torque Value: 34 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-YAX-6 at PLC register 0x10450.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-BSW-NUT-7306) with seal kit (KIT-GRO250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 34 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-YAX-6 on Grob-Aerospace-G250 by installing PN-GRO250-BSW-NUT-7306 & kit KIT-GRO250-YAX-6. Fasteners torqued to 34 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-ZAX-7

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO250-7
- PLC Hex Register Address: DB32.DBW_0x10458
- Nominal Operating Range: 33 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-CYL-SEAL-7307
- Overhaul Seal & Consumable Kit: KIT-GRO250-ZAX-7
- Critical Fastener Torque Value: 43 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-ZAX-7 at PLC register 0x10458.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-CYL-SEAL-7307) with seal kit (KIT-GRO250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 43 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-ZAX-7 on Grob-Aerospace-G250 by installing PN-GRO250-CYL-SEAL-7307 & kit KIT-GRO250-ZAX-7. Fasteners torqued to 43 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-BAX-8

Target Machine Model: Grob-Aerospace-G250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO250-8
- PLC Hex Register Address: DB32.DBW_0x10460
- Nominal Operating Range: 36 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-TRQ-STAT-7308
- Overhaul Seal & Consumable Kit: KIT-GRO250-BAX-8
- Critical Fastener Torque Value: 52 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-BAX-8 at PLC register 0x10460.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-TRQ-STAT-7308) with seal kit (KIT-GRO250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 52 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-BAX-8 on Grob-Aerospace-G250 by installing PN-GRO250-TRQ-STAT-7308 & kit KIT-GRO250-BAX-8. Fasteners torqued to 52 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-CAX-9

Target Machine Model: Grob-Aerospace-G250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO250-9
- PLC Hex Register Address: DB32.DBW_0x10468
- Nominal Operating Range: 39 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-BRK-DISC-7309
- Overhaul Seal & Consumable Kit: KIT-GRO250-CAX-9
- Critical Fastener Torque Value: 61 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-CAX-9 at PLC register 0x10468.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-BRK-DISC-7309) with seal kit (KIT-GRO250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 61 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-CAX-9 on Grob-Aerospace-G250 by installing PN-GRO250-BRK-DISC-7309 & kit KIT-GRO250-CAX-9. Fasteners torqued to 61 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-OPT-10

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO250-10
- PLC Hex Register Address: DB32.DBW_0x10470
- Nominal Operating Range: 42 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-GLASS-RD-7310
- Overhaul Seal & Consumable Kit: KIT-GRO250-OPT-10
- Critical Fastener Torque Value: 70 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-OPT-10 at PLC register 0x10470.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-GLASS-RD-7310) with seal kit (KIT-GRO250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 70 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-OPT-10 on Grob-Aerospace-G250 by installing PN-GRO250-GLASS-RD-7310 & kit KIT-GRO250-OPT-10. Fasteners torqued to 70 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-PLC-11

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO250-11
- PLC Hex Register Address: DB32.DBW_0x10478
- Nominal Operating Range: 45 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-IGBT-MOD-7311
- Overhaul Seal & Consumable Kit: KIT-GRO250-PLC-11
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-PLC-11 at PLC register 0x10478.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-IGBT-MOD-7311) with seal kit (KIT-GRO250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-PLC-11 on Grob-Aerospace-G250 by installing PN-GRO250-IGBT-MOD-7311 & kit KIT-GRO250-PLC-11. Fasteners torqued to 79 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-LUB-12

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO250-12
- PLC Hex Register Address: DB32.DBW_0x10480
- Nominal Operating Range: 48 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-INJ-MET-7312
- Overhaul Seal & Consumable Kit: KIT-GRO250-LUB-12
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-LUB-12 at PLC register 0x10480.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-INJ-MET-7312) with seal kit (KIT-GRO250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-LUB-12 on Grob-Aerospace-G250 by installing PN-GRO250-INJ-MET-7312 & kit KIT-GRO250-LUB-12. Fasteners torqued to 88 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-VAC-13

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO250-13
- PLC Hex Register Address: DB32.DBW_0x10488
- Nominal Operating Range: 51 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-GASK-VIT-7313
- Overhaul Seal & Consumable Kit: KIT-GRO250-VAC-13
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-VAC-13 at PLC register 0x10488.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-GASK-VIT-7313) with seal kit (KIT-GRO250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-VAC-13 on Grob-Aerospace-G250 by installing PN-GRO250-GASK-VIT-7313 & kit KIT-GRO250-VAC-13. Fasteners torqued to 97 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO250-THM-14

Target Machine Model: Grob-Aerospace-G250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G250 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO250-14
- PLC Hex Register Address: DB32.DBW_0x10490
- Nominal Operating Range: 54 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO250-HEX-CORE-7314
- Overhaul Seal & Consumable Kit: KIT-GRO250-THM-14
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G250 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO250-THM-14 at PLC register 0x10490.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO250-HEX-CORE-7314) with seal kit (KIT-GRO250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G250, technician must log: "Resolved ERR-GRO250-THM-14 on Grob-Aerospace-G250 by installing PN-GRO250-HEX-CORE-7314 & kit KIT-GRO250-THM-14. Fasteners torqued to 106 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-SPN-1

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO300-1
- PLC Hex Register Address: DB33.DBW_0x10748
- Nominal Operating Range: 15 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-BRG-CER-7401
- Overhaul Seal & Consumable Kit: KIT-GRO300-SPN-1
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-SPN-1 at PLC register 0x10748.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-BRG-CER-7401) with seal kit (KIT-GRO300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-SPN-1 on Grob-Aerospace-G300 by installing PN-GRO300-BRG-CER-7401 & kit KIT-GRO300-SPN-1. Fasteners torqued to 129 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-ATC-2

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO300-2
- PLC Hex Register Address: DB33.DBW_0x10750
- Nominal Operating Range: 18 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-CAM-GRIP-7402
- Overhaul Seal & Consumable Kit: KIT-GRO300-ATC-2
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-ATC-2 at PLC register 0x10750.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-CAM-GRIP-7402) with seal kit (KIT-GRO300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-ATC-2 on Grob-Aerospace-G300 by installing PN-GRO300-CAM-GRIP-7402 & kit KIT-GRO300-ATC-2. Fasteners torqued to 138 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-HYD-3

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO300-3
- PLC Hex Register Address: DB33.DBW_0x10758
- Nominal Operating Range: 21 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-VLV-PROP-7403
- Overhaul Seal & Consumable Kit: KIT-GRO300-HYD-3
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-HYD-3 at PLC register 0x10758.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-VLV-PROP-7403) with seal kit (KIT-GRO300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-HYD-3 on Grob-Aerospace-G300 by installing PN-GRO300-VLV-PROP-7403 & kit KIT-GRO300-HYD-3. Fasteners torqued to 147 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-CLN-4

Target Machine Model: Grob-Aerospace-G300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO300-4
- PLC Hex Register Address: DB33.DBW_0x10760
- Nominal Operating Range: 24 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-PMP-PLG-7404
- Overhaul Seal & Consumable Kit: KIT-GRO300-CLN-4
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-CLN-4 at PLC register 0x10760.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-PMP-PLG-7404) with seal kit (KIT-GRO300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-CLN-4 on Grob-Aerospace-G300 by installing PN-GRO300-PMP-PLG-7404 & kit KIT-GRO300-CLN-4. Fasteners torqued to 156 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-XAX-5

Target Machine Model: Grob-Aerospace-G300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO300-5
- PLC Hex Register Address: DB33.DBW_0x10768
- Nominal Operating Range: 27 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-LIN-COIL-7405
- Overhaul Seal & Consumable Kit: KIT-GRO300-XAX-5
- Critical Fastener Torque Value: 165 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-XAX-5 at PLC register 0x10768.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-LIN-COIL-7405) with seal kit (KIT-GRO300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 165 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-XAX-5 on Grob-Aerospace-G300 by installing PN-GRO300-LIN-COIL-7405 & kit KIT-GRO300-XAX-5. Fasteners torqued to 165 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-YAX-6

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO300-6
- PLC Hex Register Address: DB33.DBW_0x10770
- Nominal Operating Range: 30 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-BSW-NUT-7406
- Overhaul Seal & Consumable Kit: KIT-GRO300-YAX-6
- Critical Fastener Torque Value: 174 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-YAX-6 at PLC register 0x10770.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-BSW-NUT-7406) with seal kit (KIT-GRO300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 174 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-YAX-6 on Grob-Aerospace-G300 by installing PN-GRO300-BSW-NUT-7406 & kit KIT-GRO300-YAX-6. Fasteners torqued to 174 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-ZAX-7

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO300-7
- PLC Hex Register Address: DB33.DBW_0x10778
- Nominal Operating Range: 33 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-CYL-SEAL-7407
- Overhaul Seal & Consumable Kit: KIT-GRO300-ZAX-7
- Critical Fastener Torque Value: 183 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-ZAX-7 at PLC register 0x10778.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-CYL-SEAL-7407) with seal kit (KIT-GRO300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 183 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-ZAX-7 on Grob-Aerospace-G300 by installing PN-GRO300-CYL-SEAL-7407 & kit KIT-GRO300-ZAX-7. Fasteners torqued to 183 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-BAX-8

Target Machine Model: Grob-Aerospace-G300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO300-8
- PLC Hex Register Address: DB33.DBW_0x10780
- Nominal Operating Range: 36 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-TRQ-STAT-7408
- Overhaul Seal & Consumable Kit: KIT-GRO300-BAX-8
- Critical Fastener Torque Value: 192 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-BAX-8 at PLC register 0x10780.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-TRQ-STAT-7408) with seal kit (KIT-GRO300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 192 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-BAX-8 on Grob-Aerospace-G300 by installing PN-GRO300-TRQ-STAT-7408 & kit KIT-GRO300-BAX-8. Fasteners torqued to 192 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-CAX-9

Target Machine Model: Grob-Aerospace-G300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO300-9
- PLC Hex Register Address: DB33.DBW_0x10788
- Nominal Operating Range: 39 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-BRK-DISC-7409
- Overhaul Seal & Consumable Kit: KIT-GRO300-CAX-9
- Critical Fastener Torque Value: 201 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-CAX-9 at PLC register 0x10788.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-BRK-DISC-7409) with seal kit (KIT-GRO300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 201 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-CAX-9 on Grob-Aerospace-G300 by installing PN-GRO300-BRK-DISC-7409 & kit KIT-GRO300-CAX-9. Fasteners torqued to 201 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-OPT-10

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO300-10
- PLC Hex Register Address: DB33.DBW_0x10790
- Nominal Operating Range: 42 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-GLASS-RD-7410
- Overhaul Seal & Consumable Kit: KIT-GRO300-OPT-10
- Critical Fastener Torque Value: 20 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-OPT-10 at PLC register 0x10790.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-GLASS-RD-7410) with seal kit (KIT-GRO300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 20 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-OPT-10 on Grob-Aerospace-G300 by installing PN-GRO300-GLASS-RD-7410 & kit KIT-GRO300-OPT-10. Fasteners torqued to 20 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-PLC-11

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO300-11
- PLC Hex Register Address: DB33.DBW_0x10798
- Nominal Operating Range: 45 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-IGBT-MOD-7411
- Overhaul Seal & Consumable Kit: KIT-GRO300-PLC-11
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-PLC-11 at PLC register 0x10798.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-IGBT-MOD-7411) with seal kit (KIT-GRO300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-PLC-11 on Grob-Aerospace-G300 by installing PN-GRO300-IGBT-MOD-7411 & kit KIT-GRO300-PLC-11. Fasteners torqued to 29 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-LUB-12

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO300-12
- PLC Hex Register Address: DB33.DBW_0x107A0
- Nominal Operating Range: 48 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-INJ-MET-7412
- Overhaul Seal & Consumable Kit: KIT-GRO300-LUB-12
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-LUB-12 at PLC register 0x107A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-INJ-MET-7412) with seal kit (KIT-GRO300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-LUB-12 on Grob-Aerospace-G300 by installing PN-GRO300-INJ-MET-7412 & kit KIT-GRO300-LUB-12. Fasteners torqued to 38 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-VAC-13

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO300-13
- PLC Hex Register Address: DB33.DBW_0x107A8
- Nominal Operating Range: 51 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-GASK-VIT-7413
- Overhaul Seal & Consumable Kit: KIT-GRO300-VAC-13
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-VAC-13 at PLC register 0x107A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-GASK-VIT-7413) with seal kit (KIT-GRO300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-VAC-13 on Grob-Aerospace-G300 by installing PN-GRO300-GASK-VIT-7413 & kit KIT-GRO300-VAC-13. Fasteners torqued to 47 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO300-THM-14

Target Machine Model: Grob-Aerospace-G300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G300 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO300-14
- PLC Hex Register Address: DB33.DBW_0x107B0
- Nominal Operating Range: 54 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO300-HEX-CORE-7414
- Overhaul Seal & Consumable Kit: KIT-GRO300-THM-14
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G300 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO300-THM-14 at PLC register 0x107B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO300-HEX-CORE-7414) with seal kit (KIT-GRO300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G300, technician must log: "Resolved ERR-GRO300-THM-14 on Grob-Aerospace-G300 by installing PN-GRO300-HEX-CORE-7414 & kit KIT-GRO300-THM-14. Fasteners torqued to 56 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-SPN-1

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: PT-SPN-GRO350-1
- PLC Hex Register Address: DB34.DBW_0x10A68
- Nominal Operating Range: 15 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-BRG-CER-7501
- Overhaul Seal & Consumable Kit: KIT-GRO350-SPN-1
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-SPN-1 at PLC register 0x10A68.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-BRG-CER-7501) with seal kit (KIT-GRO350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-SPN-1 on Grob-Aerospace-G350 by installing PN-GRO350-BRG-CER-7501 & kit KIT-GRO350-SPN-1. Fasteners torqued to 79 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-ATC-2

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: PX-ATC-GRO350-2
- PLC Hex Register Address: DB34.DBW_0x10A70
- Nominal Operating Range: 18 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-CAM-GRIP-7502
- Overhaul Seal & Consumable Kit: KIT-GRO350-ATC-2
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-ATC-2 at PLC register 0x10A70.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-CAM-GRIP-7502) with seal kit (KIT-GRO350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-ATC-2 on Grob-Aerospace-G350 by installing PN-GRO350-CAM-GRIP-7502 & kit KIT-GRO350-ATC-2. Fasteners torqued to 88 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-HYD-3

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-HYD-GRO350-3
- PLC Hex Register Address: DB34.DBW_0x10A78
- Nominal Operating Range: 21 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-VLV-PROP-7503
- Overhaul Seal & Consumable Kit: KIT-GRO350-HYD-3
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-HYD-3 at PLC register 0x10A78.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-VLV-PROP-7503) with seal kit (KIT-GRO350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-HYD-3 on Grob-Aerospace-G350 by installing PN-GRO350-VLV-PROP-7503 & kit KIT-GRO350-HYD-3. Fasteners torqued to 97 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-CLN-4

Target Machine Model: Grob-Aerospace-G350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-CLN-GRO350-4
- PLC Hex Register Address: DB34.DBW_0x10A80
- Nominal Operating Range: 24 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-PMP-PLG-7504
- Overhaul Seal & Consumable Kit: KIT-GRO350-CLN-4
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-CLN-4 at PLC register 0x10A80.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-PMP-PLG-7504) with seal kit (KIT-GRO350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-CLN-4 on Grob-Aerospace-G350 by installing PN-GRO350-PMP-PLG-7504 & kit KIT-GRO350-CLN-4. Fasteners torqued to 106 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-XAX-5

Target Machine Model: Grob-Aerospace-G350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-XAX-GRO350-5
- PLC Hex Register Address: DB34.DBW_0x10A88
- Nominal Operating Range: 27 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-LIN-COIL-7505
- Overhaul Seal & Consumable Kit: KIT-GRO350-XAX-5
- Critical Fastener Torque Value: 115 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-XAX-5 at PLC register 0x10A88.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-LIN-COIL-7505) with seal kit (KIT-GRO350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 115 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-XAX-5 on Grob-Aerospace-G350 by installing PN-GRO350-LIN-COIL-7505 & kit KIT-GRO350-XAX-5. Fasteners torqued to 115 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-YAX-6

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-YAX-GRO350-6
- PLC Hex Register Address: DB34.DBW_0x10A90
- Nominal Operating Range: 30 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-BSW-NUT-7506
- Overhaul Seal & Consumable Kit: KIT-GRO350-YAX-6
- Critical Fastener Torque Value: 124 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-YAX-6 at PLC register 0x10A90.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-BSW-NUT-7506) with seal kit (KIT-GRO350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 124 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-YAX-6 on Grob-Aerospace-G350 by installing PN-GRO350-BSW-NUT-7506 & kit KIT-GRO350-YAX-6. Fasteners torqued to 124 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-ZAX-7

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-ZAX-GRO350-7
- PLC Hex Register Address: DB34.DBW_0x10A98
- Nominal Operating Range: 33 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-CYL-SEAL-7507
- Overhaul Seal & Consumable Kit: KIT-GRO350-ZAX-7
- Critical Fastener Torque Value: 133 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-ZAX-7 at PLC register 0x10A98.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-CYL-SEAL-7507) with seal kit (KIT-GRO350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 133 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-ZAX-7 on Grob-Aerospace-G350 by installing PN-GRO350-CYL-SEAL-7507 & kit KIT-GRO350-ZAX-7. Fasteners torqued to 133 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-BAX-8

Target Machine Model: Grob-Aerospace-G350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: EN-BAX-GRO350-8
- PLC Hex Register Address: DB34.DBW_0x10AA0
- Nominal Operating Range: 36 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-TRQ-STAT-7508
- Overhaul Seal & Consumable Kit: KIT-GRO350-BAX-8
- Critical Fastener Torque Value: 142 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-BAX-8 at PLC register 0x10AA0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-TRQ-STAT-7508) with seal kit (KIT-GRO350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 142 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-BAX-8 on Grob-Aerospace-G350 by installing PN-GRO350-TRQ-STAT-7508 & kit KIT-GRO350-BAX-8. Fasteners torqued to 142 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-CAX-9

Target Machine Model: Grob-Aerospace-G350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: PR-CAX-GRO350-9
- PLC Hex Register Address: DB34.DBW_0x10AA8
- Nominal Operating Range: 39 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-BRK-DISC-7509
- Overhaul Seal & Consumable Kit: KIT-GRO350-CAX-9
- Critical Fastener Torque Value: 151 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-CAX-9 at PLC register 0x10AA8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-BRK-DISC-7509) with seal kit (KIT-GRO350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 151 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-CAX-9 on Grob-Aerospace-G350 by installing PN-GRO350-BRK-DISC-7509 & kit KIT-GRO350-CAX-9. Fasteners torqued to 151 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-OPT-10

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: OP-SCL-GRO350-10
- PLC Hex Register Address: DB34.DBW_0x10AB0
- Nominal Operating Range: 42 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-GLASS-RD-7510
- Overhaul Seal & Consumable Kit: KIT-GRO350-OPT-10
- Critical Fastener Torque Value: 160 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-OPT-10 at PLC register 0x10AB0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-GLASS-RD-7510) with seal kit (KIT-GRO350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 160 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-OPT-10 on Grob-Aerospace-G350 by installing PN-GRO350-GLASS-RD-7510 & kit KIT-GRO350-OPT-10. Fasteners torqued to 160 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-PLC-11

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: VT-DRV-GRO350-11
- PLC Hex Register Address: DB34.DBW_0x10AB8
- Nominal Operating Range: 45 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-IGBT-MOD-7511
- Overhaul Seal & Consumable Kit: KIT-GRO350-PLC-11
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-PLC-11 at PLC register 0x10AB8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-IGBT-MOD-7511) with seal kit (KIT-GRO350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-PLC-11 on Grob-Aerospace-G350 by installing PN-GRO350-IGBT-MOD-7511 & kit KIT-GRO350-PLC-11. Fasteners torqued to 169 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-LUB-12

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: FL-LUB-GRO350-12
- PLC Hex Register Address: DB34.DBW_0x10AC0
- Nominal Operating Range: 48 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-INJ-MET-7512
- Overhaul Seal & Consumable Kit: KIT-GRO350-LUB-12
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-LUB-12 at PLC register 0x10AC0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-INJ-MET-7512) with seal kit (KIT-GRO350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-LUB-12 on Grob-Aerospace-G350 by installing PN-GRO350-INJ-MET-7512 & kit KIT-GRO350-LUB-12. Fasteners torqued to 178 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-VAC-13

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: VC-MNF-GRO350-13
- PLC Hex Register Address: DB34.DBW_0x10AC8
- Nominal Operating Range: 51 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-GASK-VIT-7513
- Overhaul Seal & Consumable Kit: KIT-GRO350-VAC-13
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-VAC-13 at PLC register 0x10AC8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-GASK-VIT-7513) with seal kit (KIT-GRO350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-VAC-13 on Grob-Aerospace-G350 by installing PN-GRO350-GASK-VIT-7513 & kit KIT-GRO350-VAC-13. Fasteners torqued to 187 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-GRO350-THM-14

Target Machine Model: Grob-Aerospace-G350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Grob-Aerospace-G350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Grob-Aerospace-G350 (Heidenhain TNC 640)
- Telemetry Sensor ID: TM-CHL-GRO350-14
- PLC Hex Register Address: DB34.DBW_0x10AD0
- Nominal Operating Range: 54 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-GRO350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-GRO350-HEX-CORE-7514
- Overhaul Seal & Consumable Kit: KIT-GRO350-THM-14
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Grob-Aerospace-G350 (Heidenhain TNC 640) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Heidenhain TNC 640 and verify fault latch ERR-GRO350-THM-14 at PLC register 0x10AD0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Grob-Aerospace-G350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-GRO350-HEX-CORE-7514) with seal kit (KIT-GRO350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-GRO350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Grob-Aerospace-G350, technician must log: "Resolved ERR-GRO350-THM-14 on Grob-Aerospace-G350 by installing PN-GRO350-HEX-CORE-7514 & kit KIT-GRO350-THM-14. Fasteners torqued to 196 Nm; alignment verified at ±0.0020 mm."

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L150 (BANK 1)

Machine Model: Trumpf-TruLaser-L150 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L150.

PRM_45121 [Trumpf-TruLaser-L150] — ATC_LOOP_GAIN_1 | Default: 58.650 | Linked Fault: ERR-TRU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L150.

PRM_45122 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_2 | Default: 117.300 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45123 [Trumpf-TruLaser-L150] — CLN_LOOP_GAIN_3 | Default: 175.950 | Linked Fault: ERR-TRU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L150.

PRM_45124 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_4 | Default: 234.600 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45125 [Trumpf-TruLaser-L150] — YAX_LOOP_GAIN_5 | Default: 43.250 | Linked Fault: ERR-TRU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L150.

PRM_45126 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_6 | Default: 101.900 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45127 [Trumpf-TruLaser-L150] — BAX_LOOP_GAIN_7 | Default: 160.550 | Linked Fault: ERR-TRU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L150.

PRM_45128 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_8 | Default: 219.200 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45129 [Trumpf-TruLaser-L150] — OPT_LOOP_GAIN_9 | Default: 27.850 | Linked Fault: ERR-TRU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L150.

PRM_45130 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_10 | Default: 86.500 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45131 [Trumpf-TruLaser-L150] — LUB_LOOP_GAIN_11 | Default: 145.150 | Linked Fault: ERR-TRU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L150.

PRM_45132 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_12 | Default: 203.800 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45133 [Trumpf-TruLaser-L150] — THM_LOOP_GAIN_13 | Default: 12.450 | Linked Fault: ERR-TRU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L150.

PRM_45134 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_14 | Default: 71.100 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45135 [Trumpf-TruLaser-L150] — ATC_LOOP_GAIN_15 | Default: 129.750 | Linked Fault: ERR-TRU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L150.

PRM_45136 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_16 | Default: 188.400 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L150 (BANK 2)

Machine Model: Trumpf-TruLaser-L150 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L150.

PRM_45141 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_1 | Default: 117.300 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45142 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_2 | Default: 234.600 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45143 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_3 | Default: 101.900 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45144 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_4 | Default: 219.200 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45145 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_5 | Default: 86.500 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45146 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_6 | Default: 203.800 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45147 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_7 | Default: 71.100 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45148 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_8 | Default: 188.400 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45149 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_9 | Default: 55.700 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45150 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_10 | Default: 173.000 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45151 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_11 | Default: 40.300 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45152 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_12 | Default: 157.600 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45153 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_13 | Default: 24.900 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45154 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_14 | Default: 142.200 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45155 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_15 | Default: 9.500 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45156 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_16 | Default: 126.800 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L150 (BANK 3)

Machine Model: Trumpf-TruLaser-L150 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L150.

PRM_45161 [Trumpf-TruLaser-L150] — CLN_LOOP_GAIN_1 | Default: 175.950 | Linked Fault: ERR-TRU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L150.

PRM_45162 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_2 | Default: 101.900 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45163 [Trumpf-TruLaser-L150] — OPT_LOOP_GAIN_3 | Default: 27.850 | Linked Fault: ERR-TRU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L150.

PRM_45164 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_4 | Default: 203.800 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45165 [Trumpf-TruLaser-L150] — ATC_LOOP_GAIN_5 | Default: 129.750 | Linked Fault: ERR-TRU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L150.

PRM_45166 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_6 | Default: 55.700 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45167 [Trumpf-TruLaser-L150] — BAX_LOOP_GAIN_7 | Default: 231.650 | Linked Fault: ERR-TRU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L150.

PRM_45168 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_8 | Default: 157.600 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45169 [Trumpf-TruLaser-L150] — THM_LOOP_GAIN_9 | Default: 83.550 | Linked Fault: ERR-TRU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L150.

PRM_45170 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_10 | Default: 9.500 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45171 [Trumpf-TruLaser-L150] — YAX_LOOP_GAIN_11 | Default: 185.450 | Linked Fault: ERR-TRU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L150.

PRM_45172 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_12 | Default: 111.400 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45173 [Trumpf-TruLaser-L150] — LUB_LOOP_GAIN_13 | Default: 37.350 | Linked Fault: ERR-TRU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L150.

PRM_45174 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_14 | Default: 213.300 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45175 [Trumpf-TruLaser-L150] — CLN_LOOP_GAIN_15 | Default: 139.250 | Linked Fault: ERR-TRU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L150.

PRM_45176 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_16 | Default: 65.200 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L150 (BANK 4)

Machine Model: Trumpf-TruLaser-L150 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L150.

PRM_45181 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_1 | Default: 234.600 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45182 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_2 | Default: 219.200 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45183 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_3 | Default: 203.800 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45184 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_4 | Default: 188.400 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45185 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_5 | Default: 173.000 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45186 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_6 | Default: 157.600 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45187 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_7 | Default: 142.200 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45188 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_8 | Default: 126.800 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45189 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_9 | Default: 111.400 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45190 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_10 | Default: 96.000 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45191 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_11 | Default: 80.600 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45192 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_12 | Default: 65.200 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45193 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_13 | Default: 49.800 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45194 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_14 | Default: 34.400 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45195 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_15 | Default: 19.000 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45196 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_16 | Default: 3.600 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L150 (BANK 5)

Machine Model: Trumpf-TruLaser-L150 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L150.

PRM_45201 [Trumpf-TruLaser-L150] — YAX_LOOP_GAIN_1 | Default: 43.250 | Linked Fault: ERR-TRU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L150.

PRM_45202 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_2 | Default: 86.500 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

PRM_45203 [Trumpf-TruLaser-L150] — ATC_LOOP_GAIN_3 | Default: 129.750 | Linked Fault: ERR-TRU150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L150.

PRM_45204 [Trumpf-TruLaser-L150] — ZAX_LOOP_GAIN_4 | Default: 173.000 | Linked Fault: ERR-TRU150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L150.

PRM_45205 [Trumpf-TruLaser-L150] — LUB_LOOP_GAIN_5 | Default: 216.250 | Linked Fault: ERR-TRU150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L150.

PRM_45206 [Trumpf-TruLaser-L150] — HYD_LOOP_GAIN_6 | Default: 9.500 | Linked Fault: ERR-TRU150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L150.

PRM_45207 [Trumpf-TruLaser-L150] — BAX_LOOP_GAIN_7 | Default: 52.750 | Linked Fault: ERR-TRU150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L150.

PRM_45208 [Trumpf-TruLaser-L150] — VAC_LOOP_GAIN_8 | Default: 96.000 | Linked Fault: ERR-TRU150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L150.

PRM_45209 [Trumpf-TruLaser-L150] — CLN_LOOP_GAIN_9 | Default: 139.250 | Linked Fault: ERR-TRU150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L150.

PRM_45210 [Trumpf-TruLaser-L150] — CAX_LOOP_GAIN_10 | Default: 182.500 | Linked Fault: ERR-TRU150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L150.

PRM_45211 [Trumpf-TruLaser-L150] — THM_LOOP_GAIN_11 | Default: 225.750 | Linked Fault: ERR-TRU150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L150.

PRM_45212 [Trumpf-TruLaser-L150] — XAX_LOOP_GAIN_12 | Default: 19.000 | Linked Fault: ERR-TRU150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L150.

PRM_45213 [Trumpf-TruLaser-L150] — OPT_LOOP_GAIN_13 | Default: 62.250 | Linked Fault: ERR-TRU150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L150.

PRM_45214 [Trumpf-TruLaser-L150] — SPN_LOOP_GAIN_14 | Default: 105.500 | Linked Fault: ERR-TRU150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L150.

PRM_45215 [Trumpf-TruLaser-L150] — YAX_LOOP_GAIN_15 | Default: 148.750 | Linked Fault: ERR-TRU150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L150.

PRM_45216 [Trumpf-TruLaser-L150] — PLC_LOOP_GAIN_16 | Default: 192.000 | Linked Fault: ERR-TRU150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L150.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L200 (BANK 1)

Machine Model: Trumpf-TruLaser-L200 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L200.

PRM_45221 [Trumpf-TruLaser-L200] — ATC_LOOP_GAIN_1 | Default: 59.800 | Linked Fault: ERR-TRU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L200.

PRM_45222 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_2 | Default: 119.600 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45223 [Trumpf-TruLaser-L200] — CLN_LOOP_GAIN_3 | Default: 179.400 | Linked Fault: ERR-TRU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L200.

PRM_45224 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_4 | Default: 239.200 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45225 [Trumpf-TruLaser-L200] — YAX_LOOP_GAIN_5 | Default: 49.000 | Linked Fault: ERR-TRU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L200.

PRM_45226 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_6 | Default: 108.800 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45227 [Trumpf-TruLaser-L200] — BAX_LOOP_GAIN_7 | Default: 168.600 | Linked Fault: ERR-TRU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L200.

PRM_45228 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_8 | Default: 228.400 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45229 [Trumpf-TruLaser-L200] — OPT_LOOP_GAIN_9 | Default: 38.200 | Linked Fault: ERR-TRU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L200.

PRM_45230 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_10 | Default: 98.000 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45231 [Trumpf-TruLaser-L200] — LUB_LOOP_GAIN_11 | Default: 157.800 | Linked Fault: ERR-TRU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L200.

PRM_45232 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_12 | Default: 217.600 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45233 [Trumpf-TruLaser-L200] — THM_LOOP_GAIN_13 | Default: 27.400 | Linked Fault: ERR-TRU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L200.

PRM_45234 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_14 | Default: 87.200 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45235 [Trumpf-TruLaser-L200] — ATC_LOOP_GAIN_15 | Default: 147.000 | Linked Fault: ERR-TRU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L200.

PRM_45236 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_16 | Default: 206.800 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L200 (BANK 2)

Machine Model: Trumpf-TruLaser-L200 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L200.

PRM_45241 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_1 | Default: 119.600 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45242 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_2 | Default: 239.200 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45243 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_3 | Default: 108.800 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45244 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_4 | Default: 228.400 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45245 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_5 | Default: 98.000 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45246 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_6 | Default: 217.600 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45247 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_7 | Default: 87.200 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45248 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_8 | Default: 206.800 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45249 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_9 | Default: 76.400 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45250 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_10 | Default: 196.000 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45251 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_11 | Default: 65.600 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45252 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_12 | Default: 185.200 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45253 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_13 | Default: 54.800 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45254 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_14 | Default: 174.400 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45255 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_15 | Default: 44.000 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45256 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_16 | Default: 163.600 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L200 (BANK 3)

Machine Model: Trumpf-TruLaser-L200 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L200.

PRM_45261 [Trumpf-TruLaser-L200] — CLN_LOOP_GAIN_1 | Default: 179.400 | Linked Fault: ERR-TRU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L200.

PRM_45262 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_2 | Default: 108.800 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45263 [Trumpf-TruLaser-L200] — OPT_LOOP_GAIN_3 | Default: 38.200 | Linked Fault: ERR-TRU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L200.

PRM_45264 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_4 | Default: 217.600 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45265 [Trumpf-TruLaser-L200] — ATC_LOOP_GAIN_5 | Default: 147.000 | Linked Fault: ERR-TRU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L200.

PRM_45266 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_6 | Default: 76.400 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45267 [Trumpf-TruLaser-L200] — BAX_LOOP_GAIN_7 | Default: 5.800 | Linked Fault: ERR-TRU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L200.

PRM_45268 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_8 | Default: 185.200 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45269 [Trumpf-TruLaser-L200] — THM_LOOP_GAIN_9 | Default: 114.600 | Linked Fault: ERR-TRU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L200.

PRM_45270 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_10 | Default: 44.000 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45271 [Trumpf-TruLaser-L200] — YAX_LOOP_GAIN_11 | Default: 223.400 | Linked Fault: ERR-TRU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L200.

PRM_45272 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_12 | Default: 152.800 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45273 [Trumpf-TruLaser-L200] — LUB_LOOP_GAIN_13 | Default: 82.200 | Linked Fault: ERR-TRU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L200.

PRM_45274 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_14 | Default: 11.600 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45275 [Trumpf-TruLaser-L200] — CLN_LOOP_GAIN_15 | Default: 191.000 | Linked Fault: ERR-TRU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L200.

PRM_45276 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_16 | Default: 120.400 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L200 (BANK 4)

Machine Model: Trumpf-TruLaser-L200 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L200.

PRM_45281 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_1 | Default: 239.200 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45282 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_2 | Default: 228.400 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45283 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_3 | Default: 217.600 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45284 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_4 | Default: 206.800 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45285 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_5 | Default: 196.000 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45286 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_6 | Default: 185.200 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45287 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_7 | Default: 174.400 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45288 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_8 | Default: 163.600 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45289 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_9 | Default: 152.800 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45290 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_10 | Default: 142.000 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45291 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_11 | Default: 131.200 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45292 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_12 | Default: 120.400 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45293 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_13 | Default: 109.600 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45294 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_14 | Default: 98.800 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45295 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_15 | Default: 88.000 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45296 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_16 | Default: 77.200 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L200 (BANK 5)

Machine Model: Trumpf-TruLaser-L200 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L200.

PRM_45301 [Trumpf-TruLaser-L200] — YAX_LOOP_GAIN_1 | Default: 49.000 | Linked Fault: ERR-TRU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L200.

PRM_45302 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_2 | Default: 98.000 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

PRM_45303 [Trumpf-TruLaser-L200] — ATC_LOOP_GAIN_3 | Default: 147.000 | Linked Fault: ERR-TRU200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L200.

PRM_45304 [Trumpf-TruLaser-L200] — ZAX_LOOP_GAIN_4 | Default: 196.000 | Linked Fault: ERR-TRU200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L200.

PRM_45305 [Trumpf-TruLaser-L200] — LUB_LOOP_GAIN_5 | Default: 245.000 | Linked Fault: ERR-TRU200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L200.

PRM_45306 [Trumpf-TruLaser-L200] — HYD_LOOP_GAIN_6 | Default: 44.000 | Linked Fault: ERR-TRU200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L200.

PRM_45307 [Trumpf-TruLaser-L200] — BAX_LOOP_GAIN_7 | Default: 93.000 | Linked Fault: ERR-TRU200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L200.

PRM_45308 [Trumpf-TruLaser-L200] — VAC_LOOP_GAIN_8 | Default: 142.000 | Linked Fault: ERR-TRU200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L200.

PRM_45309 [Trumpf-TruLaser-L200] — CLN_LOOP_GAIN_9 | Default: 191.000 | Linked Fault: ERR-TRU200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L200.

PRM_45310 [Trumpf-TruLaser-L200] — CAX_LOOP_GAIN_10 | Default: 240.000 | Linked Fault: ERR-TRU200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L200.

PRM_45311 [Trumpf-TruLaser-L200] — THM_LOOP_GAIN_11 | Default: 39.000 | Linked Fault: ERR-TRU200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L200.

PRM_45312 [Trumpf-TruLaser-L200] — XAX_LOOP_GAIN_12 | Default: 88.000 | Linked Fault: ERR-TRU200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L200.

PRM_45313 [Trumpf-TruLaser-L200] — OPT_LOOP_GAIN_13 | Default: 137.000 | Linked Fault: ERR-TRU200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L200.

PRM_45314 [Trumpf-TruLaser-L200] — SPN_LOOP_GAIN_14 | Default: 186.000 | Linked Fault: ERR-TRU200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L200.

PRM_45315 [Trumpf-TruLaser-L200] — YAX_LOOP_GAIN_15 | Default: 235.000 | Linked Fault: ERR-TRU200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L200.

PRM_45316 [Trumpf-TruLaser-L200] — PLC_LOOP_GAIN_16 | Default: 34.000 | Linked Fault: ERR-TRU200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L200.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L250 (BANK 1)

Machine Model: Trumpf-TruLaser-L250 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L250.

PRM_45321 [Trumpf-TruLaser-L250] — ATC_LOOP_GAIN_1 | Default: 60.950 | Linked Fault: ERR-TRU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L250.

PRM_45322 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_2 | Default: 121.900 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45323 [Trumpf-TruLaser-L250] — CLN_LOOP_GAIN_3 | Default: 182.850 | Linked Fault: ERR-TRU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L250.

PRM_45324 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_4 | Default: 243.800 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45325 [Trumpf-TruLaser-L250] — YAX_LOOP_GAIN_5 | Default: 54.750 | Linked Fault: ERR-TRU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L250.

PRM_45326 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_6 | Default: 115.700 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45327 [Trumpf-TruLaser-L250] — BAX_LOOP_GAIN_7 | Default: 176.650 | Linked Fault: ERR-TRU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L250.

PRM_45328 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_8 | Default: 237.600 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45329 [Trumpf-TruLaser-L250] — OPT_LOOP_GAIN_9 | Default: 48.550 | Linked Fault: ERR-TRU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L250.

PRM_45330 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_10 | Default: 109.500 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45331 [Trumpf-TruLaser-L250] — LUB_LOOP_GAIN_11 | Default: 170.450 | Linked Fault: ERR-TRU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L250.

PRM_45332 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_12 | Default: 231.400 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45333 [Trumpf-TruLaser-L250] — THM_LOOP_GAIN_13 | Default: 42.350 | Linked Fault: ERR-TRU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L250.

PRM_45334 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_14 | Default: 103.300 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45335 [Trumpf-TruLaser-L250] — ATC_LOOP_GAIN_15 | Default: 164.250 | Linked Fault: ERR-TRU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L250.

PRM_45336 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_16 | Default: 225.200 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L250 (BANK 2)

Machine Model: Trumpf-TruLaser-L250 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L250.

PRM_45341 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_1 | Default: 121.900 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45342 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_2 | Default: 243.800 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45343 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_3 | Default: 115.700 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45344 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_4 | Default: 237.600 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45345 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_5 | Default: 109.500 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45346 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_6 | Default: 231.400 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45347 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_7 | Default: 103.300 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45348 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_8 | Default: 225.200 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45349 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_9 | Default: 97.100 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45350 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_10 | Default: 219.000 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45351 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_11 | Default: 90.900 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45352 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_12 | Default: 212.800 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45353 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_13 | Default: 84.700 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45354 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_14 | Default: 206.600 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45355 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_15 | Default: 78.500 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45356 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_16 | Default: 200.400 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L250 (BANK 3)

Machine Model: Trumpf-TruLaser-L250 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L250.

PRM_45361 [Trumpf-TruLaser-L250] — CLN_LOOP_GAIN_1 | Default: 182.850 | Linked Fault: ERR-TRU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L250.

PRM_45362 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_2 | Default: 115.700 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45363 [Trumpf-TruLaser-L250] — OPT_LOOP_GAIN_3 | Default: 48.550 | Linked Fault: ERR-TRU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L250.

PRM_45364 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_4 | Default: 231.400 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45365 [Trumpf-TruLaser-L250] — ATC_LOOP_GAIN_5 | Default: 164.250 | Linked Fault: ERR-TRU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L250.

PRM_45366 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_6 | Default: 97.100 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45367 [Trumpf-TruLaser-L250] — BAX_LOOP_GAIN_7 | Default: 29.950 | Linked Fault: ERR-TRU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L250.

PRM_45368 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_8 | Default: 212.800 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45369 [Trumpf-TruLaser-L250] — THM_LOOP_GAIN_9 | Default: 145.650 | Linked Fault: ERR-TRU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L250.

PRM_45370 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_10 | Default: 78.500 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45371 [Trumpf-TruLaser-L250] — YAX_LOOP_GAIN_11 | Default: 11.350 | Linked Fault: ERR-TRU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L250.

PRM_45372 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_12 | Default: 194.200 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45373 [Trumpf-TruLaser-L250] — LUB_LOOP_GAIN_13 | Default: 127.050 | Linked Fault: ERR-TRU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L250.

PRM_45374 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_14 | Default: 59.900 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45375 [Trumpf-TruLaser-L250] — CLN_LOOP_GAIN_15 | Default: 242.750 | Linked Fault: ERR-TRU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L250.

PRM_45376 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_16 | Default: 175.600 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L250 (BANK 4)

Machine Model: Trumpf-TruLaser-L250 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L250.

PRM_45381 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_1 | Default: 243.800 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45382 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_2 | Default: 237.600 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45383 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_3 | Default: 231.400 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45384 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_4 | Default: 225.200 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45385 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_5 | Default: 219.000 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45386 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_6 | Default: 212.800 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45387 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_7 | Default: 206.600 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45388 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_8 | Default: 200.400 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45389 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_9 | Default: 194.200 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45390 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_10 | Default: 188.000 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45391 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_11 | Default: 181.800 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45392 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_12 | Default: 175.600 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45393 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_13 | Default: 169.400 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45394 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_14 | Default: 163.200 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45395 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_15 | Default: 157.000 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45396 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_16 | Default: 150.800 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L250 (BANK 5)

Machine Model: Trumpf-TruLaser-L250 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L250.

PRM_45401 [Trumpf-TruLaser-L250] — YAX_LOOP_GAIN_1 | Default: 54.750 | Linked Fault: ERR-TRU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L250.

PRM_45402 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_2 | Default: 109.500 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

PRM_45403 [Trumpf-TruLaser-L250] — ATC_LOOP_GAIN_3 | Default: 164.250 | Linked Fault: ERR-TRU250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L250.

PRM_45404 [Trumpf-TruLaser-L250] — ZAX_LOOP_GAIN_4 | Default: 219.000 | Linked Fault: ERR-TRU250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L250.

PRM_45405 [Trumpf-TruLaser-L250] — LUB_LOOP_GAIN_5 | Default: 23.750 | Linked Fault: ERR-TRU250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L250.

PRM_45406 [Trumpf-TruLaser-L250] — HYD_LOOP_GAIN_6 | Default: 78.500 | Linked Fault: ERR-TRU250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L250.

PRM_45407 [Trumpf-TruLaser-L250] — BAX_LOOP_GAIN_7 | Default: 133.250 | Linked Fault: ERR-TRU250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L250.

PRM_45408 [Trumpf-TruLaser-L250] — VAC_LOOP_GAIN_8 | Default: 188.000 | Linked Fault: ERR-TRU250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L250.

PRM_45409 [Trumpf-TruLaser-L250] — CLN_LOOP_GAIN_9 | Default: 242.750 | Linked Fault: ERR-TRU250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L250.

PRM_45410 [Trumpf-TruLaser-L250] — CAX_LOOP_GAIN_10 | Default: 47.500 | Linked Fault: ERR-TRU250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L250.

PRM_45411 [Trumpf-TruLaser-L250] — THM_LOOP_GAIN_11 | Default: 102.250 | Linked Fault: ERR-TRU250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L250.

PRM_45412 [Trumpf-TruLaser-L250] — XAX_LOOP_GAIN_12 | Default: 157.000 | Linked Fault: ERR-TRU250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L250.

PRM_45413 [Trumpf-TruLaser-L250] — OPT_LOOP_GAIN_13 | Default: 211.750 | Linked Fault: ERR-TRU250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L250.

PRM_45414 [Trumpf-TruLaser-L250] — SPN_LOOP_GAIN_14 | Default: 16.500 | Linked Fault: ERR-TRU250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L250.

PRM_45415 [Trumpf-TruLaser-L250] — YAX_LOOP_GAIN_15 | Default: 71.250 | Linked Fault: ERR-TRU250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L250.

PRM_45416 [Trumpf-TruLaser-L250] — PLC_LOOP_GAIN_16 | Default: 126.000 | Linked Fault: ERR-TRU250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L250.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L300 (BANK 1)

Machine Model: Trumpf-TruLaser-L300 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L300.

PRM_45421 [Trumpf-TruLaser-L300] — ATC_LOOP_GAIN_1 | Default: 62.100 | Linked Fault: ERR-TRU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L300.

PRM_45422 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_2 | Default: 124.200 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45423 [Trumpf-TruLaser-L300] — CLN_LOOP_GAIN_3 | Default: 186.300 | Linked Fault: ERR-TRU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L300.

PRM_45424 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_4 | Default: 248.400 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45425 [Trumpf-TruLaser-L300] — YAX_LOOP_GAIN_5 | Default: 60.500 | Linked Fault: ERR-TRU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L300.

PRM_45426 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_6 | Default: 122.600 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45427 [Trumpf-TruLaser-L300] — BAX_LOOP_GAIN_7 | Default: 184.700 | Linked Fault: ERR-TRU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L300.

PRM_45428 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_8 | Default: 246.800 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45429 [Trumpf-TruLaser-L300] — OPT_LOOP_GAIN_9 | Default: 58.900 | Linked Fault: ERR-TRU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L300.

PRM_45430 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_10 | Default: 121.000 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45431 [Trumpf-TruLaser-L300] — LUB_LOOP_GAIN_11 | Default: 183.100 | Linked Fault: ERR-TRU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L300.

PRM_45432 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_12 | Default: 245.200 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45433 [Trumpf-TruLaser-L300] — THM_LOOP_GAIN_13 | Default: 57.300 | Linked Fault: ERR-TRU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L300.

PRM_45434 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_14 | Default: 119.400 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45435 [Trumpf-TruLaser-L300] — ATC_LOOP_GAIN_15 | Default: 181.500 | Linked Fault: ERR-TRU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L300.

PRM_45436 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_16 | Default: 243.600 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L300 (BANK 2)

Machine Model: Trumpf-TruLaser-L300 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L300.

PRM_45441 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_1 | Default: 124.200 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45442 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_2 | Default: 248.400 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45443 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_3 | Default: 122.600 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45444 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_4 | Default: 246.800 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45445 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_5 | Default: 121.000 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45446 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_6 | Default: 245.200 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45447 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_7 | Default: 119.400 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45448 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_8 | Default: 243.600 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45449 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_9 | Default: 117.800 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45450 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_10 | Default: 242.000 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45451 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_11 | Default: 116.200 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45452 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_12 | Default: 240.400 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45453 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_13 | Default: 114.600 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45454 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_14 | Default: 238.800 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45455 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_15 | Default: 113.000 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45456 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_16 | Default: 237.200 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L300 (BANK 3)

Machine Model: Trumpf-TruLaser-L300 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L300.

PRM_45461 [Trumpf-TruLaser-L300] — CLN_LOOP_GAIN_1 | Default: 186.300 | Linked Fault: ERR-TRU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L300.

PRM_45462 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_2 | Default: 122.600 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45463 [Trumpf-TruLaser-L300] — OPT_LOOP_GAIN_3 | Default: 58.900 | Linked Fault: ERR-TRU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L300.

PRM_45464 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_4 | Default: 245.200 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45465 [Trumpf-TruLaser-L300] — ATC_LOOP_GAIN_5 | Default: 181.500 | Linked Fault: ERR-TRU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L300.

PRM_45466 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_6 | Default: 117.800 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45467 [Trumpf-TruLaser-L300] — BAX_LOOP_GAIN_7 | Default: 54.100 | Linked Fault: ERR-TRU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L300.

PRM_45468 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_8 | Default: 240.400 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45469 [Trumpf-TruLaser-L300] — THM_LOOP_GAIN_9 | Default: 176.700 | Linked Fault: ERR-TRU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L300.

PRM_45470 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_10 | Default: 113.000 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45471 [Trumpf-TruLaser-L300] — YAX_LOOP_GAIN_11 | Default: 49.300 | Linked Fault: ERR-TRU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L300.

PRM_45472 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_12 | Default: 235.600 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45473 [Trumpf-TruLaser-L300] — LUB_LOOP_GAIN_13 | Default: 171.900 | Linked Fault: ERR-TRU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L300.

PRM_45474 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_14 | Default: 108.200 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45475 [Trumpf-TruLaser-L300] — CLN_LOOP_GAIN_15 | Default: 44.500 | Linked Fault: ERR-TRU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L300.

PRM_45476 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_16 | Default: 230.800 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L300 (BANK 4)

Machine Model: Trumpf-TruLaser-L300 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L300.

PRM_45481 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_1 | Default: 248.400 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45482 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_2 | Default: 246.800 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45483 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_3 | Default: 245.200 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45484 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_4 | Default: 243.600 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45485 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_5 | Default: 242.000 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45486 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_6 | Default: 240.400 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45487 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_7 | Default: 238.800 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45488 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_8 | Default: 237.200 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45489 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_9 | Default: 235.600 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45490 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_10 | Default: 234.000 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45491 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_11 | Default: 232.400 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45492 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_12 | Default: 230.800 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45493 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_13 | Default: 229.200 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45494 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_14 | Default: 227.600 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45495 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_15 | Default: 226.000 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45496 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_16 | Default: 224.400 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L300 (BANK 5)

Machine Model: Trumpf-TruLaser-L300 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L300.

PRM_45501 [Trumpf-TruLaser-L300] — YAX_LOOP_GAIN_1 | Default: 60.500 | Linked Fault: ERR-TRU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L300.

PRM_45502 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_2 | Default: 121.000 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

PRM_45503 [Trumpf-TruLaser-L300] — ATC_LOOP_GAIN_3 | Default: 181.500 | Linked Fault: ERR-TRU300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L300.

PRM_45504 [Trumpf-TruLaser-L300] — ZAX_LOOP_GAIN_4 | Default: 242.000 | Linked Fault: ERR-TRU300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L300.

PRM_45505 [Trumpf-TruLaser-L300] — LUB_LOOP_GAIN_5 | Default: 52.500 | Linked Fault: ERR-TRU300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L300.

PRM_45506 [Trumpf-TruLaser-L300] — HYD_LOOP_GAIN_6 | Default: 113.000 | Linked Fault: ERR-TRU300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L300.

PRM_45507 [Trumpf-TruLaser-L300] — BAX_LOOP_GAIN_7 | Default: 173.500 | Linked Fault: ERR-TRU300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L300.

PRM_45508 [Trumpf-TruLaser-L300] — VAC_LOOP_GAIN_8 | Default: 234.000 | Linked Fault: ERR-TRU300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L300.

PRM_45509 [Trumpf-TruLaser-L300] — CLN_LOOP_GAIN_9 | Default: 44.500 | Linked Fault: ERR-TRU300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L300.

PRM_45510 [Trumpf-TruLaser-L300] — CAX_LOOP_GAIN_10 | Default: 105.000 | Linked Fault: ERR-TRU300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L300.

PRM_45511 [Trumpf-TruLaser-L300] — THM_LOOP_GAIN_11 | Default: 165.500 | Linked Fault: ERR-TRU300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L300.

PRM_45512 [Trumpf-TruLaser-L300] — XAX_LOOP_GAIN_12 | Default: 226.000 | Linked Fault: ERR-TRU300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L300.

PRM_45513 [Trumpf-TruLaser-L300] — OPT_LOOP_GAIN_13 | Default: 36.500 | Linked Fault: ERR-TRU300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L300.

PRM_45514 [Trumpf-TruLaser-L300] — SPN_LOOP_GAIN_14 | Default: 97.000 | Linked Fault: ERR-TRU300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L300.

PRM_45515 [Trumpf-TruLaser-L300] — YAX_LOOP_GAIN_15 | Default: 157.500 | Linked Fault: ERR-TRU300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L300.

PRM_45516 [Trumpf-TruLaser-L300] — PLC_LOOP_GAIN_16 | Default: 218.000 | Linked Fault: ERR-TRU300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L300.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L350 (BANK 1)

Machine Model: Trumpf-TruLaser-L350 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L350.

PRM_45521 [Trumpf-TruLaser-L350] — ATC_LOOP_GAIN_1 | Default: 63.250 | Linked Fault: ERR-TRU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L350.

PRM_45522 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_2 | Default: 126.500 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45523 [Trumpf-TruLaser-L350] — CLN_LOOP_GAIN_3 | Default: 189.750 | Linked Fault: ERR-TRU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L350.

PRM_45524 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_4 | Default: 3.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45525 [Trumpf-TruLaser-L350] — YAX_LOOP_GAIN_5 | Default: 66.250 | Linked Fault: ERR-TRU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L350.

PRM_45526 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_6 | Default: 129.500 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45527 [Trumpf-TruLaser-L350] — BAX_LOOP_GAIN_7 | Default: 192.750 | Linked Fault: ERR-TRU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L350.

PRM_45528 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_8 | Default: 6.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45529 [Trumpf-TruLaser-L350] — OPT_LOOP_GAIN_9 | Default: 69.250 | Linked Fault: ERR-TRU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L350.

PRM_45530 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_10 | Default: 132.500 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45531 [Trumpf-TruLaser-L350] — LUB_LOOP_GAIN_11 | Default: 195.750 | Linked Fault: ERR-TRU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L350.

PRM_45532 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_12 | Default: 9.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45533 [Trumpf-TruLaser-L350] — THM_LOOP_GAIN_13 | Default: 72.250 | Linked Fault: ERR-TRU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L350.

PRM_45534 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_14 | Default: 135.500 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45535 [Trumpf-TruLaser-L350] — ATC_LOOP_GAIN_15 | Default: 198.750 | Linked Fault: ERR-TRU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L350.

PRM_45536 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_16 | Default: 12.000 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L350 (BANK 2)

Machine Model: Trumpf-TruLaser-L350 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L350.

PRM_45541 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_1 | Default: 126.500 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45542 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_2 | Default: 3.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45543 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_3 | Default: 129.500 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45544 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_4 | Default: 6.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45545 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_5 | Default: 132.500 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45546 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_6 | Default: 9.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45547 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_7 | Default: 135.500 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45548 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_8 | Default: 12.000 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45549 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_9 | Default: 138.500 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45550 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_10 | Default: 15.000 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45551 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_11 | Default: 141.500 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45552 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_12 | Default: 18.000 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45553 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_13 | Default: 144.500 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45554 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_14 | Default: 21.000 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45555 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_15 | Default: 147.500 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45556 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_16 | Default: 24.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L350 (BANK 3)

Machine Model: Trumpf-TruLaser-L350 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L350.

PRM_45561 [Trumpf-TruLaser-L350] — CLN_LOOP_GAIN_1 | Default: 189.750 | Linked Fault: ERR-TRU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L350.

PRM_45562 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_2 | Default: 129.500 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45563 [Trumpf-TruLaser-L350] — OPT_LOOP_GAIN_3 | Default: 69.250 | Linked Fault: ERR-TRU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L350.

PRM_45564 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_4 | Default: 9.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45565 [Trumpf-TruLaser-L350] — ATC_LOOP_GAIN_5 | Default: 198.750 | Linked Fault: ERR-TRU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L350.

PRM_45566 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_6 | Default: 138.500 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45567 [Trumpf-TruLaser-L350] — BAX_LOOP_GAIN_7 | Default: 78.250 | Linked Fault: ERR-TRU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L350.

PRM_45568 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_8 | Default: 18.000 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45569 [Trumpf-TruLaser-L350] — THM_LOOP_GAIN_9 | Default: 207.750 | Linked Fault: ERR-TRU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L350.

PRM_45570 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_10 | Default: 147.500 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45571 [Trumpf-TruLaser-L350] — YAX_LOOP_GAIN_11 | Default: 87.250 | Linked Fault: ERR-TRU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L350.

PRM_45572 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_12 | Default: 27.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45573 [Trumpf-TruLaser-L350] — LUB_LOOP_GAIN_13 | Default: 216.750 | Linked Fault: ERR-TRU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L350.

PRM_45574 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_14 | Default: 156.500 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45575 [Trumpf-TruLaser-L350] — CLN_LOOP_GAIN_15 | Default: 96.250 | Linked Fault: ERR-TRU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L350.

PRM_45576 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_16 | Default: 36.000 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L350 (BANK 4)

Machine Model: Trumpf-TruLaser-L350 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L350.

PRM_45581 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_1 | Default: 3.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45582 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_2 | Default: 6.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45583 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_3 | Default: 9.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45584 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_4 | Default: 12.000 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45585 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_5 | Default: 15.000 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45586 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_6 | Default: 18.000 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45587 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_7 | Default: 21.000 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45588 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_8 | Default: 24.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45589 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_9 | Default: 27.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45590 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_10 | Default: 30.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45591 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_11 | Default: 33.000 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45592 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_12 | Default: 36.000 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45593 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_13 | Default: 39.000 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45594 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_14 | Default: 42.000 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45595 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_15 | Default: 45.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45596 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_16 | Default: 48.000 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L350 (BANK 5)

Machine Model: Trumpf-TruLaser-L350 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L350.

PRM_45601 [Trumpf-TruLaser-L350] — YAX_LOOP_GAIN_1 | Default: 66.250 | Linked Fault: ERR-TRU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L350.

PRM_45602 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_2 | Default: 132.500 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

PRM_45603 [Trumpf-TruLaser-L350] — ATC_LOOP_GAIN_3 | Default: 198.750 | Linked Fault: ERR-TRU350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L350.

PRM_45604 [Trumpf-TruLaser-L350] — ZAX_LOOP_GAIN_4 | Default: 15.000 | Linked Fault: ERR-TRU350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L350.

PRM_45605 [Trumpf-TruLaser-L350] — LUB_LOOP_GAIN_5 | Default: 81.250 | Linked Fault: ERR-TRU350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L350.

PRM_45606 [Trumpf-TruLaser-L350] — HYD_LOOP_GAIN_6 | Default: 147.500 | Linked Fault: ERR-TRU350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L350.

PRM_45607 [Trumpf-TruLaser-L350] — BAX_LOOP_GAIN_7 | Default: 213.750 | Linked Fault: ERR-TRU350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L350.

PRM_45608 [Trumpf-TruLaser-L350] — VAC_LOOP_GAIN_8 | Default: 30.000 | Linked Fault: ERR-TRU350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L350.

PRM_45609 [Trumpf-TruLaser-L350] — CLN_LOOP_GAIN_9 | Default: 96.250 | Linked Fault: ERR-TRU350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L350.

PRM_45610 [Trumpf-TruLaser-L350] — CAX_LOOP_GAIN_10 | Default: 162.500 | Linked Fault: ERR-TRU350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L350.

PRM_45611 [Trumpf-TruLaser-L350] — THM_LOOP_GAIN_11 | Default: 228.750 | Linked Fault: ERR-TRU350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L350.

PRM_45612 [Trumpf-TruLaser-L350] — XAX_LOOP_GAIN_12 | Default: 45.000 | Linked Fault: ERR-TRU350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L350.

PRM_45613 [Trumpf-TruLaser-L350] — OPT_LOOP_GAIN_13 | Default: 111.250 | Linked Fault: ERR-TRU350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L350.

PRM_45614 [Trumpf-TruLaser-L350] — SPN_LOOP_GAIN_14 | Default: 177.500 | Linked Fault: ERR-TRU350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L350.

PRM_45615 [Trumpf-TruLaser-L350] — YAX_LOOP_GAIN_15 | Default: 243.750 | Linked Fault: ERR-TRU350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L350.

PRM_45616 [Trumpf-TruLaser-L350] — PLC_LOOP_GAIN_16 | Default: 60.000 | Linked Fault: ERR-TRU350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L350.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L400 (BANK 1)

Machine Model: Trumpf-TruLaser-L400 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L400.

PRM_45621 [Trumpf-TruLaser-L400] — ATC_LOOP_GAIN_1 | Default: 64.400 | Linked Fault: ERR-TRU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L400.

PRM_45622 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_2 | Default: 128.800 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45623 [Trumpf-TruLaser-L400] — CLN_LOOP_GAIN_3 | Default: 193.200 | Linked Fault: ERR-TRU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L400.

PRM_45624 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_4 | Default: 7.600 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45625 [Trumpf-TruLaser-L400] — YAX_LOOP_GAIN_5 | Default: 72.000 | Linked Fault: ERR-TRU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L400.

PRM_45626 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_6 | Default: 136.400 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45627 [Trumpf-TruLaser-L400] — BAX_LOOP_GAIN_7 | Default: 200.800 | Linked Fault: ERR-TRU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L400.

PRM_45628 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_8 | Default: 15.200 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45629 [Trumpf-TruLaser-L400] — OPT_LOOP_GAIN_9 | Default: 79.600 | Linked Fault: ERR-TRU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L400.

PRM_45630 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_10 | Default: 144.000 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45631 [Trumpf-TruLaser-L400] — LUB_LOOP_GAIN_11 | Default: 208.400 | Linked Fault: ERR-TRU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L400.

PRM_45632 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_12 | Default: 22.800 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45633 [Trumpf-TruLaser-L400] — THM_LOOP_GAIN_13 | Default: 87.200 | Linked Fault: ERR-TRU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L400.

PRM_45634 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_14 | Default: 151.600 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45635 [Trumpf-TruLaser-L400] — ATC_LOOP_GAIN_15 | Default: 216.000 | Linked Fault: ERR-TRU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L400.

PRM_45636 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_16 | Default: 30.400 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L400 (BANK 2)

Machine Model: Trumpf-TruLaser-L400 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L400.

PRM_45641 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_1 | Default: 128.800 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45642 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_2 | Default: 7.600 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45643 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_3 | Default: 136.400 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45644 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_4 | Default: 15.200 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45645 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_5 | Default: 144.000 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45646 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_6 | Default: 22.800 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45647 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_7 | Default: 151.600 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45648 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_8 | Default: 30.400 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45649 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_9 | Default: 159.200 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45650 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_10 | Default: 38.000 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45651 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_11 | Default: 166.800 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45652 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_12 | Default: 45.600 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45653 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_13 | Default: 174.400 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45654 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_14 | Default: 53.200 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45655 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_15 | Default: 182.000 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45656 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_16 | Default: 60.800 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L400 (BANK 3)

Machine Model: Trumpf-TruLaser-L400 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L400.

PRM_45661 [Trumpf-TruLaser-L400] — CLN_LOOP_GAIN_1 | Default: 193.200 | Linked Fault: ERR-TRU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L400.

PRM_45662 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_2 | Default: 136.400 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45663 [Trumpf-TruLaser-L400] — OPT_LOOP_GAIN_3 | Default: 79.600 | Linked Fault: ERR-TRU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L400.

PRM_45664 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_4 | Default: 22.800 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45665 [Trumpf-TruLaser-L400] — ATC_LOOP_GAIN_5 | Default: 216.000 | Linked Fault: ERR-TRU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L400.

PRM_45666 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_6 | Default: 159.200 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45667 [Trumpf-TruLaser-L400] — BAX_LOOP_GAIN_7 | Default: 102.400 | Linked Fault: ERR-TRU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L400.

PRM_45668 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_8 | Default: 45.600 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45669 [Trumpf-TruLaser-L400] — THM_LOOP_GAIN_9 | Default: 238.800 | Linked Fault: ERR-TRU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L400.

PRM_45670 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_10 | Default: 182.000 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45671 [Trumpf-TruLaser-L400] — YAX_LOOP_GAIN_11 | Default: 125.200 | Linked Fault: ERR-TRU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L400.

PRM_45672 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_12 | Default: 68.400 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45673 [Trumpf-TruLaser-L400] — LUB_LOOP_GAIN_13 | Default: 11.600 | Linked Fault: ERR-TRU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L400.

PRM_45674 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_14 | Default: 204.800 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45675 [Trumpf-TruLaser-L400] — CLN_LOOP_GAIN_15 | Default: 148.000 | Linked Fault: ERR-TRU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L400.

PRM_45676 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_16 | Default: 91.200 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L400 (BANK 4)

Machine Model: Trumpf-TruLaser-L400 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L400.

PRM_45681 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_1 | Default: 7.600 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45682 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_2 | Default: 15.200 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45683 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_3 | Default: 22.800 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45684 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_4 | Default: 30.400 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45685 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_5 | Default: 38.000 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45686 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_6 | Default: 45.600 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45687 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_7 | Default: 53.200 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45688 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_8 | Default: 60.800 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45689 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_9 | Default: 68.400 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45690 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_10 | Default: 76.000 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45691 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_11 | Default: 83.600 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45692 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_12 | Default: 91.200 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45693 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_13 | Default: 98.800 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45694 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_14 | Default: 106.400 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45695 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_15 | Default: 114.000 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45696 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_16 | Default: 121.600 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L400 (BANK 5)

Machine Model: Trumpf-TruLaser-L400 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L400.

PRM_45701 [Trumpf-TruLaser-L400] — YAX_LOOP_GAIN_1 | Default: 72.000 | Linked Fault: ERR-TRU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L400.

PRM_45702 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_2 | Default: 144.000 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

PRM_45703 [Trumpf-TruLaser-L400] — ATC_LOOP_GAIN_3 | Default: 216.000 | Linked Fault: ERR-TRU400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L400.

PRM_45704 [Trumpf-TruLaser-L400] — ZAX_LOOP_GAIN_4 | Default: 38.000 | Linked Fault: ERR-TRU400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L400.

PRM_45705 [Trumpf-TruLaser-L400] — LUB_LOOP_GAIN_5 | Default: 110.000 | Linked Fault: ERR-TRU400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L400.

PRM_45706 [Trumpf-TruLaser-L400] — HYD_LOOP_GAIN_6 | Default: 182.000 | Linked Fault: ERR-TRU400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L400.

PRM_45707 [Trumpf-TruLaser-L400] — BAX_LOOP_GAIN_7 | Default: 4.000 | Linked Fault: ERR-TRU400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L400.

PRM_45708 [Trumpf-TruLaser-L400] — VAC_LOOP_GAIN_8 | Default: 76.000 | Linked Fault: ERR-TRU400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L400.

PRM_45709 [Trumpf-TruLaser-L400] — CLN_LOOP_GAIN_9 | Default: 148.000 | Linked Fault: ERR-TRU400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L400.

PRM_45710 [Trumpf-TruLaser-L400] — CAX_LOOP_GAIN_10 | Default: 220.000 | Linked Fault: ERR-TRU400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L400.

PRM_45711 [Trumpf-TruLaser-L400] — THM_LOOP_GAIN_11 | Default: 42.000 | Linked Fault: ERR-TRU400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L400.

PRM_45712 [Trumpf-TruLaser-L400] — XAX_LOOP_GAIN_12 | Default: 114.000 | Linked Fault: ERR-TRU400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L400.

PRM_45713 [Trumpf-TruLaser-L400] — OPT_LOOP_GAIN_13 | Default: 186.000 | Linked Fault: ERR-TRU400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L400.

PRM_45714 [Trumpf-TruLaser-L400] — SPN_LOOP_GAIN_14 | Default: 8.000 | Linked Fault: ERR-TRU400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L400.

PRM_45715 [Trumpf-TruLaser-L400] — YAX_LOOP_GAIN_15 | Default: 80.000 | Linked Fault: ERR-TRU400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L400.

PRM_45716 [Trumpf-TruLaser-L400] — PLC_LOOP_GAIN_16 | Default: 152.000 | Linked Fault: ERR-TRU400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L400.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L450 (BANK 1)

Machine Model: Trumpf-TruLaser-L450 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L450.

PRM_45721 [Trumpf-TruLaser-L450] — ATC_LOOP_GAIN_1 | Default: 65.550 | Linked Fault: ERR-TRU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L450.

PRM_45722 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_2 | Default: 131.100 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45723 [Trumpf-TruLaser-L450] — CLN_LOOP_GAIN_3 | Default: 196.650 | Linked Fault: ERR-TRU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L450.

PRM_45724 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_4 | Default: 12.200 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45725 [Trumpf-TruLaser-L450] — YAX_LOOP_GAIN_5 | Default: 77.750 | Linked Fault: ERR-TRU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L450.

PRM_45726 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_6 | Default: 143.300 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45727 [Trumpf-TruLaser-L450] — BAX_LOOP_GAIN_7 | Default: 208.850 | Linked Fault: ERR-TRU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L450.

PRM_45728 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_8 | Default: 24.400 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45729 [Trumpf-TruLaser-L450] — OPT_LOOP_GAIN_9 | Default: 89.950 | Linked Fault: ERR-TRU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L450.

PRM_45730 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_10 | Default: 155.500 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45731 [Trumpf-TruLaser-L450] — LUB_LOOP_GAIN_11 | Default: 221.050 | Linked Fault: ERR-TRU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L450.

PRM_45732 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_12 | Default: 36.600 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45733 [Trumpf-TruLaser-L450] — THM_LOOP_GAIN_13 | Default: 102.150 | Linked Fault: ERR-TRU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L450.

PRM_45734 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_14 | Default: 167.700 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45735 [Trumpf-TruLaser-L450] — ATC_LOOP_GAIN_15 | Default: 233.250 | Linked Fault: ERR-TRU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L450.

PRM_45736 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_16 | Default: 48.800 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L450 (BANK 2)

Machine Model: Trumpf-TruLaser-L450 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L450.

PRM_45741 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_1 | Default: 131.100 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45742 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_2 | Default: 12.200 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45743 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_3 | Default: 143.300 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45744 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_4 | Default: 24.400 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45745 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_5 | Default: 155.500 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45746 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_6 | Default: 36.600 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45747 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_7 | Default: 167.700 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45748 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_8 | Default: 48.800 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45749 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_9 | Default: 179.900 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45750 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_10 | Default: 61.000 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45751 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_11 | Default: 192.100 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45752 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_12 | Default: 73.200 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45753 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_13 | Default: 204.300 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45754 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_14 | Default: 85.400 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45755 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_15 | Default: 216.500 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45756 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_16 | Default: 97.600 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L450 (BANK 3)

Machine Model: Trumpf-TruLaser-L450 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L450.

PRM_45761 [Trumpf-TruLaser-L450] — CLN_LOOP_GAIN_1 | Default: 196.650 | Linked Fault: ERR-TRU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L450.

PRM_45762 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_2 | Default: 143.300 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45763 [Trumpf-TruLaser-L450] — OPT_LOOP_GAIN_3 | Default: 89.950 | Linked Fault: ERR-TRU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L450.

PRM_45764 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_4 | Default: 36.600 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45765 [Trumpf-TruLaser-L450] — ATC_LOOP_GAIN_5 | Default: 233.250 | Linked Fault: ERR-TRU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L450.

PRM_45766 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_6 | Default: 179.900 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45767 [Trumpf-TruLaser-L450] — BAX_LOOP_GAIN_7 | Default: 126.550 | Linked Fault: ERR-TRU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L450.

PRM_45768 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_8 | Default: 73.200 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45769 [Trumpf-TruLaser-L450] — THM_LOOP_GAIN_9 | Default: 19.850 | Linked Fault: ERR-TRU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L450.

PRM_45770 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_10 | Default: 216.500 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45771 [Trumpf-TruLaser-L450] — YAX_LOOP_GAIN_11 | Default: 163.150 | Linked Fault: ERR-TRU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L450.

PRM_45772 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_12 | Default: 109.800 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45773 [Trumpf-TruLaser-L450] — LUB_LOOP_GAIN_13 | Default: 56.450 | Linked Fault: ERR-TRU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L450.

PRM_45774 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_14 | Default: 3.100 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45775 [Trumpf-TruLaser-L450] — CLN_LOOP_GAIN_15 | Default: 199.750 | Linked Fault: ERR-TRU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L450.

PRM_45776 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_16 | Default: 146.400 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L450 (BANK 4)

Machine Model: Trumpf-TruLaser-L450 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L450.

PRM_45781 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_1 | Default: 12.200 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45782 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_2 | Default: 24.400 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45783 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_3 | Default: 36.600 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45784 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_4 | Default: 48.800 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45785 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_5 | Default: 61.000 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45786 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_6 | Default: 73.200 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45787 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_7 | Default: 85.400 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45788 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_8 | Default: 97.600 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45789 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_9 | Default: 109.800 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45790 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_10 | Default: 122.000 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45791 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_11 | Default: 134.200 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45792 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_12 | Default: 146.400 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45793 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_13 | Default: 158.600 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45794 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_14 | Default: 170.800 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45795 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_15 | Default: 183.000 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45796 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_16 | Default: 195.200 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L450 (BANK 5)

Machine Model: Trumpf-TruLaser-L450 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L450.

PRM_45801 [Trumpf-TruLaser-L450] — YAX_LOOP_GAIN_1 | Default: 77.750 | Linked Fault: ERR-TRU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L450.

PRM_45802 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_2 | Default: 155.500 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

PRM_45803 [Trumpf-TruLaser-L450] — ATC_LOOP_GAIN_3 | Default: 233.250 | Linked Fault: ERR-TRU450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L450.

PRM_45804 [Trumpf-TruLaser-L450] — ZAX_LOOP_GAIN_4 | Default: 61.000 | Linked Fault: ERR-TRU450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L450.

PRM_45805 [Trumpf-TruLaser-L450] — LUB_LOOP_GAIN_5 | Default: 138.750 | Linked Fault: ERR-TRU450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L450.

PRM_45806 [Trumpf-TruLaser-L450] — HYD_LOOP_GAIN_6 | Default: 216.500 | Linked Fault: ERR-TRU450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L450.

PRM_45807 [Trumpf-TruLaser-L450] — BAX_LOOP_GAIN_7 | Default: 44.250 | Linked Fault: ERR-TRU450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L450.

PRM_45808 [Trumpf-TruLaser-L450] — VAC_LOOP_GAIN_8 | Default: 122.000 | Linked Fault: ERR-TRU450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L450.

PRM_45809 [Trumpf-TruLaser-L450] — CLN_LOOP_GAIN_9 | Default: 199.750 | Linked Fault: ERR-TRU450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L450.

PRM_45810 [Trumpf-TruLaser-L450] — CAX_LOOP_GAIN_10 | Default: 27.500 | Linked Fault: ERR-TRU450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L450.

PRM_45811 [Trumpf-TruLaser-L450] — THM_LOOP_GAIN_11 | Default: 105.250 | Linked Fault: ERR-TRU450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L450.

PRM_45812 [Trumpf-TruLaser-L450] — XAX_LOOP_GAIN_12 | Default: 183.000 | Linked Fault: ERR-TRU450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L450.

PRM_45813 [Trumpf-TruLaser-L450] — OPT_LOOP_GAIN_13 | Default: 10.750 | Linked Fault: ERR-TRU450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L450.

PRM_45814 [Trumpf-TruLaser-L450] — SPN_LOOP_GAIN_14 | Default: 88.500 | Linked Fault: ERR-TRU450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L450.

PRM_45815 [Trumpf-TruLaser-L450] — YAX_LOOP_GAIN_15 | Default: 166.250 | Linked Fault: ERR-TRU450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L450.

PRM_45816 [Trumpf-TruLaser-L450] — PLC_LOOP_GAIN_16 | Default: 244.000 | Linked Fault: ERR-TRU450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L450.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L500 (BANK 1)

Machine Model: Trumpf-TruLaser-L500 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L500.

PRM_45821 [Trumpf-TruLaser-L500] — ATC_LOOP_GAIN_1 | Default: 66.700 | Linked Fault: ERR-TRU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L500.

PRM_45822 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_2 | Default: 133.400 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45823 [Trumpf-TruLaser-L500] — CLN_LOOP_GAIN_3 | Default: 200.100 | Linked Fault: ERR-TRU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L500.

PRM_45824 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_4 | Default: 16.800 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45825 [Trumpf-TruLaser-L500] — YAX_LOOP_GAIN_5 | Default: 83.500 | Linked Fault: ERR-TRU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L500.

PRM_45826 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_6 | Default: 150.200 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45827 [Trumpf-TruLaser-L500] — BAX_LOOP_GAIN_7 | Default: 216.900 | Linked Fault: ERR-TRU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L500.

PRM_45828 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_8 | Default: 33.600 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45829 [Trumpf-TruLaser-L500] — OPT_LOOP_GAIN_9 | Default: 100.300 | Linked Fault: ERR-TRU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L500.

PRM_45830 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_10 | Default: 167.000 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45831 [Trumpf-TruLaser-L500] — LUB_LOOP_GAIN_11 | Default: 233.700 | Linked Fault: ERR-TRU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L500.

PRM_45832 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_12 | Default: 50.400 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45833 [Trumpf-TruLaser-L500] — THM_LOOP_GAIN_13 | Default: 117.100 | Linked Fault: ERR-TRU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L500.

PRM_45834 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_14 | Default: 183.800 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45835 [Trumpf-TruLaser-L500] — ATC_LOOP_GAIN_15 | Default: 0.500 | Linked Fault: ERR-TRU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L500.

PRM_45836 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_16 | Default: 67.200 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L500 (BANK 2)

Machine Model: Trumpf-TruLaser-L500 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L500.

PRM_45841 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_1 | Default: 133.400 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45842 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_2 | Default: 16.800 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45843 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_3 | Default: 150.200 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45844 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_4 | Default: 33.600 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45845 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_5 | Default: 167.000 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45846 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_6 | Default: 50.400 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45847 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_7 | Default: 183.800 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45848 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_8 | Default: 67.200 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45849 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_9 | Default: 200.600 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45850 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_10 | Default: 84.000 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45851 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_11 | Default: 217.400 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45852 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_12 | Default: 100.800 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45853 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_13 | Default: 234.200 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45854 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_14 | Default: 117.600 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45855 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_15 | Default: 1.000 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45856 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_16 | Default: 134.400 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L500 (BANK 3)

Machine Model: Trumpf-TruLaser-L500 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L500.

PRM_45861 [Trumpf-TruLaser-L500] — CLN_LOOP_GAIN_1 | Default: 200.100 | Linked Fault: ERR-TRU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L500.

PRM_45862 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_2 | Default: 150.200 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45863 [Trumpf-TruLaser-L500] — OPT_LOOP_GAIN_3 | Default: 100.300 | Linked Fault: ERR-TRU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L500.

PRM_45864 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_4 | Default: 50.400 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45865 [Trumpf-TruLaser-L500] — ATC_LOOP_GAIN_5 | Default: 0.500 | Linked Fault: ERR-TRU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L500.

PRM_45866 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_6 | Default: 200.600 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45867 [Trumpf-TruLaser-L500] — BAX_LOOP_GAIN_7 | Default: 150.700 | Linked Fault: ERR-TRU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L500.

PRM_45868 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_8 | Default: 100.800 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45869 [Trumpf-TruLaser-L500] — THM_LOOP_GAIN_9 | Default: 50.900 | Linked Fault: ERR-TRU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L500.

PRM_45870 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_10 | Default: 1.000 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45871 [Trumpf-TruLaser-L500] — YAX_LOOP_GAIN_11 | Default: 201.100 | Linked Fault: ERR-TRU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L500.

PRM_45872 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_12 | Default: 151.200 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45873 [Trumpf-TruLaser-L500] — LUB_LOOP_GAIN_13 | Default: 101.300 | Linked Fault: ERR-TRU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L500.

PRM_45874 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_14 | Default: 51.400 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45875 [Trumpf-TruLaser-L500] — CLN_LOOP_GAIN_15 | Default: 1.500 | Linked Fault: ERR-TRU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L500.

PRM_45876 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_16 | Default: 201.600 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L500 (BANK 4)

Machine Model: Trumpf-TruLaser-L500 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L500.

PRM_45881 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_1 | Default: 16.800 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45882 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_2 | Default: 33.600 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45883 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_3 | Default: 50.400 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45884 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_4 | Default: 67.200 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45885 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_5 | Default: 84.000 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45886 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_6 | Default: 100.800 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45887 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_7 | Default: 117.600 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45888 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_8 | Default: 134.400 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45889 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_9 | Default: 151.200 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45890 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_10 | Default: 168.000 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45891 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_11 | Default: 184.800 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45892 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_12 | Default: 201.600 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45893 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_13 | Default: 218.400 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45894 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_14 | Default: 235.200 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45895 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_15 | Default: 2.000 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45896 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_16 | Default: 18.800 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L500 (BANK 5)

Machine Model: Trumpf-TruLaser-L500 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L500.

PRM_45901 [Trumpf-TruLaser-L500] — YAX_LOOP_GAIN_1 | Default: 83.500 | Linked Fault: ERR-TRU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L500.

PRM_45902 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_2 | Default: 167.000 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

PRM_45903 [Trumpf-TruLaser-L500] — ATC_LOOP_GAIN_3 | Default: 0.500 | Linked Fault: ERR-TRU500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L500.

PRM_45904 [Trumpf-TruLaser-L500] — ZAX_LOOP_GAIN_4 | Default: 84.000 | Linked Fault: ERR-TRU500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L500.

PRM_45905 [Trumpf-TruLaser-L500] — LUB_LOOP_GAIN_5 | Default: 167.500 | Linked Fault: ERR-TRU500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L500.

PRM_45906 [Trumpf-TruLaser-L500] — HYD_LOOP_GAIN_6 | Default: 1.000 | Linked Fault: ERR-TRU500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L500.

PRM_45907 [Trumpf-TruLaser-L500] — BAX_LOOP_GAIN_7 | Default: 84.500 | Linked Fault: ERR-TRU500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L500.

PRM_45908 [Trumpf-TruLaser-L500] — VAC_LOOP_GAIN_8 | Default: 168.000 | Linked Fault: ERR-TRU500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L500.

PRM_45909 [Trumpf-TruLaser-L500] — CLN_LOOP_GAIN_9 | Default: 1.500 | Linked Fault: ERR-TRU500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L500.

PRM_45910 [Trumpf-TruLaser-L500] — CAX_LOOP_GAIN_10 | Default: 85.000 | Linked Fault: ERR-TRU500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L500.

PRM_45911 [Trumpf-TruLaser-L500] — THM_LOOP_GAIN_11 | Default: 168.500 | Linked Fault: ERR-TRU500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L500.

PRM_45912 [Trumpf-TruLaser-L500] — XAX_LOOP_GAIN_12 | Default: 2.000 | Linked Fault: ERR-TRU500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L500.

PRM_45913 [Trumpf-TruLaser-L500] — OPT_LOOP_GAIN_13 | Default: 85.500 | Linked Fault: ERR-TRU500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L500.

PRM_45914 [Trumpf-TruLaser-L500] — SPN_LOOP_GAIN_14 | Default: 169.000 | Linked Fault: ERR-TRU500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L500.

PRM_45915 [Trumpf-TruLaser-L500] — YAX_LOOP_GAIN_15 | Default: 2.500 | Linked Fault: ERR-TRU500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L500.

PRM_45916 [Trumpf-TruLaser-L500] — PLC_LOOP_GAIN_16 | Default: 86.000 | Linked Fault: ERR-TRU500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L500.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L550 (BANK 1)

Machine Model: Trumpf-TruLaser-L550 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L550.

PRM_45921 [Trumpf-TruLaser-L550] — ATC_LOOP_GAIN_1 | Default: 67.850 | Linked Fault: ERR-TRU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L550.

PRM_45922 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_2 | Default: 135.700 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45923 [Trumpf-TruLaser-L550] — CLN_LOOP_GAIN_3 | Default: 203.550 | Linked Fault: ERR-TRU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L550.

PRM_45924 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_4 | Default: 21.400 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45925 [Trumpf-TruLaser-L550] — YAX_LOOP_GAIN_5 | Default: 89.250 | Linked Fault: ERR-TRU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L550.

PRM_45926 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_6 | Default: 157.100 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45927 [Trumpf-TruLaser-L550] — BAX_LOOP_GAIN_7 | Default: 224.950 | Linked Fault: ERR-TRU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L550.

PRM_45928 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_8 | Default: 42.800 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45929 [Trumpf-TruLaser-L550] — OPT_LOOP_GAIN_9 | Default: 110.650 | Linked Fault: ERR-TRU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L550.

PRM_45930 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_10 | Default: 178.500 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45931 [Trumpf-TruLaser-L550] — LUB_LOOP_GAIN_11 | Default: 246.350 | Linked Fault: ERR-TRU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L550.

PRM_45932 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_12 | Default: 64.200 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45933 [Trumpf-TruLaser-L550] — THM_LOOP_GAIN_13 | Default: 132.050 | Linked Fault: ERR-TRU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L550.

PRM_45934 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_14 | Default: 199.900 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45935 [Trumpf-TruLaser-L550] — ATC_LOOP_GAIN_15 | Default: 17.750 | Linked Fault: ERR-TRU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L550.

PRM_45936 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_16 | Default: 85.600 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L550 (BANK 2)

Machine Model: Trumpf-TruLaser-L550 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L550.

PRM_45941 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_1 | Default: 135.700 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45942 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_2 | Default: 21.400 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45943 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_3 | Default: 157.100 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45944 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_4 | Default: 42.800 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45945 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_5 | Default: 178.500 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45946 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_6 | Default: 64.200 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45947 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_7 | Default: 199.900 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45948 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_8 | Default: 85.600 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45949 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_9 | Default: 221.300 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45950 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_10 | Default: 107.000 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45951 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_11 | Default: 242.700 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45952 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_12 | Default: 128.400 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45953 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_13 | Default: 14.100 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45954 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_14 | Default: 149.800 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45955 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_15 | Default: 35.500 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45956 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_16 | Default: 171.200 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L550 (BANK 3)

Machine Model: Trumpf-TruLaser-L550 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L550.

PRM_45961 [Trumpf-TruLaser-L550] — CLN_LOOP_GAIN_1 | Default: 203.550 | Linked Fault: ERR-TRU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L550.

PRM_45962 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_2 | Default: 157.100 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45963 [Trumpf-TruLaser-L550] — OPT_LOOP_GAIN_3 | Default: 110.650 | Linked Fault: ERR-TRU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L550.

PRM_45964 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_4 | Default: 64.200 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45965 [Trumpf-TruLaser-L550] — ATC_LOOP_GAIN_5 | Default: 17.750 | Linked Fault: ERR-TRU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L550.

PRM_45966 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_6 | Default: 221.300 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45967 [Trumpf-TruLaser-L550] — BAX_LOOP_GAIN_7 | Default: 174.850 | Linked Fault: ERR-TRU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L550.

PRM_45968 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_8 | Default: 128.400 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45969 [Trumpf-TruLaser-L550] — THM_LOOP_GAIN_9 | Default: 81.950 | Linked Fault: ERR-TRU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L550.

PRM_45970 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_10 | Default: 35.500 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45971 [Trumpf-TruLaser-L550] — YAX_LOOP_GAIN_11 | Default: 239.050 | Linked Fault: ERR-TRU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L550.

PRM_45972 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_12 | Default: 192.600 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45973 [Trumpf-TruLaser-L550] — LUB_LOOP_GAIN_13 | Default: 146.150 | Linked Fault: ERR-TRU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L550.

PRM_45974 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_14 | Default: 99.700 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45975 [Trumpf-TruLaser-L550] — CLN_LOOP_GAIN_15 | Default: 53.250 | Linked Fault: ERR-TRU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L550.

PRM_45976 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_16 | Default: 6.800 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L550 (BANK 4)

Machine Model: Trumpf-TruLaser-L550 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L550.

PRM_45981 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_1 | Default: 21.400 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45982 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_2 | Default: 42.800 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45983 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_3 | Default: 64.200 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45984 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_4 | Default: 85.600 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45985 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_5 | Default: 107.000 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45986 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_6 | Default: 128.400 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45987 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_7 | Default: 149.800 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45988 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_8 | Default: 171.200 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45989 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_9 | Default: 192.600 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_45990 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_10 | Default: 214.000 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_45991 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_11 | Default: 235.400 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_45992 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_12 | Default: 6.800 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_45993 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_13 | Default: 28.200 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_45994 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_14 | Default: 49.600 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_45995 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_15 | Default: 71.000 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_45996 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_16 | Default: 92.400 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L550 (BANK 5)

Machine Model: Trumpf-TruLaser-L550 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L550.

PRM_46001 [Trumpf-TruLaser-L550] — YAX_LOOP_GAIN_1 | Default: 89.250 | Linked Fault: ERR-TRU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L550.

PRM_46002 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_2 | Default: 178.500 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

PRM_46003 [Trumpf-TruLaser-L550] — ATC_LOOP_GAIN_3 | Default: 17.750 | Linked Fault: ERR-TRU550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L550.

PRM_46004 [Trumpf-TruLaser-L550] — ZAX_LOOP_GAIN_4 | Default: 107.000 | Linked Fault: ERR-TRU550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L550.

PRM_46005 [Trumpf-TruLaser-L550] — LUB_LOOP_GAIN_5 | Default: 196.250 | Linked Fault: ERR-TRU550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L550.

PRM_46006 [Trumpf-TruLaser-L550] — HYD_LOOP_GAIN_6 | Default: 35.500 | Linked Fault: ERR-TRU550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L550.

PRM_46007 [Trumpf-TruLaser-L550] — BAX_LOOP_GAIN_7 | Default: 124.750 | Linked Fault: ERR-TRU550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L550.

PRM_46008 [Trumpf-TruLaser-L550] — VAC_LOOP_GAIN_8 | Default: 214.000 | Linked Fault: ERR-TRU550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L550.

PRM_46009 [Trumpf-TruLaser-L550] — CLN_LOOP_GAIN_9 | Default: 53.250 | Linked Fault: ERR-TRU550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L550.

PRM_46010 [Trumpf-TruLaser-L550] — CAX_LOOP_GAIN_10 | Default: 142.500 | Linked Fault: ERR-TRU550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L550.

PRM_46011 [Trumpf-TruLaser-L550] — THM_LOOP_GAIN_11 | Default: 231.750 | Linked Fault: ERR-TRU550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L550.

PRM_46012 [Trumpf-TruLaser-L550] — XAX_LOOP_GAIN_12 | Default: 71.000 | Linked Fault: ERR-TRU550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L550.

PRM_46013 [Trumpf-TruLaser-L550] — OPT_LOOP_GAIN_13 | Default: 160.250 | Linked Fault: ERR-TRU550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L550.

PRM_46014 [Trumpf-TruLaser-L550] — SPN_LOOP_GAIN_14 | Default: 249.500 | Linked Fault: ERR-TRU550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L550.

PRM_46015 [Trumpf-TruLaser-L550] — YAX_LOOP_GAIN_15 | Default: 88.750 | Linked Fault: ERR-TRU550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L550.

PRM_46016 [Trumpf-TruLaser-L550] — PLC_LOOP_GAIN_16 | Default: 178.000 | Linked Fault: ERR-TRU550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L550.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L600 (BANK 1)

Machine Model: Trumpf-TruLaser-L600 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L600.

PRM_46021 [Trumpf-TruLaser-L600] — ATC_LOOP_GAIN_1 | Default: 69.000 | Linked Fault: ERR-TRU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L600.

PRM_46022 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_2 | Default: 138.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46023 [Trumpf-TruLaser-L600] — CLN_LOOP_GAIN_3 | Default: 207.000 | Linked Fault: ERR-TRU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L600.

PRM_46024 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_4 | Default: 26.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46025 [Trumpf-TruLaser-L600] — YAX_LOOP_GAIN_5 | Default: 95.000 | Linked Fault: ERR-TRU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L600.

PRM_46026 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_6 | Default: 164.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46027 [Trumpf-TruLaser-L600] — BAX_LOOP_GAIN_7 | Default: 233.000 | Linked Fault: ERR-TRU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L600.

PRM_46028 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_8 | Default: 52.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46029 [Trumpf-TruLaser-L600] — OPT_LOOP_GAIN_9 | Default: 121.000 | Linked Fault: ERR-TRU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L600.

PRM_46030 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_10 | Default: 190.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46031 [Trumpf-TruLaser-L600] — LUB_LOOP_GAIN_11 | Default: 9.000 | Linked Fault: ERR-TRU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L600.

PRM_46032 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_12 | Default: 78.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46033 [Trumpf-TruLaser-L600] — THM_LOOP_GAIN_13 | Default: 147.000 | Linked Fault: ERR-TRU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L600.

PRM_46034 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_14 | Default: 216.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46035 [Trumpf-TruLaser-L600] — ATC_LOOP_GAIN_15 | Default: 35.000 | Linked Fault: ERR-TRU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L600.

PRM_46036 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_16 | Default: 104.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L600 (BANK 2)

Machine Model: Trumpf-TruLaser-L600 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L600.

PRM_46041 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_1 | Default: 138.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46042 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_2 | Default: 26.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46043 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_3 | Default: 164.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46044 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_4 | Default: 52.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46045 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_5 | Default: 190.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46046 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_6 | Default: 78.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46047 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_7 | Default: 216.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46048 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_8 | Default: 104.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46049 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_9 | Default: 242.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46050 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_10 | Default: 130.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46051 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_11 | Default: 18.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46052 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_12 | Default: 156.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46053 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_13 | Default: 44.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46054 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_14 | Default: 182.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46055 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_15 | Default: 70.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46056 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_16 | Default: 208.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L600 (BANK 3)

Machine Model: Trumpf-TruLaser-L600 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L600.

PRM_46061 [Trumpf-TruLaser-L600] — CLN_LOOP_GAIN_1 | Default: 207.000 | Linked Fault: ERR-TRU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L600.

PRM_46062 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_2 | Default: 164.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46063 [Trumpf-TruLaser-L600] — OPT_LOOP_GAIN_3 | Default: 121.000 | Linked Fault: ERR-TRU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L600.

PRM_46064 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_4 | Default: 78.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46065 [Trumpf-TruLaser-L600] — ATC_LOOP_GAIN_5 | Default: 35.000 | Linked Fault: ERR-TRU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L600.

PRM_46066 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_6 | Default: 242.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46067 [Trumpf-TruLaser-L600] — BAX_LOOP_GAIN_7 | Default: 199.000 | Linked Fault: ERR-TRU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L600.

PRM_46068 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_8 | Default: 156.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46069 [Trumpf-TruLaser-L600] — THM_LOOP_GAIN_9 | Default: 113.000 | Linked Fault: ERR-TRU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L600.

PRM_46070 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_10 | Default: 70.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46071 [Trumpf-TruLaser-L600] — YAX_LOOP_GAIN_11 | Default: 27.000 | Linked Fault: ERR-TRU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L600.

PRM_46072 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_12 | Default: 234.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46073 [Trumpf-TruLaser-L600] — LUB_LOOP_GAIN_13 | Default: 191.000 | Linked Fault: ERR-TRU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L600.

PRM_46074 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_14 | Default: 148.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46075 [Trumpf-TruLaser-L600] — CLN_LOOP_GAIN_15 | Default: 105.000 | Linked Fault: ERR-TRU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L600.

PRM_46076 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_16 | Default: 62.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L600 (BANK 4)

Machine Model: Trumpf-TruLaser-L600 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L600.

PRM_46081 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_1 | Default: 26.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46082 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_2 | Default: 52.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46083 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_3 | Default: 78.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46084 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_4 | Default: 104.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46085 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_5 | Default: 130.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46086 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_6 | Default: 156.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46087 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_7 | Default: 182.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46088 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_8 | Default: 208.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46089 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_9 | Default: 234.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46090 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_10 | Default: 10.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46091 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_11 | Default: 36.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46092 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_12 | Default: 62.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46093 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_13 | Default: 88.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46094 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_14 | Default: 114.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46095 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_15 | Default: 140.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46096 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_16 | Default: 166.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

CNC/PLC SERVO PARAMETER DICTIONARY — Trumpf-TruLaser-L600 (BANK 5)

Machine Model: Trumpf-TruLaser-L600 | Controller: Siemens 840D sl Laser. Use these parameter registers when diagnosing servo following errors or thermal drift on Trumpf-TruLaser-L600.

PRM_46101 [Trumpf-TruLaser-L600] — YAX_LOOP_GAIN_1 | Default: 95.000 | Linked Fault: ERR-TRU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L600.

PRM_46102 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_2 | Default: 190.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

PRM_46103 [Trumpf-TruLaser-L600] — ATC_LOOP_GAIN_3 | Default: 35.000 | Linked Fault: ERR-TRU600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Trumpf-TruLaser-L600.

PRM_46104 [Trumpf-TruLaser-L600] — ZAX_LOOP_GAIN_4 | Default: 130.000 | Linked Fault: ERR-TRU600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Trumpf-TruLaser-L600.

PRM_46105 [Trumpf-TruLaser-L600] — LUB_LOOP_GAIN_5 | Default: 225.000 | Linked Fault: ERR-TRU600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Trumpf-TruLaser-L600.

PRM_46106 [Trumpf-TruLaser-L600] — HYD_LOOP_GAIN_6 | Default: 70.000 | Linked Fault: ERR-TRU600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Trumpf-TruLaser-L600.

PRM_46107 [Trumpf-TruLaser-L600] — BAX_LOOP_GAIN_7 | Default: 165.000 | Linked Fault: ERR-TRU600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Trumpf-TruLaser-L600.

PRM_46108 [Trumpf-TruLaser-L600] — VAC_LOOP_GAIN_8 | Default: 10.000 | Linked Fault: ERR-TRU600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Trumpf-TruLaser-L600.

PRM_46109 [Trumpf-TruLaser-L600] — CLN_LOOP_GAIN_9 | Default: 105.000 | Linked Fault: ERR-TRU600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Trumpf-TruLaser-L600.

PRM_46110 [Trumpf-TruLaser-L600] — CAX_LOOP_GAIN_10 | Default: 200.000 | Linked Fault: ERR-TRU600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Trumpf-TruLaser-L600.

PRM_46111 [Trumpf-TruLaser-L600] — THM_LOOP_GAIN_11 | Default: 45.000 | Linked Fault: ERR-TRU600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Trumpf-TruLaser-L600.

PRM_46112 [Trumpf-TruLaser-L600] — XAX_LOOP_GAIN_12 | Default: 140.000 | Linked Fault: ERR-TRU600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Trumpf-TruLaser-L600.

PRM_46113 [Trumpf-TruLaser-L600] — OPT_LOOP_GAIN_13 | Default: 235.000 | Linked Fault: ERR-TRU600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Trumpf-TruLaser-L600.

PRM_46114 [Trumpf-TruLaser-L600] — SPN_LOOP_GAIN_14 | Default: 80.000 | Linked Fault: ERR-TRU600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Trumpf-TruLaser-L600.

PRM_46115 [Trumpf-TruLaser-L600] — YAX_LOOP_GAIN_15 | Default: 175.000 | Linked Fault: ERR-TRU600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Trumpf-TruLaser-L600.

PRM_46116 [Trumpf-TruLaser-L600] — PLC_LOOP_GAIN_16 | Default: 20.000 | Linked Fault: ERR-TRU600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Trumpf-TruLaser-L600.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P150 (BANK 1)

Machine Model: Schuler-HPD-P150 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P150.

PRM_46121 [Schuler-HPD-P150] — ATC_LOOP_GAIN_1 | Default: 70.150 | Linked Fault: ERR-SCH150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P150.

PRM_46122 [Schuler-HPD-P150] — HYD_LOOP_GAIN_2 | Default: 140.300 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46123 [Schuler-HPD-P150] — CLN_LOOP_GAIN_3 | Default: 210.450 | Linked Fault: ERR-SCH150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P150.

PRM_46124 [Schuler-HPD-P150] — XAX_LOOP_GAIN_4 | Default: 30.600 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46125 [Schuler-HPD-P150] — YAX_LOOP_GAIN_5 | Default: 100.750 | Linked Fault: ERR-SCH150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P150.

PRM_46126 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_6 | Default: 170.900 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46127 [Schuler-HPD-P150] — BAX_LOOP_GAIN_7 | Default: 241.050 | Linked Fault: ERR-SCH150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P150.

PRM_46128 [Schuler-HPD-P150] — CAX_LOOP_GAIN_8 | Default: 61.200 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46129 [Schuler-HPD-P150] — OPT_LOOP_GAIN_9 | Default: 131.350 | Linked Fault: ERR-SCH150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P150.

PRM_46130 [Schuler-HPD-P150] — PLC_LOOP_GAIN_10 | Default: 201.500 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46131 [Schuler-HPD-P150] — LUB_LOOP_GAIN_11 | Default: 21.650 | Linked Fault: ERR-SCH150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P150.

PRM_46132 [Schuler-HPD-P150] — VAC_LOOP_GAIN_12 | Default: 91.800 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46133 [Schuler-HPD-P150] — THM_LOOP_GAIN_13 | Default: 161.950 | Linked Fault: ERR-SCH150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P150.

PRM_46134 [Schuler-HPD-P150] — SPN_LOOP_GAIN_14 | Default: 232.100 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46135 [Schuler-HPD-P150] — ATC_LOOP_GAIN_15 | Default: 52.250 | Linked Fault: ERR-SCH150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P150.

PRM_46136 [Schuler-HPD-P150] — HYD_LOOP_GAIN_16 | Default: 122.400 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P150 (BANK 2)

Machine Model: Schuler-HPD-P150 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P150.

PRM_46141 [Schuler-HPD-P150] — HYD_LOOP_GAIN_1 | Default: 140.300 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46142 [Schuler-HPD-P150] — XAX_LOOP_GAIN_2 | Default: 30.600 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46143 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_3 | Default: 170.900 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46144 [Schuler-HPD-P150] — CAX_LOOP_GAIN_4 | Default: 61.200 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46145 [Schuler-HPD-P150] — PLC_LOOP_GAIN_5 | Default: 201.500 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46146 [Schuler-HPD-P150] — VAC_LOOP_GAIN_6 | Default: 91.800 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46147 [Schuler-HPD-P150] — SPN_LOOP_GAIN_7 | Default: 232.100 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46148 [Schuler-HPD-P150] — HYD_LOOP_GAIN_8 | Default: 122.400 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46149 [Schuler-HPD-P150] — XAX_LOOP_GAIN_9 | Default: 12.700 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46150 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_10 | Default: 153.000 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46151 [Schuler-HPD-P150] — CAX_LOOP_GAIN_11 | Default: 43.300 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46152 [Schuler-HPD-P150] — PLC_LOOP_GAIN_12 | Default: 183.600 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46153 [Schuler-HPD-P150] — VAC_LOOP_GAIN_13 | Default: 73.900 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46154 [Schuler-HPD-P150] — SPN_LOOP_GAIN_14 | Default: 214.200 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46155 [Schuler-HPD-P150] — HYD_LOOP_GAIN_15 | Default: 104.500 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46156 [Schuler-HPD-P150] — XAX_LOOP_GAIN_16 | Default: 244.800 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P150 (BANK 3)

Machine Model: Schuler-HPD-P150 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P150.

PRM_46161 [Schuler-HPD-P150] — CLN_LOOP_GAIN_1 | Default: 210.450 | Linked Fault: ERR-SCH150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P150.

PRM_46162 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_2 | Default: 170.900 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46163 [Schuler-HPD-P150] — OPT_LOOP_GAIN_3 | Default: 131.350 | Linked Fault: ERR-SCH150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P150.

PRM_46164 [Schuler-HPD-P150] — VAC_LOOP_GAIN_4 | Default: 91.800 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46165 [Schuler-HPD-P150] — ATC_LOOP_GAIN_5 | Default: 52.250 | Linked Fault: ERR-SCH150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P150.

PRM_46166 [Schuler-HPD-P150] — XAX_LOOP_GAIN_6 | Default: 12.700 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46167 [Schuler-HPD-P150] — BAX_LOOP_GAIN_7 | Default: 223.150 | Linked Fault: ERR-SCH150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P150.

PRM_46168 [Schuler-HPD-P150] — PLC_LOOP_GAIN_8 | Default: 183.600 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46169 [Schuler-HPD-P150] — THM_LOOP_GAIN_9 | Default: 144.050 | Linked Fault: ERR-SCH150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P150.

PRM_46170 [Schuler-HPD-P150] — HYD_LOOP_GAIN_10 | Default: 104.500 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46171 [Schuler-HPD-P150] — YAX_LOOP_GAIN_11 | Default: 64.950 | Linked Fault: ERR-SCH150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P150.

PRM_46172 [Schuler-HPD-P150] — CAX_LOOP_GAIN_12 | Default: 25.400 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46173 [Schuler-HPD-P150] — LUB_LOOP_GAIN_13 | Default: 235.850 | Linked Fault: ERR-SCH150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P150.

PRM_46174 [Schuler-HPD-P150] — SPN_LOOP_GAIN_14 | Default: 196.300 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46175 [Schuler-HPD-P150] — CLN_LOOP_GAIN_15 | Default: 156.750 | Linked Fault: ERR-SCH150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P150.

PRM_46176 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_16 | Default: 117.200 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P150 (BANK 4)

Machine Model: Schuler-HPD-P150 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P150.

PRM_46181 [Schuler-HPD-P150] — XAX_LOOP_GAIN_1 | Default: 30.600 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46182 [Schuler-HPD-P150] — CAX_LOOP_GAIN_2 | Default: 61.200 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46183 [Schuler-HPD-P150] — VAC_LOOP_GAIN_3 | Default: 91.800 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46184 [Schuler-HPD-P150] — HYD_LOOP_GAIN_4 | Default: 122.400 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46185 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_5 | Default: 153.000 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46186 [Schuler-HPD-P150] — PLC_LOOP_GAIN_6 | Default: 183.600 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46187 [Schuler-HPD-P150] — SPN_LOOP_GAIN_7 | Default: 214.200 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46188 [Schuler-HPD-P150] — XAX_LOOP_GAIN_8 | Default: 244.800 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46189 [Schuler-HPD-P150] — CAX_LOOP_GAIN_9 | Default: 25.400 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46190 [Schuler-HPD-P150] — VAC_LOOP_GAIN_10 | Default: 56.000 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46191 [Schuler-HPD-P150] — HYD_LOOP_GAIN_11 | Default: 86.600 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46192 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_12 | Default: 117.200 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46193 [Schuler-HPD-P150] — PLC_LOOP_GAIN_13 | Default: 147.800 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46194 [Schuler-HPD-P150] — SPN_LOOP_GAIN_14 | Default: 178.400 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46195 [Schuler-HPD-P150] — XAX_LOOP_GAIN_15 | Default: 209.000 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46196 [Schuler-HPD-P150] — CAX_LOOP_GAIN_16 | Default: 239.600 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P150 (BANK 5)

Machine Model: Schuler-HPD-P150 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P150.

PRM_46201 [Schuler-HPD-P150] — YAX_LOOP_GAIN_1 | Default: 100.750 | Linked Fault: ERR-SCH150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P150.

PRM_46202 [Schuler-HPD-P150] — PLC_LOOP_GAIN_2 | Default: 201.500 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

PRM_46203 [Schuler-HPD-P150] — ATC_LOOP_GAIN_3 | Default: 52.250 | Linked Fault: ERR-SCH150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P150.

PRM_46204 [Schuler-HPD-P150] — ZAX_LOOP_GAIN_4 | Default: 153.000 | Linked Fault: ERR-SCH150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P150.

PRM_46205 [Schuler-HPD-P150] — LUB_LOOP_GAIN_5 | Default: 3.750 | Linked Fault: ERR-SCH150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P150.

PRM_46206 [Schuler-HPD-P150] — HYD_LOOP_GAIN_6 | Default: 104.500 | Linked Fault: ERR-SCH150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P150.

PRM_46207 [Schuler-HPD-P150] — BAX_LOOP_GAIN_7 | Default: 205.250 | Linked Fault: ERR-SCH150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P150.

PRM_46208 [Schuler-HPD-P150] — VAC_LOOP_GAIN_8 | Default: 56.000 | Linked Fault: ERR-SCH150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P150.

PRM_46209 [Schuler-HPD-P150] — CLN_LOOP_GAIN_9 | Default: 156.750 | Linked Fault: ERR-SCH150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P150.

PRM_46210 [Schuler-HPD-P150] — CAX_LOOP_GAIN_10 | Default: 7.500 | Linked Fault: ERR-SCH150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P150.

PRM_46211 [Schuler-HPD-P150] — THM_LOOP_GAIN_11 | Default: 108.250 | Linked Fault: ERR-SCH150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P150.

PRM_46212 [Schuler-HPD-P150] — XAX_LOOP_GAIN_12 | Default: 209.000 | Linked Fault: ERR-SCH150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P150.

PRM_46213 [Schuler-HPD-P150] — OPT_LOOP_GAIN_13 | Default: 59.750 | Linked Fault: ERR-SCH150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P150.

PRM_46214 [Schuler-HPD-P150] — SPN_LOOP_GAIN_14 | Default: 160.500 | Linked Fault: ERR-SCH150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P150.

PRM_46215 [Schuler-HPD-P150] — YAX_LOOP_GAIN_15 | Default: 11.250 | Linked Fault: ERR-SCH150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P150.

PRM_46216 [Schuler-HPD-P150] — PLC_LOOP_GAIN_16 | Default: 112.000 | Linked Fault: ERR-SCH150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P150.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P200 (BANK 1)

Machine Model: Schuler-HPD-P200 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P200.

PRM_46221 [Schuler-HPD-P200] — ATC_LOOP_GAIN_1 | Default: 71.300 | Linked Fault: ERR-SCH200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P200.

PRM_46222 [Schuler-HPD-P200] — HYD_LOOP_GAIN_2 | Default: 142.600 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46223 [Schuler-HPD-P200] — CLN_LOOP_GAIN_3 | Default: 213.900 | Linked Fault: ERR-SCH200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P200.

PRM_46224 [Schuler-HPD-P200] — XAX_LOOP_GAIN_4 | Default: 35.200 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46225 [Schuler-HPD-P200] — YAX_LOOP_GAIN_5 | Default: 106.500 | Linked Fault: ERR-SCH200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P200.

PRM_46226 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_6 | Default: 177.800 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46227 [Schuler-HPD-P200] — BAX_LOOP_GAIN_7 | Default: 249.100 | Linked Fault: ERR-SCH200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P200.

PRM_46228 [Schuler-HPD-P200] — CAX_LOOP_GAIN_8 | Default: 70.400 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46229 [Schuler-HPD-P200] — OPT_LOOP_GAIN_9 | Default: 141.700 | Linked Fault: ERR-SCH200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P200.

PRM_46230 [Schuler-HPD-P200] — PLC_LOOP_GAIN_10 | Default: 213.000 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46231 [Schuler-HPD-P200] — LUB_LOOP_GAIN_11 | Default: 34.300 | Linked Fault: ERR-SCH200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P200.

PRM_46232 [Schuler-HPD-P200] — VAC_LOOP_GAIN_12 | Default: 105.600 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46233 [Schuler-HPD-P200] — THM_LOOP_GAIN_13 | Default: 176.900 | Linked Fault: ERR-SCH200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P200.

PRM_46234 [Schuler-HPD-P200] — SPN_LOOP_GAIN_14 | Default: 248.200 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46235 [Schuler-HPD-P200] — ATC_LOOP_GAIN_15 | Default: 69.500 | Linked Fault: ERR-SCH200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P200.

PRM_46236 [Schuler-HPD-P200] — HYD_LOOP_GAIN_16 | Default: 140.800 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P200 (BANK 2)

Machine Model: Schuler-HPD-P200 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P200.

PRM_46241 [Schuler-HPD-P200] — HYD_LOOP_GAIN_1 | Default: 142.600 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46242 [Schuler-HPD-P200] — XAX_LOOP_GAIN_2 | Default: 35.200 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46243 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_3 | Default: 177.800 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46244 [Schuler-HPD-P200] — CAX_LOOP_GAIN_4 | Default: 70.400 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46245 [Schuler-HPD-P200] — PLC_LOOP_GAIN_5 | Default: 213.000 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46246 [Schuler-HPD-P200] — VAC_LOOP_GAIN_6 | Default: 105.600 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46247 [Schuler-HPD-P200] — SPN_LOOP_GAIN_7 | Default: 248.200 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46248 [Schuler-HPD-P200] — HYD_LOOP_GAIN_8 | Default: 140.800 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46249 [Schuler-HPD-P200] — XAX_LOOP_GAIN_9 | Default: 33.400 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46250 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_10 | Default: 176.000 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46251 [Schuler-HPD-P200] — CAX_LOOP_GAIN_11 | Default: 68.600 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46252 [Schuler-HPD-P200] — PLC_LOOP_GAIN_12 | Default: 211.200 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46253 [Schuler-HPD-P200] — VAC_LOOP_GAIN_13 | Default: 103.800 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46254 [Schuler-HPD-P200] — SPN_LOOP_GAIN_14 | Default: 246.400 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46255 [Schuler-HPD-P200] — HYD_LOOP_GAIN_15 | Default: 139.000 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46256 [Schuler-HPD-P200] — XAX_LOOP_GAIN_16 | Default: 31.600 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P200 (BANK 3)

Machine Model: Schuler-HPD-P200 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P200.

PRM_46261 [Schuler-HPD-P200] — CLN_LOOP_GAIN_1 | Default: 213.900 | Linked Fault: ERR-SCH200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P200.

PRM_46262 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_2 | Default: 177.800 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46263 [Schuler-HPD-P200] — OPT_LOOP_GAIN_3 | Default: 141.700 | Linked Fault: ERR-SCH200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P200.

PRM_46264 [Schuler-HPD-P200] — VAC_LOOP_GAIN_4 | Default: 105.600 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46265 [Schuler-HPD-P200] — ATC_LOOP_GAIN_5 | Default: 69.500 | Linked Fault: ERR-SCH200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P200.

PRM_46266 [Schuler-HPD-P200] — XAX_LOOP_GAIN_6 | Default: 33.400 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46267 [Schuler-HPD-P200] — BAX_LOOP_GAIN_7 | Default: 247.300 | Linked Fault: ERR-SCH200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P200.

PRM_46268 [Schuler-HPD-P200] — PLC_LOOP_GAIN_8 | Default: 211.200 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46269 [Schuler-HPD-P200] — THM_LOOP_GAIN_9 | Default: 175.100 | Linked Fault: ERR-SCH200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P200.

PRM_46270 [Schuler-HPD-P200] — HYD_LOOP_GAIN_10 | Default: 139.000 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46271 [Schuler-HPD-P200] — YAX_LOOP_GAIN_11 | Default: 102.900 | Linked Fault: ERR-SCH200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P200.

PRM_46272 [Schuler-HPD-P200] — CAX_LOOP_GAIN_12 | Default: 66.800 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46273 [Schuler-HPD-P200] — LUB_LOOP_GAIN_13 | Default: 30.700 | Linked Fault: ERR-SCH200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P200.

PRM_46274 [Schuler-HPD-P200] — SPN_LOOP_GAIN_14 | Default: 244.600 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46275 [Schuler-HPD-P200] — CLN_LOOP_GAIN_15 | Default: 208.500 | Linked Fault: ERR-SCH200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P200.

PRM_46276 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_16 | Default: 172.400 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P200 (BANK 4)

Machine Model: Schuler-HPD-P200 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P200.

PRM_46281 [Schuler-HPD-P200] — XAX_LOOP_GAIN_1 | Default: 35.200 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46282 [Schuler-HPD-P200] — CAX_LOOP_GAIN_2 | Default: 70.400 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46283 [Schuler-HPD-P200] — VAC_LOOP_GAIN_3 | Default: 105.600 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46284 [Schuler-HPD-P200] — HYD_LOOP_GAIN_4 | Default: 140.800 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46285 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_5 | Default: 176.000 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46286 [Schuler-HPD-P200] — PLC_LOOP_GAIN_6 | Default: 211.200 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46287 [Schuler-HPD-P200] — SPN_LOOP_GAIN_7 | Default: 246.400 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46288 [Schuler-HPD-P200] — XAX_LOOP_GAIN_8 | Default: 31.600 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46289 [Schuler-HPD-P200] — CAX_LOOP_GAIN_9 | Default: 66.800 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46290 [Schuler-HPD-P200] — VAC_LOOP_GAIN_10 | Default: 102.000 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46291 [Schuler-HPD-P200] — HYD_LOOP_GAIN_11 | Default: 137.200 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46292 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_12 | Default: 172.400 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46293 [Schuler-HPD-P200] — PLC_LOOP_GAIN_13 | Default: 207.600 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46294 [Schuler-HPD-P200] — SPN_LOOP_GAIN_14 | Default: 242.800 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46295 [Schuler-HPD-P200] — XAX_LOOP_GAIN_15 | Default: 28.000 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46296 [Schuler-HPD-P200] — CAX_LOOP_GAIN_16 | Default: 63.200 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P200 (BANK 5)

Machine Model: Schuler-HPD-P200 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P200.

PRM_46301 [Schuler-HPD-P200] — YAX_LOOP_GAIN_1 | Default: 106.500 | Linked Fault: ERR-SCH200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P200.

PRM_46302 [Schuler-HPD-P200] — PLC_LOOP_GAIN_2 | Default: 213.000 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

PRM_46303 [Schuler-HPD-P200] — ATC_LOOP_GAIN_3 | Default: 69.500 | Linked Fault: ERR-SCH200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P200.

PRM_46304 [Schuler-HPD-P200] — ZAX_LOOP_GAIN_4 | Default: 176.000 | Linked Fault: ERR-SCH200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P200.

PRM_46305 [Schuler-HPD-P200] — LUB_LOOP_GAIN_5 | Default: 32.500 | Linked Fault: ERR-SCH200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P200.

PRM_46306 [Schuler-HPD-P200] — HYD_LOOP_GAIN_6 | Default: 139.000 | Linked Fault: ERR-SCH200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P200.

PRM_46307 [Schuler-HPD-P200] — BAX_LOOP_GAIN_7 | Default: 245.500 | Linked Fault: ERR-SCH200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P200.

PRM_46308 [Schuler-HPD-P200] — VAC_LOOP_GAIN_8 | Default: 102.000 | Linked Fault: ERR-SCH200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P200.

PRM_46309 [Schuler-HPD-P200] — CLN_LOOP_GAIN_9 | Default: 208.500 | Linked Fault: ERR-SCH200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P200.

PRM_46310 [Schuler-HPD-P200] — CAX_LOOP_GAIN_10 | Default: 65.000 | Linked Fault: ERR-SCH200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P200.

PRM_46311 [Schuler-HPD-P200] — THM_LOOP_GAIN_11 | Default: 171.500 | Linked Fault: ERR-SCH200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P200.

PRM_46312 [Schuler-HPD-P200] — XAX_LOOP_GAIN_12 | Default: 28.000 | Linked Fault: ERR-SCH200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P200.

PRM_46313 [Schuler-HPD-P200] — OPT_LOOP_GAIN_13 | Default: 134.500 | Linked Fault: ERR-SCH200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P200.

PRM_46314 [Schuler-HPD-P200] — SPN_LOOP_GAIN_14 | Default: 241.000 | Linked Fault: ERR-SCH200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P200.

PRM_46315 [Schuler-HPD-P200] — YAX_LOOP_GAIN_15 | Default: 97.500 | Linked Fault: ERR-SCH200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P200.

PRM_46316 [Schuler-HPD-P200] — PLC_LOOP_GAIN_16 | Default: 204.000 | Linked Fault: ERR-SCH200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P200.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P250 (BANK 1)

Machine Model: Schuler-HPD-P250 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P250.

PRM_46321 [Schuler-HPD-P250] — ATC_LOOP_GAIN_1 | Default: 72.450 | Linked Fault: ERR-SCH250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P250.

PRM_46322 [Schuler-HPD-P250] — HYD_LOOP_GAIN_2 | Default: 144.900 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46323 [Schuler-HPD-P250] — CLN_LOOP_GAIN_3 | Default: 217.350 | Linked Fault: ERR-SCH250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P250.

PRM_46324 [Schuler-HPD-P250] — XAX_LOOP_GAIN_4 | Default: 39.800 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46325 [Schuler-HPD-P250] — YAX_LOOP_GAIN_5 | Default: 112.250 | Linked Fault: ERR-SCH250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P250.

PRM_46326 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_6 | Default: 184.700 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46327 [Schuler-HPD-P250] — BAX_LOOP_GAIN_7 | Default: 7.150 | Linked Fault: ERR-SCH250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P250.

PRM_46328 [Schuler-HPD-P250] — CAX_LOOP_GAIN_8 | Default: 79.600 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46329 [Schuler-HPD-P250] — OPT_LOOP_GAIN_9 | Default: 152.050 | Linked Fault: ERR-SCH250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P250.

PRM_46330 [Schuler-HPD-P250] — PLC_LOOP_GAIN_10 | Default: 224.500 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46331 [Schuler-HPD-P250] — LUB_LOOP_GAIN_11 | Default: 46.950 | Linked Fault: ERR-SCH250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P250.

PRM_46332 [Schuler-HPD-P250] — VAC_LOOP_GAIN_12 | Default: 119.400 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46333 [Schuler-HPD-P250] — THM_LOOP_GAIN_13 | Default: 191.850 | Linked Fault: ERR-SCH250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P250.

PRM_46334 [Schuler-HPD-P250] — SPN_LOOP_GAIN_14 | Default: 14.300 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46335 [Schuler-HPD-P250] — ATC_LOOP_GAIN_15 | Default: 86.750 | Linked Fault: ERR-SCH250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P250.

PRM_46336 [Schuler-HPD-P250] — HYD_LOOP_GAIN_16 | Default: 159.200 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P250 (BANK 2)

Machine Model: Schuler-HPD-P250 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P250.

PRM_46341 [Schuler-HPD-P250] — HYD_LOOP_GAIN_1 | Default: 144.900 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46342 [Schuler-HPD-P250] — XAX_LOOP_GAIN_2 | Default: 39.800 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46343 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_3 | Default: 184.700 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46344 [Schuler-HPD-P250] — CAX_LOOP_GAIN_4 | Default: 79.600 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46345 [Schuler-HPD-P250] — PLC_LOOP_GAIN_5 | Default: 224.500 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46346 [Schuler-HPD-P250] — VAC_LOOP_GAIN_6 | Default: 119.400 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46347 [Schuler-HPD-P250] — SPN_LOOP_GAIN_7 | Default: 14.300 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46348 [Schuler-HPD-P250] — HYD_LOOP_GAIN_8 | Default: 159.200 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46349 [Schuler-HPD-P250] — XAX_LOOP_GAIN_9 | Default: 54.100 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46350 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_10 | Default: 199.000 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46351 [Schuler-HPD-P250] — CAX_LOOP_GAIN_11 | Default: 93.900 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46352 [Schuler-HPD-P250] — PLC_LOOP_GAIN_12 | Default: 238.800 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46353 [Schuler-HPD-P250] — VAC_LOOP_GAIN_13 | Default: 133.700 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46354 [Schuler-HPD-P250] — SPN_LOOP_GAIN_14 | Default: 28.600 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46355 [Schuler-HPD-P250] — HYD_LOOP_GAIN_15 | Default: 173.500 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46356 [Schuler-HPD-P250] — XAX_LOOP_GAIN_16 | Default: 68.400 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P250 (BANK 3)

Machine Model: Schuler-HPD-P250 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P250.

PRM_46361 [Schuler-HPD-P250] — CLN_LOOP_GAIN_1 | Default: 217.350 | Linked Fault: ERR-SCH250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P250.

PRM_46362 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_2 | Default: 184.700 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46363 [Schuler-HPD-P250] — OPT_LOOP_GAIN_3 | Default: 152.050 | Linked Fault: ERR-SCH250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P250.

PRM_46364 [Schuler-HPD-P250] — VAC_LOOP_GAIN_4 | Default: 119.400 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46365 [Schuler-HPD-P250] — ATC_LOOP_GAIN_5 | Default: 86.750 | Linked Fault: ERR-SCH250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P250.

PRM_46366 [Schuler-HPD-P250] — XAX_LOOP_GAIN_6 | Default: 54.100 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46367 [Schuler-HPD-P250] — BAX_LOOP_GAIN_7 | Default: 21.450 | Linked Fault: ERR-SCH250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P250.

PRM_46368 [Schuler-HPD-P250] — PLC_LOOP_GAIN_8 | Default: 238.800 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46369 [Schuler-HPD-P250] — THM_LOOP_GAIN_9 | Default: 206.150 | Linked Fault: ERR-SCH250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P250.

PRM_46370 [Schuler-HPD-P250] — HYD_LOOP_GAIN_10 | Default: 173.500 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46371 [Schuler-HPD-P250] — YAX_LOOP_GAIN_11 | Default: 140.850 | Linked Fault: ERR-SCH250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P250.

PRM_46372 [Schuler-HPD-P250] — CAX_LOOP_GAIN_12 | Default: 108.200 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46373 [Schuler-HPD-P250] — LUB_LOOP_GAIN_13 | Default: 75.550 | Linked Fault: ERR-SCH250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P250.

PRM_46374 [Schuler-HPD-P250] — SPN_LOOP_GAIN_14 | Default: 42.900 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46375 [Schuler-HPD-P250] — CLN_LOOP_GAIN_15 | Default: 10.250 | Linked Fault: ERR-SCH250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P250.

PRM_46376 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_16 | Default: 227.600 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P250 (BANK 4)

Machine Model: Schuler-HPD-P250 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P250.

PRM_46381 [Schuler-HPD-P250] — XAX_LOOP_GAIN_1 | Default: 39.800 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46382 [Schuler-HPD-P250] — CAX_LOOP_GAIN_2 | Default: 79.600 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46383 [Schuler-HPD-P250] — VAC_LOOP_GAIN_3 | Default: 119.400 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46384 [Schuler-HPD-P250] — HYD_LOOP_GAIN_4 | Default: 159.200 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46385 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_5 | Default: 199.000 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46386 [Schuler-HPD-P250] — PLC_LOOP_GAIN_6 | Default: 238.800 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46387 [Schuler-HPD-P250] — SPN_LOOP_GAIN_7 | Default: 28.600 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46388 [Schuler-HPD-P250] — XAX_LOOP_GAIN_8 | Default: 68.400 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46389 [Schuler-HPD-P250] — CAX_LOOP_GAIN_9 | Default: 108.200 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46390 [Schuler-HPD-P250] — VAC_LOOP_GAIN_10 | Default: 148.000 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46391 [Schuler-HPD-P250] — HYD_LOOP_GAIN_11 | Default: 187.800 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46392 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_12 | Default: 227.600 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46393 [Schuler-HPD-P250] — PLC_LOOP_GAIN_13 | Default: 17.400 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46394 [Schuler-HPD-P250] — SPN_LOOP_GAIN_14 | Default: 57.200 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46395 [Schuler-HPD-P250] — XAX_LOOP_GAIN_15 | Default: 97.000 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46396 [Schuler-HPD-P250] — CAX_LOOP_GAIN_16 | Default: 136.800 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P250 (BANK 5)

Machine Model: Schuler-HPD-P250 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P250.

PRM_46401 [Schuler-HPD-P250] — YAX_LOOP_GAIN_1 | Default: 112.250 | Linked Fault: ERR-SCH250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P250.

PRM_46402 [Schuler-HPD-P250] — PLC_LOOP_GAIN_2 | Default: 224.500 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

PRM_46403 [Schuler-HPD-P250] — ATC_LOOP_GAIN_3 | Default: 86.750 | Linked Fault: ERR-SCH250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P250.

PRM_46404 [Schuler-HPD-P250] — ZAX_LOOP_GAIN_4 | Default: 199.000 | Linked Fault: ERR-SCH250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P250.

PRM_46405 [Schuler-HPD-P250] — LUB_LOOP_GAIN_5 | Default: 61.250 | Linked Fault: ERR-SCH250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P250.

PRM_46406 [Schuler-HPD-P250] — HYD_LOOP_GAIN_6 | Default: 173.500 | Linked Fault: ERR-SCH250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P250.

PRM_46407 [Schuler-HPD-P250] — BAX_LOOP_GAIN_7 | Default: 35.750 | Linked Fault: ERR-SCH250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P250.

PRM_46408 [Schuler-HPD-P250] — VAC_LOOP_GAIN_8 | Default: 148.000 | Linked Fault: ERR-SCH250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P250.

PRM_46409 [Schuler-HPD-P250] — CLN_LOOP_GAIN_9 | Default: 10.250 | Linked Fault: ERR-SCH250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P250.

PRM_46410 [Schuler-HPD-P250] — CAX_LOOP_GAIN_10 | Default: 122.500 | Linked Fault: ERR-SCH250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P250.

PRM_46411 [Schuler-HPD-P250] — THM_LOOP_GAIN_11 | Default: 234.750 | Linked Fault: ERR-SCH250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P250.

PRM_46412 [Schuler-HPD-P250] — XAX_LOOP_GAIN_12 | Default: 97.000 | Linked Fault: ERR-SCH250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P250.

PRM_46413 [Schuler-HPD-P250] — OPT_LOOP_GAIN_13 | Default: 209.250 | Linked Fault: ERR-SCH250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P250.

PRM_46414 [Schuler-HPD-P250] — SPN_LOOP_GAIN_14 | Default: 71.500 | Linked Fault: ERR-SCH250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P250.

PRM_46415 [Schuler-HPD-P250] — YAX_LOOP_GAIN_15 | Default: 183.750 | Linked Fault: ERR-SCH250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P250.

PRM_46416 [Schuler-HPD-P250] — PLC_LOOP_GAIN_16 | Default: 46.000 | Linked Fault: ERR-SCH250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P250.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P300 (BANK 1)

Machine Model: Schuler-HPD-P300 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P300.

PRM_46421 [Schuler-HPD-P300] — ATC_LOOP_GAIN_1 | Default: 73.600 | Linked Fault: ERR-SCH300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P300.

PRM_46422 [Schuler-HPD-P300] — HYD_LOOP_GAIN_2 | Default: 147.200 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46423 [Schuler-HPD-P300] — CLN_LOOP_GAIN_3 | Default: 220.800 | Linked Fault: ERR-SCH300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P300.

PRM_46424 [Schuler-HPD-P300] — XAX_LOOP_GAIN_4 | Default: 44.400 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46425 [Schuler-HPD-P300] — YAX_LOOP_GAIN_5 | Default: 118.000 | Linked Fault: ERR-SCH300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P300.

PRM_46426 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_6 | Default: 191.600 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46427 [Schuler-HPD-P300] — BAX_LOOP_GAIN_7 | Default: 15.200 | Linked Fault: ERR-SCH300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P300.

PRM_46428 [Schuler-HPD-P300] — CAX_LOOP_GAIN_8 | Default: 88.800 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46429 [Schuler-HPD-P300] — OPT_LOOP_GAIN_9 | Default: 162.400 | Linked Fault: ERR-SCH300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P300.

PRM_46430 [Schuler-HPD-P300] — PLC_LOOP_GAIN_10 | Default: 236.000 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46431 [Schuler-HPD-P300] — LUB_LOOP_GAIN_11 | Default: 59.600 | Linked Fault: ERR-SCH300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P300.

PRM_46432 [Schuler-HPD-P300] — VAC_LOOP_GAIN_12 | Default: 133.200 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46433 [Schuler-HPD-P300] — THM_LOOP_GAIN_13 | Default: 206.800 | Linked Fault: ERR-SCH300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P300.

PRM_46434 [Schuler-HPD-P300] — SPN_LOOP_GAIN_14 | Default: 30.400 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46435 [Schuler-HPD-P300] — ATC_LOOP_GAIN_15 | Default: 104.000 | Linked Fault: ERR-SCH300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P300.

PRM_46436 [Schuler-HPD-P300] — HYD_LOOP_GAIN_16 | Default: 177.600 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P300 (BANK 2)

Machine Model: Schuler-HPD-P300 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P300.

PRM_46441 [Schuler-HPD-P300] — HYD_LOOP_GAIN_1 | Default: 147.200 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46442 [Schuler-HPD-P300] — XAX_LOOP_GAIN_2 | Default: 44.400 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46443 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_3 | Default: 191.600 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46444 [Schuler-HPD-P300] — CAX_LOOP_GAIN_4 | Default: 88.800 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46445 [Schuler-HPD-P300] — PLC_LOOP_GAIN_5 | Default: 236.000 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46446 [Schuler-HPD-P300] — VAC_LOOP_GAIN_6 | Default: 133.200 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46447 [Schuler-HPD-P300] — SPN_LOOP_GAIN_7 | Default: 30.400 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46448 [Schuler-HPD-P300] — HYD_LOOP_GAIN_8 | Default: 177.600 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46449 [Schuler-HPD-P300] — XAX_LOOP_GAIN_9 | Default: 74.800 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46450 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_10 | Default: 222.000 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46451 [Schuler-HPD-P300] — CAX_LOOP_GAIN_11 | Default: 119.200 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46452 [Schuler-HPD-P300] — PLC_LOOP_GAIN_12 | Default: 16.400 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46453 [Schuler-HPD-P300] — VAC_LOOP_GAIN_13 | Default: 163.600 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46454 [Schuler-HPD-P300] — SPN_LOOP_GAIN_14 | Default: 60.800 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46455 [Schuler-HPD-P300] — HYD_LOOP_GAIN_15 | Default: 208.000 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46456 [Schuler-HPD-P300] — XAX_LOOP_GAIN_16 | Default: 105.200 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P300 (BANK 3)

Machine Model: Schuler-HPD-P300 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P300.

PRM_46461 [Schuler-HPD-P300] — CLN_LOOP_GAIN_1 | Default: 220.800 | Linked Fault: ERR-SCH300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P300.

PRM_46462 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_2 | Default: 191.600 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46463 [Schuler-HPD-P300] — OPT_LOOP_GAIN_3 | Default: 162.400 | Linked Fault: ERR-SCH300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P300.

PRM_46464 [Schuler-HPD-P300] — VAC_LOOP_GAIN_4 | Default: 133.200 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46465 [Schuler-HPD-P300] — ATC_LOOP_GAIN_5 | Default: 104.000 | Linked Fault: ERR-SCH300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P300.

PRM_46466 [Schuler-HPD-P300] — XAX_LOOP_GAIN_6 | Default: 74.800 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46467 [Schuler-HPD-P300] — BAX_LOOP_GAIN_7 | Default: 45.600 | Linked Fault: ERR-SCH300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P300.

PRM_46468 [Schuler-HPD-P300] — PLC_LOOP_GAIN_8 | Default: 16.400 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46469 [Schuler-HPD-P300] — THM_LOOP_GAIN_9 | Default: 237.200 | Linked Fault: ERR-SCH300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P300.

PRM_46470 [Schuler-HPD-P300] — HYD_LOOP_GAIN_10 | Default: 208.000 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46471 [Schuler-HPD-P300] — YAX_LOOP_GAIN_11 | Default: 178.800 | Linked Fault: ERR-SCH300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P300.

PRM_46472 [Schuler-HPD-P300] — CAX_LOOP_GAIN_12 | Default: 149.600 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46473 [Schuler-HPD-P300] — LUB_LOOP_GAIN_13 | Default: 120.400 | Linked Fault: ERR-SCH300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P300.

PRM_46474 [Schuler-HPD-P300] — SPN_LOOP_GAIN_14 | Default: 91.200 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46475 [Schuler-HPD-P300] — CLN_LOOP_GAIN_15 | Default: 62.000 | Linked Fault: ERR-SCH300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P300.

PRM_46476 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_16 | Default: 32.800 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P300 (BANK 4)

Machine Model: Schuler-HPD-P300 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P300.

PRM_46481 [Schuler-HPD-P300] — XAX_LOOP_GAIN_1 | Default: 44.400 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46482 [Schuler-HPD-P300] — CAX_LOOP_GAIN_2 | Default: 88.800 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46483 [Schuler-HPD-P300] — VAC_LOOP_GAIN_3 | Default: 133.200 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46484 [Schuler-HPD-P300] — HYD_LOOP_GAIN_4 | Default: 177.600 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46485 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_5 | Default: 222.000 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46486 [Schuler-HPD-P300] — PLC_LOOP_GAIN_6 | Default: 16.400 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46487 [Schuler-HPD-P300] — SPN_LOOP_GAIN_7 | Default: 60.800 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46488 [Schuler-HPD-P300] — XAX_LOOP_GAIN_8 | Default: 105.200 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46489 [Schuler-HPD-P300] — CAX_LOOP_GAIN_9 | Default: 149.600 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46490 [Schuler-HPD-P300] — VAC_LOOP_GAIN_10 | Default: 194.000 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46491 [Schuler-HPD-P300] — HYD_LOOP_GAIN_11 | Default: 238.400 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46492 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_12 | Default: 32.800 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46493 [Schuler-HPD-P300] — PLC_LOOP_GAIN_13 | Default: 77.200 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46494 [Schuler-HPD-P300] — SPN_LOOP_GAIN_14 | Default: 121.600 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46495 [Schuler-HPD-P300] — XAX_LOOP_GAIN_15 | Default: 166.000 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46496 [Schuler-HPD-P300] — CAX_LOOP_GAIN_16 | Default: 210.400 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P300 (BANK 5)

Machine Model: Schuler-HPD-P300 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P300.

PRM_46501 [Schuler-HPD-P300] — YAX_LOOP_GAIN_1 | Default: 118.000 | Linked Fault: ERR-SCH300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P300.

PRM_46502 [Schuler-HPD-P300] — PLC_LOOP_GAIN_2 | Default: 236.000 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

PRM_46503 [Schuler-HPD-P300] — ATC_LOOP_GAIN_3 | Default: 104.000 | Linked Fault: ERR-SCH300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P300.

PRM_46504 [Schuler-HPD-P300] — ZAX_LOOP_GAIN_4 | Default: 222.000 | Linked Fault: ERR-SCH300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P300.

PRM_46505 [Schuler-HPD-P300] — LUB_LOOP_GAIN_5 | Default: 90.000 | Linked Fault: ERR-SCH300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P300.

PRM_46506 [Schuler-HPD-P300] — HYD_LOOP_GAIN_6 | Default: 208.000 | Linked Fault: ERR-SCH300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P300.

PRM_46507 [Schuler-HPD-P300] — BAX_LOOP_GAIN_7 | Default: 76.000 | Linked Fault: ERR-SCH300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P300.

PRM_46508 [Schuler-HPD-P300] — VAC_LOOP_GAIN_8 | Default: 194.000 | Linked Fault: ERR-SCH300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P300.

PRM_46509 [Schuler-HPD-P300] — CLN_LOOP_GAIN_9 | Default: 62.000 | Linked Fault: ERR-SCH300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P300.

PRM_46510 [Schuler-HPD-P300] — CAX_LOOP_GAIN_10 | Default: 180.000 | Linked Fault: ERR-SCH300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P300.

PRM_46511 [Schuler-HPD-P300] — THM_LOOP_GAIN_11 | Default: 48.000 | Linked Fault: ERR-SCH300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P300.

PRM_46512 [Schuler-HPD-P300] — XAX_LOOP_GAIN_12 | Default: 166.000 | Linked Fault: ERR-SCH300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P300.

PRM_46513 [Schuler-HPD-P300] — OPT_LOOP_GAIN_13 | Default: 34.000 | Linked Fault: ERR-SCH300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P300.

PRM_46514 [Schuler-HPD-P300] — SPN_LOOP_GAIN_14 | Default: 152.000 | Linked Fault: ERR-SCH300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P300.

PRM_46515 [Schuler-HPD-P300] — YAX_LOOP_GAIN_15 | Default: 20.000 | Linked Fault: ERR-SCH300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P300.

PRM_46516 [Schuler-HPD-P300] — PLC_LOOP_GAIN_16 | Default: 138.000 | Linked Fault: ERR-SCH300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P300.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P350 (BANK 1)

Machine Model: Schuler-HPD-P350 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P350.

PRM_46521 [Schuler-HPD-P350] — ATC_LOOP_GAIN_1 | Default: 74.750 | Linked Fault: ERR-SCH350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P350.

PRM_46522 [Schuler-HPD-P350] — HYD_LOOP_GAIN_2 | Default: 149.500 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46523 [Schuler-HPD-P350] — CLN_LOOP_GAIN_3 | Default: 224.250 | Linked Fault: ERR-SCH350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P350.

PRM_46524 [Schuler-HPD-P350] — XAX_LOOP_GAIN_4 | Default: 49.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46525 [Schuler-HPD-P350] — YAX_LOOP_GAIN_5 | Default: 123.750 | Linked Fault: ERR-SCH350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P350.

PRM_46526 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_6 | Default: 198.500 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46527 [Schuler-HPD-P350] — BAX_LOOP_GAIN_7 | Default: 23.250 | Linked Fault: ERR-SCH350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P350.

PRM_46528 [Schuler-HPD-P350] — CAX_LOOP_GAIN_8 | Default: 98.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46529 [Schuler-HPD-P350] — OPT_LOOP_GAIN_9 | Default: 172.750 | Linked Fault: ERR-SCH350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P350.

PRM_46530 [Schuler-HPD-P350] — PLC_LOOP_GAIN_10 | Default: 247.500 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46531 [Schuler-HPD-P350] — LUB_LOOP_GAIN_11 | Default: 72.250 | Linked Fault: ERR-SCH350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P350.

PRM_46532 [Schuler-HPD-P350] — VAC_LOOP_GAIN_12 | Default: 147.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46533 [Schuler-HPD-P350] — THM_LOOP_GAIN_13 | Default: 221.750 | Linked Fault: ERR-SCH350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P350.

PRM_46534 [Schuler-HPD-P350] — SPN_LOOP_GAIN_14 | Default: 46.500 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46535 [Schuler-HPD-P350] — ATC_LOOP_GAIN_15 | Default: 121.250 | Linked Fault: ERR-SCH350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P350.

PRM_46536 [Schuler-HPD-P350] — HYD_LOOP_GAIN_16 | Default: 196.000 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P350 (BANK 2)

Machine Model: Schuler-HPD-P350 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P350.

PRM_46541 [Schuler-HPD-P350] — HYD_LOOP_GAIN_1 | Default: 149.500 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46542 [Schuler-HPD-P350] — XAX_LOOP_GAIN_2 | Default: 49.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46543 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_3 | Default: 198.500 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46544 [Schuler-HPD-P350] — CAX_LOOP_GAIN_4 | Default: 98.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46545 [Schuler-HPD-P350] — PLC_LOOP_GAIN_5 | Default: 247.500 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46546 [Schuler-HPD-P350] — VAC_LOOP_GAIN_6 | Default: 147.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46547 [Schuler-HPD-P350] — SPN_LOOP_GAIN_7 | Default: 46.500 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46548 [Schuler-HPD-P350] — HYD_LOOP_GAIN_8 | Default: 196.000 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46549 [Schuler-HPD-P350] — XAX_LOOP_GAIN_9 | Default: 95.500 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46550 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_10 | Default: 245.000 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46551 [Schuler-HPD-P350] — CAX_LOOP_GAIN_11 | Default: 144.500 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46552 [Schuler-HPD-P350] — PLC_LOOP_GAIN_12 | Default: 44.000 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46553 [Schuler-HPD-P350] — VAC_LOOP_GAIN_13 | Default: 193.500 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46554 [Schuler-HPD-P350] — SPN_LOOP_GAIN_14 | Default: 93.000 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46555 [Schuler-HPD-P350] — HYD_LOOP_GAIN_15 | Default: 242.500 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46556 [Schuler-HPD-P350] — XAX_LOOP_GAIN_16 | Default: 142.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P350 (BANK 3)

Machine Model: Schuler-HPD-P350 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P350.

PRM_46561 [Schuler-HPD-P350] — CLN_LOOP_GAIN_1 | Default: 224.250 | Linked Fault: ERR-SCH350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P350.

PRM_46562 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_2 | Default: 198.500 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46563 [Schuler-HPD-P350] — OPT_LOOP_GAIN_3 | Default: 172.750 | Linked Fault: ERR-SCH350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P350.

PRM_46564 [Schuler-HPD-P350] — VAC_LOOP_GAIN_4 | Default: 147.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46565 [Schuler-HPD-P350] — ATC_LOOP_GAIN_5 | Default: 121.250 | Linked Fault: ERR-SCH350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P350.

PRM_46566 [Schuler-HPD-P350] — XAX_LOOP_GAIN_6 | Default: 95.500 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46567 [Schuler-HPD-P350] — BAX_LOOP_GAIN_7 | Default: 69.750 | Linked Fault: ERR-SCH350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P350.

PRM_46568 [Schuler-HPD-P350] — PLC_LOOP_GAIN_8 | Default: 44.000 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46569 [Schuler-HPD-P350] — THM_LOOP_GAIN_9 | Default: 18.250 | Linked Fault: ERR-SCH350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P350.

PRM_46570 [Schuler-HPD-P350] — HYD_LOOP_GAIN_10 | Default: 242.500 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46571 [Schuler-HPD-P350] — YAX_LOOP_GAIN_11 | Default: 216.750 | Linked Fault: ERR-SCH350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P350.

PRM_46572 [Schuler-HPD-P350] — CAX_LOOP_GAIN_12 | Default: 191.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46573 [Schuler-HPD-P350] — LUB_LOOP_GAIN_13 | Default: 165.250 | Linked Fault: ERR-SCH350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P350.

PRM_46574 [Schuler-HPD-P350] — SPN_LOOP_GAIN_14 | Default: 139.500 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46575 [Schuler-HPD-P350] — CLN_LOOP_GAIN_15 | Default: 113.750 | Linked Fault: ERR-SCH350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P350.

PRM_46576 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_16 | Default: 88.000 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P350 (BANK 4)

Machine Model: Schuler-HPD-P350 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P350.

PRM_46581 [Schuler-HPD-P350] — XAX_LOOP_GAIN_1 | Default: 49.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46582 [Schuler-HPD-P350] — CAX_LOOP_GAIN_2 | Default: 98.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46583 [Schuler-HPD-P350] — VAC_LOOP_GAIN_3 | Default: 147.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46584 [Schuler-HPD-P350] — HYD_LOOP_GAIN_4 | Default: 196.000 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46585 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_5 | Default: 245.000 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46586 [Schuler-HPD-P350] — PLC_LOOP_GAIN_6 | Default: 44.000 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46587 [Schuler-HPD-P350] — SPN_LOOP_GAIN_7 | Default: 93.000 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46588 [Schuler-HPD-P350] — XAX_LOOP_GAIN_8 | Default: 142.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46589 [Schuler-HPD-P350] — CAX_LOOP_GAIN_9 | Default: 191.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46590 [Schuler-HPD-P350] — VAC_LOOP_GAIN_10 | Default: 240.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46591 [Schuler-HPD-P350] — HYD_LOOP_GAIN_11 | Default: 39.000 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46592 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_12 | Default: 88.000 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46593 [Schuler-HPD-P350] — PLC_LOOP_GAIN_13 | Default: 137.000 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46594 [Schuler-HPD-P350] — SPN_LOOP_GAIN_14 | Default: 186.000 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46595 [Schuler-HPD-P350] — XAX_LOOP_GAIN_15 | Default: 235.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46596 [Schuler-HPD-P350] — CAX_LOOP_GAIN_16 | Default: 34.000 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P350 (BANK 5)

Machine Model: Schuler-HPD-P350 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P350.

PRM_46601 [Schuler-HPD-P350] — YAX_LOOP_GAIN_1 | Default: 123.750 | Linked Fault: ERR-SCH350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P350.

PRM_46602 [Schuler-HPD-P350] — PLC_LOOP_GAIN_2 | Default: 247.500 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

PRM_46603 [Schuler-HPD-P350] — ATC_LOOP_GAIN_3 | Default: 121.250 | Linked Fault: ERR-SCH350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P350.

PRM_46604 [Schuler-HPD-P350] — ZAX_LOOP_GAIN_4 | Default: 245.000 | Linked Fault: ERR-SCH350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P350.

PRM_46605 [Schuler-HPD-P350] — LUB_LOOP_GAIN_5 | Default: 118.750 | Linked Fault: ERR-SCH350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P350.

PRM_46606 [Schuler-HPD-P350] — HYD_LOOP_GAIN_6 | Default: 242.500 | Linked Fault: ERR-SCH350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P350.

PRM_46607 [Schuler-HPD-P350] — BAX_LOOP_GAIN_7 | Default: 116.250 | Linked Fault: ERR-SCH350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P350.

PRM_46608 [Schuler-HPD-P350] — VAC_LOOP_GAIN_8 | Default: 240.000 | Linked Fault: ERR-SCH350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P350.

PRM_46609 [Schuler-HPD-P350] — CLN_LOOP_GAIN_9 | Default: 113.750 | Linked Fault: ERR-SCH350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P350.

PRM_46610 [Schuler-HPD-P350] — CAX_LOOP_GAIN_10 | Default: 237.500 | Linked Fault: ERR-SCH350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P350.

PRM_46611 [Schuler-HPD-P350] — THM_LOOP_GAIN_11 | Default: 111.250 | Linked Fault: ERR-SCH350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P350.

PRM_46612 [Schuler-HPD-P350] — XAX_LOOP_GAIN_12 | Default: 235.000 | Linked Fault: ERR-SCH350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P350.

PRM_46613 [Schuler-HPD-P350] — OPT_LOOP_GAIN_13 | Default: 108.750 | Linked Fault: ERR-SCH350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P350.

PRM_46614 [Schuler-HPD-P350] — SPN_LOOP_GAIN_14 | Default: 232.500 | Linked Fault: ERR-SCH350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P350.

PRM_46615 [Schuler-HPD-P350] — YAX_LOOP_GAIN_15 | Default: 106.250 | Linked Fault: ERR-SCH350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P350.

PRM_46616 [Schuler-HPD-P350] — PLC_LOOP_GAIN_16 | Default: 230.000 | Linked Fault: ERR-SCH350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P350.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P400 (BANK 1)

Machine Model: Schuler-HPD-P400 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P400.

PRM_46621 [Schuler-HPD-P400] — ATC_LOOP_GAIN_1 | Default: 75.900 | Linked Fault: ERR-SCH400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P400.

PRM_46622 [Schuler-HPD-P400] — HYD_LOOP_GAIN_2 | Default: 151.800 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46623 [Schuler-HPD-P400] — CLN_LOOP_GAIN_3 | Default: 227.700 | Linked Fault: ERR-SCH400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P400.

PRM_46624 [Schuler-HPD-P400] — XAX_LOOP_GAIN_4 | Default: 53.600 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46625 [Schuler-HPD-P400] — YAX_LOOP_GAIN_5 | Default: 129.500 | Linked Fault: ERR-SCH400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P400.

PRM_46626 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_6 | Default: 205.400 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46627 [Schuler-HPD-P400] — BAX_LOOP_GAIN_7 | Default: 31.300 | Linked Fault: ERR-SCH400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P400.

PRM_46628 [Schuler-HPD-P400] — CAX_LOOP_GAIN_8 | Default: 107.200 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46629 [Schuler-HPD-P400] — OPT_LOOP_GAIN_9 | Default: 183.100 | Linked Fault: ERR-SCH400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P400.

PRM_46630 [Schuler-HPD-P400] — PLC_LOOP_GAIN_10 | Default: 9.000 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46631 [Schuler-HPD-P400] — LUB_LOOP_GAIN_11 | Default: 84.900 | Linked Fault: ERR-SCH400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P400.

PRM_46632 [Schuler-HPD-P400] — VAC_LOOP_GAIN_12 | Default: 160.800 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46633 [Schuler-HPD-P400] — THM_LOOP_GAIN_13 | Default: 236.700 | Linked Fault: ERR-SCH400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P400.

PRM_46634 [Schuler-HPD-P400] — SPN_LOOP_GAIN_14 | Default: 62.600 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46635 [Schuler-HPD-P400] — ATC_LOOP_GAIN_15 | Default: 138.500 | Linked Fault: ERR-SCH400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P400.

PRM_46636 [Schuler-HPD-P400] — HYD_LOOP_GAIN_16 | Default: 214.400 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P400 (BANK 2)

Machine Model: Schuler-HPD-P400 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P400.

PRM_46641 [Schuler-HPD-P400] — HYD_LOOP_GAIN_1 | Default: 151.800 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46642 [Schuler-HPD-P400] — XAX_LOOP_GAIN_2 | Default: 53.600 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46643 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_3 | Default: 205.400 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46644 [Schuler-HPD-P400] — CAX_LOOP_GAIN_4 | Default: 107.200 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46645 [Schuler-HPD-P400] — PLC_LOOP_GAIN_5 | Default: 9.000 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46646 [Schuler-HPD-P400] — VAC_LOOP_GAIN_6 | Default: 160.800 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46647 [Schuler-HPD-P400] — SPN_LOOP_GAIN_7 | Default: 62.600 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46648 [Schuler-HPD-P400] — HYD_LOOP_GAIN_8 | Default: 214.400 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46649 [Schuler-HPD-P400] — XAX_LOOP_GAIN_9 | Default: 116.200 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46650 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_10 | Default: 18.000 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46651 [Schuler-HPD-P400] — CAX_LOOP_GAIN_11 | Default: 169.800 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46652 [Schuler-HPD-P400] — PLC_LOOP_GAIN_12 | Default: 71.600 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46653 [Schuler-HPD-P400] — VAC_LOOP_GAIN_13 | Default: 223.400 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46654 [Schuler-HPD-P400] — SPN_LOOP_GAIN_14 | Default: 125.200 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46655 [Schuler-HPD-P400] — HYD_LOOP_GAIN_15 | Default: 27.000 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46656 [Schuler-HPD-P400] — XAX_LOOP_GAIN_16 | Default: 178.800 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P400 (BANK 3)

Machine Model: Schuler-HPD-P400 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P400.

PRM_46661 [Schuler-HPD-P400] — CLN_LOOP_GAIN_1 | Default: 227.700 | Linked Fault: ERR-SCH400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P400.

PRM_46662 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_2 | Default: 205.400 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46663 [Schuler-HPD-P400] — OPT_LOOP_GAIN_3 | Default: 183.100 | Linked Fault: ERR-SCH400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P400.

PRM_46664 [Schuler-HPD-P400] — VAC_LOOP_GAIN_4 | Default: 160.800 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46665 [Schuler-HPD-P400] — ATC_LOOP_GAIN_5 | Default: 138.500 | Linked Fault: ERR-SCH400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P400.

PRM_46666 [Schuler-HPD-P400] — XAX_LOOP_GAIN_6 | Default: 116.200 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46667 [Schuler-HPD-P400] — BAX_LOOP_GAIN_7 | Default: 93.900 | Linked Fault: ERR-SCH400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P400.

PRM_46668 [Schuler-HPD-P400] — PLC_LOOP_GAIN_8 | Default: 71.600 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46669 [Schuler-HPD-P400] — THM_LOOP_GAIN_9 | Default: 49.300 | Linked Fault: ERR-SCH400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P400.

PRM_46670 [Schuler-HPD-P400] — HYD_LOOP_GAIN_10 | Default: 27.000 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46671 [Schuler-HPD-P400] — YAX_LOOP_GAIN_11 | Default: 4.700 | Linked Fault: ERR-SCH400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P400.

PRM_46672 [Schuler-HPD-P400] — CAX_LOOP_GAIN_12 | Default: 232.400 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46673 [Schuler-HPD-P400] — LUB_LOOP_GAIN_13 | Default: 210.100 | Linked Fault: ERR-SCH400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P400.

PRM_46674 [Schuler-HPD-P400] — SPN_LOOP_GAIN_14 | Default: 187.800 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46675 [Schuler-HPD-P400] — CLN_LOOP_GAIN_15 | Default: 165.500 | Linked Fault: ERR-SCH400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P400.

PRM_46676 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_16 | Default: 143.200 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P400 (BANK 4)

Machine Model: Schuler-HPD-P400 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P400.

PRM_46681 [Schuler-HPD-P400] — XAX_LOOP_GAIN_1 | Default: 53.600 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46682 [Schuler-HPD-P400] — CAX_LOOP_GAIN_2 | Default: 107.200 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46683 [Schuler-HPD-P400] — VAC_LOOP_GAIN_3 | Default: 160.800 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46684 [Schuler-HPD-P400] — HYD_LOOP_GAIN_4 | Default: 214.400 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46685 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_5 | Default: 18.000 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46686 [Schuler-HPD-P400] — PLC_LOOP_GAIN_6 | Default: 71.600 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46687 [Schuler-HPD-P400] — SPN_LOOP_GAIN_7 | Default: 125.200 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46688 [Schuler-HPD-P400] — XAX_LOOP_GAIN_8 | Default: 178.800 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46689 [Schuler-HPD-P400] — CAX_LOOP_GAIN_9 | Default: 232.400 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46690 [Schuler-HPD-P400] — VAC_LOOP_GAIN_10 | Default: 36.000 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46691 [Schuler-HPD-P400] — HYD_LOOP_GAIN_11 | Default: 89.600 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46692 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_12 | Default: 143.200 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46693 [Schuler-HPD-P400] — PLC_LOOP_GAIN_13 | Default: 196.800 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46694 [Schuler-HPD-P400] — SPN_LOOP_GAIN_14 | Default: 0.400 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46695 [Schuler-HPD-P400] — XAX_LOOP_GAIN_15 | Default: 54.000 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46696 [Schuler-HPD-P400] — CAX_LOOP_GAIN_16 | Default: 107.600 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P400 (BANK 5)

Machine Model: Schuler-HPD-P400 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P400.

PRM_46701 [Schuler-HPD-P400] — YAX_LOOP_GAIN_1 | Default: 129.500 | Linked Fault: ERR-SCH400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P400.

PRM_46702 [Schuler-HPD-P400] — PLC_LOOP_GAIN_2 | Default: 9.000 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

PRM_46703 [Schuler-HPD-P400] — ATC_LOOP_GAIN_3 | Default: 138.500 | Linked Fault: ERR-SCH400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P400.

PRM_46704 [Schuler-HPD-P400] — ZAX_LOOP_GAIN_4 | Default: 18.000 | Linked Fault: ERR-SCH400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P400.

PRM_46705 [Schuler-HPD-P400] — LUB_LOOP_GAIN_5 | Default: 147.500 | Linked Fault: ERR-SCH400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P400.

PRM_46706 [Schuler-HPD-P400] — HYD_LOOP_GAIN_6 | Default: 27.000 | Linked Fault: ERR-SCH400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P400.

PRM_46707 [Schuler-HPD-P400] — BAX_LOOP_GAIN_7 | Default: 156.500 | Linked Fault: ERR-SCH400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P400.

PRM_46708 [Schuler-HPD-P400] — VAC_LOOP_GAIN_8 | Default: 36.000 | Linked Fault: ERR-SCH400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P400.

PRM_46709 [Schuler-HPD-P400] — CLN_LOOP_GAIN_9 | Default: 165.500 | Linked Fault: ERR-SCH400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P400.

PRM_46710 [Schuler-HPD-P400] — CAX_LOOP_GAIN_10 | Default: 45.000 | Linked Fault: ERR-SCH400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P400.

PRM_46711 [Schuler-HPD-P400] — THM_LOOP_GAIN_11 | Default: 174.500 | Linked Fault: ERR-SCH400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P400.

PRM_46712 [Schuler-HPD-P400] — XAX_LOOP_GAIN_12 | Default: 54.000 | Linked Fault: ERR-SCH400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P400.

PRM_46713 [Schuler-HPD-P400] — OPT_LOOP_GAIN_13 | Default: 183.500 | Linked Fault: ERR-SCH400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P400.

PRM_46714 [Schuler-HPD-P400] — SPN_LOOP_GAIN_14 | Default: 63.000 | Linked Fault: ERR-SCH400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P400.

PRM_46715 [Schuler-HPD-P400] — YAX_LOOP_GAIN_15 | Default: 192.500 | Linked Fault: ERR-SCH400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P400.

PRM_46716 [Schuler-HPD-P400] — PLC_LOOP_GAIN_16 | Default: 72.000 | Linked Fault: ERR-SCH400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P400.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P450 (BANK 1)

Machine Model: Schuler-HPD-P450 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P450.

PRM_46721 [Schuler-HPD-P450] — ATC_LOOP_GAIN_1 | Default: 77.050 | Linked Fault: ERR-SCH450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P450.

PRM_46722 [Schuler-HPD-P450] — HYD_LOOP_GAIN_2 | Default: 154.100 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46723 [Schuler-HPD-P450] — CLN_LOOP_GAIN_3 | Default: 231.150 | Linked Fault: ERR-SCH450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P450.

PRM_46724 [Schuler-HPD-P450] — XAX_LOOP_GAIN_4 | Default: 58.200 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46725 [Schuler-HPD-P450] — YAX_LOOP_GAIN_5 | Default: 135.250 | Linked Fault: ERR-SCH450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P450.

PRM_46726 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_6 | Default: 212.300 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46727 [Schuler-HPD-P450] — BAX_LOOP_GAIN_7 | Default: 39.350 | Linked Fault: ERR-SCH450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P450.

PRM_46728 [Schuler-HPD-P450] — CAX_LOOP_GAIN_8 | Default: 116.400 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46729 [Schuler-HPD-P450] — OPT_LOOP_GAIN_9 | Default: 193.450 | Linked Fault: ERR-SCH450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P450.

PRM_46730 [Schuler-HPD-P450] — PLC_LOOP_GAIN_10 | Default: 20.500 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46731 [Schuler-HPD-P450] — LUB_LOOP_GAIN_11 | Default: 97.550 | Linked Fault: ERR-SCH450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P450.

PRM_46732 [Schuler-HPD-P450] — VAC_LOOP_GAIN_12 | Default: 174.600 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46733 [Schuler-HPD-P450] — THM_LOOP_GAIN_13 | Default: 1.650 | Linked Fault: ERR-SCH450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P450.

PRM_46734 [Schuler-HPD-P450] — SPN_LOOP_GAIN_14 | Default: 78.700 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46735 [Schuler-HPD-P450] — ATC_LOOP_GAIN_15 | Default: 155.750 | Linked Fault: ERR-SCH450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P450.

PRM_46736 [Schuler-HPD-P450] — HYD_LOOP_GAIN_16 | Default: 232.800 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P450 (BANK 2)

Machine Model: Schuler-HPD-P450 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P450.

PRM_46741 [Schuler-HPD-P450] — HYD_LOOP_GAIN_1 | Default: 154.100 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46742 [Schuler-HPD-P450] — XAX_LOOP_GAIN_2 | Default: 58.200 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46743 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_3 | Default: 212.300 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46744 [Schuler-HPD-P450] — CAX_LOOP_GAIN_4 | Default: 116.400 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46745 [Schuler-HPD-P450] — PLC_LOOP_GAIN_5 | Default: 20.500 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46746 [Schuler-HPD-P450] — VAC_LOOP_GAIN_6 | Default: 174.600 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46747 [Schuler-HPD-P450] — SPN_LOOP_GAIN_7 | Default: 78.700 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46748 [Schuler-HPD-P450] — HYD_LOOP_GAIN_8 | Default: 232.800 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46749 [Schuler-HPD-P450] — XAX_LOOP_GAIN_9 | Default: 136.900 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46750 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_10 | Default: 41.000 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46751 [Schuler-HPD-P450] — CAX_LOOP_GAIN_11 | Default: 195.100 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46752 [Schuler-HPD-P450] — PLC_LOOP_GAIN_12 | Default: 99.200 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46753 [Schuler-HPD-P450] — VAC_LOOP_GAIN_13 | Default: 3.300 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46754 [Schuler-HPD-P450] — SPN_LOOP_GAIN_14 | Default: 157.400 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46755 [Schuler-HPD-P450] — HYD_LOOP_GAIN_15 | Default: 61.500 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46756 [Schuler-HPD-P450] — XAX_LOOP_GAIN_16 | Default: 215.600 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P450 (BANK 3)

Machine Model: Schuler-HPD-P450 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P450.

PRM_46761 [Schuler-HPD-P450] — CLN_LOOP_GAIN_1 | Default: 231.150 | Linked Fault: ERR-SCH450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P450.

PRM_46762 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_2 | Default: 212.300 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46763 [Schuler-HPD-P450] — OPT_LOOP_GAIN_3 | Default: 193.450 | Linked Fault: ERR-SCH450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P450.

PRM_46764 [Schuler-HPD-P450] — VAC_LOOP_GAIN_4 | Default: 174.600 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46765 [Schuler-HPD-P450] — ATC_LOOP_GAIN_5 | Default: 155.750 | Linked Fault: ERR-SCH450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P450.

PRM_46766 [Schuler-HPD-P450] — XAX_LOOP_GAIN_6 | Default: 136.900 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46767 [Schuler-HPD-P450] — BAX_LOOP_GAIN_7 | Default: 118.050 | Linked Fault: ERR-SCH450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P450.

PRM_46768 [Schuler-HPD-P450] — PLC_LOOP_GAIN_8 | Default: 99.200 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46769 [Schuler-HPD-P450] — THM_LOOP_GAIN_9 | Default: 80.350 | Linked Fault: ERR-SCH450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P450.

PRM_46770 [Schuler-HPD-P450] — HYD_LOOP_GAIN_10 | Default: 61.500 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46771 [Schuler-HPD-P450] — YAX_LOOP_GAIN_11 | Default: 42.650 | Linked Fault: ERR-SCH450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P450.

PRM_46772 [Schuler-HPD-P450] — CAX_LOOP_GAIN_12 | Default: 23.800 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46773 [Schuler-HPD-P450] — LUB_LOOP_GAIN_13 | Default: 4.950 | Linked Fault: ERR-SCH450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P450.

PRM_46774 [Schuler-HPD-P450] — SPN_LOOP_GAIN_14 | Default: 236.100 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46775 [Schuler-HPD-P450] — CLN_LOOP_GAIN_15 | Default: 217.250 | Linked Fault: ERR-SCH450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P450.

PRM_46776 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_16 | Default: 198.400 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P450 (BANK 4)

Machine Model: Schuler-HPD-P450 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P450.

PRM_46781 [Schuler-HPD-P450] — XAX_LOOP_GAIN_1 | Default: 58.200 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46782 [Schuler-HPD-P450] — CAX_LOOP_GAIN_2 | Default: 116.400 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46783 [Schuler-HPD-P450] — VAC_LOOP_GAIN_3 | Default: 174.600 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46784 [Schuler-HPD-P450] — HYD_LOOP_GAIN_4 | Default: 232.800 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46785 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_5 | Default: 41.000 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46786 [Schuler-HPD-P450] — PLC_LOOP_GAIN_6 | Default: 99.200 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46787 [Schuler-HPD-P450] — SPN_LOOP_GAIN_7 | Default: 157.400 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46788 [Schuler-HPD-P450] — XAX_LOOP_GAIN_8 | Default: 215.600 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46789 [Schuler-HPD-P450] — CAX_LOOP_GAIN_9 | Default: 23.800 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46790 [Schuler-HPD-P450] — VAC_LOOP_GAIN_10 | Default: 82.000 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46791 [Schuler-HPD-P450] — HYD_LOOP_GAIN_11 | Default: 140.200 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46792 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_12 | Default: 198.400 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46793 [Schuler-HPD-P450] — PLC_LOOP_GAIN_13 | Default: 6.600 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46794 [Schuler-HPD-P450] — SPN_LOOP_GAIN_14 | Default: 64.800 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46795 [Schuler-HPD-P450] — XAX_LOOP_GAIN_15 | Default: 123.000 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46796 [Schuler-HPD-P450] — CAX_LOOP_GAIN_16 | Default: 181.200 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P450 (BANK 5)

Machine Model: Schuler-HPD-P450 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P450.

PRM_46801 [Schuler-HPD-P450] — YAX_LOOP_GAIN_1 | Default: 135.250 | Linked Fault: ERR-SCH450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P450.

PRM_46802 [Schuler-HPD-P450] — PLC_LOOP_GAIN_2 | Default: 20.500 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

PRM_46803 [Schuler-HPD-P450] — ATC_LOOP_GAIN_3 | Default: 155.750 | Linked Fault: ERR-SCH450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P450.

PRM_46804 [Schuler-HPD-P450] — ZAX_LOOP_GAIN_4 | Default: 41.000 | Linked Fault: ERR-SCH450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P450.

PRM_46805 [Schuler-HPD-P450] — LUB_LOOP_GAIN_5 | Default: 176.250 | Linked Fault: ERR-SCH450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P450.

PRM_46806 [Schuler-HPD-P450] — HYD_LOOP_GAIN_6 | Default: 61.500 | Linked Fault: ERR-SCH450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P450.

PRM_46807 [Schuler-HPD-P450] — BAX_LOOP_GAIN_7 | Default: 196.750 | Linked Fault: ERR-SCH450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P450.

PRM_46808 [Schuler-HPD-P450] — VAC_LOOP_GAIN_8 | Default: 82.000 | Linked Fault: ERR-SCH450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P450.

PRM_46809 [Schuler-HPD-P450] — CLN_LOOP_GAIN_9 | Default: 217.250 | Linked Fault: ERR-SCH450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P450.

PRM_46810 [Schuler-HPD-P450] — CAX_LOOP_GAIN_10 | Default: 102.500 | Linked Fault: ERR-SCH450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P450.

PRM_46811 [Schuler-HPD-P450] — THM_LOOP_GAIN_11 | Default: 237.750 | Linked Fault: ERR-SCH450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P450.

PRM_46812 [Schuler-HPD-P450] — XAX_LOOP_GAIN_12 | Default: 123.000 | Linked Fault: ERR-SCH450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P450.

PRM_46813 [Schuler-HPD-P450] — OPT_LOOP_GAIN_13 | Default: 8.250 | Linked Fault: ERR-SCH450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P450.

PRM_46814 [Schuler-HPD-P450] — SPN_LOOP_GAIN_14 | Default: 143.500 | Linked Fault: ERR-SCH450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P450.

PRM_46815 [Schuler-HPD-P450] — YAX_LOOP_GAIN_15 | Default: 28.750 | Linked Fault: ERR-SCH450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P450.

PRM_46816 [Schuler-HPD-P450] — PLC_LOOP_GAIN_16 | Default: 164.000 | Linked Fault: ERR-SCH450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P450.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P500 (BANK 1)

Machine Model: Schuler-HPD-P500 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P500.

PRM_46821 [Schuler-HPD-P500] — ATC_LOOP_GAIN_1 | Default: 78.200 | Linked Fault: ERR-SCH500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P500.

PRM_46822 [Schuler-HPD-P500] — HYD_LOOP_GAIN_2 | Default: 156.400 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46823 [Schuler-HPD-P500] — CLN_LOOP_GAIN_3 | Default: 234.600 | Linked Fault: ERR-SCH500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P500.

PRM_46824 [Schuler-HPD-P500] — XAX_LOOP_GAIN_4 | Default: 62.800 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46825 [Schuler-HPD-P500] — YAX_LOOP_GAIN_5 | Default: 141.000 | Linked Fault: ERR-SCH500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P500.

PRM_46826 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_6 | Default: 219.200 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46827 [Schuler-HPD-P500] — BAX_LOOP_GAIN_7 | Default: 47.400 | Linked Fault: ERR-SCH500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P500.

PRM_46828 [Schuler-HPD-P500] — CAX_LOOP_GAIN_8 | Default: 125.600 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46829 [Schuler-HPD-P500] — OPT_LOOP_GAIN_9 | Default: 203.800 | Linked Fault: ERR-SCH500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P500.

PRM_46830 [Schuler-HPD-P500] — PLC_LOOP_GAIN_10 | Default: 32.000 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46831 [Schuler-HPD-P500] — LUB_LOOP_GAIN_11 | Default: 110.200 | Linked Fault: ERR-SCH500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P500.

PRM_46832 [Schuler-HPD-P500] — VAC_LOOP_GAIN_12 | Default: 188.400 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46833 [Schuler-HPD-P500] — THM_LOOP_GAIN_13 | Default: 16.600 | Linked Fault: ERR-SCH500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P500.

PRM_46834 [Schuler-HPD-P500] — SPN_LOOP_GAIN_14 | Default: 94.800 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46835 [Schuler-HPD-P500] — ATC_LOOP_GAIN_15 | Default: 173.000 | Linked Fault: ERR-SCH500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P500.

PRM_46836 [Schuler-HPD-P500] — HYD_LOOP_GAIN_16 | Default: 1.200 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P500 (BANK 2)

Machine Model: Schuler-HPD-P500 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P500.

PRM_46841 [Schuler-HPD-P500] — HYD_LOOP_GAIN_1 | Default: 156.400 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46842 [Schuler-HPD-P500] — XAX_LOOP_GAIN_2 | Default: 62.800 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46843 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_3 | Default: 219.200 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46844 [Schuler-HPD-P500] — CAX_LOOP_GAIN_4 | Default: 125.600 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46845 [Schuler-HPD-P500] — PLC_LOOP_GAIN_5 | Default: 32.000 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46846 [Schuler-HPD-P500] — VAC_LOOP_GAIN_6 | Default: 188.400 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46847 [Schuler-HPD-P500] — SPN_LOOP_GAIN_7 | Default: 94.800 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46848 [Schuler-HPD-P500] — HYD_LOOP_GAIN_8 | Default: 1.200 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46849 [Schuler-HPD-P500] — XAX_LOOP_GAIN_9 | Default: 157.600 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46850 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_10 | Default: 64.000 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46851 [Schuler-HPD-P500] — CAX_LOOP_GAIN_11 | Default: 220.400 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46852 [Schuler-HPD-P500] — PLC_LOOP_GAIN_12 | Default: 126.800 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46853 [Schuler-HPD-P500] — VAC_LOOP_GAIN_13 | Default: 33.200 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46854 [Schuler-HPD-P500] — SPN_LOOP_GAIN_14 | Default: 189.600 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46855 [Schuler-HPD-P500] — HYD_LOOP_GAIN_15 | Default: 96.000 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46856 [Schuler-HPD-P500] — XAX_LOOP_GAIN_16 | Default: 2.400 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P500 (BANK 3)

Machine Model: Schuler-HPD-P500 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P500.

PRM_46861 [Schuler-HPD-P500] — CLN_LOOP_GAIN_1 | Default: 234.600 | Linked Fault: ERR-SCH500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P500.

PRM_46862 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_2 | Default: 219.200 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46863 [Schuler-HPD-P500] — OPT_LOOP_GAIN_3 | Default: 203.800 | Linked Fault: ERR-SCH500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P500.

PRM_46864 [Schuler-HPD-P500] — VAC_LOOP_GAIN_4 | Default: 188.400 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46865 [Schuler-HPD-P500] — ATC_LOOP_GAIN_5 | Default: 173.000 | Linked Fault: ERR-SCH500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P500.

PRM_46866 [Schuler-HPD-P500] — XAX_LOOP_GAIN_6 | Default: 157.600 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46867 [Schuler-HPD-P500] — BAX_LOOP_GAIN_7 | Default: 142.200 | Linked Fault: ERR-SCH500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P500.

PRM_46868 [Schuler-HPD-P500] — PLC_LOOP_GAIN_8 | Default: 126.800 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46869 [Schuler-HPD-P500] — THM_LOOP_GAIN_9 | Default: 111.400 | Linked Fault: ERR-SCH500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P500.

PRM_46870 [Schuler-HPD-P500] — HYD_LOOP_GAIN_10 | Default: 96.000 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46871 [Schuler-HPD-P500] — YAX_LOOP_GAIN_11 | Default: 80.600 | Linked Fault: ERR-SCH500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P500.

PRM_46872 [Schuler-HPD-P500] — CAX_LOOP_GAIN_12 | Default: 65.200 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46873 [Schuler-HPD-P500] — LUB_LOOP_GAIN_13 | Default: 49.800 | Linked Fault: ERR-SCH500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P500.

PRM_46874 [Schuler-HPD-P500] — SPN_LOOP_GAIN_14 | Default: 34.400 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46875 [Schuler-HPD-P500] — CLN_LOOP_GAIN_15 | Default: 19.000 | Linked Fault: ERR-SCH500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P500.

PRM_46876 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_16 | Default: 3.600 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P500 (BANK 4)

Machine Model: Schuler-HPD-P500 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P500.

PRM_46881 [Schuler-HPD-P500] — XAX_LOOP_GAIN_1 | Default: 62.800 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46882 [Schuler-HPD-P500] — CAX_LOOP_GAIN_2 | Default: 125.600 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46883 [Schuler-HPD-P500] — VAC_LOOP_GAIN_3 | Default: 188.400 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46884 [Schuler-HPD-P500] — HYD_LOOP_GAIN_4 | Default: 1.200 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46885 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_5 | Default: 64.000 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46886 [Schuler-HPD-P500] — PLC_LOOP_GAIN_6 | Default: 126.800 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46887 [Schuler-HPD-P500] — SPN_LOOP_GAIN_7 | Default: 189.600 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46888 [Schuler-HPD-P500] — XAX_LOOP_GAIN_8 | Default: 2.400 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46889 [Schuler-HPD-P500] — CAX_LOOP_GAIN_9 | Default: 65.200 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46890 [Schuler-HPD-P500] — VAC_LOOP_GAIN_10 | Default: 128.000 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46891 [Schuler-HPD-P500] — HYD_LOOP_GAIN_11 | Default: 190.800 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46892 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_12 | Default: 3.600 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46893 [Schuler-HPD-P500] — PLC_LOOP_GAIN_13 | Default: 66.400 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46894 [Schuler-HPD-P500] — SPN_LOOP_GAIN_14 | Default: 129.200 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46895 [Schuler-HPD-P500] — XAX_LOOP_GAIN_15 | Default: 192.000 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46896 [Schuler-HPD-P500] — CAX_LOOP_GAIN_16 | Default: 4.800 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P500 (BANK 5)

Machine Model: Schuler-HPD-P500 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P500.

PRM_46901 [Schuler-HPD-P500] — YAX_LOOP_GAIN_1 | Default: 141.000 | Linked Fault: ERR-SCH500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P500.

PRM_46902 [Schuler-HPD-P500] — PLC_LOOP_GAIN_2 | Default: 32.000 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

PRM_46903 [Schuler-HPD-P500] — ATC_LOOP_GAIN_3 | Default: 173.000 | Linked Fault: ERR-SCH500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P500.

PRM_46904 [Schuler-HPD-P500] — ZAX_LOOP_GAIN_4 | Default: 64.000 | Linked Fault: ERR-SCH500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P500.

PRM_46905 [Schuler-HPD-P500] — LUB_LOOP_GAIN_5 | Default: 205.000 | Linked Fault: ERR-SCH500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P500.

PRM_46906 [Schuler-HPD-P500] — HYD_LOOP_GAIN_6 | Default: 96.000 | Linked Fault: ERR-SCH500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P500.

PRM_46907 [Schuler-HPD-P500] — BAX_LOOP_GAIN_7 | Default: 237.000 | Linked Fault: ERR-SCH500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P500.

PRM_46908 [Schuler-HPD-P500] — VAC_LOOP_GAIN_8 | Default: 128.000 | Linked Fault: ERR-SCH500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P500.

PRM_46909 [Schuler-HPD-P500] — CLN_LOOP_GAIN_9 | Default: 19.000 | Linked Fault: ERR-SCH500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P500.

PRM_46910 [Schuler-HPD-P500] — CAX_LOOP_GAIN_10 | Default: 160.000 | Linked Fault: ERR-SCH500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P500.

PRM_46911 [Schuler-HPD-P500] — THM_LOOP_GAIN_11 | Default: 51.000 | Linked Fault: ERR-SCH500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P500.

PRM_46912 [Schuler-HPD-P500] — XAX_LOOP_GAIN_12 | Default: 192.000 | Linked Fault: ERR-SCH500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P500.

PRM_46913 [Schuler-HPD-P500] — OPT_LOOP_GAIN_13 | Default: 83.000 | Linked Fault: ERR-SCH500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P500.

PRM_46914 [Schuler-HPD-P500] — SPN_LOOP_GAIN_14 | Default: 224.000 | Linked Fault: ERR-SCH500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P500.

PRM_46915 [Schuler-HPD-P500] — YAX_LOOP_GAIN_15 | Default: 115.000 | Linked Fault: ERR-SCH500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P500.

PRM_46916 [Schuler-HPD-P500] — PLC_LOOP_GAIN_16 | Default: 6.000 | Linked Fault: ERR-SCH500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P500.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P550 (BANK 1)

Machine Model: Schuler-HPD-P550 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P550.

PRM_46921 [Schuler-HPD-P550] — ATC_LOOP_GAIN_1 | Default: 79.350 | Linked Fault: ERR-SCH550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P550.

PRM_46922 [Schuler-HPD-P550] — HYD_LOOP_GAIN_2 | Default: 158.700 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46923 [Schuler-HPD-P550] — CLN_LOOP_GAIN_3 | Default: 238.050 | Linked Fault: ERR-SCH550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P550.

PRM_46924 [Schuler-HPD-P550] — XAX_LOOP_GAIN_4 | Default: 67.400 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46925 [Schuler-HPD-P550] — YAX_LOOP_GAIN_5 | Default: 146.750 | Linked Fault: ERR-SCH550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P550.

PRM_46926 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_6 | Default: 226.100 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46927 [Schuler-HPD-P550] — BAX_LOOP_GAIN_7 | Default: 55.450 | Linked Fault: ERR-SCH550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P550.

PRM_46928 [Schuler-HPD-P550] — CAX_LOOP_GAIN_8 | Default: 134.800 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46929 [Schuler-HPD-P550] — OPT_LOOP_GAIN_9 | Default: 214.150 | Linked Fault: ERR-SCH550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P550.

PRM_46930 [Schuler-HPD-P550] — PLC_LOOP_GAIN_10 | Default: 43.500 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46931 [Schuler-HPD-P550] — LUB_LOOP_GAIN_11 | Default: 122.850 | Linked Fault: ERR-SCH550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P550.

PRM_46932 [Schuler-HPD-P550] — VAC_LOOP_GAIN_12 | Default: 202.200 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46933 [Schuler-HPD-P550] — THM_LOOP_GAIN_13 | Default: 31.550 | Linked Fault: ERR-SCH550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P550.

PRM_46934 [Schuler-HPD-P550] — SPN_LOOP_GAIN_14 | Default: 110.900 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46935 [Schuler-HPD-P550] — ATC_LOOP_GAIN_15 | Default: 190.250 | Linked Fault: ERR-SCH550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P550.

PRM_46936 [Schuler-HPD-P550] — HYD_LOOP_GAIN_16 | Default: 19.600 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P550 (BANK 2)

Machine Model: Schuler-HPD-P550 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P550.

PRM_46941 [Schuler-HPD-P550] — HYD_LOOP_GAIN_1 | Default: 158.700 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46942 [Schuler-HPD-P550] — XAX_LOOP_GAIN_2 | Default: 67.400 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46943 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_3 | Default: 226.100 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46944 [Schuler-HPD-P550] — CAX_LOOP_GAIN_4 | Default: 134.800 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46945 [Schuler-HPD-P550] — PLC_LOOP_GAIN_5 | Default: 43.500 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46946 [Schuler-HPD-P550] — VAC_LOOP_GAIN_6 | Default: 202.200 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46947 [Schuler-HPD-P550] — SPN_LOOP_GAIN_7 | Default: 110.900 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46948 [Schuler-HPD-P550] — HYD_LOOP_GAIN_8 | Default: 19.600 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46949 [Schuler-HPD-P550] — XAX_LOOP_GAIN_9 | Default: 178.300 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46950 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_10 | Default: 87.000 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46951 [Schuler-HPD-P550] — CAX_LOOP_GAIN_11 | Default: 245.700 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46952 [Schuler-HPD-P550] — PLC_LOOP_GAIN_12 | Default: 154.400 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46953 [Schuler-HPD-P550] — VAC_LOOP_GAIN_13 | Default: 63.100 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46954 [Schuler-HPD-P550] — SPN_LOOP_GAIN_14 | Default: 221.800 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46955 [Schuler-HPD-P550] — HYD_LOOP_GAIN_15 | Default: 130.500 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46956 [Schuler-HPD-P550] — XAX_LOOP_GAIN_16 | Default: 39.200 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P550 (BANK 3)

Machine Model: Schuler-HPD-P550 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P550.

PRM_46961 [Schuler-HPD-P550] — CLN_LOOP_GAIN_1 | Default: 238.050 | Linked Fault: ERR-SCH550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P550.

PRM_46962 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_2 | Default: 226.100 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46963 [Schuler-HPD-P550] — OPT_LOOP_GAIN_3 | Default: 214.150 | Linked Fault: ERR-SCH550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P550.

PRM_46964 [Schuler-HPD-P550] — VAC_LOOP_GAIN_4 | Default: 202.200 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46965 [Schuler-HPD-P550] — ATC_LOOP_GAIN_5 | Default: 190.250 | Linked Fault: ERR-SCH550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P550.

PRM_46966 [Schuler-HPD-P550] — XAX_LOOP_GAIN_6 | Default: 178.300 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46967 [Schuler-HPD-P550] — BAX_LOOP_GAIN_7 | Default: 166.350 | Linked Fault: ERR-SCH550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P550.

PRM_46968 [Schuler-HPD-P550] — PLC_LOOP_GAIN_8 | Default: 154.400 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46969 [Schuler-HPD-P550] — THM_LOOP_GAIN_9 | Default: 142.450 | Linked Fault: ERR-SCH550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P550.

PRM_46970 [Schuler-HPD-P550] — HYD_LOOP_GAIN_10 | Default: 130.500 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46971 [Schuler-HPD-P550] — YAX_LOOP_GAIN_11 | Default: 118.550 | Linked Fault: ERR-SCH550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P550.

PRM_46972 [Schuler-HPD-P550] — CAX_LOOP_GAIN_12 | Default: 106.600 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46973 [Schuler-HPD-P550] — LUB_LOOP_GAIN_13 | Default: 94.650 | Linked Fault: ERR-SCH550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P550.

PRM_46974 [Schuler-HPD-P550] — SPN_LOOP_GAIN_14 | Default: 82.700 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46975 [Schuler-HPD-P550] — CLN_LOOP_GAIN_15 | Default: 70.750 | Linked Fault: ERR-SCH550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P550.

PRM_46976 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_16 | Default: 58.800 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P550 (BANK 4)

Machine Model: Schuler-HPD-P550 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P550.

PRM_46981 [Schuler-HPD-P550] — XAX_LOOP_GAIN_1 | Default: 67.400 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46982 [Schuler-HPD-P550] — CAX_LOOP_GAIN_2 | Default: 134.800 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46983 [Schuler-HPD-P550] — VAC_LOOP_GAIN_3 | Default: 202.200 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46984 [Schuler-HPD-P550] — HYD_LOOP_GAIN_4 | Default: 19.600 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46985 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_5 | Default: 87.000 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46986 [Schuler-HPD-P550] — PLC_LOOP_GAIN_6 | Default: 154.400 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46987 [Schuler-HPD-P550] — SPN_LOOP_GAIN_7 | Default: 221.800 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46988 [Schuler-HPD-P550] — XAX_LOOP_GAIN_8 | Default: 39.200 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46989 [Schuler-HPD-P550] — CAX_LOOP_GAIN_9 | Default: 106.600 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_46990 [Schuler-HPD-P550] — VAC_LOOP_GAIN_10 | Default: 174.000 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_46991 [Schuler-HPD-P550] — HYD_LOOP_GAIN_11 | Default: 241.400 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_46992 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_12 | Default: 58.800 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_46993 [Schuler-HPD-P550] — PLC_LOOP_GAIN_13 | Default: 126.200 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_46994 [Schuler-HPD-P550] — SPN_LOOP_GAIN_14 | Default: 193.600 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_46995 [Schuler-HPD-P550] — XAX_LOOP_GAIN_15 | Default: 11.000 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_46996 [Schuler-HPD-P550] — CAX_LOOP_GAIN_16 | Default: 78.400 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P550 (BANK 5)

Machine Model: Schuler-HPD-P550 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P550.

PRM_47001 [Schuler-HPD-P550] — YAX_LOOP_GAIN_1 | Default: 146.750 | Linked Fault: ERR-SCH550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P550.

PRM_47002 [Schuler-HPD-P550] — PLC_LOOP_GAIN_2 | Default: 43.500 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

PRM_47003 [Schuler-HPD-P550] — ATC_LOOP_GAIN_3 | Default: 190.250 | Linked Fault: ERR-SCH550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P550.

PRM_47004 [Schuler-HPD-P550] — ZAX_LOOP_GAIN_4 | Default: 87.000 | Linked Fault: ERR-SCH550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P550.

PRM_47005 [Schuler-HPD-P550] — LUB_LOOP_GAIN_5 | Default: 233.750 | Linked Fault: ERR-SCH550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P550.

PRM_47006 [Schuler-HPD-P550] — HYD_LOOP_GAIN_6 | Default: 130.500 | Linked Fault: ERR-SCH550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P550.

PRM_47007 [Schuler-HPD-P550] — BAX_LOOP_GAIN_7 | Default: 27.250 | Linked Fault: ERR-SCH550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P550.

PRM_47008 [Schuler-HPD-P550] — VAC_LOOP_GAIN_8 | Default: 174.000 | Linked Fault: ERR-SCH550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P550.

PRM_47009 [Schuler-HPD-P550] — CLN_LOOP_GAIN_9 | Default: 70.750 | Linked Fault: ERR-SCH550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P550.

PRM_47010 [Schuler-HPD-P550] — CAX_LOOP_GAIN_10 | Default: 217.500 | Linked Fault: ERR-SCH550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P550.

PRM_47011 [Schuler-HPD-P550] — THM_LOOP_GAIN_11 | Default: 114.250 | Linked Fault: ERR-SCH550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P550.

PRM_47012 [Schuler-HPD-P550] — XAX_LOOP_GAIN_12 | Default: 11.000 | Linked Fault: ERR-SCH550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P550.

PRM_47013 [Schuler-HPD-P550] — OPT_LOOP_GAIN_13 | Default: 157.750 | Linked Fault: ERR-SCH550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P550.

PRM_47014 [Schuler-HPD-P550] — SPN_LOOP_GAIN_14 | Default: 54.500 | Linked Fault: ERR-SCH550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P550.

PRM_47015 [Schuler-HPD-P550] — YAX_LOOP_GAIN_15 | Default: 201.250 | Linked Fault: ERR-SCH550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P550.

PRM_47016 [Schuler-HPD-P550] — PLC_LOOP_GAIN_16 | Default: 98.000 | Linked Fault: ERR-SCH550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P550.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P600 (BANK 1)

Machine Model: Schuler-HPD-P600 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P600.

PRM_47021 [Schuler-HPD-P600] — ATC_LOOP_GAIN_1 | Default: 80.500 | Linked Fault: ERR-SCH600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P600.

PRM_47022 [Schuler-HPD-P600] — HYD_LOOP_GAIN_2 | Default: 161.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47023 [Schuler-HPD-P600] — CLN_LOOP_GAIN_3 | Default: 241.500 | Linked Fault: ERR-SCH600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P600.

PRM_47024 [Schuler-HPD-P600] — XAX_LOOP_GAIN_4 | Default: 72.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47025 [Schuler-HPD-P600] — YAX_LOOP_GAIN_5 | Default: 152.500 | Linked Fault: ERR-SCH600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P600.

PRM_47026 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_6 | Default: 233.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47027 [Schuler-HPD-P600] — BAX_LOOP_GAIN_7 | Default: 63.500 | Linked Fault: ERR-SCH600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P600.

PRM_47028 [Schuler-HPD-P600] — CAX_LOOP_GAIN_8 | Default: 144.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47029 [Schuler-HPD-P600] — OPT_LOOP_GAIN_9 | Default: 224.500 | Linked Fault: ERR-SCH600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P600.

PRM_47030 [Schuler-HPD-P600] — PLC_LOOP_GAIN_10 | Default: 55.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47031 [Schuler-HPD-P600] — LUB_LOOP_GAIN_11 | Default: 135.500 | Linked Fault: ERR-SCH600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P600.

PRM_47032 [Schuler-HPD-P600] — VAC_LOOP_GAIN_12 | Default: 216.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47033 [Schuler-HPD-P600] — THM_LOOP_GAIN_13 | Default: 46.500 | Linked Fault: ERR-SCH600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P600.

PRM_47034 [Schuler-HPD-P600] — SPN_LOOP_GAIN_14 | Default: 127.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47035 [Schuler-HPD-P600] — ATC_LOOP_GAIN_15 | Default: 207.500 | Linked Fault: ERR-SCH600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P600.

PRM_47036 [Schuler-HPD-P600] — HYD_LOOP_GAIN_16 | Default: 38.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P600 (BANK 2)

Machine Model: Schuler-HPD-P600 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P600.

PRM_47041 [Schuler-HPD-P600] — HYD_LOOP_GAIN_1 | Default: 161.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47042 [Schuler-HPD-P600] — XAX_LOOP_GAIN_2 | Default: 72.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47043 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_3 | Default: 233.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47044 [Schuler-HPD-P600] — CAX_LOOP_GAIN_4 | Default: 144.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47045 [Schuler-HPD-P600] — PLC_LOOP_GAIN_5 | Default: 55.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47046 [Schuler-HPD-P600] — VAC_LOOP_GAIN_6 | Default: 216.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47047 [Schuler-HPD-P600] — SPN_LOOP_GAIN_7 | Default: 127.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47048 [Schuler-HPD-P600] — HYD_LOOP_GAIN_8 | Default: 38.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47049 [Schuler-HPD-P600] — XAX_LOOP_GAIN_9 | Default: 199.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47050 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_10 | Default: 110.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47051 [Schuler-HPD-P600] — CAX_LOOP_GAIN_11 | Default: 21.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47052 [Schuler-HPD-P600] — PLC_LOOP_GAIN_12 | Default: 182.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47053 [Schuler-HPD-P600] — VAC_LOOP_GAIN_13 | Default: 93.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47054 [Schuler-HPD-P600] — SPN_LOOP_GAIN_14 | Default: 4.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47055 [Schuler-HPD-P600] — HYD_LOOP_GAIN_15 | Default: 165.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47056 [Schuler-HPD-P600] — XAX_LOOP_GAIN_16 | Default: 76.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P600 (BANK 3)

Machine Model: Schuler-HPD-P600 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P600.

PRM_47061 [Schuler-HPD-P600] — CLN_LOOP_GAIN_1 | Default: 241.500 | Linked Fault: ERR-SCH600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P600.

PRM_47062 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_2 | Default: 233.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47063 [Schuler-HPD-P600] — OPT_LOOP_GAIN_3 | Default: 224.500 | Linked Fault: ERR-SCH600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P600.

PRM_47064 [Schuler-HPD-P600] — VAC_LOOP_GAIN_4 | Default: 216.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47065 [Schuler-HPD-P600] — ATC_LOOP_GAIN_5 | Default: 207.500 | Linked Fault: ERR-SCH600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P600.

PRM_47066 [Schuler-HPD-P600] — XAX_LOOP_GAIN_6 | Default: 199.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47067 [Schuler-HPD-P600] — BAX_LOOP_GAIN_7 | Default: 190.500 | Linked Fault: ERR-SCH600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P600.

PRM_47068 [Schuler-HPD-P600] — PLC_LOOP_GAIN_8 | Default: 182.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47069 [Schuler-HPD-P600] — THM_LOOP_GAIN_9 | Default: 173.500 | Linked Fault: ERR-SCH600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P600.

PRM_47070 [Schuler-HPD-P600] — HYD_LOOP_GAIN_10 | Default: 165.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47071 [Schuler-HPD-P600] — YAX_LOOP_GAIN_11 | Default: 156.500 | Linked Fault: ERR-SCH600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P600.

PRM_47072 [Schuler-HPD-P600] — CAX_LOOP_GAIN_12 | Default: 148.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47073 [Schuler-HPD-P600] — LUB_LOOP_GAIN_13 | Default: 139.500 | Linked Fault: ERR-SCH600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P600.

PRM_47074 [Schuler-HPD-P600] — SPN_LOOP_GAIN_14 | Default: 131.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47075 [Schuler-HPD-P600] — CLN_LOOP_GAIN_15 | Default: 122.500 | Linked Fault: ERR-SCH600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P600.

PRM_47076 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_16 | Default: 114.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P600 (BANK 4)

Machine Model: Schuler-HPD-P600 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P600.

PRM_47081 [Schuler-HPD-P600] — XAX_LOOP_GAIN_1 | Default: 72.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47082 [Schuler-HPD-P600] — CAX_LOOP_GAIN_2 | Default: 144.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47083 [Schuler-HPD-P600] — VAC_LOOP_GAIN_3 | Default: 216.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47084 [Schuler-HPD-P600] — HYD_LOOP_GAIN_4 | Default: 38.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47085 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_5 | Default: 110.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47086 [Schuler-HPD-P600] — PLC_LOOP_GAIN_6 | Default: 182.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47087 [Schuler-HPD-P600] — SPN_LOOP_GAIN_7 | Default: 4.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47088 [Schuler-HPD-P600] — XAX_LOOP_GAIN_8 | Default: 76.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47089 [Schuler-HPD-P600] — CAX_LOOP_GAIN_9 | Default: 148.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47090 [Schuler-HPD-P600] — VAC_LOOP_GAIN_10 | Default: 220.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47091 [Schuler-HPD-P600] — HYD_LOOP_GAIN_11 | Default: 42.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47092 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_12 | Default: 114.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47093 [Schuler-HPD-P600] — PLC_LOOP_GAIN_13 | Default: 186.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47094 [Schuler-HPD-P600] — SPN_LOOP_GAIN_14 | Default: 8.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47095 [Schuler-HPD-P600] — XAX_LOOP_GAIN_15 | Default: 80.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47096 [Schuler-HPD-P600] — CAX_LOOP_GAIN_16 | Default: 152.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

CNC/PLC SERVO PARAMETER DICTIONARY — Schuler-HPD-P600 (BANK 5)

Machine Model: Schuler-HPD-P600 | Controller: Beckhoff TwinCAT 3. Use these parameter registers when diagnosing servo following errors or thermal drift on Schuler-HPD-P600.

PRM_47101 [Schuler-HPD-P600] — YAX_LOOP_GAIN_1 | Default: 152.500 | Linked Fault: ERR-SCH600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P600.

PRM_47102 [Schuler-HPD-P600] — PLC_LOOP_GAIN_2 | Default: 55.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

PRM_47103 [Schuler-HPD-P600] — ATC_LOOP_GAIN_3 | Default: 207.500 | Linked Fault: ERR-SCH600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Schuler-HPD-P600.

PRM_47104 [Schuler-HPD-P600] — ZAX_LOOP_GAIN_4 | Default: 110.000 | Linked Fault: ERR-SCH600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Schuler-HPD-P600.

PRM_47105 [Schuler-HPD-P600] — LUB_LOOP_GAIN_5 | Default: 12.500 | Linked Fault: ERR-SCH600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Schuler-HPD-P600.

PRM_47106 [Schuler-HPD-P600] — HYD_LOOP_GAIN_6 | Default: 165.000 | Linked Fault: ERR-SCH600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Schuler-HPD-P600.

PRM_47107 [Schuler-HPD-P600] — BAX_LOOP_GAIN_7 | Default: 67.500 | Linked Fault: ERR-SCH600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Schuler-HPD-P600.

PRM_47108 [Schuler-HPD-P600] — VAC_LOOP_GAIN_8 | Default: 220.000 | Linked Fault: ERR-SCH600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Schuler-HPD-P600.

PRM_47109 [Schuler-HPD-P600] — CLN_LOOP_GAIN_9 | Default: 122.500 | Linked Fault: ERR-SCH600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Schuler-HPD-P600.

PRM_47110 [Schuler-HPD-P600] — CAX_LOOP_GAIN_10 | Default: 25.000 | Linked Fault: ERR-SCH600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Schuler-HPD-P600.

PRM_47111 [Schuler-HPD-P600] — THM_LOOP_GAIN_11 | Default: 177.500 | Linked Fault: ERR-SCH600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Schuler-HPD-P600.

PRM_47112 [Schuler-HPD-P600] — XAX_LOOP_GAIN_12 | Default: 80.000 | Linked Fault: ERR-SCH600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Schuler-HPD-P600.

PRM_47113 [Schuler-HPD-P600] — OPT_LOOP_GAIN_13 | Default: 232.500 | Linked Fault: ERR-SCH600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Schuler-HPD-P600.

PRM_47114 [Schuler-HPD-P600] — SPN_LOOP_GAIN_14 | Default: 135.000 | Linked Fault: ERR-SCH600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Schuler-HPD-P600.

PRM_47115 [Schuler-HPD-P600] — YAX_LOOP_GAIN_15 | Default: 37.500 | Linked Fault: ERR-SCH600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Schuler-HPD-P600.

PRM_47116 [Schuler-HPD-P600] — PLC_LOOP_GAIN_16 | Default: 190.000 | Linked Fault: ERR-SCH600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Schuler-HPD-P600.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G150 (BANK 1)

Machine Model: Grob-Aerospace-G150 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G150.

PRM_47121 [Grob-Aerospace-G150] — ATC_LOOP_GAIN_1 | Default: 81.650 | Linked Fault: ERR-GRO150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G150.

PRM_47122 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_2 | Default: 163.300 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47123 [Grob-Aerospace-G150] — CLN_LOOP_GAIN_3 | Default: 244.950 | Linked Fault: ERR-GRO150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G150.

PRM_47124 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_4 | Default: 76.600 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47125 [Grob-Aerospace-G150] — YAX_LOOP_GAIN_5 | Default: 158.250 | Linked Fault: ERR-GRO150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G150.

PRM_47126 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_6 | Default: 239.900 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47127 [Grob-Aerospace-G150] — BAX_LOOP_GAIN_7 | Default: 71.550 | Linked Fault: ERR-GRO150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G150.

PRM_47128 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_8 | Default: 153.200 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47129 [Grob-Aerospace-G150] — OPT_LOOP_GAIN_9 | Default: 234.850 | Linked Fault: ERR-GRO150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G150.

PRM_47130 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_10 | Default: 66.500 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47131 [Grob-Aerospace-G150] — LUB_LOOP_GAIN_11 | Default: 148.150 | Linked Fault: ERR-GRO150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G150.

PRM_47132 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_12 | Default: 229.800 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47133 [Grob-Aerospace-G150] — THM_LOOP_GAIN_13 | Default: 61.450 | Linked Fault: ERR-GRO150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G150.

PRM_47134 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_14 | Default: 143.100 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47135 [Grob-Aerospace-G150] — ATC_LOOP_GAIN_15 | Default: 224.750 | Linked Fault: ERR-GRO150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G150.

PRM_47136 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_16 | Default: 56.400 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G150 (BANK 2)

Machine Model: Grob-Aerospace-G150 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G150.

PRM_47141 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_1 | Default: 163.300 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47142 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_2 | Default: 76.600 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47143 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_3 | Default: 239.900 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47144 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_4 | Default: 153.200 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47145 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_5 | Default: 66.500 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47146 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_6 | Default: 229.800 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47147 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_7 | Default: 143.100 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47148 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_8 | Default: 56.400 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47149 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_9 | Default: 219.700 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47150 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_10 | Default: 133.000 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47151 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_11 | Default: 46.300 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47152 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_12 | Default: 209.600 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47153 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_13 | Default: 122.900 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47154 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_14 | Default: 36.200 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47155 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_15 | Default: 199.500 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47156 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_16 | Default: 112.800 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G150 (BANK 3)

Machine Model: Grob-Aerospace-G150 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G150.

PRM_47161 [Grob-Aerospace-G150] — CLN_LOOP_GAIN_1 | Default: 244.950 | Linked Fault: ERR-GRO150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G150.

PRM_47162 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_2 | Default: 239.900 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47163 [Grob-Aerospace-G150] — OPT_LOOP_GAIN_3 | Default: 234.850 | Linked Fault: ERR-GRO150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G150.

PRM_47164 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_4 | Default: 229.800 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47165 [Grob-Aerospace-G150] — ATC_LOOP_GAIN_5 | Default: 224.750 | Linked Fault: ERR-GRO150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G150.

PRM_47166 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_6 | Default: 219.700 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47167 [Grob-Aerospace-G150] — BAX_LOOP_GAIN_7 | Default: 214.650 | Linked Fault: ERR-GRO150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G150.

PRM_47168 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_8 | Default: 209.600 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47169 [Grob-Aerospace-G150] — THM_LOOP_GAIN_9 | Default: 204.550 | Linked Fault: ERR-GRO150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G150.

PRM_47170 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_10 | Default: 199.500 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47171 [Grob-Aerospace-G150] — YAX_LOOP_GAIN_11 | Default: 194.450 | Linked Fault: ERR-GRO150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G150.

PRM_47172 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_12 | Default: 189.400 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47173 [Grob-Aerospace-G150] — LUB_LOOP_GAIN_13 | Default: 184.350 | Linked Fault: ERR-GRO150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G150.

PRM_47174 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_14 | Default: 179.300 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47175 [Grob-Aerospace-G150] — CLN_LOOP_GAIN_15 | Default: 174.250 | Linked Fault: ERR-GRO150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G150.

PRM_47176 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_16 | Default: 169.200 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G150 (BANK 4)

Machine Model: Grob-Aerospace-G150 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G150.

PRM_47181 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_1 | Default: 76.600 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47182 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_2 | Default: 153.200 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47183 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_3 | Default: 229.800 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47184 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_4 | Default: 56.400 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47185 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_5 | Default: 133.000 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47186 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_6 | Default: 209.600 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47187 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_7 | Default: 36.200 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47188 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_8 | Default: 112.800 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47189 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_9 | Default: 189.400 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47190 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_10 | Default: 16.000 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47191 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_11 | Default: 92.600 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47192 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_12 | Default: 169.200 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47193 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_13 | Default: 245.800 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47194 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_14 | Default: 72.400 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47195 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_15 | Default: 149.000 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47196 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_16 | Default: 225.600 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G150 (BANK 5)

Machine Model: Grob-Aerospace-G150 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G150.

PRM_47201 [Grob-Aerospace-G150] — YAX_LOOP_GAIN_1 | Default: 158.250 | Linked Fault: ERR-GRO150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G150.

PRM_47202 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_2 | Default: 66.500 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

PRM_47203 [Grob-Aerospace-G150] — ATC_LOOP_GAIN_3 | Default: 224.750 | Linked Fault: ERR-GRO150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G150.

PRM_47204 [Grob-Aerospace-G150] — ZAX_LOOP_GAIN_4 | Default: 133.000 | Linked Fault: ERR-GRO150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G150.

PRM_47205 [Grob-Aerospace-G150] — LUB_LOOP_GAIN_5 | Default: 41.250 | Linked Fault: ERR-GRO150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G150.

PRM_47206 [Grob-Aerospace-G150] — HYD_LOOP_GAIN_6 | Default: 199.500 | Linked Fault: ERR-GRO150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G150.

PRM_47207 [Grob-Aerospace-G150] — BAX_LOOP_GAIN_7 | Default: 107.750 | Linked Fault: ERR-GRO150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G150.

PRM_47208 [Grob-Aerospace-G150] — VAC_LOOP_GAIN_8 | Default: 16.000 | Linked Fault: ERR-GRO150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G150.

PRM_47209 [Grob-Aerospace-G150] — CLN_LOOP_GAIN_9 | Default: 174.250 | Linked Fault: ERR-GRO150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G150.

PRM_47210 [Grob-Aerospace-G150] — CAX_LOOP_GAIN_10 | Default: 82.500 | Linked Fault: ERR-GRO150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G150.

PRM_47211 [Grob-Aerospace-G150] — THM_LOOP_GAIN_11 | Default: 240.750 | Linked Fault: ERR-GRO150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G150.

PRM_47212 [Grob-Aerospace-G150] — XAX_LOOP_GAIN_12 | Default: 149.000 | Linked Fault: ERR-GRO150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G150.

PRM_47213 [Grob-Aerospace-G150] — OPT_LOOP_GAIN_13 | Default: 57.250 | Linked Fault: ERR-GRO150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G150.

PRM_47214 [Grob-Aerospace-G150] — SPN_LOOP_GAIN_14 | Default: 215.500 | Linked Fault: ERR-GRO150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G150.

PRM_47215 [Grob-Aerospace-G150] — YAX_LOOP_GAIN_15 | Default: 123.750 | Linked Fault: ERR-GRO150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G150.

PRM_47216 [Grob-Aerospace-G150] — PLC_LOOP_GAIN_16 | Default: 32.000 | Linked Fault: ERR-GRO150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G150.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G200 (BANK 1)

Machine Model: Grob-Aerospace-G200 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G200.

PRM_47221 [Grob-Aerospace-G200] — ATC_LOOP_GAIN_1 | Default: 82.800 | Linked Fault: ERR-GRO200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G200.

PRM_47222 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_2 | Default: 165.600 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47223 [Grob-Aerospace-G200] — CLN_LOOP_GAIN_3 | Default: 248.400 | Linked Fault: ERR-GRO200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G200.

PRM_47224 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_4 | Default: 81.200 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47225 [Grob-Aerospace-G200] — YAX_LOOP_GAIN_5 | Default: 164.000 | Linked Fault: ERR-GRO200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G200.

PRM_47226 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_6 | Default: 246.800 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47227 [Grob-Aerospace-G200] — BAX_LOOP_GAIN_7 | Default: 79.600 | Linked Fault: ERR-GRO200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G200.

PRM_47228 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_8 | Default: 162.400 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47229 [Grob-Aerospace-G200] — OPT_LOOP_GAIN_9 | Default: 245.200 | Linked Fault: ERR-GRO200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G200.

PRM_47230 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_10 | Default: 78.000 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47231 [Grob-Aerospace-G200] — LUB_LOOP_GAIN_11 | Default: 160.800 | Linked Fault: ERR-GRO200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G200.

PRM_47232 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_12 | Default: 243.600 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47233 [Grob-Aerospace-G200] — THM_LOOP_GAIN_13 | Default: 76.400 | Linked Fault: ERR-GRO200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G200.

PRM_47234 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_14 | Default: 159.200 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47235 [Grob-Aerospace-G200] — ATC_LOOP_GAIN_15 | Default: 242.000 | Linked Fault: ERR-GRO200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G200.

PRM_47236 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_16 | Default: 74.800 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G200 (BANK 2)

Machine Model: Grob-Aerospace-G200 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G200.

PRM_47241 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_1 | Default: 165.600 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47242 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_2 | Default: 81.200 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47243 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_3 | Default: 246.800 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47244 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_4 | Default: 162.400 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47245 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_5 | Default: 78.000 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47246 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_6 | Default: 243.600 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47247 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_7 | Default: 159.200 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47248 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_8 | Default: 74.800 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47249 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_9 | Default: 240.400 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47250 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_10 | Default: 156.000 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47251 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_11 | Default: 71.600 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47252 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_12 | Default: 237.200 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47253 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_13 | Default: 152.800 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47254 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_14 | Default: 68.400 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47255 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_15 | Default: 234.000 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47256 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_16 | Default: 149.600 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G200 (BANK 3)

Machine Model: Grob-Aerospace-G200 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G200.

PRM_47261 [Grob-Aerospace-G200] — CLN_LOOP_GAIN_1 | Default: 248.400 | Linked Fault: ERR-GRO200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G200.

PRM_47262 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_2 | Default: 246.800 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47263 [Grob-Aerospace-G200] — OPT_LOOP_GAIN_3 | Default: 245.200 | Linked Fault: ERR-GRO200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G200.

PRM_47264 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_4 | Default: 243.600 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47265 [Grob-Aerospace-G200] — ATC_LOOP_GAIN_5 | Default: 242.000 | Linked Fault: ERR-GRO200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G200.

PRM_47266 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_6 | Default: 240.400 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47267 [Grob-Aerospace-G200] — BAX_LOOP_GAIN_7 | Default: 238.800 | Linked Fault: ERR-GRO200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G200.

PRM_47268 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_8 | Default: 237.200 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47269 [Grob-Aerospace-G200] — THM_LOOP_GAIN_9 | Default: 235.600 | Linked Fault: ERR-GRO200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G200.

PRM_47270 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_10 | Default: 234.000 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47271 [Grob-Aerospace-G200] — YAX_LOOP_GAIN_11 | Default: 232.400 | Linked Fault: ERR-GRO200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G200.

PRM_47272 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_12 | Default: 230.800 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47273 [Grob-Aerospace-G200] — LUB_LOOP_GAIN_13 | Default: 229.200 | Linked Fault: ERR-GRO200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G200.

PRM_47274 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_14 | Default: 227.600 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47275 [Grob-Aerospace-G200] — CLN_LOOP_GAIN_15 | Default: 226.000 | Linked Fault: ERR-GRO200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G200.

PRM_47276 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_16 | Default: 224.400 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G200 (BANK 4)

Machine Model: Grob-Aerospace-G200 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G200.

PRM_47281 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_1 | Default: 81.200 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47282 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_2 | Default: 162.400 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47283 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_3 | Default: 243.600 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47284 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_4 | Default: 74.800 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47285 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_5 | Default: 156.000 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47286 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_6 | Default: 237.200 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47287 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_7 | Default: 68.400 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47288 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_8 | Default: 149.600 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47289 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_9 | Default: 230.800 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47290 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_10 | Default: 62.000 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47291 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_11 | Default: 143.200 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47292 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_12 | Default: 224.400 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47293 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_13 | Default: 55.600 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47294 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_14 | Default: 136.800 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47295 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_15 | Default: 218.000 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47296 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_16 | Default: 49.200 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G200 (BANK 5)

Machine Model: Grob-Aerospace-G200 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G200.

PRM_47301 [Grob-Aerospace-G200] — YAX_LOOP_GAIN_1 | Default: 164.000 | Linked Fault: ERR-GRO200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G200.

PRM_47302 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_2 | Default: 78.000 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

PRM_47303 [Grob-Aerospace-G200] — ATC_LOOP_GAIN_3 | Default: 242.000 | Linked Fault: ERR-GRO200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G200.

PRM_47304 [Grob-Aerospace-G200] — ZAX_LOOP_GAIN_4 | Default: 156.000 | Linked Fault: ERR-GRO200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G200.

PRM_47305 [Grob-Aerospace-G200] — LUB_LOOP_GAIN_5 | Default: 70.000 | Linked Fault: ERR-GRO200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G200.

PRM_47306 [Grob-Aerospace-G200] — HYD_LOOP_GAIN_6 | Default: 234.000 | Linked Fault: ERR-GRO200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G200.

PRM_47307 [Grob-Aerospace-G200] — BAX_LOOP_GAIN_7 | Default: 148.000 | Linked Fault: ERR-GRO200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G200.

PRM_47308 [Grob-Aerospace-G200] — VAC_LOOP_GAIN_8 | Default: 62.000 | Linked Fault: ERR-GRO200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G200.

PRM_47309 [Grob-Aerospace-G200] — CLN_LOOP_GAIN_9 | Default: 226.000 | Linked Fault: ERR-GRO200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G200.

PRM_47310 [Grob-Aerospace-G200] — CAX_LOOP_GAIN_10 | Default: 140.000 | Linked Fault: ERR-GRO200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G200.

PRM_47311 [Grob-Aerospace-G200] — THM_LOOP_GAIN_11 | Default: 54.000 | Linked Fault: ERR-GRO200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G200.

PRM_47312 [Grob-Aerospace-G200] — XAX_LOOP_GAIN_12 | Default: 218.000 | Linked Fault: ERR-GRO200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G200.

PRM_47313 [Grob-Aerospace-G200] — OPT_LOOP_GAIN_13 | Default: 132.000 | Linked Fault: ERR-GRO200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G200.

PRM_47314 [Grob-Aerospace-G200] — SPN_LOOP_GAIN_14 | Default: 46.000 | Linked Fault: ERR-GRO200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G200.

PRM_47315 [Grob-Aerospace-G200] — YAX_LOOP_GAIN_15 | Default: 210.000 | Linked Fault: ERR-GRO200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G200.

PRM_47316 [Grob-Aerospace-G200] — PLC_LOOP_GAIN_16 | Default: 124.000 | Linked Fault: ERR-GRO200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G200.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G250 (BANK 1)

Machine Model: Grob-Aerospace-G250 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G250.

PRM_47321 [Grob-Aerospace-G250] — ATC_LOOP_GAIN_1 | Default: 83.950 | Linked Fault: ERR-GRO250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G250.

PRM_47322 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_2 | Default: 167.900 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47323 [Grob-Aerospace-G250] — CLN_LOOP_GAIN_3 | Default: 1.850 | Linked Fault: ERR-GRO250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G250.

PRM_47324 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_4 | Default: 85.800 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47325 [Grob-Aerospace-G250] — YAX_LOOP_GAIN_5 | Default: 169.750 | Linked Fault: ERR-GRO250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G250.

PRM_47326 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_6 | Default: 3.700 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47327 [Grob-Aerospace-G250] — BAX_LOOP_GAIN_7 | Default: 87.650 | Linked Fault: ERR-GRO250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G250.

PRM_47328 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_8 | Default: 171.600 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47329 [Grob-Aerospace-G250] — OPT_LOOP_GAIN_9 | Default: 5.550 | Linked Fault: ERR-GRO250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G250.

PRM_47330 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_10 | Default: 89.500 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47331 [Grob-Aerospace-G250] — LUB_LOOP_GAIN_11 | Default: 173.450 | Linked Fault: ERR-GRO250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G250.

PRM_47332 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_12 | Default: 7.400 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47333 [Grob-Aerospace-G250] — THM_LOOP_GAIN_13 | Default: 91.350 | Linked Fault: ERR-GRO250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G250.

PRM_47334 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_14 | Default: 175.300 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47335 [Grob-Aerospace-G250] — ATC_LOOP_GAIN_15 | Default: 9.250 | Linked Fault: ERR-GRO250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G250.

PRM_47336 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_16 | Default: 93.200 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G250 (BANK 2)

Machine Model: Grob-Aerospace-G250 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G250.

PRM_47341 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_1 | Default: 167.900 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47342 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_2 | Default: 85.800 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47343 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_3 | Default: 3.700 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47344 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_4 | Default: 171.600 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47345 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_5 | Default: 89.500 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47346 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_6 | Default: 7.400 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47347 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_7 | Default: 175.300 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47348 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_8 | Default: 93.200 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47349 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_9 | Default: 11.100 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47350 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_10 | Default: 179.000 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47351 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_11 | Default: 96.900 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47352 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_12 | Default: 14.800 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47353 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_13 | Default: 182.700 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47354 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_14 | Default: 100.600 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47355 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_15 | Default: 18.500 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47356 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_16 | Default: 186.400 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G250 (BANK 3)

Machine Model: Grob-Aerospace-G250 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G250.

PRM_47361 [Grob-Aerospace-G250] — CLN_LOOP_GAIN_1 | Default: 1.850 | Linked Fault: ERR-GRO250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G250.

PRM_47362 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_2 | Default: 3.700 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47363 [Grob-Aerospace-G250] — OPT_LOOP_GAIN_3 | Default: 5.550 | Linked Fault: ERR-GRO250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G250.

PRM_47364 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_4 | Default: 7.400 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47365 [Grob-Aerospace-G250] — ATC_LOOP_GAIN_5 | Default: 9.250 | Linked Fault: ERR-GRO250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G250.

PRM_47366 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_6 | Default: 11.100 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47367 [Grob-Aerospace-G250] — BAX_LOOP_GAIN_7 | Default: 12.950 | Linked Fault: ERR-GRO250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G250.

PRM_47368 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_8 | Default: 14.800 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47369 [Grob-Aerospace-G250] — THM_LOOP_GAIN_9 | Default: 16.650 | Linked Fault: ERR-GRO250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G250.

PRM_47370 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_10 | Default: 18.500 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47371 [Grob-Aerospace-G250] — YAX_LOOP_GAIN_11 | Default: 20.350 | Linked Fault: ERR-GRO250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G250.

PRM_47372 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_12 | Default: 22.200 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47373 [Grob-Aerospace-G250] — LUB_LOOP_GAIN_13 | Default: 24.050 | Linked Fault: ERR-GRO250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G250.

PRM_47374 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_14 | Default: 25.900 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47375 [Grob-Aerospace-G250] — CLN_LOOP_GAIN_15 | Default: 27.750 | Linked Fault: ERR-GRO250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G250.

PRM_47376 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_16 | Default: 29.600 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G250 (BANK 4)

Machine Model: Grob-Aerospace-G250 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G250.

PRM_47381 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_1 | Default: 85.800 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47382 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_2 | Default: 171.600 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47383 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_3 | Default: 7.400 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47384 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_4 | Default: 93.200 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47385 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_5 | Default: 179.000 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47386 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_6 | Default: 14.800 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47387 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_7 | Default: 100.600 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47388 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_8 | Default: 186.400 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47389 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_9 | Default: 22.200 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47390 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_10 | Default: 108.000 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47391 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_11 | Default: 193.800 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47392 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_12 | Default: 29.600 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47393 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_13 | Default: 115.400 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47394 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_14 | Default: 201.200 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47395 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_15 | Default: 37.000 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47396 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_16 | Default: 122.800 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G250 (BANK 5)

Machine Model: Grob-Aerospace-G250 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G250.

PRM_47401 [Grob-Aerospace-G250] — YAX_LOOP_GAIN_1 | Default: 169.750 | Linked Fault: ERR-GRO250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G250.

PRM_47402 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_2 | Default: 89.500 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

PRM_47403 [Grob-Aerospace-G250] — ATC_LOOP_GAIN_3 | Default: 9.250 | Linked Fault: ERR-GRO250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G250.

PRM_47404 [Grob-Aerospace-G250] — ZAX_LOOP_GAIN_4 | Default: 179.000 | Linked Fault: ERR-GRO250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G250.

PRM_47405 [Grob-Aerospace-G250] — LUB_LOOP_GAIN_5 | Default: 98.750 | Linked Fault: ERR-GRO250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G250.

PRM_47406 [Grob-Aerospace-G250] — HYD_LOOP_GAIN_6 | Default: 18.500 | Linked Fault: ERR-GRO250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G250.

PRM_47407 [Grob-Aerospace-G250] — BAX_LOOP_GAIN_7 | Default: 188.250 | Linked Fault: ERR-GRO250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G250.

PRM_47408 [Grob-Aerospace-G250] — VAC_LOOP_GAIN_8 | Default: 108.000 | Linked Fault: ERR-GRO250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G250.

PRM_47409 [Grob-Aerospace-G250] — CLN_LOOP_GAIN_9 | Default: 27.750 | Linked Fault: ERR-GRO250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G250.

PRM_47410 [Grob-Aerospace-G250] — CAX_LOOP_GAIN_10 | Default: 197.500 | Linked Fault: ERR-GRO250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G250.

PRM_47411 [Grob-Aerospace-G250] — THM_LOOP_GAIN_11 | Default: 117.250 | Linked Fault: ERR-GRO250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G250.

PRM_47412 [Grob-Aerospace-G250] — XAX_LOOP_GAIN_12 | Default: 37.000 | Linked Fault: ERR-GRO250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G250.

PRM_47413 [Grob-Aerospace-G250] — OPT_LOOP_GAIN_13 | Default: 206.750 | Linked Fault: ERR-GRO250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G250.

PRM_47414 [Grob-Aerospace-G250] — SPN_LOOP_GAIN_14 | Default: 126.500 | Linked Fault: ERR-GRO250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G250.

PRM_47415 [Grob-Aerospace-G250] — YAX_LOOP_GAIN_15 | Default: 46.250 | Linked Fault: ERR-GRO250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G250.

PRM_47416 [Grob-Aerospace-G250] — PLC_LOOP_GAIN_16 | Default: 216.000 | Linked Fault: ERR-GRO250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G250.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G300 (BANK 1)

Machine Model: Grob-Aerospace-G300 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G300.

PRM_47421 [Grob-Aerospace-G300] — ATC_LOOP_GAIN_1 | Default: 85.100 | Linked Fault: ERR-GRO300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G300.

PRM_47422 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_2 | Default: 170.200 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47423 [Grob-Aerospace-G300] — CLN_LOOP_GAIN_3 | Default: 5.300 | Linked Fault: ERR-GRO300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G300.

PRM_47424 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_4 | Default: 90.400 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47425 [Grob-Aerospace-G300] — YAX_LOOP_GAIN_5 | Default: 175.500 | Linked Fault: ERR-GRO300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G300.

PRM_47426 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_6 | Default: 10.600 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47427 [Grob-Aerospace-G300] — BAX_LOOP_GAIN_7 | Default: 95.700 | Linked Fault: ERR-GRO300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G300.

PRM_47428 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_8 | Default: 180.800 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47429 [Grob-Aerospace-G300] — OPT_LOOP_GAIN_9 | Default: 15.900 | Linked Fault: ERR-GRO300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G300.

PRM_47430 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_10 | Default: 101.000 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47431 [Grob-Aerospace-G300] — LUB_LOOP_GAIN_11 | Default: 186.100 | Linked Fault: ERR-GRO300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G300.

PRM_47432 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_12 | Default: 21.200 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47433 [Grob-Aerospace-G300] — THM_LOOP_GAIN_13 | Default: 106.300 | Linked Fault: ERR-GRO300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G300.

PRM_47434 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_14 | Default: 191.400 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47435 [Grob-Aerospace-G300] — ATC_LOOP_GAIN_15 | Default: 26.500 | Linked Fault: ERR-GRO300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G300.

PRM_47436 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_16 | Default: 111.600 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G300 (BANK 2)

Machine Model: Grob-Aerospace-G300 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G300.

PRM_47441 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_1 | Default: 170.200 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47442 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_2 | Default: 90.400 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47443 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_3 | Default: 10.600 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47444 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_4 | Default: 180.800 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47445 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_5 | Default: 101.000 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47446 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_6 | Default: 21.200 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47447 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_7 | Default: 191.400 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47448 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_8 | Default: 111.600 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47449 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_9 | Default: 31.800 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47450 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_10 | Default: 202.000 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47451 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_11 | Default: 122.200 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47452 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_12 | Default: 42.400 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47453 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_13 | Default: 212.600 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47454 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_14 | Default: 132.800 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47455 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_15 | Default: 53.000 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47456 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_16 | Default: 223.200 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G300 (BANK 3)

Machine Model: Grob-Aerospace-G300 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G300.

PRM_47461 [Grob-Aerospace-G300] — CLN_LOOP_GAIN_1 | Default: 5.300 | Linked Fault: ERR-GRO300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G300.

PRM_47462 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_2 | Default: 10.600 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47463 [Grob-Aerospace-G300] — OPT_LOOP_GAIN_3 | Default: 15.900 | Linked Fault: ERR-GRO300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G300.

PRM_47464 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_4 | Default: 21.200 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47465 [Grob-Aerospace-G300] — ATC_LOOP_GAIN_5 | Default: 26.500 | Linked Fault: ERR-GRO300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G300.

PRM_47466 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_6 | Default: 31.800 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47467 [Grob-Aerospace-G300] — BAX_LOOP_GAIN_7 | Default: 37.100 | Linked Fault: ERR-GRO300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G300.

PRM_47468 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_8 | Default: 42.400 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47469 [Grob-Aerospace-G300] — THM_LOOP_GAIN_9 | Default: 47.700 | Linked Fault: ERR-GRO300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G300.

PRM_47470 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_10 | Default: 53.000 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47471 [Grob-Aerospace-G300] — YAX_LOOP_GAIN_11 | Default: 58.300 | Linked Fault: ERR-GRO300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G300.

PRM_47472 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_12 | Default: 63.600 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47473 [Grob-Aerospace-G300] — LUB_LOOP_GAIN_13 | Default: 68.900 | Linked Fault: ERR-GRO300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G300.

PRM_47474 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_14 | Default: 74.200 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47475 [Grob-Aerospace-G300] — CLN_LOOP_GAIN_15 | Default: 79.500 | Linked Fault: ERR-GRO300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G300.

PRM_47476 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_16 | Default: 84.800 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G300 (BANK 4)

Machine Model: Grob-Aerospace-G300 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G300.

PRM_47481 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_1 | Default: 90.400 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47482 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_2 | Default: 180.800 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47483 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_3 | Default: 21.200 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47484 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_4 | Default: 111.600 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47485 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_5 | Default: 202.000 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47486 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_6 | Default: 42.400 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47487 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_7 | Default: 132.800 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47488 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_8 | Default: 223.200 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47489 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_9 | Default: 63.600 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47490 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_10 | Default: 154.000 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47491 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_11 | Default: 244.400 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47492 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_12 | Default: 84.800 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47493 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_13 | Default: 175.200 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47494 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_14 | Default: 15.600 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47495 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_15 | Default: 106.000 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47496 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_16 | Default: 196.400 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G300 (BANK 5)

Machine Model: Grob-Aerospace-G300 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G300.

PRM_47501 [Grob-Aerospace-G300] — YAX_LOOP_GAIN_1 | Default: 175.500 | Linked Fault: ERR-GRO300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G300.

PRM_47502 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_2 | Default: 101.000 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

PRM_47503 [Grob-Aerospace-G300] — ATC_LOOP_GAIN_3 | Default: 26.500 | Linked Fault: ERR-GRO300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G300.

PRM_47504 [Grob-Aerospace-G300] — ZAX_LOOP_GAIN_4 | Default: 202.000 | Linked Fault: ERR-GRO300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G300.

PRM_47505 [Grob-Aerospace-G300] — LUB_LOOP_GAIN_5 | Default: 127.500 | Linked Fault: ERR-GRO300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G300.

PRM_47506 [Grob-Aerospace-G300] — HYD_LOOP_GAIN_6 | Default: 53.000 | Linked Fault: ERR-GRO300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G300.

PRM_47507 [Grob-Aerospace-G300] — BAX_LOOP_GAIN_7 | Default: 228.500 | Linked Fault: ERR-GRO300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G300.

PRM_47508 [Grob-Aerospace-G300] — VAC_LOOP_GAIN_8 | Default: 154.000 | Linked Fault: ERR-GRO300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G300.

PRM_47509 [Grob-Aerospace-G300] — CLN_LOOP_GAIN_9 | Default: 79.500 | Linked Fault: ERR-GRO300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G300.

PRM_47510 [Grob-Aerospace-G300] — CAX_LOOP_GAIN_10 | Default: 5.000 | Linked Fault: ERR-GRO300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G300.

PRM_47511 [Grob-Aerospace-G300] — THM_LOOP_GAIN_11 | Default: 180.500 | Linked Fault: ERR-GRO300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G300.

PRM_47512 [Grob-Aerospace-G300] — XAX_LOOP_GAIN_12 | Default: 106.000 | Linked Fault: ERR-GRO300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G300.

PRM_47513 [Grob-Aerospace-G300] — OPT_LOOP_GAIN_13 | Default: 31.500 | Linked Fault: ERR-GRO300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G300.

PRM_47514 [Grob-Aerospace-G300] — SPN_LOOP_GAIN_14 | Default: 207.000 | Linked Fault: ERR-GRO300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G300.

PRM_47515 [Grob-Aerospace-G300] — YAX_LOOP_GAIN_15 | Default: 132.500 | Linked Fault: ERR-GRO300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G300.

PRM_47516 [Grob-Aerospace-G300] — PLC_LOOP_GAIN_16 | Default: 58.000 | Linked Fault: ERR-GRO300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G300.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G350 (BANK 1)

Machine Model: Grob-Aerospace-G350 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G350.

PRM_47521 [Grob-Aerospace-G350] — ATC_LOOP_GAIN_1 | Default: 86.250 | Linked Fault: ERR-GRO350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G350.

PRM_47522 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_2 | Default: 172.500 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47523 [Grob-Aerospace-G350] — CLN_LOOP_GAIN_3 | Default: 8.750 | Linked Fault: ERR-GRO350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G350.

PRM_47524 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_4 | Default: 95.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47525 [Grob-Aerospace-G350] — YAX_LOOP_GAIN_5 | Default: 181.250 | Linked Fault: ERR-GRO350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G350.

PRM_47526 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_6 | Default: 17.500 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47527 [Grob-Aerospace-G350] — BAX_LOOP_GAIN_7 | Default: 103.750 | Linked Fault: ERR-GRO350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G350.

PRM_47528 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_8 | Default: 190.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47529 [Grob-Aerospace-G350] — OPT_LOOP_GAIN_9 | Default: 26.250 | Linked Fault: ERR-GRO350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G350.

PRM_47530 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_10 | Default: 112.500 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47531 [Grob-Aerospace-G350] — LUB_LOOP_GAIN_11 | Default: 198.750 | Linked Fault: ERR-GRO350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G350.

PRM_47532 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_12 | Default: 35.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47533 [Grob-Aerospace-G350] — THM_LOOP_GAIN_13 | Default: 121.250 | Linked Fault: ERR-GRO350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G350.

PRM_47534 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_14 | Default: 207.500 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47535 [Grob-Aerospace-G350] — ATC_LOOP_GAIN_15 | Default: 43.750 | Linked Fault: ERR-GRO350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G350.

PRM_47536 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_16 | Default: 130.000 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G350 (BANK 2)

Machine Model: Grob-Aerospace-G350 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G350.

PRM_47541 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_1 | Default: 172.500 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47542 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_2 | Default: 95.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47543 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_3 | Default: 17.500 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47544 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_4 | Default: 190.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47545 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_5 | Default: 112.500 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47546 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_6 | Default: 35.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47547 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_7 | Default: 207.500 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47548 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_8 | Default: 130.000 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47549 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_9 | Default: 52.500 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47550 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_10 | Default: 225.000 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47551 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_11 | Default: 147.500 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47552 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_12 | Default: 70.000 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47553 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_13 | Default: 242.500 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47554 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_14 | Default: 165.000 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47555 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_15 | Default: 87.500 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47556 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_16 | Default: 10.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G350 (BANK 3)

Machine Model: Grob-Aerospace-G350 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G350.

PRM_47561 [Grob-Aerospace-G350] — CLN_LOOP_GAIN_1 | Default: 8.750 | Linked Fault: ERR-GRO350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G350.

PRM_47562 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_2 | Default: 17.500 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47563 [Grob-Aerospace-G350] — OPT_LOOP_GAIN_3 | Default: 26.250 | Linked Fault: ERR-GRO350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G350.

PRM_47564 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_4 | Default: 35.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47565 [Grob-Aerospace-G350] — ATC_LOOP_GAIN_5 | Default: 43.750 | Linked Fault: ERR-GRO350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G350.

PRM_47566 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_6 | Default: 52.500 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47567 [Grob-Aerospace-G350] — BAX_LOOP_GAIN_7 | Default: 61.250 | Linked Fault: ERR-GRO350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G350.

PRM_47568 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_8 | Default: 70.000 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47569 [Grob-Aerospace-G350] — THM_LOOP_GAIN_9 | Default: 78.750 | Linked Fault: ERR-GRO350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G350.

PRM_47570 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_10 | Default: 87.500 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47571 [Grob-Aerospace-G350] — YAX_LOOP_GAIN_11 | Default: 96.250 | Linked Fault: ERR-GRO350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G350.

PRM_47572 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_12 | Default: 105.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47573 [Grob-Aerospace-G350] — LUB_LOOP_GAIN_13 | Default: 113.750 | Linked Fault: ERR-GRO350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G350.

PRM_47574 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_14 | Default: 122.500 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47575 [Grob-Aerospace-G350] — CLN_LOOP_GAIN_15 | Default: 131.250 | Linked Fault: ERR-GRO350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G350.

PRM_47576 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_16 | Default: 140.000 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G350 (BANK 4)

Machine Model: Grob-Aerospace-G350 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G350.

PRM_47581 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_1 | Default: 95.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47582 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_2 | Default: 190.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47583 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_3 | Default: 35.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47584 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_4 | Default: 130.000 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47585 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_5 | Default: 225.000 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47586 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_6 | Default: 70.000 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47587 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_7 | Default: 165.000 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47588 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_8 | Default: 10.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47589 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_9 | Default: 105.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47590 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_10 | Default: 200.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47591 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_11 | Default: 45.000 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47592 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_12 | Default: 140.000 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47593 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_13 | Default: 235.000 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47594 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_14 | Default: 80.000 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47595 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_15 | Default: 175.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47596 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_16 | Default: 20.000 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

CNC/PLC SERVO PARAMETER DICTIONARY — Grob-Aerospace-G350 (BANK 5)

Machine Model: Grob-Aerospace-G350 | Controller: Heidenhain TNC 640. Use these parameter registers when diagnosing servo following errors or thermal drift on Grob-Aerospace-G350.

PRM_47601 [Grob-Aerospace-G350] — YAX_LOOP_GAIN_1 | Default: 181.250 | Linked Fault: ERR-GRO350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G350.

PRM_47602 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_2 | Default: 112.500 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

PRM_47603 [Grob-Aerospace-G350] — ATC_LOOP_GAIN_3 | Default: 43.750 | Linked Fault: ERR-GRO350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Grob-Aerospace-G350.

PRM_47604 [Grob-Aerospace-G350] — ZAX_LOOP_GAIN_4 | Default: 225.000 | Linked Fault: ERR-GRO350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Grob-Aerospace-G350.

PRM_47605 [Grob-Aerospace-G350] — LUB_LOOP_GAIN_5 | Default: 156.250 | Linked Fault: ERR-GRO350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Grob-Aerospace-G350.

PRM_47606 [Grob-Aerospace-G350] — HYD_LOOP_GAIN_6 | Default: 87.500 | Linked Fault: ERR-GRO350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Grob-Aerospace-G350.

PRM_47607 [Grob-Aerospace-G350] — BAX_LOOP_GAIN_7 | Default: 18.750 | Linked Fault: ERR-GRO350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Grob-Aerospace-G350.

PRM_47608 [Grob-Aerospace-G350] — VAC_LOOP_GAIN_8 | Default: 200.000 | Linked Fault: ERR-GRO350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Grob-Aerospace-G350.

PRM_47609 [Grob-Aerospace-G350] — CLN_LOOP_GAIN_9 | Default: 131.250 | Linked Fault: ERR-GRO350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Grob-Aerospace-G350.

PRM_47610 [Grob-Aerospace-G350] — CAX_LOOP_GAIN_10 | Default: 62.500 | Linked Fault: ERR-GRO350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Grob-Aerospace-G350.

PRM_47611 [Grob-Aerospace-G350] — THM_LOOP_GAIN_11 | Default: 243.750 | Linked Fault: ERR-GRO350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Grob-Aerospace-G350.

PRM_47612 [Grob-Aerospace-G350] — XAX_LOOP_GAIN_12 | Default: 175.000 | Linked Fault: ERR-GRO350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Grob-Aerospace-G350.

PRM_47613 [Grob-Aerospace-G350] — OPT_LOOP_GAIN_13 | Default: 106.250 | Linked Fault: ERR-GRO350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Grob-Aerospace-G350.

PRM_47614 [Grob-Aerospace-G350] — SPN_LOOP_GAIN_14 | Default: 37.500 | Linked Fault: ERR-GRO350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Grob-Aerospace-G350.

PRM_47615 [Grob-Aerospace-G350] — YAX_LOOP_GAIN_15 | Default: 218.750 | Linked Fault: ERR-GRO350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Grob-Aerospace-G350.

PRM_47616 [Grob-Aerospace-G350] — PLC_LOOP_GAIN_16 | Default: 150.000 | Linked Fault: ERR-GRO350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Grob-Aerospace-G350.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L150 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L150. Verify part numbers before closing Work Orders.

BOM-51-1.1: PN-TRU150-CAM-GRIP-5102 / Kit KIT-TRU150-ATC-2 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-51-1.2: PN-TRU150-VLV-PROP-5103 / Kit KIT-TRU150-HYD-3 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-51-1.3: PN-TRU150-PMP-PLG-5104 / Kit KIT-TRU150-CLN-4 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-51-1.4: PN-TRU150-LIN-COIL-5105 / Kit KIT-TRU150-XAX-5 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-51-1.5: PN-TRU150-BSW-NUT-5106 / Kit KIT-TRU150-YAX-6 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-51-1.6: PN-TRU150-CYL-SEAL-5107 / Kit KIT-TRU150-ZAX-7 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-51-1.7: PN-TRU150-TRQ-STAT-5108 / Kit KIT-TRU150-BAX-8 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-51-1.8: PN-TRU150-BRK-DISC-5109 / Kit KIT-TRU150-CAX-9 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-51-1.9: PN-TRU150-GLASS-RD-5110 / Kit KIT-TRU150-OPT-10 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-51-1.10: PN-TRU150-IGBT-MOD-5111 / Kit KIT-TRU150-PLC-11 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-51-1.11: PN-TRU150-INJ-MET-5112 / Kit KIT-TRU150-LUB-12 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-51-1.12: PN-TRU150-GASK-VIT-5113 / Kit KIT-TRU150-VAC-13 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-51-1.13: PN-TRU150-HEX-CORE-5114 / Kit KIT-TRU150-THM-14 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-51-1.14: PN-TRU150-BRG-CER-5101 / Kit KIT-TRU150-SPN-1 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-51-1.15: PN-TRU150-CAM-GRIP-5102 / Kit KIT-TRU150-ATC-2 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU150-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L150 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L150. Verify part numbers before closing Work Orders.

BOM-51-2.1: PN-TRU150-VLV-PROP-5103 / Kit KIT-TRU150-HYD-3 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-51-2.2: PN-TRU150-LIN-COIL-5105 / Kit KIT-TRU150-XAX-5 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-51-2.3: PN-TRU150-CYL-SEAL-5107 / Kit KIT-TRU150-ZAX-7 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-51-2.4: PN-TRU150-BRK-DISC-5109 / Kit KIT-TRU150-CAX-9 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-51-2.5: PN-TRU150-IGBT-MOD-5111 / Kit KIT-TRU150-PLC-11 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-51-2.6: PN-TRU150-GASK-VIT-5113 / Kit KIT-TRU150-VAC-13 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-51-2.7: PN-TRU150-BRG-CER-5101 / Kit KIT-TRU150-SPN-1 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-51-2.8: PN-TRU150-VLV-PROP-5103 / Kit KIT-TRU150-HYD-3 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-51-2.9: PN-TRU150-LIN-COIL-5105 / Kit KIT-TRU150-XAX-5 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-51-2.10: PN-TRU150-CYL-SEAL-5107 / Kit KIT-TRU150-ZAX-7 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-51-2.11: PN-TRU150-BRK-DISC-5109 / Kit KIT-TRU150-CAX-9 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-51-2.12: PN-TRU150-IGBT-MOD-5111 / Kit KIT-TRU150-PLC-11 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-51-2.13: PN-TRU150-GASK-VIT-5113 / Kit KIT-TRU150-VAC-13 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-51-2.14: PN-TRU150-BRG-CER-5101 / Kit KIT-TRU150-SPN-1 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-51-2.15: PN-TRU150-VLV-PROP-5103 / Kit KIT-TRU150-HYD-3 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU150-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L150 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L150. Verify part numbers before closing Work Orders.

BOM-51-3.1: PN-TRU150-PMP-PLG-5104 / Kit KIT-TRU150-CLN-4 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-51-3.2: PN-TRU150-CYL-SEAL-5107 / Kit KIT-TRU150-ZAX-7 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-51-3.3: PN-TRU150-GLASS-RD-5110 / Kit KIT-TRU150-OPT-10 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-51-3.4: PN-TRU150-GASK-VIT-5113 / Kit KIT-TRU150-VAC-13 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-51-3.5: PN-TRU150-CAM-GRIP-5102 / Kit KIT-TRU150-ATC-2 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-51-3.6: PN-TRU150-LIN-COIL-5105 / Kit KIT-TRU150-XAX-5 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-51-3.7: PN-TRU150-TRQ-STAT-5108 / Kit KIT-TRU150-BAX-8 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-51-3.8: PN-TRU150-IGBT-MOD-5111 / Kit KIT-TRU150-PLC-11 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-51-3.9: PN-TRU150-HEX-CORE-5114 / Kit KIT-TRU150-THM-14 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-51-3.10: PN-TRU150-VLV-PROP-5103 / Kit KIT-TRU150-HYD-3 (Trumpf-TruLaser-L150) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-51-3.11: PN-TRU150-BSW-NUT-5106 / Kit KIT-TRU150-YAX-6 (Trumpf-TruLaser-L150) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-51-3.12: PN-TRU150-BRK-DISC-5109 / Kit KIT-TRU150-CAX-9 (Trumpf-TruLaser-L150) | PM Interval: 250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-51-3.13: PN-TRU150-INJ-MET-5112 / Kit KIT-TRU150-LUB-12 (Trumpf-TruLaser-L150) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-51-3.14: PN-TRU150-BRG-CER-5101 / Kit KIT-TRU150-SPN-1 (Trumpf-TruLaser-L150) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-51-3.15: PN-TRU150-PMP-PLG-5104 / Kit KIT-TRU150-CLN-4 (Trumpf-TruLaser-L150) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU150-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L200 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L200. Verify part numbers before closing Work Orders.

BOM-52-1.1: PN-TRU200-CAM-GRIP-5202 / Kit KIT-TRU200-ATC-2 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-52-1.2: PN-TRU200-VLV-PROP-5203 / Kit KIT-TRU200-HYD-3 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-52-1.3: PN-TRU200-PMP-PLG-5204 / Kit KIT-TRU200-CLN-4 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-52-1.4: PN-TRU200-LIN-COIL-5205 / Kit KIT-TRU200-XAX-5 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-52-1.5: PN-TRU200-BSW-NUT-5206 / Kit KIT-TRU200-YAX-6 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-52-1.6: PN-TRU200-CYL-SEAL-5207 / Kit KIT-TRU200-ZAX-7 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-52-1.7: PN-TRU200-TRQ-STAT-5208 / Kit KIT-TRU200-BAX-8 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-52-1.8: PN-TRU200-BRK-DISC-5209 / Kit KIT-TRU200-CAX-9 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-52-1.9: PN-TRU200-GLASS-RD-5210 / Kit KIT-TRU200-OPT-10 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-52-1.10: PN-TRU200-IGBT-MOD-5211 / Kit KIT-TRU200-PLC-11 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-52-1.11: PN-TRU200-INJ-MET-5212 / Kit KIT-TRU200-LUB-12 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-52-1.12: PN-TRU200-GASK-VIT-5213 / Kit KIT-TRU200-VAC-13 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-52-1.13: PN-TRU200-HEX-CORE-5214 / Kit KIT-TRU200-THM-14 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-52-1.14: PN-TRU200-BRG-CER-5201 / Kit KIT-TRU200-SPN-1 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-52-1.15: PN-TRU200-CAM-GRIP-5202 / Kit KIT-TRU200-ATC-2 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU200-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L200 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L200. Verify part numbers before closing Work Orders.

BOM-52-2.1: PN-TRU200-VLV-PROP-5203 / Kit KIT-TRU200-HYD-3 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-52-2.2: PN-TRU200-LIN-COIL-5205 / Kit KIT-TRU200-XAX-5 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-52-2.3: PN-TRU200-CYL-SEAL-5207 / Kit KIT-TRU200-ZAX-7 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-52-2.4: PN-TRU200-BRK-DISC-5209 / Kit KIT-TRU200-CAX-9 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-52-2.5: PN-TRU200-IGBT-MOD-5211 / Kit KIT-TRU200-PLC-11 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-52-2.6: PN-TRU200-GASK-VIT-5213 / Kit KIT-TRU200-VAC-13 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-52-2.7: PN-TRU200-BRG-CER-5201 / Kit KIT-TRU200-SPN-1 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-52-2.8: PN-TRU200-VLV-PROP-5203 / Kit KIT-TRU200-HYD-3 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-52-2.9: PN-TRU200-LIN-COIL-5205 / Kit KIT-TRU200-XAX-5 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-52-2.10: PN-TRU200-CYL-SEAL-5207 / Kit KIT-TRU200-ZAX-7 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-52-2.11: PN-TRU200-BRK-DISC-5209 / Kit KIT-TRU200-CAX-9 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-52-2.12: PN-TRU200-IGBT-MOD-5211 / Kit KIT-TRU200-PLC-11 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-52-2.13: PN-TRU200-GASK-VIT-5213 / Kit KIT-TRU200-VAC-13 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-52-2.14: PN-TRU200-BRG-CER-5201 / Kit KIT-TRU200-SPN-1 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-52-2.15: PN-TRU200-VLV-PROP-5203 / Kit KIT-TRU200-HYD-3 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU200-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L200 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L200. Verify part numbers before closing Work Orders.

BOM-52-3.1: PN-TRU200-PMP-PLG-5204 / Kit KIT-TRU200-CLN-4 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-52-3.2: PN-TRU200-CYL-SEAL-5207 / Kit KIT-TRU200-ZAX-7 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-52-3.3: PN-TRU200-GLASS-RD-5210 / Kit KIT-TRU200-OPT-10 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-52-3.4: PN-TRU200-GASK-VIT-5213 / Kit KIT-TRU200-VAC-13 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-52-3.5: PN-TRU200-CAM-GRIP-5202 / Kit KIT-TRU200-ATC-2 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-52-3.6: PN-TRU200-LIN-COIL-5205 / Kit KIT-TRU200-XAX-5 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-52-3.7: PN-TRU200-TRQ-STAT-5208 / Kit KIT-TRU200-BAX-8 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-52-3.8: PN-TRU200-IGBT-MOD-5211 / Kit KIT-TRU200-PLC-11 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-52-3.9: PN-TRU200-HEX-CORE-5214 / Kit KIT-TRU200-THM-14 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-52-3.10: PN-TRU200-VLV-PROP-5203 / Kit KIT-TRU200-HYD-3 (Trumpf-TruLaser-L200) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-52-3.11: PN-TRU200-BSW-NUT-5206 / Kit KIT-TRU200-YAX-6 (Trumpf-TruLaser-L200) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-52-3.12: PN-TRU200-BRK-DISC-5209 / Kit KIT-TRU200-CAX-9 (Trumpf-TruLaser-L200) | PM Interval: 250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-52-3.13: PN-TRU200-INJ-MET-5212 / Kit KIT-TRU200-LUB-12 (Trumpf-TruLaser-L200) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-52-3.14: PN-TRU200-BRG-CER-5201 / Kit KIT-TRU200-SPN-1 (Trumpf-TruLaser-L200) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-52-3.15: PN-TRU200-PMP-PLG-5204 / Kit KIT-TRU200-CLN-4 (Trumpf-TruLaser-L200) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU200-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L250 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L250. Verify part numbers before closing Work Orders.

BOM-53-1.1: PN-TRU250-CAM-GRIP-5302 / Kit KIT-TRU250-ATC-2 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-53-1.2: PN-TRU250-VLV-PROP-5303 / Kit KIT-TRU250-HYD-3 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-53-1.3: PN-TRU250-PMP-PLG-5304 / Kit KIT-TRU250-CLN-4 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-53-1.4: PN-TRU250-LIN-COIL-5305 / Kit KIT-TRU250-XAX-5 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-53-1.5: PN-TRU250-BSW-NUT-5306 / Kit KIT-TRU250-YAX-6 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-53-1.6: PN-TRU250-CYL-SEAL-5307 / Kit KIT-TRU250-ZAX-7 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-53-1.7: PN-TRU250-TRQ-STAT-5308 / Kit KIT-TRU250-BAX-8 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-53-1.8: PN-TRU250-BRK-DISC-5309 / Kit KIT-TRU250-CAX-9 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-53-1.9: PN-TRU250-GLASS-RD-5310 / Kit KIT-TRU250-OPT-10 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-53-1.10: PN-TRU250-IGBT-MOD-5311 / Kit KIT-TRU250-PLC-11 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-53-1.11: PN-TRU250-INJ-MET-5312 / Kit KIT-TRU250-LUB-12 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-53-1.12: PN-TRU250-GASK-VIT-5313 / Kit KIT-TRU250-VAC-13 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-53-1.13: PN-TRU250-HEX-CORE-5314 / Kit KIT-TRU250-THM-14 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-53-1.14: PN-TRU250-BRG-CER-5301 / Kit KIT-TRU250-SPN-1 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-53-1.15: PN-TRU250-CAM-GRIP-5302 / Kit KIT-TRU250-ATC-2 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU250-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L250 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L250. Verify part numbers before closing Work Orders.

BOM-53-2.1: PN-TRU250-VLV-PROP-5303 / Kit KIT-TRU250-HYD-3 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-53-2.2: PN-TRU250-LIN-COIL-5305 / Kit KIT-TRU250-XAX-5 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-53-2.3: PN-TRU250-CYL-SEAL-5307 / Kit KIT-TRU250-ZAX-7 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-53-2.4: PN-TRU250-BRK-DISC-5309 / Kit KIT-TRU250-CAX-9 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-53-2.5: PN-TRU250-IGBT-MOD-5311 / Kit KIT-TRU250-PLC-11 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-53-2.6: PN-TRU250-GASK-VIT-5313 / Kit KIT-TRU250-VAC-13 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-53-2.7: PN-TRU250-BRG-CER-5301 / Kit KIT-TRU250-SPN-1 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-53-2.8: PN-TRU250-VLV-PROP-5303 / Kit KIT-TRU250-HYD-3 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-53-2.9: PN-TRU250-LIN-COIL-5305 / Kit KIT-TRU250-XAX-5 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-53-2.10: PN-TRU250-CYL-SEAL-5307 / Kit KIT-TRU250-ZAX-7 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-53-2.11: PN-TRU250-BRK-DISC-5309 / Kit KIT-TRU250-CAX-9 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-53-2.12: PN-TRU250-IGBT-MOD-5311 / Kit KIT-TRU250-PLC-11 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-53-2.13: PN-TRU250-GASK-VIT-5313 / Kit KIT-TRU250-VAC-13 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-53-2.14: PN-TRU250-BRG-CER-5301 / Kit KIT-TRU250-SPN-1 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-53-2.15: PN-TRU250-VLV-PROP-5303 / Kit KIT-TRU250-HYD-3 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU250-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L250 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L250. Verify part numbers before closing Work Orders.

BOM-53-3.1: PN-TRU250-PMP-PLG-5304 / Kit KIT-TRU250-CLN-4 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-53-3.2: PN-TRU250-CYL-SEAL-5307 / Kit KIT-TRU250-ZAX-7 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-53-3.3: PN-TRU250-GLASS-RD-5310 / Kit KIT-TRU250-OPT-10 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-53-3.4: PN-TRU250-GASK-VIT-5313 / Kit KIT-TRU250-VAC-13 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-53-3.5: PN-TRU250-CAM-GRIP-5302 / Kit KIT-TRU250-ATC-2 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-53-3.6: PN-TRU250-LIN-COIL-5305 / Kit KIT-TRU250-XAX-5 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-53-3.7: PN-TRU250-TRQ-STAT-5308 / Kit KIT-TRU250-BAX-8 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-53-3.8: PN-TRU250-IGBT-MOD-5311 / Kit KIT-TRU250-PLC-11 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-53-3.9: PN-TRU250-HEX-CORE-5314 / Kit KIT-TRU250-THM-14 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-53-3.10: PN-TRU250-VLV-PROP-5303 / Kit KIT-TRU250-HYD-3 (Trumpf-TruLaser-L250) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-53-3.11: PN-TRU250-BSW-NUT-5306 / Kit KIT-TRU250-YAX-6 (Trumpf-TruLaser-L250) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-53-3.12: PN-TRU250-BRK-DISC-5309 / Kit KIT-TRU250-CAX-9 (Trumpf-TruLaser-L250) | PM Interval: 250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-53-3.13: PN-TRU250-INJ-MET-5312 / Kit KIT-TRU250-LUB-12 (Trumpf-TruLaser-L250) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-53-3.14: PN-TRU250-BRG-CER-5301 / Kit KIT-TRU250-SPN-1 (Trumpf-TruLaser-L250) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-53-3.15: PN-TRU250-PMP-PLG-5304 / Kit KIT-TRU250-CLN-4 (Trumpf-TruLaser-L250) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU250-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L300 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L300. Verify part numbers before closing Work Orders.

BOM-54-1.1: PN-TRU300-CAM-GRIP-5402 / Kit KIT-TRU300-ATC-2 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-54-1.2: PN-TRU300-VLV-PROP-5403 / Kit KIT-TRU300-HYD-3 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-54-1.3: PN-TRU300-PMP-PLG-5404 / Kit KIT-TRU300-CLN-4 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-54-1.4: PN-TRU300-LIN-COIL-5405 / Kit KIT-TRU300-XAX-5 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-54-1.5: PN-TRU300-BSW-NUT-5406 / Kit KIT-TRU300-YAX-6 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-54-1.6: PN-TRU300-CYL-SEAL-5407 / Kit KIT-TRU300-ZAX-7 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-54-1.7: PN-TRU300-TRQ-STAT-5408 / Kit KIT-TRU300-BAX-8 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-54-1.8: PN-TRU300-BRK-DISC-5409 / Kit KIT-TRU300-CAX-9 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-54-1.9: PN-TRU300-GLASS-RD-5410 / Kit KIT-TRU300-OPT-10 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-54-1.10: PN-TRU300-IGBT-MOD-5411 / Kit KIT-TRU300-PLC-11 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-54-1.11: PN-TRU300-INJ-MET-5412 / Kit KIT-TRU300-LUB-12 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-54-1.12: PN-TRU300-GASK-VIT-5413 / Kit KIT-TRU300-VAC-13 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-54-1.13: PN-TRU300-HEX-CORE-5414 / Kit KIT-TRU300-THM-14 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-54-1.14: PN-TRU300-BRG-CER-5401 / Kit KIT-TRU300-SPN-1 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-54-1.15: PN-TRU300-CAM-GRIP-5402 / Kit KIT-TRU300-ATC-2 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU300-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L300 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L300. Verify part numbers before closing Work Orders.

BOM-54-2.1: PN-TRU300-VLV-PROP-5403 / Kit KIT-TRU300-HYD-3 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-54-2.2: PN-TRU300-LIN-COIL-5405 / Kit KIT-TRU300-XAX-5 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-54-2.3: PN-TRU300-CYL-SEAL-5407 / Kit KIT-TRU300-ZAX-7 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-54-2.4: PN-TRU300-BRK-DISC-5409 / Kit KIT-TRU300-CAX-9 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-54-2.5: PN-TRU300-IGBT-MOD-5411 / Kit KIT-TRU300-PLC-11 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-54-2.6: PN-TRU300-GASK-VIT-5413 / Kit KIT-TRU300-VAC-13 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-54-2.7: PN-TRU300-BRG-CER-5401 / Kit KIT-TRU300-SPN-1 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-54-2.8: PN-TRU300-VLV-PROP-5403 / Kit KIT-TRU300-HYD-3 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-54-2.9: PN-TRU300-LIN-COIL-5405 / Kit KIT-TRU300-XAX-5 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-54-2.10: PN-TRU300-CYL-SEAL-5407 / Kit KIT-TRU300-ZAX-7 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-54-2.11: PN-TRU300-BRK-DISC-5409 / Kit KIT-TRU300-CAX-9 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-54-2.12: PN-TRU300-IGBT-MOD-5411 / Kit KIT-TRU300-PLC-11 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-54-2.13: PN-TRU300-GASK-VIT-5413 / Kit KIT-TRU300-VAC-13 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-54-2.14: PN-TRU300-BRG-CER-5401 / Kit KIT-TRU300-SPN-1 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-54-2.15: PN-TRU300-VLV-PROP-5403 / Kit KIT-TRU300-HYD-3 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU300-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L300 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L300. Verify part numbers before closing Work Orders.

BOM-54-3.1: PN-TRU300-PMP-PLG-5404 / Kit KIT-TRU300-CLN-4 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-54-3.2: PN-TRU300-CYL-SEAL-5407 / Kit KIT-TRU300-ZAX-7 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-54-3.3: PN-TRU300-GLASS-RD-5410 / Kit KIT-TRU300-OPT-10 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-54-3.4: PN-TRU300-GASK-VIT-5413 / Kit KIT-TRU300-VAC-13 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-54-3.5: PN-TRU300-CAM-GRIP-5402 / Kit KIT-TRU300-ATC-2 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-54-3.6: PN-TRU300-LIN-COIL-5405 / Kit KIT-TRU300-XAX-5 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-54-3.7: PN-TRU300-TRQ-STAT-5408 / Kit KIT-TRU300-BAX-8 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-54-3.8: PN-TRU300-IGBT-MOD-5411 / Kit KIT-TRU300-PLC-11 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-54-3.9: PN-TRU300-HEX-CORE-5414 / Kit KIT-TRU300-THM-14 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-54-3.10: PN-TRU300-VLV-PROP-5403 / Kit KIT-TRU300-HYD-3 (Trumpf-TruLaser-L300) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-54-3.11: PN-TRU300-BSW-NUT-5406 / Kit KIT-TRU300-YAX-6 (Trumpf-TruLaser-L300) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-54-3.12: PN-TRU300-BRK-DISC-5409 / Kit KIT-TRU300-CAX-9 (Trumpf-TruLaser-L300) | PM Interval: 250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-54-3.13: PN-TRU300-INJ-MET-5412 / Kit KIT-TRU300-LUB-12 (Trumpf-TruLaser-L300) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-54-3.14: PN-TRU300-BRG-CER-5401 / Kit KIT-TRU300-SPN-1 (Trumpf-TruLaser-L300) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-54-3.15: PN-TRU300-PMP-PLG-5404 / Kit KIT-TRU300-CLN-4 (Trumpf-TruLaser-L300) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU300-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L350 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L350. Verify part numbers before closing Work Orders.

BOM-55-1.1: PN-TRU350-CAM-GRIP-5502 / Kit KIT-TRU350-ATC-2 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-55-1.2: PN-TRU350-VLV-PROP-5503 / Kit KIT-TRU350-HYD-3 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-55-1.3: PN-TRU350-PMP-PLG-5504 / Kit KIT-TRU350-CLN-4 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-55-1.4: PN-TRU350-LIN-COIL-5505 / Kit KIT-TRU350-XAX-5 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-55-1.5: PN-TRU350-BSW-NUT-5506 / Kit KIT-TRU350-YAX-6 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-55-1.6: PN-TRU350-CYL-SEAL-5507 / Kit KIT-TRU350-ZAX-7 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-55-1.7: PN-TRU350-TRQ-STAT-5508 / Kit KIT-TRU350-BAX-8 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-55-1.8: PN-TRU350-BRK-DISC-5509 / Kit KIT-TRU350-CAX-9 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-55-1.9: PN-TRU350-GLASS-RD-5510 / Kit KIT-TRU350-OPT-10 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-55-1.10: PN-TRU350-IGBT-MOD-5511 / Kit KIT-TRU350-PLC-11 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-55-1.11: PN-TRU350-INJ-MET-5512 / Kit KIT-TRU350-LUB-12 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-55-1.12: PN-TRU350-GASK-VIT-5513 / Kit KIT-TRU350-VAC-13 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-55-1.13: PN-TRU350-HEX-CORE-5514 / Kit KIT-TRU350-THM-14 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-55-1.14: PN-TRU350-BRG-CER-5501 / Kit KIT-TRU350-SPN-1 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-55-1.15: PN-TRU350-CAM-GRIP-5502 / Kit KIT-TRU350-ATC-2 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU350-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L350 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L350. Verify part numbers before closing Work Orders.

BOM-55-2.1: PN-TRU350-VLV-PROP-5503 / Kit KIT-TRU350-HYD-3 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-55-2.2: PN-TRU350-LIN-COIL-5505 / Kit KIT-TRU350-XAX-5 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-55-2.3: PN-TRU350-CYL-SEAL-5507 / Kit KIT-TRU350-ZAX-7 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-55-2.4: PN-TRU350-BRK-DISC-5509 / Kit KIT-TRU350-CAX-9 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-55-2.5: PN-TRU350-IGBT-MOD-5511 / Kit KIT-TRU350-PLC-11 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-55-2.6: PN-TRU350-GASK-VIT-5513 / Kit KIT-TRU350-VAC-13 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-55-2.7: PN-TRU350-BRG-CER-5501 / Kit KIT-TRU350-SPN-1 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-55-2.8: PN-TRU350-VLV-PROP-5503 / Kit KIT-TRU350-HYD-3 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-55-2.9: PN-TRU350-LIN-COIL-5505 / Kit KIT-TRU350-XAX-5 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-55-2.10: PN-TRU350-CYL-SEAL-5507 / Kit KIT-TRU350-ZAX-7 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-55-2.11: PN-TRU350-BRK-DISC-5509 / Kit KIT-TRU350-CAX-9 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-55-2.12: PN-TRU350-IGBT-MOD-5511 / Kit KIT-TRU350-PLC-11 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-55-2.13: PN-TRU350-GASK-VIT-5513 / Kit KIT-TRU350-VAC-13 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-55-2.14: PN-TRU350-BRG-CER-5501 / Kit KIT-TRU350-SPN-1 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-55-2.15: PN-TRU350-VLV-PROP-5503 / Kit KIT-TRU350-HYD-3 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU350-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L350 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L350. Verify part numbers before closing Work Orders.

BOM-55-3.1: PN-TRU350-PMP-PLG-5504 / Kit KIT-TRU350-CLN-4 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-55-3.2: PN-TRU350-CYL-SEAL-5507 / Kit KIT-TRU350-ZAX-7 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-55-3.3: PN-TRU350-GLASS-RD-5510 / Kit KIT-TRU350-OPT-10 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-55-3.4: PN-TRU350-GASK-VIT-5513 / Kit KIT-TRU350-VAC-13 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-55-3.5: PN-TRU350-CAM-GRIP-5502 / Kit KIT-TRU350-ATC-2 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-55-3.6: PN-TRU350-LIN-COIL-5505 / Kit KIT-TRU350-XAX-5 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-55-3.7: PN-TRU350-TRQ-STAT-5508 / Kit KIT-TRU350-BAX-8 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-55-3.8: PN-TRU350-IGBT-MOD-5511 / Kit KIT-TRU350-PLC-11 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-55-3.9: PN-TRU350-HEX-CORE-5514 / Kit KIT-TRU350-THM-14 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-55-3.10: PN-TRU350-VLV-PROP-5503 / Kit KIT-TRU350-HYD-3 (Trumpf-TruLaser-L350) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-55-3.11: PN-TRU350-BSW-NUT-5506 / Kit KIT-TRU350-YAX-6 (Trumpf-TruLaser-L350) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-55-3.12: PN-TRU350-BRK-DISC-5509 / Kit KIT-TRU350-CAX-9 (Trumpf-TruLaser-L350) | PM Interval: 250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-55-3.13: PN-TRU350-INJ-MET-5512 / Kit KIT-TRU350-LUB-12 (Trumpf-TruLaser-L350) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-55-3.14: PN-TRU350-BRG-CER-5501 / Kit KIT-TRU350-SPN-1 (Trumpf-TruLaser-L350) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-55-3.15: PN-TRU350-PMP-PLG-5504 / Kit KIT-TRU350-CLN-4 (Trumpf-TruLaser-L350) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU350-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L400 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L400. Verify part numbers before closing Work Orders.

BOM-56-1.1: PN-TRU400-CAM-GRIP-5602 / Kit KIT-TRU400-ATC-2 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-56-1.2: PN-TRU400-VLV-PROP-5603 / Kit KIT-TRU400-HYD-3 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-56-1.3: PN-TRU400-PMP-PLG-5604 / Kit KIT-TRU400-CLN-4 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-56-1.4: PN-TRU400-LIN-COIL-5605 / Kit KIT-TRU400-XAX-5 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-56-1.5: PN-TRU400-BSW-NUT-5606 / Kit KIT-TRU400-YAX-6 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-56-1.6: PN-TRU400-CYL-SEAL-5607 / Kit KIT-TRU400-ZAX-7 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-56-1.7: PN-TRU400-TRQ-STAT-5608 / Kit KIT-TRU400-BAX-8 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-56-1.8: PN-TRU400-BRK-DISC-5609 / Kit KIT-TRU400-CAX-9 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-56-1.9: PN-TRU400-GLASS-RD-5610 / Kit KIT-TRU400-OPT-10 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-56-1.10: PN-TRU400-IGBT-MOD-5611 / Kit KIT-TRU400-PLC-11 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-56-1.11: PN-TRU400-INJ-MET-5612 / Kit KIT-TRU400-LUB-12 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-56-1.12: PN-TRU400-GASK-VIT-5613 / Kit KIT-TRU400-VAC-13 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-56-1.13: PN-TRU400-HEX-CORE-5614 / Kit KIT-TRU400-THM-14 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-56-1.14: PN-TRU400-BRG-CER-5601 / Kit KIT-TRU400-SPN-1 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-56-1.15: PN-TRU400-CAM-GRIP-5602 / Kit KIT-TRU400-ATC-2 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU400-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L400 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L400. Verify part numbers before closing Work Orders.

BOM-56-2.1: PN-TRU400-VLV-PROP-5603 / Kit KIT-TRU400-HYD-3 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-56-2.2: PN-TRU400-LIN-COIL-5605 / Kit KIT-TRU400-XAX-5 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-56-2.3: PN-TRU400-CYL-SEAL-5607 / Kit KIT-TRU400-ZAX-7 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-56-2.4: PN-TRU400-BRK-DISC-5609 / Kit KIT-TRU400-CAX-9 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-56-2.5: PN-TRU400-IGBT-MOD-5611 / Kit KIT-TRU400-PLC-11 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-56-2.6: PN-TRU400-GASK-VIT-5613 / Kit KIT-TRU400-VAC-13 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-56-2.7: PN-TRU400-BRG-CER-5601 / Kit KIT-TRU400-SPN-1 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-56-2.8: PN-TRU400-VLV-PROP-5603 / Kit KIT-TRU400-HYD-3 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-56-2.9: PN-TRU400-LIN-COIL-5605 / Kit KIT-TRU400-XAX-5 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-56-2.10: PN-TRU400-CYL-SEAL-5607 / Kit KIT-TRU400-ZAX-7 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-56-2.11: PN-TRU400-BRK-DISC-5609 / Kit KIT-TRU400-CAX-9 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-56-2.12: PN-TRU400-IGBT-MOD-5611 / Kit KIT-TRU400-PLC-11 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-56-2.13: PN-TRU400-GASK-VIT-5613 / Kit KIT-TRU400-VAC-13 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-56-2.14: PN-TRU400-BRG-CER-5601 / Kit KIT-TRU400-SPN-1 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-56-2.15: PN-TRU400-VLV-PROP-5603 / Kit KIT-TRU400-HYD-3 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU400-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L400 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L400. Verify part numbers before closing Work Orders.

BOM-56-3.1: PN-TRU400-PMP-PLG-5604 / Kit KIT-TRU400-CLN-4 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU400-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-56-3.2: PN-TRU400-CYL-SEAL-5607 / Kit KIT-TRU400-ZAX-7 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU400-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-56-3.3: PN-TRU400-GLASS-RD-5610 / Kit KIT-TRU400-OPT-10 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU400-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-56-3.4: PN-TRU400-GASK-VIT-5613 / Kit KIT-TRU400-VAC-13 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU400-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-56-3.5: PN-TRU400-CAM-GRIP-5602 / Kit KIT-TRU400-ATC-2 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU400-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-56-3.6: PN-TRU400-LIN-COIL-5605 / Kit KIT-TRU400-XAX-5 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU400-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-56-3.7: PN-TRU400-TRQ-STAT-5608 / Kit KIT-TRU400-BAX-8 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU400-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-56-3.8: PN-TRU400-IGBT-MOD-5611 / Kit KIT-TRU400-PLC-11 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU400-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-56-3.9: PN-TRU400-HEX-CORE-5614 / Kit KIT-TRU400-THM-14 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU400-THM-14 within tolerance ±0.0060 mm TIR.

BOM-56-3.10: PN-TRU400-VLV-PROP-5603 / Kit KIT-TRU400-HYD-3 (Trumpf-TruLaser-L400) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU400-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-56-3.11: PN-TRU400-BSW-NUT-5606 / Kit KIT-TRU400-YAX-6 (Trumpf-TruLaser-L400) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU400-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-56-3.12: PN-TRU400-BRK-DISC-5609 / Kit KIT-TRU400-CAX-9 (Trumpf-TruLaser-L400) | PM Interval: 250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU400-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-56-3.13: PN-TRU400-INJ-MET-5612 / Kit KIT-TRU400-LUB-12 (Trumpf-TruLaser-L400) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU400-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-56-3.14: PN-TRU400-BRG-CER-5601 / Kit KIT-TRU400-SPN-1 (Trumpf-TruLaser-L400) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU400-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-56-3.15: PN-TRU400-PMP-PLG-5604 / Kit KIT-TRU400-CLN-4 (Trumpf-TruLaser-L400) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU400-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L450 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L450. Verify part numbers before closing Work Orders.

BOM-57-1.1: PN-TRU450-CAM-GRIP-5702 / Kit KIT-TRU450-ATC-2 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-57-1.2: PN-TRU450-VLV-PROP-5703 / Kit KIT-TRU450-HYD-3 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-57-1.3: PN-TRU450-PMP-PLG-5704 / Kit KIT-TRU450-CLN-4 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-57-1.4: PN-TRU450-LIN-COIL-5705 / Kit KIT-TRU450-XAX-5 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-57-1.5: PN-TRU450-BSW-NUT-5706 / Kit KIT-TRU450-YAX-6 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-57-1.6: PN-TRU450-CYL-SEAL-5707 / Kit KIT-TRU450-ZAX-7 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-57-1.7: PN-TRU450-TRQ-STAT-5708 / Kit KIT-TRU450-BAX-8 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-57-1.8: PN-TRU450-BRK-DISC-5709 / Kit KIT-TRU450-CAX-9 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-57-1.9: PN-TRU450-GLASS-RD-5710 / Kit KIT-TRU450-OPT-10 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-57-1.10: PN-TRU450-IGBT-MOD-5711 / Kit KIT-TRU450-PLC-11 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-57-1.11: PN-TRU450-INJ-MET-5712 / Kit KIT-TRU450-LUB-12 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-57-1.12: PN-TRU450-GASK-VIT-5713 / Kit KIT-TRU450-VAC-13 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-57-1.13: PN-TRU450-HEX-CORE-5714 / Kit KIT-TRU450-THM-14 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-57-1.14: PN-TRU450-BRG-CER-5701 / Kit KIT-TRU450-SPN-1 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-57-1.15: PN-TRU450-CAM-GRIP-5702 / Kit KIT-TRU450-ATC-2 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU450-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L450 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L450. Verify part numbers before closing Work Orders.

BOM-57-2.1: PN-TRU450-VLV-PROP-5703 / Kit KIT-TRU450-HYD-3 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-57-2.2: PN-TRU450-LIN-COIL-5705 / Kit KIT-TRU450-XAX-5 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-57-2.3: PN-TRU450-CYL-SEAL-5707 / Kit KIT-TRU450-ZAX-7 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-57-2.4: PN-TRU450-BRK-DISC-5709 / Kit KIT-TRU450-CAX-9 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-57-2.5: PN-TRU450-IGBT-MOD-5711 / Kit KIT-TRU450-PLC-11 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-57-2.6: PN-TRU450-GASK-VIT-5713 / Kit KIT-TRU450-VAC-13 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-57-2.7: PN-TRU450-BRG-CER-5701 / Kit KIT-TRU450-SPN-1 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-57-2.8: PN-TRU450-VLV-PROP-5703 / Kit KIT-TRU450-HYD-3 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-57-2.9: PN-TRU450-LIN-COIL-5705 / Kit KIT-TRU450-XAX-5 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-57-2.10: PN-TRU450-CYL-SEAL-5707 / Kit KIT-TRU450-ZAX-7 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-57-2.11: PN-TRU450-BRK-DISC-5709 / Kit KIT-TRU450-CAX-9 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-57-2.12: PN-TRU450-IGBT-MOD-5711 / Kit KIT-TRU450-PLC-11 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-57-2.13: PN-TRU450-GASK-VIT-5713 / Kit KIT-TRU450-VAC-13 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-57-2.14: PN-TRU450-BRG-CER-5701 / Kit KIT-TRU450-SPN-1 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-57-2.15: PN-TRU450-VLV-PROP-5703 / Kit KIT-TRU450-HYD-3 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU450-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L450 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L450. Verify part numbers before closing Work Orders.

BOM-57-3.1: PN-TRU450-PMP-PLG-5704 / Kit KIT-TRU450-CLN-4 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU450-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-57-3.2: PN-TRU450-CYL-SEAL-5707 / Kit KIT-TRU450-ZAX-7 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU450-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-57-3.3: PN-TRU450-GLASS-RD-5710 / Kit KIT-TRU450-OPT-10 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU450-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-57-3.4: PN-TRU450-GASK-VIT-5713 / Kit KIT-TRU450-VAC-13 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU450-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-57-3.5: PN-TRU450-CAM-GRIP-5702 / Kit KIT-TRU450-ATC-2 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU450-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-57-3.6: PN-TRU450-LIN-COIL-5705 / Kit KIT-TRU450-XAX-5 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU450-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-57-3.7: PN-TRU450-TRQ-STAT-5708 / Kit KIT-TRU450-BAX-8 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU450-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-57-3.8: PN-TRU450-IGBT-MOD-5711 / Kit KIT-TRU450-PLC-11 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU450-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-57-3.9: PN-TRU450-HEX-CORE-5714 / Kit KIT-TRU450-THM-14 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU450-THM-14 within tolerance ±0.0020 mm TIR.

BOM-57-3.10: PN-TRU450-VLV-PROP-5703 / Kit KIT-TRU450-HYD-3 (Trumpf-TruLaser-L450) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU450-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-57-3.11: PN-TRU450-BSW-NUT-5706 / Kit KIT-TRU450-YAX-6 (Trumpf-TruLaser-L450) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU450-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-57-3.12: PN-TRU450-BRK-DISC-5709 / Kit KIT-TRU450-CAX-9 (Trumpf-TruLaser-L450) | PM Interval: 250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU450-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-57-3.13: PN-TRU450-INJ-MET-5712 / Kit KIT-TRU450-LUB-12 (Trumpf-TruLaser-L450) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU450-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-57-3.14: PN-TRU450-BRG-CER-5701 / Kit KIT-TRU450-SPN-1 (Trumpf-TruLaser-L450) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU450-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-57-3.15: PN-TRU450-PMP-PLG-5704 / Kit KIT-TRU450-CLN-4 (Trumpf-TruLaser-L450) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU450-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L500 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L500. Verify part numbers before closing Work Orders.

BOM-58-1.1: PN-TRU500-CAM-GRIP-5802 / Kit KIT-TRU500-ATC-2 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-58-1.2: PN-TRU500-VLV-PROP-5803 / Kit KIT-TRU500-HYD-3 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-58-1.3: PN-TRU500-PMP-PLG-5804 / Kit KIT-TRU500-CLN-4 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-58-1.4: PN-TRU500-LIN-COIL-5805 / Kit KIT-TRU500-XAX-5 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-58-1.5: PN-TRU500-BSW-NUT-5806 / Kit KIT-TRU500-YAX-6 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-58-1.6: PN-TRU500-CYL-SEAL-5807 / Kit KIT-TRU500-ZAX-7 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-58-1.7: PN-TRU500-TRQ-STAT-5808 / Kit KIT-TRU500-BAX-8 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-58-1.8: PN-TRU500-BRK-DISC-5809 / Kit KIT-TRU500-CAX-9 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-58-1.9: PN-TRU500-GLASS-RD-5810 / Kit KIT-TRU500-OPT-10 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-58-1.10: PN-TRU500-IGBT-MOD-5811 / Kit KIT-TRU500-PLC-11 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-58-1.11: PN-TRU500-INJ-MET-5812 / Kit KIT-TRU500-LUB-12 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-58-1.12: PN-TRU500-GASK-VIT-5813 / Kit KIT-TRU500-VAC-13 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-58-1.13: PN-TRU500-HEX-CORE-5814 / Kit KIT-TRU500-THM-14 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-58-1.14: PN-TRU500-BRG-CER-5801 / Kit KIT-TRU500-SPN-1 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-58-1.15: PN-TRU500-CAM-GRIP-5802 / Kit KIT-TRU500-ATC-2 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU500-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L500 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L500. Verify part numbers before closing Work Orders.

BOM-58-2.1: PN-TRU500-VLV-PROP-5803 / Kit KIT-TRU500-HYD-3 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-58-2.2: PN-TRU500-LIN-COIL-5805 / Kit KIT-TRU500-XAX-5 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-58-2.3: PN-TRU500-CYL-SEAL-5807 / Kit KIT-TRU500-ZAX-7 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-58-2.4: PN-TRU500-BRK-DISC-5809 / Kit KIT-TRU500-CAX-9 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-58-2.5: PN-TRU500-IGBT-MOD-5811 / Kit KIT-TRU500-PLC-11 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-58-2.6: PN-TRU500-GASK-VIT-5813 / Kit KIT-TRU500-VAC-13 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-58-2.7: PN-TRU500-BRG-CER-5801 / Kit KIT-TRU500-SPN-1 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-58-2.8: PN-TRU500-VLV-PROP-5803 / Kit KIT-TRU500-HYD-3 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-58-2.9: PN-TRU500-LIN-COIL-5805 / Kit KIT-TRU500-XAX-5 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-58-2.10: PN-TRU500-CYL-SEAL-5807 / Kit KIT-TRU500-ZAX-7 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-58-2.11: PN-TRU500-BRK-DISC-5809 / Kit KIT-TRU500-CAX-9 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-58-2.12: PN-TRU500-IGBT-MOD-5811 / Kit KIT-TRU500-PLC-11 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-58-2.13: PN-TRU500-GASK-VIT-5813 / Kit KIT-TRU500-VAC-13 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-58-2.14: PN-TRU500-BRG-CER-5801 / Kit KIT-TRU500-SPN-1 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-58-2.15: PN-TRU500-VLV-PROP-5803 / Kit KIT-TRU500-HYD-3 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU500-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L500 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L500. Verify part numbers before closing Work Orders.

BOM-58-3.1: PN-TRU500-PMP-PLG-5804 / Kit KIT-TRU500-CLN-4 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU500-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-58-3.2: PN-TRU500-CYL-SEAL-5807 / Kit KIT-TRU500-ZAX-7 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU500-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-58-3.3: PN-TRU500-GLASS-RD-5810 / Kit KIT-TRU500-OPT-10 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU500-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-58-3.4: PN-TRU500-GASK-VIT-5813 / Kit KIT-TRU500-VAC-13 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU500-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-58-3.5: PN-TRU500-CAM-GRIP-5802 / Kit KIT-TRU500-ATC-2 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU500-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-58-3.6: PN-TRU500-LIN-COIL-5805 / Kit KIT-TRU500-XAX-5 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU500-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-58-3.7: PN-TRU500-TRQ-STAT-5808 / Kit KIT-TRU500-BAX-8 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU500-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-58-3.8: PN-TRU500-IGBT-MOD-5811 / Kit KIT-TRU500-PLC-11 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU500-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-58-3.9: PN-TRU500-HEX-CORE-5814 / Kit KIT-TRU500-THM-14 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU500-THM-14 within tolerance ±0.0040 mm TIR.

BOM-58-3.10: PN-TRU500-VLV-PROP-5803 / Kit KIT-TRU500-HYD-3 (Trumpf-TruLaser-L500) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU500-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-58-3.11: PN-TRU500-BSW-NUT-5806 / Kit KIT-TRU500-YAX-6 (Trumpf-TruLaser-L500) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU500-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-58-3.12: PN-TRU500-BRK-DISC-5809 / Kit KIT-TRU500-CAX-9 (Trumpf-TruLaser-L500) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU500-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-58-3.13: PN-TRU500-INJ-MET-5812 / Kit KIT-TRU500-LUB-12 (Trumpf-TruLaser-L500) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU500-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-58-3.14: PN-TRU500-BRG-CER-5801 / Kit KIT-TRU500-SPN-1 (Trumpf-TruLaser-L500) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU500-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-58-3.15: PN-TRU500-PMP-PLG-5804 / Kit KIT-TRU500-CLN-4 (Trumpf-TruLaser-L500) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU500-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L550 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L550. Verify part numbers before closing Work Orders.

BOM-59-1.1: PN-TRU550-CAM-GRIP-5902 / Kit KIT-TRU550-ATC-2 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-59-1.2: PN-TRU550-VLV-PROP-5903 / Kit KIT-TRU550-HYD-3 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-59-1.3: PN-TRU550-PMP-PLG-5904 / Kit KIT-TRU550-CLN-4 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-59-1.4: PN-TRU550-LIN-COIL-5905 / Kit KIT-TRU550-XAX-5 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-59-1.5: PN-TRU550-BSW-NUT-5906 / Kit KIT-TRU550-YAX-6 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-59-1.6: PN-TRU550-CYL-SEAL-5907 / Kit KIT-TRU550-ZAX-7 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-59-1.7: PN-TRU550-TRQ-STAT-5908 / Kit KIT-TRU550-BAX-8 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-59-1.8: PN-TRU550-BRK-DISC-5909 / Kit KIT-TRU550-CAX-9 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-59-1.9: PN-TRU550-GLASS-RD-5910 / Kit KIT-TRU550-OPT-10 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-59-1.10: PN-TRU550-IGBT-MOD-5911 / Kit KIT-TRU550-PLC-11 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-59-1.11: PN-TRU550-INJ-MET-5912 / Kit KIT-TRU550-LUB-12 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-59-1.12: PN-TRU550-GASK-VIT-5913 / Kit KIT-TRU550-VAC-13 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-59-1.13: PN-TRU550-HEX-CORE-5914 / Kit KIT-TRU550-THM-14 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-59-1.14: PN-TRU550-BRG-CER-5901 / Kit KIT-TRU550-SPN-1 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-59-1.15: PN-TRU550-CAM-GRIP-5902 / Kit KIT-TRU550-ATC-2 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU550-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L550 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L550. Verify part numbers before closing Work Orders.

BOM-59-2.1: PN-TRU550-VLV-PROP-5903 / Kit KIT-TRU550-HYD-3 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-59-2.2: PN-TRU550-LIN-COIL-5905 / Kit KIT-TRU550-XAX-5 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-59-2.3: PN-TRU550-CYL-SEAL-5907 / Kit KIT-TRU550-ZAX-7 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-59-2.4: PN-TRU550-BRK-DISC-5909 / Kit KIT-TRU550-CAX-9 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-59-2.5: PN-TRU550-IGBT-MOD-5911 / Kit KIT-TRU550-PLC-11 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-59-2.6: PN-TRU550-GASK-VIT-5913 / Kit KIT-TRU550-VAC-13 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-59-2.7: PN-TRU550-BRG-CER-5901 / Kit KIT-TRU550-SPN-1 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-59-2.8: PN-TRU550-VLV-PROP-5903 / Kit KIT-TRU550-HYD-3 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-59-2.9: PN-TRU550-LIN-COIL-5905 / Kit KIT-TRU550-XAX-5 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-59-2.10: PN-TRU550-CYL-SEAL-5907 / Kit KIT-TRU550-ZAX-7 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-59-2.11: PN-TRU550-BRK-DISC-5909 / Kit KIT-TRU550-CAX-9 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-59-2.12: PN-TRU550-IGBT-MOD-5911 / Kit KIT-TRU550-PLC-11 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-59-2.13: PN-TRU550-GASK-VIT-5913 / Kit KIT-TRU550-VAC-13 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-59-2.14: PN-TRU550-BRG-CER-5901 / Kit KIT-TRU550-SPN-1 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-59-2.15: PN-TRU550-VLV-PROP-5903 / Kit KIT-TRU550-HYD-3 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU550-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L550 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L550. Verify part numbers before closing Work Orders.

BOM-59-3.1: PN-TRU550-PMP-PLG-5904 / Kit KIT-TRU550-CLN-4 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU550-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-59-3.2: PN-TRU550-CYL-SEAL-5907 / Kit KIT-TRU550-ZAX-7 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU550-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-59-3.3: PN-TRU550-GLASS-RD-5910 / Kit KIT-TRU550-OPT-10 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU550-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-59-3.4: PN-TRU550-GASK-VIT-5913 / Kit KIT-TRU550-VAC-13 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU550-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-59-3.5: PN-TRU550-CAM-GRIP-5902 / Kit KIT-TRU550-ATC-2 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU550-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-59-3.6: PN-TRU550-LIN-COIL-5905 / Kit KIT-TRU550-XAX-5 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU550-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-59-3.7: PN-TRU550-TRQ-STAT-5908 / Kit KIT-TRU550-BAX-8 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU550-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-59-3.8: PN-TRU550-IGBT-MOD-5911 / Kit KIT-TRU550-PLC-11 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU550-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-59-3.9: PN-TRU550-HEX-CORE-5914 / Kit KIT-TRU550-THM-14 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU550-THM-14 within tolerance ±0.0060 mm TIR.

BOM-59-3.10: PN-TRU550-VLV-PROP-5903 / Kit KIT-TRU550-HYD-3 (Trumpf-TruLaser-L550) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU550-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-59-3.11: PN-TRU550-BSW-NUT-5906 / Kit KIT-TRU550-YAX-6 (Trumpf-TruLaser-L550) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU550-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-59-3.12: PN-TRU550-BRK-DISC-5909 / Kit KIT-TRU550-CAX-9 (Trumpf-TruLaser-L550) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU550-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-59-3.13: PN-TRU550-INJ-MET-5912 / Kit KIT-TRU550-LUB-12 (Trumpf-TruLaser-L550) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU550-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-59-3.14: PN-TRU550-BRG-CER-5901 / Kit KIT-TRU550-SPN-1 (Trumpf-TruLaser-L550) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU550-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-59-3.15: PN-TRU550-PMP-PLG-5904 / Kit KIT-TRU550-CLN-4 (Trumpf-TruLaser-L550) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU550-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L600 (GROUP 1)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L600. Verify part numbers before closing Work Orders.

BOM-60-1.1: PN-TRU600-CAM-GRIP-6002 / Kit KIT-TRU600-ATC-2 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-60-1.2: PN-TRU600-VLV-PROP-6003 / Kit KIT-TRU600-HYD-3 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-60-1.3: PN-TRU600-PMP-PLG-6004 / Kit KIT-TRU600-CLN-4 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-60-1.4: PN-TRU600-LIN-COIL-6005 / Kit KIT-TRU600-XAX-5 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-60-1.5: PN-TRU600-BSW-NUT-6006 / Kit KIT-TRU600-YAX-6 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-60-1.6: PN-TRU600-CYL-SEAL-6007 / Kit KIT-TRU600-ZAX-7 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-60-1.7: PN-TRU600-TRQ-STAT-6008 / Kit KIT-TRU600-BAX-8 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-60-1.8: PN-TRU600-BRK-DISC-6009 / Kit KIT-TRU600-CAX-9 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-60-1.9: PN-TRU600-GLASS-RD-6010 / Kit KIT-TRU600-OPT-10 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-60-1.10: PN-TRU600-IGBT-MOD-6011 / Kit KIT-TRU600-PLC-11 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-60-1.11: PN-TRU600-INJ-MET-6012 / Kit KIT-TRU600-LUB-12 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-60-1.12: PN-TRU600-GASK-VIT-6013 / Kit KIT-TRU600-VAC-13 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-60-1.13: PN-TRU600-HEX-CORE-6014 / Kit KIT-TRU600-THM-14 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-60-1.14: PN-TRU600-BRG-CER-6001 / Kit KIT-TRU600-SPN-1 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-60-1.15: PN-TRU600-CAM-GRIP-6002 / Kit KIT-TRU600-ATC-2 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU600-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L600 (GROUP 2)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L600. Verify part numbers before closing Work Orders.

BOM-60-2.1: PN-TRU600-VLV-PROP-6003 / Kit KIT-TRU600-HYD-3 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-60-2.2: PN-TRU600-LIN-COIL-6005 / Kit KIT-TRU600-XAX-5 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-60-2.3: PN-TRU600-CYL-SEAL-6007 / Kit KIT-TRU600-ZAX-7 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-60-2.4: PN-TRU600-BRK-DISC-6009 / Kit KIT-TRU600-CAX-9 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-60-2.5: PN-TRU600-IGBT-MOD-6011 / Kit KIT-TRU600-PLC-11 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-60-2.6: PN-TRU600-GASK-VIT-6013 / Kit KIT-TRU600-VAC-13 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-60-2.7: PN-TRU600-BRG-CER-6001 / Kit KIT-TRU600-SPN-1 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-60-2.8: PN-TRU600-VLV-PROP-6003 / Kit KIT-TRU600-HYD-3 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-60-2.9: PN-TRU600-LIN-COIL-6005 / Kit KIT-TRU600-XAX-5 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-60-2.10: PN-TRU600-CYL-SEAL-6007 / Kit KIT-TRU600-ZAX-7 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-60-2.11: PN-TRU600-BRK-DISC-6009 / Kit KIT-TRU600-CAX-9 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-60-2.12: PN-TRU600-IGBT-MOD-6011 / Kit KIT-TRU600-PLC-11 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-60-2.13: PN-TRU600-GASK-VIT-6013 / Kit KIT-TRU600-VAC-13 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-60-2.14: PN-TRU600-BRG-CER-6001 / Kit KIT-TRU600-SPN-1 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-60-2.15: PN-TRU600-VLV-PROP-6003 / Kit KIT-TRU600-HYD-3 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU600-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Trumpf-TruLaser-L600 (GROUP 3)

Certified Aerospace Bill of Materials for Trumpf-TruLaser-L600. Verify part numbers before closing Work Orders.

BOM-60-3.1: PN-TRU600-PMP-PLG-6004 / Kit KIT-TRU600-CLN-4 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU600-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-60-3.2: PN-TRU600-CYL-SEAL-6007 / Kit KIT-TRU600-ZAX-7 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-TRU600-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-60-3.3: PN-TRU600-GLASS-RD-6010 / Kit KIT-TRU600-OPT-10 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-TRU600-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-60-3.4: PN-TRU600-GASK-VIT-6013 / Kit KIT-TRU600-VAC-13 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-TRU600-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-60-3.5: PN-TRU600-CAM-GRIP-6002 / Kit KIT-TRU600-ATC-2 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-TRU600-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-60-3.6: PN-TRU600-LIN-COIL-6005 / Kit KIT-TRU600-XAX-5 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-TRU600-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-60-3.7: PN-TRU600-TRQ-STAT-6008 / Kit KIT-TRU600-BAX-8 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-TRU600-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-60-3.8: PN-TRU600-IGBT-MOD-6011 / Kit KIT-TRU600-PLC-11 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-TRU600-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-60-3.9: PN-TRU600-HEX-CORE-6014 / Kit KIT-TRU600-THM-14 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-TRU600-THM-14 within tolerance ±0.0020 mm TIR.

BOM-60-3.10: PN-TRU600-VLV-PROP-6003 / Kit KIT-TRU600-HYD-3 (Trumpf-TruLaser-L600) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-TRU600-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-60-3.11: PN-TRU600-BSW-NUT-6006 / Kit KIT-TRU600-YAX-6 (Trumpf-TruLaser-L600) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-TRU600-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-60-3.12: PN-TRU600-BRK-DISC-6009 / Kit KIT-TRU600-CAX-9 (Trumpf-TruLaser-L600) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-TRU600-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-60-3.13: PN-TRU600-INJ-MET-6012 / Kit KIT-TRU600-LUB-12 (Trumpf-TruLaser-L600) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-TRU600-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-60-3.14: PN-TRU600-BRG-CER-6001 / Kit KIT-TRU600-SPN-1 (Trumpf-TruLaser-L600) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-TRU600-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-60-3.15: PN-TRU600-PMP-PLG-6004 / Kit KIT-TRU600-CLN-4 (Trumpf-TruLaser-L600) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-TRU600-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P150 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P150. Verify part numbers before closing Work Orders.

BOM-61-1.1: PN-SCH150-CAM-GRIP-6102 / Kit KIT-SCH150-ATC-2 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-61-1.2: PN-SCH150-VLV-PROP-6103 / Kit KIT-SCH150-HYD-3 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-61-1.3: PN-SCH150-PMP-PLG-6104 / Kit KIT-SCH150-CLN-4 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-61-1.4: PN-SCH150-LIN-COIL-6105 / Kit KIT-SCH150-XAX-5 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-61-1.5: PN-SCH150-BSW-NUT-6106 / Kit KIT-SCH150-YAX-6 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-61-1.6: PN-SCH150-CYL-SEAL-6107 / Kit KIT-SCH150-ZAX-7 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-61-1.7: PN-SCH150-TRQ-STAT-6108 / Kit KIT-SCH150-BAX-8 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-61-1.8: PN-SCH150-BRK-DISC-6109 / Kit KIT-SCH150-CAX-9 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-61-1.9: PN-SCH150-GLASS-RD-6110 / Kit KIT-SCH150-OPT-10 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-61-1.10: PN-SCH150-IGBT-MOD-6111 / Kit KIT-SCH150-PLC-11 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-61-1.11: PN-SCH150-INJ-MET-6112 / Kit KIT-SCH150-LUB-12 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-61-1.12: PN-SCH150-GASK-VIT-6113 / Kit KIT-SCH150-VAC-13 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-61-1.13: PN-SCH150-HEX-CORE-6114 / Kit KIT-SCH150-THM-14 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-61-1.14: PN-SCH150-BRG-CER-6101 / Kit KIT-SCH150-SPN-1 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-61-1.15: PN-SCH150-CAM-GRIP-6102 / Kit KIT-SCH150-ATC-2 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH150-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P150 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P150. Verify part numbers before closing Work Orders.

BOM-61-2.1: PN-SCH150-VLV-PROP-6103 / Kit KIT-SCH150-HYD-3 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-61-2.2: PN-SCH150-LIN-COIL-6105 / Kit KIT-SCH150-XAX-5 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-61-2.3: PN-SCH150-CYL-SEAL-6107 / Kit KIT-SCH150-ZAX-7 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-61-2.4: PN-SCH150-BRK-DISC-6109 / Kit KIT-SCH150-CAX-9 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-61-2.5: PN-SCH150-IGBT-MOD-6111 / Kit KIT-SCH150-PLC-11 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-61-2.6: PN-SCH150-GASK-VIT-6113 / Kit KIT-SCH150-VAC-13 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-61-2.7: PN-SCH150-BRG-CER-6101 / Kit KIT-SCH150-SPN-1 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-61-2.8: PN-SCH150-VLV-PROP-6103 / Kit KIT-SCH150-HYD-3 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-61-2.9: PN-SCH150-LIN-COIL-6105 / Kit KIT-SCH150-XAX-5 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-61-2.10: PN-SCH150-CYL-SEAL-6107 / Kit KIT-SCH150-ZAX-7 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-61-2.11: PN-SCH150-BRK-DISC-6109 / Kit KIT-SCH150-CAX-9 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-61-2.12: PN-SCH150-IGBT-MOD-6111 / Kit KIT-SCH150-PLC-11 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-61-2.13: PN-SCH150-GASK-VIT-6113 / Kit KIT-SCH150-VAC-13 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-61-2.14: PN-SCH150-BRG-CER-6101 / Kit KIT-SCH150-SPN-1 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-61-2.15: PN-SCH150-VLV-PROP-6103 / Kit KIT-SCH150-HYD-3 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH150-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P150 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P150. Verify part numbers before closing Work Orders.

BOM-61-3.1: PN-SCH150-PMP-PLG-6104 / Kit KIT-SCH150-CLN-4 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-61-3.2: PN-SCH150-CYL-SEAL-6107 / Kit KIT-SCH150-ZAX-7 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-61-3.3: PN-SCH150-GLASS-RD-6110 / Kit KIT-SCH150-OPT-10 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-61-3.4: PN-SCH150-GASK-VIT-6113 / Kit KIT-SCH150-VAC-13 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-61-3.5: PN-SCH150-CAM-GRIP-6102 / Kit KIT-SCH150-ATC-2 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-61-3.6: PN-SCH150-LIN-COIL-6105 / Kit KIT-SCH150-XAX-5 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-61-3.7: PN-SCH150-TRQ-STAT-6108 / Kit KIT-SCH150-BAX-8 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-61-3.8: PN-SCH150-IGBT-MOD-6111 / Kit KIT-SCH150-PLC-11 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-61-3.9: PN-SCH150-HEX-CORE-6114 / Kit KIT-SCH150-THM-14 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-61-3.10: PN-SCH150-VLV-PROP-6103 / Kit KIT-SCH150-HYD-3 (Schuler-HPD-P150) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-61-3.11: PN-SCH150-BSW-NUT-6106 / Kit KIT-SCH150-YAX-6 (Schuler-HPD-P150) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-61-3.12: PN-SCH150-BRK-DISC-6109 / Kit KIT-SCH150-CAX-9 (Schuler-HPD-P150) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-61-3.13: PN-SCH150-INJ-MET-6112 / Kit KIT-SCH150-LUB-12 (Schuler-HPD-P150) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-61-3.14: PN-SCH150-BRG-CER-6101 / Kit KIT-SCH150-SPN-1 (Schuler-HPD-P150) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-61-3.15: PN-SCH150-PMP-PLG-6104 / Kit KIT-SCH150-CLN-4 (Schuler-HPD-P150) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH150-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P200 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P200. Verify part numbers before closing Work Orders.

BOM-62-1.1: PN-SCH200-CAM-GRIP-6202 / Kit KIT-SCH200-ATC-2 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-62-1.2: PN-SCH200-VLV-PROP-6203 / Kit KIT-SCH200-HYD-3 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-62-1.3: PN-SCH200-PMP-PLG-6204 / Kit KIT-SCH200-CLN-4 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-62-1.4: PN-SCH200-LIN-COIL-6205 / Kit KIT-SCH200-XAX-5 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-62-1.5: PN-SCH200-BSW-NUT-6206 / Kit KIT-SCH200-YAX-6 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-62-1.6: PN-SCH200-CYL-SEAL-6207 / Kit KIT-SCH200-ZAX-7 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-62-1.7: PN-SCH200-TRQ-STAT-6208 / Kit KIT-SCH200-BAX-8 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-62-1.8: PN-SCH200-BRK-DISC-6209 / Kit KIT-SCH200-CAX-9 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-62-1.9: PN-SCH200-GLASS-RD-6210 / Kit KIT-SCH200-OPT-10 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-62-1.10: PN-SCH200-IGBT-MOD-6211 / Kit KIT-SCH200-PLC-11 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-62-1.11: PN-SCH200-INJ-MET-6212 / Kit KIT-SCH200-LUB-12 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-62-1.12: PN-SCH200-GASK-VIT-6213 / Kit KIT-SCH200-VAC-13 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-62-1.13: PN-SCH200-HEX-CORE-6214 / Kit KIT-SCH200-THM-14 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-62-1.14: PN-SCH200-BRG-CER-6201 / Kit KIT-SCH200-SPN-1 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-62-1.15: PN-SCH200-CAM-GRIP-6202 / Kit KIT-SCH200-ATC-2 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH200-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P200 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P200. Verify part numbers before closing Work Orders.

BOM-62-2.1: PN-SCH200-VLV-PROP-6203 / Kit KIT-SCH200-HYD-3 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-62-2.2: PN-SCH200-LIN-COIL-6205 / Kit KIT-SCH200-XAX-5 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-62-2.3: PN-SCH200-CYL-SEAL-6207 / Kit KIT-SCH200-ZAX-7 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-62-2.4: PN-SCH200-BRK-DISC-6209 / Kit KIT-SCH200-CAX-9 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-62-2.5: PN-SCH200-IGBT-MOD-6211 / Kit KIT-SCH200-PLC-11 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-62-2.6: PN-SCH200-GASK-VIT-6213 / Kit KIT-SCH200-VAC-13 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-62-2.7: PN-SCH200-BRG-CER-6201 / Kit KIT-SCH200-SPN-1 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-62-2.8: PN-SCH200-VLV-PROP-6203 / Kit KIT-SCH200-HYD-3 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-62-2.9: PN-SCH200-LIN-COIL-6205 / Kit KIT-SCH200-XAX-5 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-62-2.10: PN-SCH200-CYL-SEAL-6207 / Kit KIT-SCH200-ZAX-7 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-62-2.11: PN-SCH200-BRK-DISC-6209 / Kit KIT-SCH200-CAX-9 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-62-2.12: PN-SCH200-IGBT-MOD-6211 / Kit KIT-SCH200-PLC-11 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-62-2.13: PN-SCH200-GASK-VIT-6213 / Kit KIT-SCH200-VAC-13 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-62-2.14: PN-SCH200-BRG-CER-6201 / Kit KIT-SCH200-SPN-1 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-62-2.15: PN-SCH200-VLV-PROP-6203 / Kit KIT-SCH200-HYD-3 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH200-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P200 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P200. Verify part numbers before closing Work Orders.

BOM-62-3.1: PN-SCH200-PMP-PLG-6204 / Kit KIT-SCH200-CLN-4 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-62-3.2: PN-SCH200-CYL-SEAL-6207 / Kit KIT-SCH200-ZAX-7 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-62-3.3: PN-SCH200-GLASS-RD-6210 / Kit KIT-SCH200-OPT-10 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-62-3.4: PN-SCH200-GASK-VIT-6213 / Kit KIT-SCH200-VAC-13 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-62-3.5: PN-SCH200-CAM-GRIP-6202 / Kit KIT-SCH200-ATC-2 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-62-3.6: PN-SCH200-LIN-COIL-6205 / Kit KIT-SCH200-XAX-5 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-62-3.7: PN-SCH200-TRQ-STAT-6208 / Kit KIT-SCH200-BAX-8 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-62-3.8: PN-SCH200-IGBT-MOD-6211 / Kit KIT-SCH200-PLC-11 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-62-3.9: PN-SCH200-HEX-CORE-6214 / Kit KIT-SCH200-THM-14 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-62-3.10: PN-SCH200-VLV-PROP-6203 / Kit KIT-SCH200-HYD-3 (Schuler-HPD-P200) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-62-3.11: PN-SCH200-BSW-NUT-6206 / Kit KIT-SCH200-YAX-6 (Schuler-HPD-P200) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-62-3.12: PN-SCH200-BRK-DISC-6209 / Kit KIT-SCH200-CAX-9 (Schuler-HPD-P200) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-62-3.13: PN-SCH200-INJ-MET-6212 / Kit KIT-SCH200-LUB-12 (Schuler-HPD-P200) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-62-3.14: PN-SCH200-BRG-CER-6201 / Kit KIT-SCH200-SPN-1 (Schuler-HPD-P200) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-62-3.15: PN-SCH200-PMP-PLG-6204 / Kit KIT-SCH200-CLN-4 (Schuler-HPD-P200) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH200-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P250 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P250. Verify part numbers before closing Work Orders.

BOM-63-1.1: PN-SCH250-CAM-GRIP-6302 / Kit KIT-SCH250-ATC-2 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-63-1.2: PN-SCH250-VLV-PROP-6303 / Kit KIT-SCH250-HYD-3 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-63-1.3: PN-SCH250-PMP-PLG-6304 / Kit KIT-SCH250-CLN-4 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-63-1.4: PN-SCH250-LIN-COIL-6305 / Kit KIT-SCH250-XAX-5 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-63-1.5: PN-SCH250-BSW-NUT-6306 / Kit KIT-SCH250-YAX-6 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-63-1.6: PN-SCH250-CYL-SEAL-6307 / Kit KIT-SCH250-ZAX-7 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-63-1.7: PN-SCH250-TRQ-STAT-6308 / Kit KIT-SCH250-BAX-8 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-63-1.8: PN-SCH250-BRK-DISC-6309 / Kit KIT-SCH250-CAX-9 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-63-1.9: PN-SCH250-GLASS-RD-6310 / Kit KIT-SCH250-OPT-10 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-63-1.10: PN-SCH250-IGBT-MOD-6311 / Kit KIT-SCH250-PLC-11 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-63-1.11: PN-SCH250-INJ-MET-6312 / Kit KIT-SCH250-LUB-12 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-63-1.12: PN-SCH250-GASK-VIT-6313 / Kit KIT-SCH250-VAC-13 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-63-1.13: PN-SCH250-HEX-CORE-6314 / Kit KIT-SCH250-THM-14 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-63-1.14: PN-SCH250-BRG-CER-6301 / Kit KIT-SCH250-SPN-1 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-63-1.15: PN-SCH250-CAM-GRIP-6302 / Kit KIT-SCH250-ATC-2 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH250-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P250 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P250. Verify part numbers before closing Work Orders.

BOM-63-2.1: PN-SCH250-VLV-PROP-6303 / Kit KIT-SCH250-HYD-3 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-63-2.2: PN-SCH250-LIN-COIL-6305 / Kit KIT-SCH250-XAX-5 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-63-2.3: PN-SCH250-CYL-SEAL-6307 / Kit KIT-SCH250-ZAX-7 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-63-2.4: PN-SCH250-BRK-DISC-6309 / Kit KIT-SCH250-CAX-9 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-63-2.5: PN-SCH250-IGBT-MOD-6311 / Kit KIT-SCH250-PLC-11 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-63-2.6: PN-SCH250-GASK-VIT-6313 / Kit KIT-SCH250-VAC-13 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-63-2.7: PN-SCH250-BRG-CER-6301 / Kit KIT-SCH250-SPN-1 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-63-2.8: PN-SCH250-VLV-PROP-6303 / Kit KIT-SCH250-HYD-3 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-63-2.9: PN-SCH250-LIN-COIL-6305 / Kit KIT-SCH250-XAX-5 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-63-2.10: PN-SCH250-CYL-SEAL-6307 / Kit KIT-SCH250-ZAX-7 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-63-2.11: PN-SCH250-BRK-DISC-6309 / Kit KIT-SCH250-CAX-9 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-63-2.12: PN-SCH250-IGBT-MOD-6311 / Kit KIT-SCH250-PLC-11 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-63-2.13: PN-SCH250-GASK-VIT-6313 / Kit KIT-SCH250-VAC-13 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-63-2.14: PN-SCH250-BRG-CER-6301 / Kit KIT-SCH250-SPN-1 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-63-2.15: PN-SCH250-VLV-PROP-6303 / Kit KIT-SCH250-HYD-3 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH250-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P250 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P250. Verify part numbers before closing Work Orders.

BOM-63-3.1: PN-SCH250-PMP-PLG-6304 / Kit KIT-SCH250-CLN-4 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-63-3.2: PN-SCH250-CYL-SEAL-6307 / Kit KIT-SCH250-ZAX-7 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-63-3.3: PN-SCH250-GLASS-RD-6310 / Kit KIT-SCH250-OPT-10 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-63-3.4: PN-SCH250-GASK-VIT-6313 / Kit KIT-SCH250-VAC-13 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-63-3.5: PN-SCH250-CAM-GRIP-6302 / Kit KIT-SCH250-ATC-2 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-63-3.6: PN-SCH250-LIN-COIL-6305 / Kit KIT-SCH250-XAX-5 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-63-3.7: PN-SCH250-TRQ-STAT-6308 / Kit KIT-SCH250-BAX-8 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-63-3.8: PN-SCH250-IGBT-MOD-6311 / Kit KIT-SCH250-PLC-11 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-63-3.9: PN-SCH250-HEX-CORE-6314 / Kit KIT-SCH250-THM-14 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-63-3.10: PN-SCH250-VLV-PROP-6303 / Kit KIT-SCH250-HYD-3 (Schuler-HPD-P250) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-63-3.11: PN-SCH250-BSW-NUT-6306 / Kit KIT-SCH250-YAX-6 (Schuler-HPD-P250) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-63-3.12: PN-SCH250-BRK-DISC-6309 / Kit KIT-SCH250-CAX-9 (Schuler-HPD-P250) | PM Interval: 250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-63-3.13: PN-SCH250-INJ-MET-6312 / Kit KIT-SCH250-LUB-12 (Schuler-HPD-P250) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-63-3.14: PN-SCH250-BRG-CER-6301 / Kit KIT-SCH250-SPN-1 (Schuler-HPD-P250) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-63-3.15: PN-SCH250-PMP-PLG-6304 / Kit KIT-SCH250-CLN-4 (Schuler-HPD-P250) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH250-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P300 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P300. Verify part numbers before closing Work Orders.

BOM-64-1.1: PN-SCH300-CAM-GRIP-6402 / Kit KIT-SCH300-ATC-2 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-64-1.2: PN-SCH300-VLV-PROP-6403 / Kit KIT-SCH300-HYD-3 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-64-1.3: PN-SCH300-PMP-PLG-6404 / Kit KIT-SCH300-CLN-4 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-64-1.4: PN-SCH300-LIN-COIL-6405 / Kit KIT-SCH300-XAX-5 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-64-1.5: PN-SCH300-BSW-NUT-6406 / Kit KIT-SCH300-YAX-6 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-64-1.6: PN-SCH300-CYL-SEAL-6407 / Kit KIT-SCH300-ZAX-7 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-64-1.7: PN-SCH300-TRQ-STAT-6408 / Kit KIT-SCH300-BAX-8 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-64-1.8: PN-SCH300-BRK-DISC-6409 / Kit KIT-SCH300-CAX-9 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-64-1.9: PN-SCH300-GLASS-RD-6410 / Kit KIT-SCH300-OPT-10 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-64-1.10: PN-SCH300-IGBT-MOD-6411 / Kit KIT-SCH300-PLC-11 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-64-1.11: PN-SCH300-INJ-MET-6412 / Kit KIT-SCH300-LUB-12 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-64-1.12: PN-SCH300-GASK-VIT-6413 / Kit KIT-SCH300-VAC-13 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-64-1.13: PN-SCH300-HEX-CORE-6414 / Kit KIT-SCH300-THM-14 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-64-1.14: PN-SCH300-BRG-CER-6401 / Kit KIT-SCH300-SPN-1 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-64-1.15: PN-SCH300-CAM-GRIP-6402 / Kit KIT-SCH300-ATC-2 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH300-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P300 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P300. Verify part numbers before closing Work Orders.

BOM-64-2.1: PN-SCH300-VLV-PROP-6403 / Kit KIT-SCH300-HYD-3 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-64-2.2: PN-SCH300-LIN-COIL-6405 / Kit KIT-SCH300-XAX-5 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-64-2.3: PN-SCH300-CYL-SEAL-6407 / Kit KIT-SCH300-ZAX-7 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-64-2.4: PN-SCH300-BRK-DISC-6409 / Kit KIT-SCH300-CAX-9 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-64-2.5: PN-SCH300-IGBT-MOD-6411 / Kit KIT-SCH300-PLC-11 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-64-2.6: PN-SCH300-GASK-VIT-6413 / Kit KIT-SCH300-VAC-13 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-64-2.7: PN-SCH300-BRG-CER-6401 / Kit KIT-SCH300-SPN-1 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-64-2.8: PN-SCH300-VLV-PROP-6403 / Kit KIT-SCH300-HYD-3 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-64-2.9: PN-SCH300-LIN-COIL-6405 / Kit KIT-SCH300-XAX-5 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-64-2.10: PN-SCH300-CYL-SEAL-6407 / Kit KIT-SCH300-ZAX-7 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-64-2.11: PN-SCH300-BRK-DISC-6409 / Kit KIT-SCH300-CAX-9 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-64-2.12: PN-SCH300-IGBT-MOD-6411 / Kit KIT-SCH300-PLC-11 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-64-2.13: PN-SCH300-GASK-VIT-6413 / Kit KIT-SCH300-VAC-13 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-64-2.14: PN-SCH300-BRG-CER-6401 / Kit KIT-SCH300-SPN-1 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-64-2.15: PN-SCH300-VLV-PROP-6403 / Kit KIT-SCH300-HYD-3 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH300-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P300 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P300. Verify part numbers before closing Work Orders.

BOM-64-3.1: PN-SCH300-PMP-PLG-6404 / Kit KIT-SCH300-CLN-4 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-64-3.2: PN-SCH300-CYL-SEAL-6407 / Kit KIT-SCH300-ZAX-7 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-64-3.3: PN-SCH300-GLASS-RD-6410 / Kit KIT-SCH300-OPT-10 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-64-3.4: PN-SCH300-GASK-VIT-6413 / Kit KIT-SCH300-VAC-13 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-64-3.5: PN-SCH300-CAM-GRIP-6402 / Kit KIT-SCH300-ATC-2 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-64-3.6: PN-SCH300-LIN-COIL-6405 / Kit KIT-SCH300-XAX-5 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-64-3.7: PN-SCH300-TRQ-STAT-6408 / Kit KIT-SCH300-BAX-8 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-64-3.8: PN-SCH300-IGBT-MOD-6411 / Kit KIT-SCH300-PLC-11 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-64-3.9: PN-SCH300-HEX-CORE-6414 / Kit KIT-SCH300-THM-14 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-64-3.10: PN-SCH300-VLV-PROP-6403 / Kit KIT-SCH300-HYD-3 (Schuler-HPD-P300) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-64-3.11: PN-SCH300-BSW-NUT-6406 / Kit KIT-SCH300-YAX-6 (Schuler-HPD-P300) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-64-3.12: PN-SCH300-BRK-DISC-6409 / Kit KIT-SCH300-CAX-9 (Schuler-HPD-P300) | PM Interval: 250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-64-3.13: PN-SCH300-INJ-MET-6412 / Kit KIT-SCH300-LUB-12 (Schuler-HPD-P300) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-64-3.14: PN-SCH300-BRG-CER-6401 / Kit KIT-SCH300-SPN-1 (Schuler-HPD-P300) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-64-3.15: PN-SCH300-PMP-PLG-6404 / Kit KIT-SCH300-CLN-4 (Schuler-HPD-P300) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH300-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P350 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P350. Verify part numbers before closing Work Orders.

BOM-65-1.1: PN-SCH350-CAM-GRIP-6502 / Kit KIT-SCH350-ATC-2 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-65-1.2: PN-SCH350-VLV-PROP-6503 / Kit KIT-SCH350-HYD-3 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-65-1.3: PN-SCH350-PMP-PLG-6504 / Kit KIT-SCH350-CLN-4 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-65-1.4: PN-SCH350-LIN-COIL-6505 / Kit KIT-SCH350-XAX-5 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-65-1.5: PN-SCH350-BSW-NUT-6506 / Kit KIT-SCH350-YAX-6 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-65-1.6: PN-SCH350-CYL-SEAL-6507 / Kit KIT-SCH350-ZAX-7 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-65-1.7: PN-SCH350-TRQ-STAT-6508 / Kit KIT-SCH350-BAX-8 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-65-1.8: PN-SCH350-BRK-DISC-6509 / Kit KIT-SCH350-CAX-9 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-65-1.9: PN-SCH350-GLASS-RD-6510 / Kit KIT-SCH350-OPT-10 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-65-1.10: PN-SCH350-IGBT-MOD-6511 / Kit KIT-SCH350-PLC-11 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-65-1.11: PN-SCH350-INJ-MET-6512 / Kit KIT-SCH350-LUB-12 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-65-1.12: PN-SCH350-GASK-VIT-6513 / Kit KIT-SCH350-VAC-13 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-65-1.13: PN-SCH350-HEX-CORE-6514 / Kit KIT-SCH350-THM-14 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-65-1.14: PN-SCH350-BRG-CER-6501 / Kit KIT-SCH350-SPN-1 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-65-1.15: PN-SCH350-CAM-GRIP-6502 / Kit KIT-SCH350-ATC-2 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH350-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P350 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P350. Verify part numbers before closing Work Orders.

BOM-65-2.1: PN-SCH350-VLV-PROP-6503 / Kit KIT-SCH350-HYD-3 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-65-2.2: PN-SCH350-LIN-COIL-6505 / Kit KIT-SCH350-XAX-5 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-65-2.3: PN-SCH350-CYL-SEAL-6507 / Kit KIT-SCH350-ZAX-7 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-65-2.4: PN-SCH350-BRK-DISC-6509 / Kit KIT-SCH350-CAX-9 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-65-2.5: PN-SCH350-IGBT-MOD-6511 / Kit KIT-SCH350-PLC-11 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-65-2.6: PN-SCH350-GASK-VIT-6513 / Kit KIT-SCH350-VAC-13 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-65-2.7: PN-SCH350-BRG-CER-6501 / Kit KIT-SCH350-SPN-1 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-65-2.8: PN-SCH350-VLV-PROP-6503 / Kit KIT-SCH350-HYD-3 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-65-2.9: PN-SCH350-LIN-COIL-6505 / Kit KIT-SCH350-XAX-5 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-65-2.10: PN-SCH350-CYL-SEAL-6507 / Kit KIT-SCH350-ZAX-7 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-65-2.11: PN-SCH350-BRK-DISC-6509 / Kit KIT-SCH350-CAX-9 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-65-2.12: PN-SCH350-IGBT-MOD-6511 / Kit KIT-SCH350-PLC-11 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-65-2.13: PN-SCH350-GASK-VIT-6513 / Kit KIT-SCH350-VAC-13 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-65-2.14: PN-SCH350-BRG-CER-6501 / Kit KIT-SCH350-SPN-1 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-65-2.15: PN-SCH350-VLV-PROP-6503 / Kit KIT-SCH350-HYD-3 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH350-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P350 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P350. Verify part numbers before closing Work Orders.

BOM-65-3.1: PN-SCH350-PMP-PLG-6504 / Kit KIT-SCH350-CLN-4 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-65-3.2: PN-SCH350-CYL-SEAL-6507 / Kit KIT-SCH350-ZAX-7 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-65-3.3: PN-SCH350-GLASS-RD-6510 / Kit KIT-SCH350-OPT-10 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-65-3.4: PN-SCH350-GASK-VIT-6513 / Kit KIT-SCH350-VAC-13 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-65-3.5: PN-SCH350-CAM-GRIP-6502 / Kit KIT-SCH350-ATC-2 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-65-3.6: PN-SCH350-LIN-COIL-6505 / Kit KIT-SCH350-XAX-5 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-65-3.7: PN-SCH350-TRQ-STAT-6508 / Kit KIT-SCH350-BAX-8 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-65-3.8: PN-SCH350-IGBT-MOD-6511 / Kit KIT-SCH350-PLC-11 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-65-3.9: PN-SCH350-HEX-CORE-6514 / Kit KIT-SCH350-THM-14 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-65-3.10: PN-SCH350-VLV-PROP-6503 / Kit KIT-SCH350-HYD-3 (Schuler-HPD-P350) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-65-3.11: PN-SCH350-BSW-NUT-6506 / Kit KIT-SCH350-YAX-6 (Schuler-HPD-P350) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-65-3.12: PN-SCH350-BRK-DISC-6509 / Kit KIT-SCH350-CAX-9 (Schuler-HPD-P350) | PM Interval: 250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-65-3.13: PN-SCH350-INJ-MET-6512 / Kit KIT-SCH350-LUB-12 (Schuler-HPD-P350) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-65-3.14: PN-SCH350-BRG-CER-6501 / Kit KIT-SCH350-SPN-1 (Schuler-HPD-P350) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-65-3.15: PN-SCH350-PMP-PLG-6504 / Kit KIT-SCH350-CLN-4 (Schuler-HPD-P350) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH350-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P400 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P400. Verify part numbers before closing Work Orders.

BOM-66-1.1: PN-SCH400-CAM-GRIP-6602 / Kit KIT-SCH400-ATC-2 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-66-1.2: PN-SCH400-VLV-PROP-6603 / Kit KIT-SCH400-HYD-3 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-66-1.3: PN-SCH400-PMP-PLG-6604 / Kit KIT-SCH400-CLN-4 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-66-1.4: PN-SCH400-LIN-COIL-6605 / Kit KIT-SCH400-XAX-5 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-66-1.5: PN-SCH400-BSW-NUT-6606 / Kit KIT-SCH400-YAX-6 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-66-1.6: PN-SCH400-CYL-SEAL-6607 / Kit KIT-SCH400-ZAX-7 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-66-1.7: PN-SCH400-TRQ-STAT-6608 / Kit KIT-SCH400-BAX-8 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-66-1.8: PN-SCH400-BRK-DISC-6609 / Kit KIT-SCH400-CAX-9 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-66-1.9: PN-SCH400-GLASS-RD-6610 / Kit KIT-SCH400-OPT-10 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-66-1.10: PN-SCH400-IGBT-MOD-6611 / Kit KIT-SCH400-PLC-11 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-66-1.11: PN-SCH400-INJ-MET-6612 / Kit KIT-SCH400-LUB-12 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-66-1.12: PN-SCH400-GASK-VIT-6613 / Kit KIT-SCH400-VAC-13 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-66-1.13: PN-SCH400-HEX-CORE-6614 / Kit KIT-SCH400-THM-14 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-66-1.14: PN-SCH400-BRG-CER-6601 / Kit KIT-SCH400-SPN-1 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-66-1.15: PN-SCH400-CAM-GRIP-6602 / Kit KIT-SCH400-ATC-2 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH400-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P400 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P400. Verify part numbers before closing Work Orders.

BOM-66-2.1: PN-SCH400-VLV-PROP-6603 / Kit KIT-SCH400-HYD-3 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-66-2.2: PN-SCH400-LIN-COIL-6605 / Kit KIT-SCH400-XAX-5 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-66-2.3: PN-SCH400-CYL-SEAL-6607 / Kit KIT-SCH400-ZAX-7 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-66-2.4: PN-SCH400-BRK-DISC-6609 / Kit KIT-SCH400-CAX-9 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-66-2.5: PN-SCH400-IGBT-MOD-6611 / Kit KIT-SCH400-PLC-11 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-66-2.6: PN-SCH400-GASK-VIT-6613 / Kit KIT-SCH400-VAC-13 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-66-2.7: PN-SCH400-BRG-CER-6601 / Kit KIT-SCH400-SPN-1 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-66-2.8: PN-SCH400-VLV-PROP-6603 / Kit KIT-SCH400-HYD-3 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-66-2.9: PN-SCH400-LIN-COIL-6605 / Kit KIT-SCH400-XAX-5 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-66-2.10: PN-SCH400-CYL-SEAL-6607 / Kit KIT-SCH400-ZAX-7 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-66-2.11: PN-SCH400-BRK-DISC-6609 / Kit KIT-SCH400-CAX-9 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-66-2.12: PN-SCH400-IGBT-MOD-6611 / Kit KIT-SCH400-PLC-11 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-66-2.13: PN-SCH400-GASK-VIT-6613 / Kit KIT-SCH400-VAC-13 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-66-2.14: PN-SCH400-BRG-CER-6601 / Kit KIT-SCH400-SPN-1 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-66-2.15: PN-SCH400-VLV-PROP-6603 / Kit KIT-SCH400-HYD-3 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH400-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P400 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P400. Verify part numbers before closing Work Orders.

BOM-66-3.1: PN-SCH400-PMP-PLG-6604 / Kit KIT-SCH400-CLN-4 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-66-3.2: PN-SCH400-CYL-SEAL-6607 / Kit KIT-SCH400-ZAX-7 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-66-3.3: PN-SCH400-GLASS-RD-6610 / Kit KIT-SCH400-OPT-10 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-66-3.4: PN-SCH400-GASK-VIT-6613 / Kit KIT-SCH400-VAC-13 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-66-3.5: PN-SCH400-CAM-GRIP-6602 / Kit KIT-SCH400-ATC-2 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-66-3.6: PN-SCH400-LIN-COIL-6605 / Kit KIT-SCH400-XAX-5 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-66-3.7: PN-SCH400-TRQ-STAT-6608 / Kit KIT-SCH400-BAX-8 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-66-3.8: PN-SCH400-IGBT-MOD-6611 / Kit KIT-SCH400-PLC-11 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-66-3.9: PN-SCH400-HEX-CORE-6614 / Kit KIT-SCH400-THM-14 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-66-3.10: PN-SCH400-VLV-PROP-6603 / Kit KIT-SCH400-HYD-3 (Schuler-HPD-P400) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-66-3.11: PN-SCH400-BSW-NUT-6606 / Kit KIT-SCH400-YAX-6 (Schuler-HPD-P400) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-66-3.12: PN-SCH400-BRK-DISC-6609 / Kit KIT-SCH400-CAX-9 (Schuler-HPD-P400) | PM Interval: 250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-66-3.13: PN-SCH400-INJ-MET-6612 / Kit KIT-SCH400-LUB-12 (Schuler-HPD-P400) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-66-3.14: PN-SCH400-BRG-CER-6601 / Kit KIT-SCH400-SPN-1 (Schuler-HPD-P400) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-66-3.15: PN-SCH400-PMP-PLG-6604 / Kit KIT-SCH400-CLN-4 (Schuler-HPD-P400) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH400-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P450 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P450. Verify part numbers before closing Work Orders.

BOM-67-1.1: PN-SCH450-CAM-GRIP-6702 / Kit KIT-SCH450-ATC-2 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-67-1.2: PN-SCH450-VLV-PROP-6703 / Kit KIT-SCH450-HYD-3 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-67-1.3: PN-SCH450-PMP-PLG-6704 / Kit KIT-SCH450-CLN-4 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-67-1.4: PN-SCH450-LIN-COIL-6705 / Kit KIT-SCH450-XAX-5 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-67-1.5: PN-SCH450-BSW-NUT-6706 / Kit KIT-SCH450-YAX-6 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-67-1.6: PN-SCH450-CYL-SEAL-6707 / Kit KIT-SCH450-ZAX-7 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-67-1.7: PN-SCH450-TRQ-STAT-6708 / Kit KIT-SCH450-BAX-8 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-67-1.8: PN-SCH450-BRK-DISC-6709 / Kit KIT-SCH450-CAX-9 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-67-1.9: PN-SCH450-GLASS-RD-6710 / Kit KIT-SCH450-OPT-10 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-67-1.10: PN-SCH450-IGBT-MOD-6711 / Kit KIT-SCH450-PLC-11 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-67-1.11: PN-SCH450-INJ-MET-6712 / Kit KIT-SCH450-LUB-12 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-67-1.12: PN-SCH450-GASK-VIT-6713 / Kit KIT-SCH450-VAC-13 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-67-1.13: PN-SCH450-HEX-CORE-6714 / Kit KIT-SCH450-THM-14 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-67-1.14: PN-SCH450-BRG-CER-6701 / Kit KIT-SCH450-SPN-1 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-67-1.15: PN-SCH450-CAM-GRIP-6702 / Kit KIT-SCH450-ATC-2 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH450-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P450 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P450. Verify part numbers before closing Work Orders.

BOM-67-2.1: PN-SCH450-VLV-PROP-6703 / Kit KIT-SCH450-HYD-3 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-67-2.2: PN-SCH450-LIN-COIL-6705 / Kit KIT-SCH450-XAX-5 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-67-2.3: PN-SCH450-CYL-SEAL-6707 / Kit KIT-SCH450-ZAX-7 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-67-2.4: PN-SCH450-BRK-DISC-6709 / Kit KIT-SCH450-CAX-9 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-67-2.5: PN-SCH450-IGBT-MOD-6711 / Kit KIT-SCH450-PLC-11 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-67-2.6: PN-SCH450-GASK-VIT-6713 / Kit KIT-SCH450-VAC-13 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-67-2.7: PN-SCH450-BRG-CER-6701 / Kit KIT-SCH450-SPN-1 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-67-2.8: PN-SCH450-VLV-PROP-6703 / Kit KIT-SCH450-HYD-3 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-67-2.9: PN-SCH450-LIN-COIL-6705 / Kit KIT-SCH450-XAX-5 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-67-2.10: PN-SCH450-CYL-SEAL-6707 / Kit KIT-SCH450-ZAX-7 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-67-2.11: PN-SCH450-BRK-DISC-6709 / Kit KIT-SCH450-CAX-9 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-67-2.12: PN-SCH450-IGBT-MOD-6711 / Kit KIT-SCH450-PLC-11 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-67-2.13: PN-SCH450-GASK-VIT-6713 / Kit KIT-SCH450-VAC-13 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-67-2.14: PN-SCH450-BRG-CER-6701 / Kit KIT-SCH450-SPN-1 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-67-2.15: PN-SCH450-VLV-PROP-6703 / Kit KIT-SCH450-HYD-3 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH450-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P450 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P450. Verify part numbers before closing Work Orders.

BOM-67-3.1: PN-SCH450-PMP-PLG-6704 / Kit KIT-SCH450-CLN-4 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-67-3.2: PN-SCH450-CYL-SEAL-6707 / Kit KIT-SCH450-ZAX-7 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-67-3.3: PN-SCH450-GLASS-RD-6710 / Kit KIT-SCH450-OPT-10 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-67-3.4: PN-SCH450-GASK-VIT-6713 / Kit KIT-SCH450-VAC-13 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-67-3.5: PN-SCH450-CAM-GRIP-6702 / Kit KIT-SCH450-ATC-2 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-67-3.6: PN-SCH450-LIN-COIL-6705 / Kit KIT-SCH450-XAX-5 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-67-3.7: PN-SCH450-TRQ-STAT-6708 / Kit KIT-SCH450-BAX-8 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-67-3.8: PN-SCH450-IGBT-MOD-6711 / Kit KIT-SCH450-PLC-11 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-67-3.9: PN-SCH450-HEX-CORE-6714 / Kit KIT-SCH450-THM-14 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-67-3.10: PN-SCH450-VLV-PROP-6703 / Kit KIT-SCH450-HYD-3 (Schuler-HPD-P450) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-67-3.11: PN-SCH450-BSW-NUT-6706 / Kit KIT-SCH450-YAX-6 (Schuler-HPD-P450) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-67-3.12: PN-SCH450-BRK-DISC-6709 / Kit KIT-SCH450-CAX-9 (Schuler-HPD-P450) | PM Interval: 250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-67-3.13: PN-SCH450-INJ-MET-6712 / Kit KIT-SCH450-LUB-12 (Schuler-HPD-P450) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-67-3.14: PN-SCH450-BRG-CER-6701 / Kit KIT-SCH450-SPN-1 (Schuler-HPD-P450) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-67-3.15: PN-SCH450-PMP-PLG-6704 / Kit KIT-SCH450-CLN-4 (Schuler-HPD-P450) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH450-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P500 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P500. Verify part numbers before closing Work Orders.

BOM-68-1.1: PN-SCH500-CAM-GRIP-6802 / Kit KIT-SCH500-ATC-2 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-68-1.2: PN-SCH500-VLV-PROP-6803 / Kit KIT-SCH500-HYD-3 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-68-1.3: PN-SCH500-PMP-PLG-6804 / Kit KIT-SCH500-CLN-4 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-68-1.4: PN-SCH500-LIN-COIL-6805 / Kit KIT-SCH500-XAX-5 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-68-1.5: PN-SCH500-BSW-NUT-6806 / Kit KIT-SCH500-YAX-6 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-68-1.6: PN-SCH500-CYL-SEAL-6807 / Kit KIT-SCH500-ZAX-7 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-68-1.7: PN-SCH500-TRQ-STAT-6808 / Kit KIT-SCH500-BAX-8 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-68-1.8: PN-SCH500-BRK-DISC-6809 / Kit KIT-SCH500-CAX-9 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-68-1.9: PN-SCH500-GLASS-RD-6810 / Kit KIT-SCH500-OPT-10 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-68-1.10: PN-SCH500-IGBT-MOD-6811 / Kit KIT-SCH500-PLC-11 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-68-1.11: PN-SCH500-INJ-MET-6812 / Kit KIT-SCH500-LUB-12 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-68-1.12: PN-SCH500-GASK-VIT-6813 / Kit KIT-SCH500-VAC-13 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-68-1.13: PN-SCH500-HEX-CORE-6814 / Kit KIT-SCH500-THM-14 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-68-1.14: PN-SCH500-BRG-CER-6801 / Kit KIT-SCH500-SPN-1 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-68-1.15: PN-SCH500-CAM-GRIP-6802 / Kit KIT-SCH500-ATC-2 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH500-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P500 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P500. Verify part numbers before closing Work Orders.

BOM-68-2.1: PN-SCH500-VLV-PROP-6803 / Kit KIT-SCH500-HYD-3 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-68-2.2: PN-SCH500-LIN-COIL-6805 / Kit KIT-SCH500-XAX-5 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-68-2.3: PN-SCH500-CYL-SEAL-6807 / Kit KIT-SCH500-ZAX-7 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-68-2.4: PN-SCH500-BRK-DISC-6809 / Kit KIT-SCH500-CAX-9 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-68-2.5: PN-SCH500-IGBT-MOD-6811 / Kit KIT-SCH500-PLC-11 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-68-2.6: PN-SCH500-GASK-VIT-6813 / Kit KIT-SCH500-VAC-13 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-68-2.7: PN-SCH500-BRG-CER-6801 / Kit KIT-SCH500-SPN-1 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-68-2.8: PN-SCH500-VLV-PROP-6803 / Kit KIT-SCH500-HYD-3 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-68-2.9: PN-SCH500-LIN-COIL-6805 / Kit KIT-SCH500-XAX-5 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-68-2.10: PN-SCH500-CYL-SEAL-6807 / Kit KIT-SCH500-ZAX-7 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-68-2.11: PN-SCH500-BRK-DISC-6809 / Kit KIT-SCH500-CAX-9 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-68-2.12: PN-SCH500-IGBT-MOD-6811 / Kit KIT-SCH500-PLC-11 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-68-2.13: PN-SCH500-GASK-VIT-6813 / Kit KIT-SCH500-VAC-13 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-68-2.14: PN-SCH500-BRG-CER-6801 / Kit KIT-SCH500-SPN-1 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-68-2.15: PN-SCH500-VLV-PROP-6803 / Kit KIT-SCH500-HYD-3 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH500-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P500 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P500. Verify part numbers before closing Work Orders.

BOM-68-3.1: PN-SCH500-PMP-PLG-6804 / Kit KIT-SCH500-CLN-4 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-68-3.2: PN-SCH500-CYL-SEAL-6807 / Kit KIT-SCH500-ZAX-7 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-68-3.3: PN-SCH500-GLASS-RD-6810 / Kit KIT-SCH500-OPT-10 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-68-3.4: PN-SCH500-GASK-VIT-6813 / Kit KIT-SCH500-VAC-13 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-68-3.5: PN-SCH500-CAM-GRIP-6802 / Kit KIT-SCH500-ATC-2 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-68-3.6: PN-SCH500-LIN-COIL-6805 / Kit KIT-SCH500-XAX-5 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-68-3.7: PN-SCH500-TRQ-STAT-6808 / Kit KIT-SCH500-BAX-8 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-68-3.8: PN-SCH500-IGBT-MOD-6811 / Kit KIT-SCH500-PLC-11 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-68-3.9: PN-SCH500-HEX-CORE-6814 / Kit KIT-SCH500-THM-14 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-68-3.10: PN-SCH500-VLV-PROP-6803 / Kit KIT-SCH500-HYD-3 (Schuler-HPD-P500) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-68-3.11: PN-SCH500-BSW-NUT-6806 / Kit KIT-SCH500-YAX-6 (Schuler-HPD-P500) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-68-3.12: PN-SCH500-BRK-DISC-6809 / Kit KIT-SCH500-CAX-9 (Schuler-HPD-P500) | PM Interval: 250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-68-3.13: PN-SCH500-INJ-MET-6812 / Kit KIT-SCH500-LUB-12 (Schuler-HPD-P500) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-68-3.14: PN-SCH500-BRG-CER-6801 / Kit KIT-SCH500-SPN-1 (Schuler-HPD-P500) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-68-3.15: PN-SCH500-PMP-PLG-6804 / Kit KIT-SCH500-CLN-4 (Schuler-HPD-P500) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH500-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P550 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P550. Verify part numbers before closing Work Orders.

BOM-69-1.1: PN-SCH550-CAM-GRIP-6902 / Kit KIT-SCH550-ATC-2 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-69-1.2: PN-SCH550-VLV-PROP-6903 / Kit KIT-SCH550-HYD-3 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-69-1.3: PN-SCH550-PMP-PLG-6904 / Kit KIT-SCH550-CLN-4 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-69-1.4: PN-SCH550-LIN-COIL-6905 / Kit KIT-SCH550-XAX-5 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-69-1.5: PN-SCH550-BSW-NUT-6906 / Kit KIT-SCH550-YAX-6 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-69-1.6: PN-SCH550-CYL-SEAL-6907 / Kit KIT-SCH550-ZAX-7 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-69-1.7: PN-SCH550-TRQ-STAT-6908 / Kit KIT-SCH550-BAX-8 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-69-1.8: PN-SCH550-BRK-DISC-6909 / Kit KIT-SCH550-CAX-9 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-69-1.9: PN-SCH550-GLASS-RD-6910 / Kit KIT-SCH550-OPT-10 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-69-1.10: PN-SCH550-IGBT-MOD-6911 / Kit KIT-SCH550-PLC-11 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-69-1.11: PN-SCH550-INJ-MET-6912 / Kit KIT-SCH550-LUB-12 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-69-1.12: PN-SCH550-GASK-VIT-6913 / Kit KIT-SCH550-VAC-13 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-69-1.13: PN-SCH550-HEX-CORE-6914 / Kit KIT-SCH550-THM-14 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-69-1.14: PN-SCH550-BRG-CER-6901 / Kit KIT-SCH550-SPN-1 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-69-1.15: PN-SCH550-CAM-GRIP-6902 / Kit KIT-SCH550-ATC-2 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH550-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P550 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P550. Verify part numbers before closing Work Orders.

BOM-69-2.1: PN-SCH550-VLV-PROP-6903 / Kit KIT-SCH550-HYD-3 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-69-2.2: PN-SCH550-LIN-COIL-6905 / Kit KIT-SCH550-XAX-5 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-69-2.3: PN-SCH550-CYL-SEAL-6907 / Kit KIT-SCH550-ZAX-7 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-69-2.4: PN-SCH550-BRK-DISC-6909 / Kit KIT-SCH550-CAX-9 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-69-2.5: PN-SCH550-IGBT-MOD-6911 / Kit KIT-SCH550-PLC-11 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-69-2.6: PN-SCH550-GASK-VIT-6913 / Kit KIT-SCH550-VAC-13 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-69-2.7: PN-SCH550-BRG-CER-6901 / Kit KIT-SCH550-SPN-1 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-69-2.8: PN-SCH550-VLV-PROP-6903 / Kit KIT-SCH550-HYD-3 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-69-2.9: PN-SCH550-LIN-COIL-6905 / Kit KIT-SCH550-XAX-5 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-69-2.10: PN-SCH550-CYL-SEAL-6907 / Kit KIT-SCH550-ZAX-7 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-69-2.11: PN-SCH550-BRK-DISC-6909 / Kit KIT-SCH550-CAX-9 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-69-2.12: PN-SCH550-IGBT-MOD-6911 / Kit KIT-SCH550-PLC-11 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-69-2.13: PN-SCH550-GASK-VIT-6913 / Kit KIT-SCH550-VAC-13 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-69-2.14: PN-SCH550-BRG-CER-6901 / Kit KIT-SCH550-SPN-1 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-69-2.15: PN-SCH550-VLV-PROP-6903 / Kit KIT-SCH550-HYD-3 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH550-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P550 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P550. Verify part numbers before closing Work Orders.

BOM-69-3.1: PN-SCH550-PMP-PLG-6904 / Kit KIT-SCH550-CLN-4 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-69-3.2: PN-SCH550-CYL-SEAL-6907 / Kit KIT-SCH550-ZAX-7 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-69-3.3: PN-SCH550-GLASS-RD-6910 / Kit KIT-SCH550-OPT-10 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-69-3.4: PN-SCH550-GASK-VIT-6913 / Kit KIT-SCH550-VAC-13 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-69-3.5: PN-SCH550-CAM-GRIP-6902 / Kit KIT-SCH550-ATC-2 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-69-3.6: PN-SCH550-LIN-COIL-6905 / Kit KIT-SCH550-XAX-5 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-69-3.7: PN-SCH550-TRQ-STAT-6908 / Kit KIT-SCH550-BAX-8 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-69-3.8: PN-SCH550-IGBT-MOD-6911 / Kit KIT-SCH550-PLC-11 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-69-3.9: PN-SCH550-HEX-CORE-6914 / Kit KIT-SCH550-THM-14 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-69-3.10: PN-SCH550-VLV-PROP-6903 / Kit KIT-SCH550-HYD-3 (Schuler-HPD-P550) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-69-3.11: PN-SCH550-BSW-NUT-6906 / Kit KIT-SCH550-YAX-6 (Schuler-HPD-P550) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-69-3.12: PN-SCH550-BRK-DISC-6909 / Kit KIT-SCH550-CAX-9 (Schuler-HPD-P550) | PM Interval: 250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-69-3.13: PN-SCH550-INJ-MET-6912 / Kit KIT-SCH550-LUB-12 (Schuler-HPD-P550) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-69-3.14: PN-SCH550-BRG-CER-6901 / Kit KIT-SCH550-SPN-1 (Schuler-HPD-P550) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-69-3.15: PN-SCH550-PMP-PLG-6904 / Kit KIT-SCH550-CLN-4 (Schuler-HPD-P550) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH550-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P600 (GROUP 1)

Certified Aerospace Bill of Materials for Schuler-HPD-P600. Verify part numbers before closing Work Orders.

BOM-70-1.1: PN-SCH600-CAM-GRIP-7002 / Kit KIT-SCH600-ATC-2 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-70-1.2: PN-SCH600-VLV-PROP-7003 / Kit KIT-SCH600-HYD-3 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-70-1.3: PN-SCH600-PMP-PLG-7004 / Kit KIT-SCH600-CLN-4 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-70-1.4: PN-SCH600-LIN-COIL-7005 / Kit KIT-SCH600-XAX-5 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-70-1.5: PN-SCH600-BSW-NUT-7006 / Kit KIT-SCH600-YAX-6 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-70-1.6: PN-SCH600-CYL-SEAL-7007 / Kit KIT-SCH600-ZAX-7 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-70-1.7: PN-SCH600-TRQ-STAT-7008 / Kit KIT-SCH600-BAX-8 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-70-1.8: PN-SCH600-BRK-DISC-7009 / Kit KIT-SCH600-CAX-9 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-70-1.9: PN-SCH600-GLASS-RD-7010 / Kit KIT-SCH600-OPT-10 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-70-1.10: PN-SCH600-IGBT-MOD-7011 / Kit KIT-SCH600-PLC-11 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-70-1.11: PN-SCH600-INJ-MET-7012 / Kit KIT-SCH600-LUB-12 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-70-1.12: PN-SCH600-GASK-VIT-7013 / Kit KIT-SCH600-VAC-13 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-70-1.13: PN-SCH600-HEX-CORE-7014 / Kit KIT-SCH600-THM-14 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-70-1.14: PN-SCH600-BRG-CER-7001 / Kit KIT-SCH600-SPN-1 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-70-1.15: PN-SCH600-CAM-GRIP-7002 / Kit KIT-SCH600-ATC-2 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH600-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P600 (GROUP 2)

Certified Aerospace Bill of Materials for Schuler-HPD-P600. Verify part numbers before closing Work Orders.

BOM-70-2.1: PN-SCH600-VLV-PROP-7003 / Kit KIT-SCH600-HYD-3 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-70-2.2: PN-SCH600-LIN-COIL-7005 / Kit KIT-SCH600-XAX-5 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-70-2.3: PN-SCH600-CYL-SEAL-7007 / Kit KIT-SCH600-ZAX-7 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-70-2.4: PN-SCH600-BRK-DISC-7009 / Kit KIT-SCH600-CAX-9 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-70-2.5: PN-SCH600-IGBT-MOD-7011 / Kit KIT-SCH600-PLC-11 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-70-2.6: PN-SCH600-GASK-VIT-7013 / Kit KIT-SCH600-VAC-13 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-70-2.7: PN-SCH600-BRG-CER-7001 / Kit KIT-SCH600-SPN-1 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-70-2.8: PN-SCH600-VLV-PROP-7003 / Kit KIT-SCH600-HYD-3 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-70-2.9: PN-SCH600-LIN-COIL-7005 / Kit KIT-SCH600-XAX-5 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-70-2.10: PN-SCH600-CYL-SEAL-7007 / Kit KIT-SCH600-ZAX-7 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-70-2.11: PN-SCH600-BRK-DISC-7009 / Kit KIT-SCH600-CAX-9 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-70-2.12: PN-SCH600-IGBT-MOD-7011 / Kit KIT-SCH600-PLC-11 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-70-2.13: PN-SCH600-GASK-VIT-7013 / Kit KIT-SCH600-VAC-13 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-70-2.14: PN-SCH600-BRG-CER-7001 / Kit KIT-SCH600-SPN-1 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-70-2.15: PN-SCH600-VLV-PROP-7003 / Kit KIT-SCH600-HYD-3 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH600-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Schuler-HPD-P600 (GROUP 3)

Certified Aerospace Bill of Materials for Schuler-HPD-P600. Verify part numbers before closing Work Orders.

BOM-70-3.1: PN-SCH600-PMP-PLG-7004 / Kit KIT-SCH600-CLN-4 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-70-3.2: PN-SCH600-CYL-SEAL-7007 / Kit KIT-SCH600-ZAX-7 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SCH600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-70-3.3: PN-SCH600-GLASS-RD-7010 / Kit KIT-SCH600-OPT-10 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SCH600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-70-3.4: PN-SCH600-GASK-VIT-7013 / Kit KIT-SCH600-VAC-13 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SCH600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-70-3.5: PN-SCH600-CAM-GRIP-7002 / Kit KIT-SCH600-ATC-2 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SCH600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-70-3.6: PN-SCH600-LIN-COIL-7005 / Kit KIT-SCH600-XAX-5 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SCH600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-70-3.7: PN-SCH600-TRQ-STAT-7008 / Kit KIT-SCH600-BAX-8 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SCH600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-70-3.8: PN-SCH600-IGBT-MOD-7011 / Kit KIT-SCH600-PLC-11 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SCH600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-70-3.9: PN-SCH600-HEX-CORE-7014 / Kit KIT-SCH600-THM-14 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SCH600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-70-3.10: PN-SCH600-VLV-PROP-7003 / Kit KIT-SCH600-HYD-3 (Schuler-HPD-P600) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SCH600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-70-3.11: PN-SCH600-BSW-NUT-7006 / Kit KIT-SCH600-YAX-6 (Schuler-HPD-P600) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SCH600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-70-3.12: PN-SCH600-BRK-DISC-7009 / Kit KIT-SCH600-CAX-9 (Schuler-HPD-P600) | PM Interval: 250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SCH600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-70-3.13: PN-SCH600-INJ-MET-7012 / Kit KIT-SCH600-LUB-12 (Schuler-HPD-P600) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SCH600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-70-3.14: PN-SCH600-BRG-CER-7001 / Kit KIT-SCH600-SPN-1 (Schuler-HPD-P600) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SCH600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-70-3.15: PN-SCH600-PMP-PLG-7004 / Kit KIT-SCH600-CLN-4 (Schuler-HPD-P600) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SCH600-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G150 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G150. Verify part numbers before closing Work Orders.

BOM-71-1.1: PN-GRO150-CAM-GRIP-7102 / Kit KIT-GRO150-ATC-2 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-71-1.2: PN-GRO150-VLV-PROP-7103 / Kit KIT-GRO150-HYD-3 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-71-1.3: PN-GRO150-PMP-PLG-7104 / Kit KIT-GRO150-CLN-4 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-71-1.4: PN-GRO150-LIN-COIL-7105 / Kit KIT-GRO150-XAX-5 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-71-1.5: PN-GRO150-BSW-NUT-7106 / Kit KIT-GRO150-YAX-6 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-71-1.6: PN-GRO150-CYL-SEAL-7107 / Kit KIT-GRO150-ZAX-7 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-71-1.7: PN-GRO150-TRQ-STAT-7108 / Kit KIT-GRO150-BAX-8 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-71-1.8: PN-GRO150-BRK-DISC-7109 / Kit KIT-GRO150-CAX-9 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-71-1.9: PN-GRO150-GLASS-RD-7110 / Kit KIT-GRO150-OPT-10 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-71-1.10: PN-GRO150-IGBT-MOD-7111 / Kit KIT-GRO150-PLC-11 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-71-1.11: PN-GRO150-INJ-MET-7112 / Kit KIT-GRO150-LUB-12 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-71-1.12: PN-GRO150-GASK-VIT-7113 / Kit KIT-GRO150-VAC-13 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-71-1.13: PN-GRO150-HEX-CORE-7114 / Kit KIT-GRO150-THM-14 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-71-1.14: PN-GRO150-BRG-CER-7101 / Kit KIT-GRO150-SPN-1 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-71-1.15: PN-GRO150-CAM-GRIP-7102 / Kit KIT-GRO150-ATC-2 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO150-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G150 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G150. Verify part numbers before closing Work Orders.

BOM-71-2.1: PN-GRO150-VLV-PROP-7103 / Kit KIT-GRO150-HYD-3 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-71-2.2: PN-GRO150-LIN-COIL-7105 / Kit KIT-GRO150-XAX-5 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-71-2.3: PN-GRO150-CYL-SEAL-7107 / Kit KIT-GRO150-ZAX-7 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-71-2.4: PN-GRO150-BRK-DISC-7109 / Kit KIT-GRO150-CAX-9 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-71-2.5: PN-GRO150-IGBT-MOD-7111 / Kit KIT-GRO150-PLC-11 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-71-2.6: PN-GRO150-GASK-VIT-7113 / Kit KIT-GRO150-VAC-13 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-71-2.7: PN-GRO150-BRG-CER-7101 / Kit KIT-GRO150-SPN-1 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-71-2.8: PN-GRO150-VLV-PROP-7103 / Kit KIT-GRO150-HYD-3 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-71-2.9: PN-GRO150-LIN-COIL-7105 / Kit KIT-GRO150-XAX-5 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-71-2.10: PN-GRO150-CYL-SEAL-7107 / Kit KIT-GRO150-ZAX-7 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-71-2.11: PN-GRO150-BRK-DISC-7109 / Kit KIT-GRO150-CAX-9 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-71-2.12: PN-GRO150-IGBT-MOD-7111 / Kit KIT-GRO150-PLC-11 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-71-2.13: PN-GRO150-GASK-VIT-7113 / Kit KIT-GRO150-VAC-13 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-71-2.14: PN-GRO150-BRG-CER-7101 / Kit KIT-GRO150-SPN-1 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-71-2.15: PN-GRO150-VLV-PROP-7103 / Kit KIT-GRO150-HYD-3 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO150-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G150 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G150. Verify part numbers before closing Work Orders.

BOM-71-3.1: PN-GRO150-PMP-PLG-7104 / Kit KIT-GRO150-CLN-4 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-71-3.2: PN-GRO150-CYL-SEAL-7107 / Kit KIT-GRO150-ZAX-7 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-71-3.3: PN-GRO150-GLASS-RD-7110 / Kit KIT-GRO150-OPT-10 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-71-3.4: PN-GRO150-GASK-VIT-7113 / Kit KIT-GRO150-VAC-13 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-71-3.5: PN-GRO150-CAM-GRIP-7102 / Kit KIT-GRO150-ATC-2 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-71-3.6: PN-GRO150-LIN-COIL-7105 / Kit KIT-GRO150-XAX-5 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-71-3.7: PN-GRO150-TRQ-STAT-7108 / Kit KIT-GRO150-BAX-8 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-71-3.8: PN-GRO150-IGBT-MOD-7111 / Kit KIT-GRO150-PLC-11 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-71-3.9: PN-GRO150-HEX-CORE-7114 / Kit KIT-GRO150-THM-14 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-71-3.10: PN-GRO150-VLV-PROP-7103 / Kit KIT-GRO150-HYD-3 (Grob-Aerospace-G150) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-71-3.11: PN-GRO150-BSW-NUT-7106 / Kit KIT-GRO150-YAX-6 (Grob-Aerospace-G150) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-71-3.12: PN-GRO150-BRK-DISC-7109 / Kit KIT-GRO150-CAX-9 (Grob-Aerospace-G150) | PM Interval: 250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-71-3.13: PN-GRO150-INJ-MET-7112 / Kit KIT-GRO150-LUB-12 (Grob-Aerospace-G150) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-71-3.14: PN-GRO150-BRG-CER-7101 / Kit KIT-GRO150-SPN-1 (Grob-Aerospace-G150) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-71-3.15: PN-GRO150-PMP-PLG-7104 / Kit KIT-GRO150-CLN-4 (Grob-Aerospace-G150) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO150-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G200 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G200. Verify part numbers before closing Work Orders.

BOM-72-1.1: PN-GRO200-CAM-GRIP-7202 / Kit KIT-GRO200-ATC-2 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-72-1.2: PN-GRO200-VLV-PROP-7203 / Kit KIT-GRO200-HYD-3 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-72-1.3: PN-GRO200-PMP-PLG-7204 / Kit KIT-GRO200-CLN-4 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-72-1.4: PN-GRO200-LIN-COIL-7205 / Kit KIT-GRO200-XAX-5 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-72-1.5: PN-GRO200-BSW-NUT-7206 / Kit KIT-GRO200-YAX-6 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-72-1.6: PN-GRO200-CYL-SEAL-7207 / Kit KIT-GRO200-ZAX-7 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-72-1.7: PN-GRO200-TRQ-STAT-7208 / Kit KIT-GRO200-BAX-8 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-72-1.8: PN-GRO200-BRK-DISC-7209 / Kit KIT-GRO200-CAX-9 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-72-1.9: PN-GRO200-GLASS-RD-7210 / Kit KIT-GRO200-OPT-10 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-72-1.10: PN-GRO200-IGBT-MOD-7211 / Kit KIT-GRO200-PLC-11 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-72-1.11: PN-GRO200-INJ-MET-7212 / Kit KIT-GRO200-LUB-12 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-72-1.12: PN-GRO200-GASK-VIT-7213 / Kit KIT-GRO200-VAC-13 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-72-1.13: PN-GRO200-HEX-CORE-7214 / Kit KIT-GRO200-THM-14 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-72-1.14: PN-GRO200-BRG-CER-7201 / Kit KIT-GRO200-SPN-1 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-72-1.15: PN-GRO200-CAM-GRIP-7202 / Kit KIT-GRO200-ATC-2 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO200-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G200 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G200. Verify part numbers before closing Work Orders.

BOM-72-2.1: PN-GRO200-VLV-PROP-7203 / Kit KIT-GRO200-HYD-3 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-72-2.2: PN-GRO200-LIN-COIL-7205 / Kit KIT-GRO200-XAX-5 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-72-2.3: PN-GRO200-CYL-SEAL-7207 / Kit KIT-GRO200-ZAX-7 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-72-2.4: PN-GRO200-BRK-DISC-7209 / Kit KIT-GRO200-CAX-9 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-72-2.5: PN-GRO200-IGBT-MOD-7211 / Kit KIT-GRO200-PLC-11 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-72-2.6: PN-GRO200-GASK-VIT-7213 / Kit KIT-GRO200-VAC-13 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-72-2.7: PN-GRO200-BRG-CER-7201 / Kit KIT-GRO200-SPN-1 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-72-2.8: PN-GRO200-VLV-PROP-7203 / Kit KIT-GRO200-HYD-3 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-72-2.9: PN-GRO200-LIN-COIL-7205 / Kit KIT-GRO200-XAX-5 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-72-2.10: PN-GRO200-CYL-SEAL-7207 / Kit KIT-GRO200-ZAX-7 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-72-2.11: PN-GRO200-BRK-DISC-7209 / Kit KIT-GRO200-CAX-9 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-72-2.12: PN-GRO200-IGBT-MOD-7211 / Kit KIT-GRO200-PLC-11 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-72-2.13: PN-GRO200-GASK-VIT-7213 / Kit KIT-GRO200-VAC-13 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-72-2.14: PN-GRO200-BRG-CER-7201 / Kit KIT-GRO200-SPN-1 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-72-2.15: PN-GRO200-VLV-PROP-7203 / Kit KIT-GRO200-HYD-3 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO200-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G200 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G200. Verify part numbers before closing Work Orders.

BOM-72-3.1: PN-GRO200-PMP-PLG-7204 / Kit KIT-GRO200-CLN-4 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-72-3.2: PN-GRO200-CYL-SEAL-7207 / Kit KIT-GRO200-ZAX-7 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-72-3.3: PN-GRO200-GLASS-RD-7210 / Kit KIT-GRO200-OPT-10 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-72-3.4: PN-GRO200-GASK-VIT-7213 / Kit KIT-GRO200-VAC-13 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-72-3.5: PN-GRO200-CAM-GRIP-7202 / Kit KIT-GRO200-ATC-2 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-72-3.6: PN-GRO200-LIN-COIL-7205 / Kit KIT-GRO200-XAX-5 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-72-3.7: PN-GRO200-TRQ-STAT-7208 / Kit KIT-GRO200-BAX-8 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-72-3.8: PN-GRO200-IGBT-MOD-7211 / Kit KIT-GRO200-PLC-11 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-72-3.9: PN-GRO200-HEX-CORE-7214 / Kit KIT-GRO200-THM-14 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-72-3.10: PN-GRO200-VLV-PROP-7203 / Kit KIT-GRO200-HYD-3 (Grob-Aerospace-G200) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-72-3.11: PN-GRO200-BSW-NUT-7206 / Kit KIT-GRO200-YAX-6 (Grob-Aerospace-G200) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-72-3.12: PN-GRO200-BRK-DISC-7209 / Kit KIT-GRO200-CAX-9 (Grob-Aerospace-G200) | PM Interval: 250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-72-3.13: PN-GRO200-INJ-MET-7212 / Kit KIT-GRO200-LUB-12 (Grob-Aerospace-G200) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-72-3.14: PN-GRO200-BRG-CER-7201 / Kit KIT-GRO200-SPN-1 (Grob-Aerospace-G200) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-72-3.15: PN-GRO200-PMP-PLG-7204 / Kit KIT-GRO200-CLN-4 (Grob-Aerospace-G200) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO200-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G250 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G250. Verify part numbers before closing Work Orders.

BOM-73-1.1: PN-GRO250-CAM-GRIP-7302 / Kit KIT-GRO250-ATC-2 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-73-1.2: PN-GRO250-VLV-PROP-7303 / Kit KIT-GRO250-HYD-3 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-73-1.3: PN-GRO250-PMP-PLG-7304 / Kit KIT-GRO250-CLN-4 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-73-1.4: PN-GRO250-LIN-COIL-7305 / Kit KIT-GRO250-XAX-5 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-73-1.5: PN-GRO250-BSW-NUT-7306 / Kit KIT-GRO250-YAX-6 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-73-1.6: PN-GRO250-CYL-SEAL-7307 / Kit KIT-GRO250-ZAX-7 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-73-1.7: PN-GRO250-TRQ-STAT-7308 / Kit KIT-GRO250-BAX-8 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-73-1.8: PN-GRO250-BRK-DISC-7309 / Kit KIT-GRO250-CAX-9 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-73-1.9: PN-GRO250-GLASS-RD-7310 / Kit KIT-GRO250-OPT-10 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-73-1.10: PN-GRO250-IGBT-MOD-7311 / Kit KIT-GRO250-PLC-11 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-73-1.11: PN-GRO250-INJ-MET-7312 / Kit KIT-GRO250-LUB-12 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-73-1.12: PN-GRO250-GASK-VIT-7313 / Kit KIT-GRO250-VAC-13 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-73-1.13: PN-GRO250-HEX-CORE-7314 / Kit KIT-GRO250-THM-14 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-73-1.14: PN-GRO250-BRG-CER-7301 / Kit KIT-GRO250-SPN-1 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-73-1.15: PN-GRO250-CAM-GRIP-7302 / Kit KIT-GRO250-ATC-2 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO250-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G250 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G250. Verify part numbers before closing Work Orders.

BOM-73-2.1: PN-GRO250-VLV-PROP-7303 / Kit KIT-GRO250-HYD-3 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-73-2.2: PN-GRO250-LIN-COIL-7305 / Kit KIT-GRO250-XAX-5 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-73-2.3: PN-GRO250-CYL-SEAL-7307 / Kit KIT-GRO250-ZAX-7 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-73-2.4: PN-GRO250-BRK-DISC-7309 / Kit KIT-GRO250-CAX-9 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-73-2.5: PN-GRO250-IGBT-MOD-7311 / Kit KIT-GRO250-PLC-11 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-73-2.6: PN-GRO250-GASK-VIT-7313 / Kit KIT-GRO250-VAC-13 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-73-2.7: PN-GRO250-BRG-CER-7301 / Kit KIT-GRO250-SPN-1 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-73-2.8: PN-GRO250-VLV-PROP-7303 / Kit KIT-GRO250-HYD-3 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-73-2.9: PN-GRO250-LIN-COIL-7305 / Kit KIT-GRO250-XAX-5 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-73-2.10: PN-GRO250-CYL-SEAL-7307 / Kit KIT-GRO250-ZAX-7 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-73-2.11: PN-GRO250-BRK-DISC-7309 / Kit KIT-GRO250-CAX-9 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-73-2.12: PN-GRO250-IGBT-MOD-7311 / Kit KIT-GRO250-PLC-11 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-73-2.13: PN-GRO250-GASK-VIT-7313 / Kit KIT-GRO250-VAC-13 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-73-2.14: PN-GRO250-BRG-CER-7301 / Kit KIT-GRO250-SPN-1 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-73-2.15: PN-GRO250-VLV-PROP-7303 / Kit KIT-GRO250-HYD-3 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO250-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G250 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G250. Verify part numbers before closing Work Orders.

BOM-73-3.1: PN-GRO250-PMP-PLG-7304 / Kit KIT-GRO250-CLN-4 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-73-3.2: PN-GRO250-CYL-SEAL-7307 / Kit KIT-GRO250-ZAX-7 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-73-3.3: PN-GRO250-GLASS-RD-7310 / Kit KIT-GRO250-OPT-10 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-73-3.4: PN-GRO250-GASK-VIT-7313 / Kit KIT-GRO250-VAC-13 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-73-3.5: PN-GRO250-CAM-GRIP-7302 / Kit KIT-GRO250-ATC-2 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-73-3.6: PN-GRO250-LIN-COIL-7305 / Kit KIT-GRO250-XAX-5 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-73-3.7: PN-GRO250-TRQ-STAT-7308 / Kit KIT-GRO250-BAX-8 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-73-3.8: PN-GRO250-IGBT-MOD-7311 / Kit KIT-GRO250-PLC-11 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-73-3.9: PN-GRO250-HEX-CORE-7314 / Kit KIT-GRO250-THM-14 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-73-3.10: PN-GRO250-VLV-PROP-7303 / Kit KIT-GRO250-HYD-3 (Grob-Aerospace-G250) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-73-3.11: PN-GRO250-BSW-NUT-7306 / Kit KIT-GRO250-YAX-6 (Grob-Aerospace-G250) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-73-3.12: PN-GRO250-BRK-DISC-7309 / Kit KIT-GRO250-CAX-9 (Grob-Aerospace-G250) | PM Interval: 250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-73-3.13: PN-GRO250-INJ-MET-7312 / Kit KIT-GRO250-LUB-12 (Grob-Aerospace-G250) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-73-3.14: PN-GRO250-BRG-CER-7301 / Kit KIT-GRO250-SPN-1 (Grob-Aerospace-G250) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-73-3.15: PN-GRO250-PMP-PLG-7304 / Kit KIT-GRO250-CLN-4 (Grob-Aerospace-G250) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO250-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G300 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G300. Verify part numbers before closing Work Orders.

BOM-74-1.1: PN-GRO300-CAM-GRIP-7402 / Kit KIT-GRO300-ATC-2 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-74-1.2: PN-GRO300-VLV-PROP-7403 / Kit KIT-GRO300-HYD-3 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-74-1.3: PN-GRO300-PMP-PLG-7404 / Kit KIT-GRO300-CLN-4 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-74-1.4: PN-GRO300-LIN-COIL-7405 / Kit KIT-GRO300-XAX-5 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-74-1.5: PN-GRO300-BSW-NUT-7406 / Kit KIT-GRO300-YAX-6 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-74-1.6: PN-GRO300-CYL-SEAL-7407 / Kit KIT-GRO300-ZAX-7 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-74-1.7: PN-GRO300-TRQ-STAT-7408 / Kit KIT-GRO300-BAX-8 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-74-1.8: PN-GRO300-BRK-DISC-7409 / Kit KIT-GRO300-CAX-9 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-74-1.9: PN-GRO300-GLASS-RD-7410 / Kit KIT-GRO300-OPT-10 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-74-1.10: PN-GRO300-IGBT-MOD-7411 / Kit KIT-GRO300-PLC-11 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-74-1.11: PN-GRO300-INJ-MET-7412 / Kit KIT-GRO300-LUB-12 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-74-1.12: PN-GRO300-GASK-VIT-7413 / Kit KIT-GRO300-VAC-13 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-74-1.13: PN-GRO300-HEX-CORE-7414 / Kit KIT-GRO300-THM-14 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-74-1.14: PN-GRO300-BRG-CER-7401 / Kit KIT-GRO300-SPN-1 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-74-1.15: PN-GRO300-CAM-GRIP-7402 / Kit KIT-GRO300-ATC-2 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO300-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G300 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G300. Verify part numbers before closing Work Orders.

BOM-74-2.1: PN-GRO300-VLV-PROP-7403 / Kit KIT-GRO300-HYD-3 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-74-2.2: PN-GRO300-LIN-COIL-7405 / Kit KIT-GRO300-XAX-5 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-74-2.3: PN-GRO300-CYL-SEAL-7407 / Kit KIT-GRO300-ZAX-7 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-74-2.4: PN-GRO300-BRK-DISC-7409 / Kit KIT-GRO300-CAX-9 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-74-2.5: PN-GRO300-IGBT-MOD-7411 / Kit KIT-GRO300-PLC-11 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-74-2.6: PN-GRO300-GASK-VIT-7413 / Kit KIT-GRO300-VAC-13 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-74-2.7: PN-GRO300-BRG-CER-7401 / Kit KIT-GRO300-SPN-1 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-74-2.8: PN-GRO300-VLV-PROP-7403 / Kit KIT-GRO300-HYD-3 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-74-2.9: PN-GRO300-LIN-COIL-7405 / Kit KIT-GRO300-XAX-5 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-74-2.10: PN-GRO300-CYL-SEAL-7407 / Kit KIT-GRO300-ZAX-7 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-74-2.11: PN-GRO300-BRK-DISC-7409 / Kit KIT-GRO300-CAX-9 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-74-2.12: PN-GRO300-IGBT-MOD-7411 / Kit KIT-GRO300-PLC-11 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-74-2.13: PN-GRO300-GASK-VIT-7413 / Kit KIT-GRO300-VAC-13 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-74-2.14: PN-GRO300-BRG-CER-7401 / Kit KIT-GRO300-SPN-1 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-74-2.15: PN-GRO300-VLV-PROP-7403 / Kit KIT-GRO300-HYD-3 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO300-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G300 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G300. Verify part numbers before closing Work Orders.

BOM-74-3.1: PN-GRO300-PMP-PLG-7404 / Kit KIT-GRO300-CLN-4 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-74-3.2: PN-GRO300-CYL-SEAL-7407 / Kit KIT-GRO300-ZAX-7 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-74-3.3: PN-GRO300-GLASS-RD-7410 / Kit KIT-GRO300-OPT-10 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-74-3.4: PN-GRO300-GASK-VIT-7413 / Kit KIT-GRO300-VAC-13 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-74-3.5: PN-GRO300-CAM-GRIP-7402 / Kit KIT-GRO300-ATC-2 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-74-3.6: PN-GRO300-LIN-COIL-7405 / Kit KIT-GRO300-XAX-5 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-74-3.7: PN-GRO300-TRQ-STAT-7408 / Kit KIT-GRO300-BAX-8 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-74-3.8: PN-GRO300-IGBT-MOD-7411 / Kit KIT-GRO300-PLC-11 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-74-3.9: PN-GRO300-HEX-CORE-7414 / Kit KIT-GRO300-THM-14 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-74-3.10: PN-GRO300-VLV-PROP-7403 / Kit KIT-GRO300-HYD-3 (Grob-Aerospace-G300) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-74-3.11: PN-GRO300-BSW-NUT-7406 / Kit KIT-GRO300-YAX-6 (Grob-Aerospace-G300) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-74-3.12: PN-GRO300-BRK-DISC-7409 / Kit KIT-GRO300-CAX-9 (Grob-Aerospace-G300) | PM Interval: 250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-74-3.13: PN-GRO300-INJ-MET-7412 / Kit KIT-GRO300-LUB-12 (Grob-Aerospace-G300) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-74-3.14: PN-GRO300-BRG-CER-7401 / Kit KIT-GRO300-SPN-1 (Grob-Aerospace-G300) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-74-3.15: PN-GRO300-PMP-PLG-7404 / Kit KIT-GRO300-CLN-4 (Grob-Aerospace-G300) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO300-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G350 (GROUP 1)

Certified Aerospace Bill of Materials for Grob-Aerospace-G350. Verify part numbers before closing Work Orders.

BOM-75-1.1: PN-GRO350-CAM-GRIP-7502 / Kit KIT-GRO350-ATC-2 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-75-1.2: PN-GRO350-VLV-PROP-7503 / Kit KIT-GRO350-HYD-3 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-75-1.3: PN-GRO350-PMP-PLG-7504 / Kit KIT-GRO350-CLN-4 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-75-1.4: PN-GRO350-LIN-COIL-7505 / Kit KIT-GRO350-XAX-5 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-75-1.5: PN-GRO350-BSW-NUT-7506 / Kit KIT-GRO350-YAX-6 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-75-1.6: PN-GRO350-CYL-SEAL-7507 / Kit KIT-GRO350-ZAX-7 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-75-1.7: PN-GRO350-TRQ-STAT-7508 / Kit KIT-GRO350-BAX-8 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-75-1.8: PN-GRO350-BRK-DISC-7509 / Kit KIT-GRO350-CAX-9 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-75-1.9: PN-GRO350-GLASS-RD-7510 / Kit KIT-GRO350-OPT-10 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-75-1.10: PN-GRO350-IGBT-MOD-7511 / Kit KIT-GRO350-PLC-11 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-75-1.11: PN-GRO350-INJ-MET-7512 / Kit KIT-GRO350-LUB-12 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-75-1.12: PN-GRO350-GASK-VIT-7513 / Kit KIT-GRO350-VAC-13 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-75-1.13: PN-GRO350-HEX-CORE-7514 / Kit KIT-GRO350-THM-14 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-75-1.14: PN-GRO350-BRG-CER-7501 / Kit KIT-GRO350-SPN-1 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-75-1.15: PN-GRO350-CAM-GRIP-7502 / Kit KIT-GRO350-ATC-2 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO350-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G350 (GROUP 2)

Certified Aerospace Bill of Materials for Grob-Aerospace-G350. Verify part numbers before closing Work Orders.

BOM-75-2.1: PN-GRO350-VLV-PROP-7503 / Kit KIT-GRO350-HYD-3 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-75-2.2: PN-GRO350-LIN-COIL-7505 / Kit KIT-GRO350-XAX-5 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-75-2.3: PN-GRO350-CYL-SEAL-7507 / Kit KIT-GRO350-ZAX-7 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-75-2.4: PN-GRO350-BRK-DISC-7509 / Kit KIT-GRO350-CAX-9 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-75-2.5: PN-GRO350-IGBT-MOD-7511 / Kit KIT-GRO350-PLC-11 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-75-2.6: PN-GRO350-GASK-VIT-7513 / Kit KIT-GRO350-VAC-13 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-75-2.7: PN-GRO350-BRG-CER-7501 / Kit KIT-GRO350-SPN-1 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-75-2.8: PN-GRO350-VLV-PROP-7503 / Kit KIT-GRO350-HYD-3 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-75-2.9: PN-GRO350-LIN-COIL-7505 / Kit KIT-GRO350-XAX-5 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-75-2.10: PN-GRO350-CYL-SEAL-7507 / Kit KIT-GRO350-ZAX-7 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-75-2.11: PN-GRO350-BRK-DISC-7509 / Kit KIT-GRO350-CAX-9 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-75-2.12: PN-GRO350-IGBT-MOD-7511 / Kit KIT-GRO350-PLC-11 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-75-2.13: PN-GRO350-GASK-VIT-7513 / Kit KIT-GRO350-VAC-13 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-75-2.14: PN-GRO350-BRG-CER-7501 / Kit KIT-GRO350-SPN-1 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-75-2.15: PN-GRO350-VLV-PROP-7503 / Kit KIT-GRO350-HYD-3 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO350-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Grob-Aerospace-G350 (GROUP 3)

Certified Aerospace Bill of Materials for Grob-Aerospace-G350. Verify part numbers before closing Work Orders.

BOM-75-3.1: PN-GRO350-PMP-PLG-7504 / Kit KIT-GRO350-CLN-4 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-75-3.2: PN-GRO350-CYL-SEAL-7507 / Kit KIT-GRO350-ZAX-7 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-GRO350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-75-3.3: PN-GRO350-GLASS-RD-7510 / Kit KIT-GRO350-OPT-10 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-GRO350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-75-3.4: PN-GRO350-GASK-VIT-7513 / Kit KIT-GRO350-VAC-13 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-GRO350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-75-3.5: PN-GRO350-CAM-GRIP-7502 / Kit KIT-GRO350-ATC-2 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-GRO350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-75-3.6: PN-GRO350-LIN-COIL-7505 / Kit KIT-GRO350-XAX-5 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-GRO350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-75-3.7: PN-GRO350-TRQ-STAT-7508 / Kit KIT-GRO350-BAX-8 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-GRO350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-75-3.8: PN-GRO350-IGBT-MOD-7511 / Kit KIT-GRO350-PLC-11 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-GRO350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-75-3.9: PN-GRO350-HEX-CORE-7514 / Kit KIT-GRO350-THM-14 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-GRO350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-75-3.10: PN-GRO350-VLV-PROP-7503 / Kit KIT-GRO350-HYD-3 (Grob-Aerospace-G350) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-GRO350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-75-3.11: PN-GRO350-BSW-NUT-7506 / Kit KIT-GRO350-YAX-6 (Grob-Aerospace-G350) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-GRO350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-75-3.12: PN-GRO350-BRK-DISC-7509 / Kit KIT-GRO350-CAX-9 (Grob-Aerospace-G350) | PM Interval: 250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-GRO350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-75-3.13: PN-GRO350-INJ-MET-7512 / Kit KIT-GRO350-LUB-12 (Grob-Aerospace-G350) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-GRO350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-75-3.14: PN-GRO350-BRG-CER-7501 / Kit KIT-GRO350-SPN-1 (Grob-Aerospace-G350) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-GRO350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-75-3.15: PN-GRO350-PMP-PLG-7504 / Kit KIT-GRO350-CLN-4 (Grob-Aerospace-G350) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-GRO350-CLN-4 within tolerance ±0.0030 mm TIR.